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Q1 Feb/Mar 2025 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Solve the equation

$$\ln(1 - e^{-2x}) + 3 = 0.$$

Give your final answer correct to 4 decimal places. [3]

Examiner comment

> "A significant number of candidates struggled with this question, demonstrating a lack of knowledge of the rules of logarithms and how to apply them when solving equations."

Mark scheme

State that $1 - e^{-2x} = e^{-3}$ **B1** OE, with $\ln$ removed.

Use correct method to solve an equation of the form $e^{\pm 2x} = a$, where $a > 0$, and a reasonable attempt at the B1, for $\pm 2x \ln e$ or $\pm x \ln e$ **M1** E.g. $e^{-2x} = 1 - e^{-3}$, $-2x = \ln(1 - e^{-3})$ OE. Can be numerical $\ln(1 - 0.049787) = \ln 0.9502$. Evidence of method must be seen.

Obtain answer **0.0255 A1** CAO. Must be 4 decimal places. No working seen scores 0.

Alternative Method for Question 1

State that $1 - e^{-2x} = e^{-3}$ **B1** OE, without $\ln$.

Rearrange to obtain an expression for e^x and solve an equation of the form $e^{\pm x} = a$, where $a > 0$, and a reasonable attempt at the B1, for x **M1** e.g. $e^x = \left(\frac{1}{1-e^{-3}}\right)^{1/2}$, $x = \ln\left(\frac{1}{\sqrt{1-e^{-3}}}\right)$. Can be numerical. Evidence of method must be seen.

Obtain answer **0.0255 A1** CAO. Must be 4 decimal places. No working seen scores 0.

Total: 3

Q2 Feb/Mar 2025 | Paper 3 (variant 32) | Question 2

Topics: Differentiation

The equation of a curve is

$$xy^2 + \ln(x + 2y) = 1.$$

Find the gradient of the curve at the point where $x = 0$. [5]

Examiner comment

> "However, many found it challenging to differentiate In(x + 2y) correctly and therefore, unless they found the correct value for y at x = 0, gained no further marks."

> "A few candidates were able to gain marks following this rearrangement."

Mark scheme

State or imply $\frac{1+2\frac{dy}{dx}}{x+2y}$ as the derivative of $\ln(x+2y)$ **B1**

State derivative of xy^2 is $x \cdot 2y\frac{dy}{dx} + y^2$ **B1**

Obtain $y^2 + 2xy\frac{dy}{dx} + \frac{1+2\frac{dy}{dx}}{x+2y} = 0$ **B1** OE. May be implied by correct final answer.

Obtain $y = \frac{1}{2}e$ when $x = 0$ **B1** OE. Allow $\frac{1}{2} \times 2.718$ or 1.36 or better e.g. 1.359... May be implied by correct final answer.

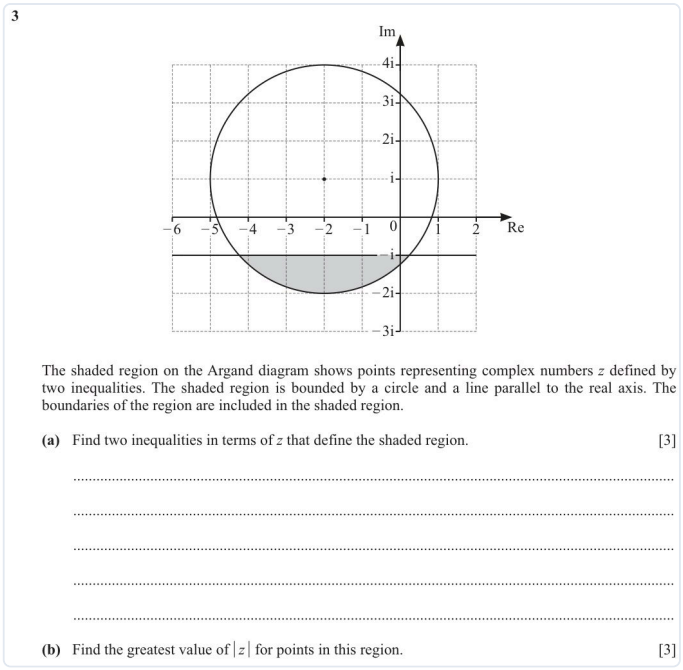
Obtain $\frac{dy}{dx} = -\frac{1}{8}(3e^3 + 4)$ **B1** OE. Accept AWRT -3.01. ISW.

Total: 5

Q3 Feb/Mar 2025 | Paper 3 (variant 32) | Question 3

Topics: Complex Numbers

3



The shaded region on the Argand diagram shows points representing complex numbers z defined by two inequalities. The shaded region is bounded by a circle and a line parallel to the real axis. The boundaries of the region are included in the shaded region.

(a) Find two inequalities in terms of z that define the shaded region. [3]

(b) Find the greatest value of $|z|$ for points in this region. [3]

Examiner comment

> "Question 3 (a) This proved to be a challenging question and relatively few candidates gained full marks."

> "A common mistake was to use Im(z) ≤ i."

> "(b) This part also proved challenging for candidates, with many not attempting a solution at all and only very few

> gaining full marks."

Mark scheme

(a)

Obtain $\mathrm{Im}(z) \leq -1$ **B1** Condone strict inequalities throughout (a).

Obtain answer of the form $|z - a| \leq b$ **M1** Accept equation or any inequality sign. $a = \pm 2 \pm i$ and $b = 3$, e.g. $|z + 2 - i| = 3$ or $|z + 2 - i| < 3$.

Obtain answer $|z + 2 - i| \leq 3$ **A1** Accept $|z - (-2 + i)| \leq 3$ as final answer. Do not ISW.

Total: 3

(b)

Identify the coordinates of correct point $(-2 - \sqrt{5}, -1)$ **M1*** From solving $(x \pm 2)^2 + (y \pm 1)^2 = 3^2$ with $y = -1$, or attempt to get $2 + \sqrt{5}$ using a right-angled triangle.

Carry out a correct method for finding the greatest value of $|z|$ **DM1**

Obtain answer **4.35** or $\sqrt{10 + 4\sqrt{5}}$ **A1** AWRT 4.35, e.g. 4.3525...

Total: 3

Q4 Feb/Mar 2025 | Paper 3 (variant 32) | Question 9

Topics: Algebra, Differentiation

The polynomial $6x^3 + ax^2 + bx + 9$ is denoted by $p(x)$, where a and b are constants. It is given that $(x - 3)$ is a factor of $p(x)$, and when the first derivative $p'(x)$ is divided by $(x - 3)$ the remainder is 72.

(a) Find the values of a and b. [5]

(b) When a and b have the values found in part (a), factorise $p(x)$ completely. [3]

(c) Hence solve the inequality $p(x) < 0$. [2]

Examiner comment

> "(c) Very few candidates were fully successful in this part, perhaps because it relied on correct answers in both parts (a) and (b)."

Mark scheme

(a)

Substitute $x = 3$ or -3 into $p(x)$ and equate to 0 or into $p'(x)$ and equate to 72 **M1***

Obtain $162 + 9a + 3b + 9 = 0$ **A1** OE

Obtain $162 + 6a + b = 72$ **A1** OE

Solve simultaneous equations to obtain either a or b after using $p(\pm 3) = 0$ and $p'(\pm 3) = 72$ **DM1**

Obtain $a = -11$ and $b = -24$ **A1**

Total: 5

(b)

Equate $(x-3)(6x^2+Ax+B)$ to $6x^3-11x^2-24x+9$ and obtain equations to solve for A and B, or divide $6x^3-11x^2-24x+9$ by $x-3$ and reach $6x^2 \pm 7x$ **M1** Using their a and their b.

$(x-3)(6x^2+7x-3)$ **A1** SOI

Obtain $(x-3)(2x+3)(3x-1)$ **A1**

Special Case: If only $(x-3)(2x+3)(3x-1)$ seen, SC B1 only, but can gain two marks in (c).

Total: 3

(c)

Obtain one correct region $x < -\frac{3}{2}$ or $\frac{1}{3} < x < 3$ **B1 FT** Must be final answer not in working. FT is on the last two brackets, not $(x-3)$.

Obtain both regions $x < -\frac{3}{2}$, $\frac{1}{3} < x < 3$ **B1 FT** If incorrect factor or factors in (b) but correct regions here, allow SC B1 only.

Total: 2

Q5 Feb/Mar 2023 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

It is given that $x = \ln(2y-3) - \ln(y+4)$.

Express y in terms of x. [3]

Examiner comment

> "Question 1 Most candidates scored full marks, however a few believed ln(a ± b) could be replaced by lna ± lnb and hence made no progress."

Mark scheme

Use law of the logarithm of a quotient or express x as $\ln e^x$ **M1** $x = \ln\left(\frac{2y-3}{y+4}\right)$ or $\ln e^x = \ln(2y-3) - \ln(y+4)$.

Remove logarithms and obtain a correct equation e.g. $e^x = \frac{2y-3}{y+4}$ **A1**

Obtain answer $y = \frac{3+4e^x}{2-e^x}$ **A1** OE ISW

Total: 3

Q6 Feb/Mar 2023 | Paper 3 (variant 32) | Question 3

Topics: Algebra

The polynomial $2x^4 + ax^3 + bx - 1$, where a and b are constants, is denoted by $p(x)$. When $p(x)$ is divided by $x^2 - x + 1$ the remainder is $3x + 2$.

Find the values of a and b. [5]

Examiner comment

> "Candidates found it challenging to deal with the unknown constants in the polynomial when attempting long division, hence they rarely had a linear expression to equate to the given remainder."

> "This is something which candidates found difficult, as seen by their work in Question 4."

Mark scheme

Commence division and reach partial quotient $2x^2 + (a \pm 2)x$ **M1** Need $2x^2 + (a \pm 2)x$.

Obtain correct quotient $2x^2 + (a + 2)x + a$ **A1** Allow sign error e.g. in $b - 2$.

Set their linear remainder equal to part of $3x + 2$ and solve for a or for b **M1** Remainder $= 3x + 2 = (b - 2)x - 1 - a$. Allow for just equating x term or constant term.

Obtain answer $a = -3$ **A1**

Obtain answer $b = 5$ **A1**

Alternative method for Question 3

State $2x^4 + ax^3 + 0x^2 + bx - 1 = (x^2 - x + 1)(2x^2 + Ax + B) + 3x + 2$ and form and solve equation(s) to obtain A or B **M1** e.g. $0 = B - A + 2$ and $-1 = B + 2$.

Obtain $A = -1$, $B = -3$ **A1**

Form and solve equations for a or for b **M1** e.g. $a = A - 2$ or $b = -B + A + 3$.

Obtain answer $a = -3$ **A1**

Obtain answer $b = 5$ **A1**

Alternative method for Question 3

Use remainder theorem with $x = \frac{1 \pm \sqrt{3}i}{2}$ **M1** Allow correct use of a reasonable attempt at either root in exact or decimal form in the remainder theorem.

Obtain an exact equation for either root **A1**

Solve simultaneous equations, or single equation, for a or for b **M1**

Obtain answer $a = -3$ from exact working **A1**

Obtain answer $b = 5$ from exact working **A1**

Total: 5

Q7 Feb/Mar 2023 | Paper 3 (variant 32) | Question 4

Topics: Complex Numbers

Solve the equation

$$\frac{5z}{1+2i} - zz^* + 30 + 10i = 0,$$

giving your answers in the form $x + iy$, where x and y are real. [5]

Examiner comment

> "Question 4 Unfortunately, this was the third question in a row which candidates found challenging."

> "Few candidates reached a correct simplified equation."

Mark scheme

Substitute $z = x + iy$ and $z^* = x - iy$ to obtain a correct equation, horizontal or with $(1-2i)/(1-2i)$ seen, in x and y **B1**

$$5(x+iy) - (x+iy)(x-iy)(1+2i) + (30+10i)(1+2i) = 0$$

or

$$\frac{5(x+iy)(1-2i)}{(1+2i)(1-2i)} - (x+iy)(x-iy) + (30+10i) = 0$$

Use $i^2 = -1$ at least once and equate real and imaginary parts to zero *M1 OE for their horizontal equation.

Obtain two correct equations e.g. $x + 2y - x^2 - y^2 + 30 = 0$ and $-2x + y + 10 = 0$ **A1**

Solve quadratic equation for x or for y **DM1** $x^2 - 9x + 18 = (x-3)(x-6) = 0$ or $y^2 + 2y - 8 = (y+4)(y-2) = 0$. DM0 if x or y imaginary.

Obtain answers $3 - 4i$ and $6 + 2i$ **A1**

Total: 5

Q8 Feb/Mar 2023 | Paper 3 (variant 32) | Question 9

Topics: Differential Equations

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = e^{3y} \sin^2 2x.$$

It is given that $y = 0$ when $x = 0$.

Solve the differential equation and find the value of y when $x = \frac{1}{2}$. [7]

Examiner comment

> "Question 9 This question proved extremely difficult for most."

Mark scheme

Separate variables correctly and obtain e^{-3y} and $\sin^2 2x$ on the opposite sides **B1**

Obtain term $-\frac{1}{3}e^{-3y}$ **B1**

Use correct double angle formula for $\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$ **M1**

Obtain terms $\frac{1}{2}x - \frac{1}{8}\sin 4x$ OE **A1**

Use $x = 0$, $y = 0$ to evaluate a constant or as limits in a solution containing terms of the form ax and $b\sin 4x$ and $ce^{\pm 3y}$ **M1**

Obtain correct answer in any form e.g. $-\frac{1}{3}e^{-3y} = \frac{1}{2}x - \frac{1}{8}\sin 4x - \frac{1}{3}$ **A1**

Substitute $x = \frac{1}{2}$ and obtain $y = 0.175$ or $-\frac{1}{3}\ln\left(1 - \frac{3}{4} + \frac{3}{8}\sin 2\right)$ **A1** OE ISW

Total: 7

Q9 Feb/Mar 2023 | Paper 3 (variant 32) | Question 10

Topics: none of the listed topics

With respect to the origin O, the points A, B, C and D have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}, \quad \overrightarrow{OD} = \begin{pmatrix} 5 \\ -6 \\ 11 \end{pmatrix}.$$

(a) Find the obtuse angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [3]

The line l passes through the points A and B.

(b) Find a vector equation for the line l. [2]

(c) Find the position vector of the point of intersection of the line l and the line passing through C and D. [4]

Examiner comment

> "However, the most common error was the incorrect notation on the left-hand side of the equation."

> "not left blank, expressed in words or just denoted as l."

Mark scheme

(a)

Carry out correct process for evaluating the scalar product of $\overrightarrow{OA}$ and $\overrightarrow{OB}$ **M1**

$\pm(3, -1, 2) \cdot (1, 2, -3) = \pm(3 - 2 - 6) = [-5]$.

Using the correct process for the moduli, divide the scalar product by the product of the moduli and obtain $\cos^{-1}\left\{\frac{\pm(3-2-6)}{\sqrt{3^2+(-1)^2+2^2}\sqrt{1^2+2^2+(-3)^2}}\right\}$ **A1**

Obtain answer $110.9°$ or 1.94^c **A1**

Total: 3

(b)

Use a correct method to form an equation for line through AB **M1**

Obtain $\mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + \mu_1(2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$ **A1** OE e.g. $\mathbf{r} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k} + \mu_2(-2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k})$. Need $\mathbf{r}$ or (x, y, z).

Total: 2

(c)

Obtain a correct equation for line through CD, e.g. $\mathbf{r} = \mathbf{i} - 2\mathbf{j} + 5\mathbf{k} + \lambda_1(-4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k})$ **B1** OE e.g. $\mathbf{r} = 5\mathbf{i} - 6\mathbf{j} + 11\mathbf{k} + \lambda_2(-4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k})$. $\mathbf{r}$ can be omitted or another symbol used.

Equate two pairs of components of general points on their l and their CD and solve for λ or for μ **M1**

Obtain e.g. $\lambda_1 = -2$ or $\mu_1 = 3$ or $\lambda_2 = -1$ or $\mu_2 = -4$ **A1**

Obtain position vector $9\mathbf{i} - 10\mathbf{j} + 17\mathbf{k}$ **A1** Condone $(9, -10, 17)$ but not $(9\mathbf{i}, -10\mathbf{j}, 17\mathbf{k})$.

Total: 4

Q10 Feb/Mar 2023 | Paper 3 (variant 32) | Question 11

Topics: Algebra, Integration

Let $f(x) = \frac{5x^2+x+11}{(4+x^2)(1+x)}$.

(a) Express $f(x)$ in partial fractions. [5]

(b) Hence show that $\int_0^2 f(x)\,dx = \ln 54 - \frac{1}{8}\pi$. [5]

Examiner comment

> "A Bx+C (b) Virtually all candidates established the correct integral for , however the integral of 1+x 4+x2 Bx C proved challenging for most."

Mark scheme

(a)

State or imply the form $\frac{Ax+B}{4+x^2} + \frac{C}{1+x}$ **B1**

Use a correct method for finding a coefficient **M1** $(Ax + B)(1 + x) + C(4 + x^2) = 5x^2 + x + 11$.

Obtain one of $A = 2$, $B = -1$ and $C = 3$ **A1** If error present in above still allow A1 for C.

Obtain a second value **A1**

Obtain the third value **A1** If $A = 0$ then max M1 A1 for C.

Total: 5

(b)

Integrate and obtain terms $\frac{A}{2}\ln(4+x^2)$, $\frac{B}{2}\tan^{-1}\left(\frac{x}{2}\right)$ and $C\ln(1+x)$ **B1FT + B1FT + B1FT** The FT is on A, B and C. Integral of $\frac{Ax+B}{4+x^2}$ is $\frac{B}{2}\tan^{-1}\left(\frac{x}{2}\right)$ or $\frac{A}{2}\ln(4+x^2)$ only. BOFT unless clearly from single term.

Substitute limits 0 and 2 correctly in an integral of the form $a\ln(4+x^2)+b\tan^{-1}\left(\frac{x}{2}\right)+c\ln(1+x)$, where $abc \neq 0$ **M1** Allow one sign or substitution error. Allow omission of $b\tan^{-1}0+c\ln 1$.

Obtain answer $\ln 54 - \frac{\pi}{8}$ after full and correct working **A1** AG. Work to combine or simplify at least 2 of $\ln$ terms is required CWO. A0 if any non-exact value(s) seen.

Total: 5

Q11 May/June 2023 | Paper 3 (variant 31) | Question 1

Topics: Indices, Logarithms & Exponents

Question 1

Solve the equation

$$3e^{2x} - 4e^{-2x} = 5$$

Give the answer correct to 3 decimal places. **[3]**

Examiner comment

> "In solutions with a correct initial method, the most common error was to try to process a negative value for ex."

Mark scheme

Question 1 Mark Scheme

$$3(e^{2x})^2 - 5(e^{2x}) - 4 = 0$$

B1 | OE Form 3 term quadratic in e^{2x}.

$$e^{2x} = \frac{5 \pm \sqrt{73}}{6}, \quad x = \frac{1}{2}\ln\left(\frac{5+\sqrt{73}}{6}\right)$$

M1 | Use correct method to solve for x.

$$x = 0.407$$

A1 | Only

Total: 3

Q12 May/June 2023 | Paper 3 (variant 31) | Question 8

Topics: Algebra, Integration

Question 8

Let $f(x) = \frac{5-5x^2}{(2x+1)(x+2)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence find the exact value of $\int_0^4 f(x)\,dx$, giving your answer in the form $a + b\ln c$, where a, b and c are integers. **[5]**

Examiner comment

> "For candidates starting with a correct form, the most common errors were due to slips in the arithmetic."

> "Candidates with only two fractions often dealt with one correctly and made no progress with the second."

Mark scheme

Question 8 Mark Scheme

(a)

State or imply the form $\frac{A}{2x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$

B1 | Accept $\frac{A}{2x+1} + \frac{Dx+E}{(x+2)^2}$.

Use a correct method for finding a constant

M1 |

Obtain one of $A = 1, B = -2, C = 3$

A1 | For alternative form: $A = 1, D = -2, E = -1$.

Obtain a second value

A1 |

Obtain the third value

A1 |

Total: 5

(b)

Integrate and obtain one of $\frac{1}{2}\ln(2x+1), -2\ln(x+2), \frac{-3}{x+2}$

B1 FT | The follow through is on **their** A, B, C.

Obtain a second term

B1 FT | If the alternative form is used, then either need to use integration by parts or split the fraction

further.

Obtain the third term

B1 FT |

Substitute limits correctly in an integral with at least two terms of the form $\frac{1}{2}\ln(2x+1), -2\ln(x+2)$ and $\frac{-3}{x+2}$ and subtract in correct order

M1 | The terms used need to have been obtained correctly. Must be exact values, not decimals.

Obtain $\mathbf{1-\ln 3}$

A1 |

Total: 5

Q13 May/June 2023 | Paper 3 (variant 31) | Question 9

Topics: Integration, Numerical Solutions/Iteration

Question 9

The constant a is such that $\int_0^a xe^{-2x}\,dx = \frac{1}{8}$.

(a) Show that $a = \frac{1}{2}\ln(4a+2)$. **[5]**

(b) Verify by calculation that a lies between 0.5 and 1. **[2]**

(c) Use an iterative formula based on the equation in **(a)** to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Examiner comment

> "Using the limits to obtain the given equation proved to be more challenging, with several candidates not attempting to rearrange their equation."

> "For candidates using the iterative process correctly, the most common errors were due to not working to the required accuracy, or drawing the incorrect conclusion from correct work."

Mark scheme

Question 9 Mark Scheme

(a)

Commence integration and reach $pxe^{-2x} + q\int e^{-2x}\,dx$

M1 | OE

Obtain $-\frac{1}{2}xe^{-2x} + \frac{1}{4}\int e^{-2x}\,dx$

A1 | OE

Complete integration and obtain $-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$

A1 |

Use limits correctly and equate to $\frac{1}{8}$, having integrated twice

DM1 | $-\frac{1}{2}ae^{-2a} - \frac{1}{4}e^{-2a} + \frac{1}{4} - \frac{1}{8}$.

Obtain $a = \frac{1}{2}\ln(4a+2)$ correctly

A1 | AG

Total: 5

(b)

Calculate the values of a relevant expression or pair of expressions at $a = 0.5$ and $a = 1$

M1 |

Justify the given statement with correct calculated values

A1 | e.g. $0.5 < 0.69\ldots, 1 > 0.89\ldots$ 0.193 > 0, -1.105 < 0. 0.066 < 0.125, 0.148 > 0.125 if put limits in the integral. Condone if they use calculator for the definite integral.

Total: 2

(c)

Use the iterative process $a_{n+1} = \frac{1}{2}\ln(4a_n+2)$ correctly at least once.

M1 |

Obtain final answer 0.84

A1 |

Show sufficient iterations to at least 4 d.p. to justify 0.84 to 2 d.p. or show that there is a sign change in (0.835, 0.845)

A1 | e.g. 0.75, 0.8047, 0.8261, 0.8343, 0.8373, 0.8385 1, 0.8959, 0.8599, 0.8469, 0.8420, 0.8402.

Total: 3

Q14 May/June 2023 | Paper 3 (variant 31) | Question 10

Topics: Algebra, Complex Numbers

Question 10

The polynomial $x^3 + 5x^2 + 31x + 75$ is denoted by $p(x)$.

(a) Show that $(x + 3)$ is a factor of $p(x)$. **[2]**

(b) Show that $z = -1 + 2\sqrt{6}i$ is a root of $p(z) = 0$. **[3]**

(c) Hence find the complex numbers z which are roots of $p(z^2) = 0$. **[7]**

Total: 12

Examiner comment

> "There were several blank responses."

Mark scheme

Question 10 - Mark Scheme

(a)

- **M1**: Substitute $x = -3$ to obtain value of $p(-3)$
- **A1**: Obtain $p(-3) = 0$ and hence given result

Alternative method for Question 10(a)

- **M1**: Divide $p(x)$ by $(x + 3)$ to obtain quotient $x^2 \pm 2x + \ldots$
- **A1**: Obtain quotient $x^2 + 2x + 25$, with zero remainder and hence given result

Total for (a): 2

(b)

- **M1**: Substitute $z = -1 + 2\sqrt{6}i$ and attempt expansions of z^2 and z^3
- **M1**: Use $i^2 = -1$
- **A1**: Obtain $p(z) = 0$ and hence given result (SC B1 if there is no evidence of working for the square or the cube. Total 1/3.)

Alternative Method 1

- **M1**: Use roots $z = -1 + 2\sqrt{6}i$ to form quadratic factor
- **M1**: Divide $p(z)$ by their quadratic factor
- **A1**: Obtain zero remainder and hence given result.

Alternative Method 2

- **M1**: Set their quadratic factor from (a) equal to zero
- **M1**: Solve for z
- **A1**: Obtain $z = -1 + 2\sqrt{6}i$ (and $z = -1 - 2\sqrt{6}i$)

Alternative Method 3

- **M1**: Substitute $z = -1 + 2\sqrt{6}i$ into their quadratic factor and attempt expansion of z^2

- **M1**: Use $i^2 = -1$

- **A1**: Obtain 0 and hence given result

Total for (b): 3

(c)

- **B1**: State $z_1 = \sqrt{3}i$ and $z_2 = -\sqrt{3}i$

- **M1**: Expand $(x + iy)^2 = -1 + 2\sqrt{6}i$ and compare real and imaginary parts

- **A1**: Obtain $x^2 - y^2 = -1$ and $xy = \sqrt{6}$

- **M1**: Solve to obtain x and y

- **A1**: Obtain $z_3 = \sqrt{2} + \sqrt{3}i$ and $z_4 = -\sqrt{2} - \sqrt{3}i$

- **M1**: Use $z^2 = -1 - 2\sqrt{6}i$ to obtain z_5 and z_6

- **A1**: Obtain $z_5 = \sqrt{2} - \sqrt{3}i$ and $z_6 = -\sqrt{2} + \sqrt{3}i$

Total for (c): 7

Question Total: 12

Q15 May/June 2023 | Paper 3 (variant 32) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

Question 6

The equation $\cot \frac{x}{2} = 3x$ has one root in the interval $0 < x < \pi$, denoted by α.

(a) Show by calculation that α lies between 0.5 and 1. **[2]**

(b) Show that, if a sequence of positive values given by the iterative formula

$$x_{n+1} = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$$

converges, then it converges to α. **[2]**

(c) Use this iterative formula to calculate α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Total: 7

Examiner comment

> "A common error was to round the final answer to 0.80 rather than 0.79."

Mark scheme

Question 6 - Mark Scheme

(a)

- **M1**: Calculate the values of a relevant expression or pair of expressions at $x = 0.5$ and $x = 1$. (Need to evaluate at both points, but **M1** still available if one value incorrect. Use of degrees is M0. Correct use of a

smaller interval is **M1**.)

- **A1**: Complete the argument correctly with conclusion about change of sign or change of inequalities and with correct calculated values. (Can all be in symbols – an explanation in words is not required. Embedded values 0.5 and 1 are not sufficient. 3.92 and 1.83 alone are not sufficient.)

Total for (a): 2

(b)

- **M1**: State $x = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$
- **A1**: Rearrange to the given equation $\cot\left(\frac{x}{2}\right) = 3x$ as far as $2y = 4\tan^{-1}\left(\frac{1}{3x}\right)$ or continue rearrangement to $x = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$ and state iterative formula of $x_{n+1} = \frac{1}{3}\left(x_n + 4\tan^{-1}\frac{1}{3x_n}\right)$. (Need intermediate step between $\frac{x}{2} = \tan^{-1}\frac{1}{3x}$ and $\cot\left(\frac{x}{2}\right) = 3x$)

Total for (b): 2

(c)

- **M1**: Use the iterative process correctly at least once
- **A1**: Obtain final answer 0.79
- **A1**: Show sufficient iterations to at least 4 d.p. to justify 0.79 to 2 d.p. or show there is a sign change in the interval (0.785, 0.795). Example iterations: 1, 0.7623, 0.8037, 0.7921, 0.7951, 0.7943, 0.7945 or 0.5, 0.9506, 0.7665, 0.8024, 0.7924, 0.7950, 0.7944, 0.7945 or 0.75, 0.8076, 0.7911, 0.7954, 0.7943, 0.7946, 0.7945. Condone truncation. Allow recovery. Condone minor differences in the final d.p.

Total for (c): 3

Question Total: 7

Q16 May/June 2023 | Paper 3 (variant 32) | Question 7

Topics: Differentiation

Question 7

The equation of a curve is $3x^2 + 4xy + 3y^2 = 5$.

(a) Show that $\frac{dy}{dx} = -\frac{3x+2y}{2x+3y}$. **[4]**

(b) Hence find the exact coordinates of the two points on the curve at which the tangent is parallel to $y + 2x = 0$. **[5]**

Total: 9

Examiner comment

> "The most common error was to substitute y =−2x into the original equation."

Mark scheme

Question 7 - Mark Scheme

(a)

- **B1**: State or imply $6y\frac{dy}{dx}$ as the derivative of $3y^2$
- **B1**: State or imply $4x\frac{dy}{dx} + 4y$ as the derivative of $4xy$
- **M1**: Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$
- **A1**: Obtain $\frac{dy}{dx} = -\frac{3x+2y}{2x+3y}$

Total for (a): 4

(b)

- ***M1**: Equate $\frac{dy}{dx}$ to -2 and solve for x in terms of y or for y in terms of x
- **A1**: Obtain $x = -4y$ or $y = -\frac{x}{4}$
- **DM1**: Substitute their $x = -4y$ or their $y = -\frac{x}{4}$ in curve equation (Allow unsimplified.)
- **A1**: Obtain $y = \pm\frac{1}{\sqrt{7}}$ or $x = \pm\frac{4}{\sqrt{7}}$
- **A1**: Obtain both pairs of values

Guidance: Or exact equivalent. Or $x = \pm\frac{4}{\sqrt{7}}$ and $y = -\frac{1}{\sqrt{7}}$ or exact equivalent. Or $x = -\frac{4}{\sqrt{7}}$ and $y = -\frac{1}{\sqrt{7}}$ or exact equivalent. A1 A0 for incorrect final pairing.

Total for (b): 5

Question Total: 9

Q17 May/June 2023 | Paper 3 (variant 32) | Question 8

Topics: Differential Equations

Question 8

(a) The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = \frac{4 + 9y^2}{e^{2x+1}}$$

It is given that $y = 0$ when $x = 1$.

Solve the differential equation, obtaining an expression for y in terms of x. **[7]**

(b) State what happens to the value of y as x tends to infinity. Give your answer in an exact form. **[1]**

Total: 8

Examiner comment

> “Common errors included writing e2x+1 as e2x +1 or obtaining e2x+1 = 4+9y2 .”

Mark scheme

Question 8 - Mark Scheme

(a)

- **B1**: Separate variables correctly as $\int \frac{1}{4+9y^2}dy = \int e^{-(2x+1)}dx$

(Condone missing integral signs or dx and dy missing.)

- **B1**: Obtain term $-\frac{1}{2}e^{-2x-1}$ (OE e.g. $-\frac{1}{2e}e^{-2x}$.)

- **M1**: Obtain term of the form $a\tan^{-1}\left(\frac{3y}{2}\right)$

- **A1**: Obtain term $\frac{1}{6}\tan^{-1}\left(\frac{3y}{2}\right)$ (OE e.g. $\frac{1}{9}\cdot\frac{3}{2}\tan^{-1}\frac{3y}{2}$.)

- **M1**: Use $x = 1, y = 0$ to evaluate a constant or as limits in a solution containing or derived from terms of the form $atan^{-1}(by)$ and $ce^{(2x+1)}$

- **A1**: Obtain correct answer in any form

- **A1**: Obtain final answer $y = \frac{2}{3}\tan\left(3e^{-3}\cdot e^{-2x+1}\right)$ (OE Allow with $3e^{-3} = 0.149...$)

Total for (a): 7

(b)

- **B1 FT**: State that y approaches $\frac{2}{3}\tan\left(3e^{-3}\right)$

(Or exact equivalent. The FT is on correct work on a solution containing e^{2x-1}. Condone $y = ...$. Accept correct answer stated with minimal wording. 0.10032... is not exact so **B0**.)

Total for (b): 1

Question Total: 8

Q18 May/June 2023 | Paper 3 (variant 32) | Question 9

Topics: Algebra, Integration

Question 9

Let $f(x) = \frac{2x^2+17x-17}{(1+2x)(2-x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence show that $\int_0^1 f(x)dx = \frac{5}{2} - \ln 72$. **[5]**

Total: 10

Examiner comment

> "(b) This part proved to be more of a challenge and there were several errors with signs and coefficients."

> "Many recognised that the two fractions with linear denominators integrated to log terms, but they struggled with k dx."

> "As mentioned in part (b), candidates with a fraction of the form Dx+E often made no progress with (2−x)2 the integration, but a few were successful in using integration by parts."

Mark scheme

Question 9 - Mark Scheme

(a)

- **B1**: State or imply the form $\frac{A}{1+2x} + \frac{B}{2-x} + \frac{C}{(2-x)^2}$

(Alternative form: $\frac{A}{1+2x} + \frac{Dx+E}{(2-x)^2}$)

- **M1**: Use a correct method for finding a coefficient (e.g. $A(2-x)^2 + B(1+2x)(2-x) + C(1+2x) = 2x^2 + 17x - 17$ and compare coefficients or substitute for x.)

- **A1**: Obtain one of $A = -4, B = -3$ and $C = 5$

- **A1**: Obtain a second value

- **A1**: Obtain the third value (Extra term in partial fractions, then **B0** unless recover at end. Allow the marks for any constants found correctly. Missing terms in partial fractions, **B0** but M1A1 is available for a correct method that obtains at least one correct constant (e.g. cover-up rule) Max 2/5. Ignore any substitution back into their original expression.)

(If alternative form used: $A = -4, D = 3$ and $E = -1$.)

Total for (a): 5

(b)

- **B1FT**: Integrate and obtain terms $-2\ln(1+2x) + 3\ln(2-x) + \frac{5}{2-x}$

- **B1FT**: [Multiple conditional marks based on form used]

- **B1FT**: [If using A, D, E form then B1 for the A term, but no further marks until partial fractions are used to split the second term or they use integration by parts]

- **M1**: Substitute limits correctly in an integral with two terms (obtained correctly) of the form $a\ln(1+2x) + b\ln(2-x) + \frac{c}{2-x}$, where $abc \neq 0$

- **A1**: Obtain answer $\frac{5}{2} - \ln 72$ after full and correct working (AG – evidence of some correct work to combine or simplify logs is required e.g. allow from $-\ln 9 + \ln \frac{1}{8}$ or $-\ln 2^3 - \ln 3^2$.)

Total for (b): 5

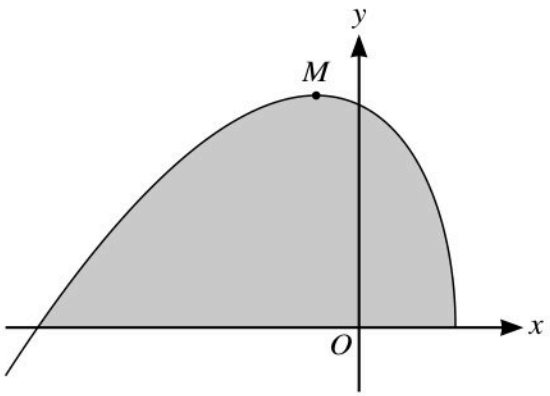
Question Total: 10

Q19 May/June 2023 | Paper 3 (variant 32) | Question 10

Topics: Differentiation, Integration

Question 10

10



The diagram shows the curve $y = (x + 5)\sqrt{3 - 2x}$ and its maximum point M.

(a) Find the exact coordinates of M. [5]

The diagram shows the curve $y = (x + 5)\sqrt{3 - 2x}$ and its maximum point M.

(a) Find the exact coordinates of M. **[5]**

(b) Using the substitution $u = 3 - 2x$, find by integration the area of the shaded region bounded by the curve and the x-axis. Give your answer in the form $a\sqrt{13}$, where a is a rational number. **[5]**

Total: 10

Examiner comment

> "The most common errors occurred in using the chain rule to differentiate the square root."

> "Some struggled with the two terms involving the square root."

> "In the resulting linear equation, the common error was to obtain –x+5 instead of –x–5, resulting from an error in removing brackets."

> "(b) This part provided a real challenge for candidates and there were very few fully correct solutions."

Mark scheme

Question 10 - Mark Scheme

(a)

- ***M1**: Use the product rule correctly to obtain $p(x + 5)(3 - 2x)'' + q(3 - 2x)^{\frac{1}{2}}$

(Allow with incorrect chain rule. BOD over sign errors unless an incorrect rule is quoted.)

- **A1**: Obtain correct derivative in any form (e.g. $-(x + 5)(3 - 2x)^{-\frac{1}{2}} + (3 - 2x)^{\frac{1}{2}}$.)

- **DM1**: Equate derivative to zero and obtain a linear equation

(Allow with surd factor e.g. $(3 - 2x)^{-\frac{1}{2}}\,[-(x + 5) + (3 - 2x)] = 0$.)

- **A1**: Obtain a correct linear equation (e.g. $-(x+5)+3-2x=0$.)
- **A1**: Obtain answer $\left(-\frac{2}{3}, \frac{13\sqrt{13}}{9}\right)$ or $\left(-\frac{2}{3}, \frac{13\sqrt{13}}{9}\right)$ (Or exact equivalent e.g. $\left(-\frac{2}{3}, \frac{2\sqrt{2197}}{27}\right)$.)

Alternative Method for Question 10(a)

- *M1*: Obtain y^2 and differentiate
- **A1**: Obtain correct derivative in any form (e.g. $-6x^2-34x-20$.)
- **DM1**: Equate derivative to zero and solve for x
- **A1**: Obtain $-\frac{2}{3}$

(Ignore -5 if seen.)
- **A1**: Obtain answer $\left(-\frac{2}{3}, \frac{13\sqrt{13}}{9}\right)$ only

(ISW)

Total for (a): 5

(b)
- ***M1**: Use the given substitution and reach $a\int\left(\frac{13-u}{2}\right)u^{\frac{1}{2}}du$

(OE Need to see -2 or $-\frac{1}{2}$ used. Condone if du missing or the integral sign is missing.)
- **A1**: Obtain correct integral $-\frac{1}{2}\int\left(\frac{3-u}{2}\sqrt{u}du+5\int\sqrt{u}du\right)$.

(OE e.g. $-\frac{1}{2}\left[\int\frac{3-u}{2}\sqrt{u}du+5\int\sqrt{u}du\right]$.)

(Ignore limits at this stage. Condone if du missing.)
- **B1**: SOI e.g. by $u=13$ and 0. In any order and at any stage.
- **DM1**: Use correct limits the right way round in an integral of the form $a\left[\frac{26}{3}u^{\frac{3}{2}}-\frac{2}{5}u^{\frac{5}{2}}\right]$
- **A1**: Obtain answer $\frac{169}{15}\sqrt{13}$ or $a=\frac{169}{15}$

(or exact equivalents.)

Total for (b): 5

Question Total: 10

Q20 May/June 2023 | Paper 3 (variant 32) | Question 11

Topics: none of the listed topics

Question 11

The points A and B have position vectors $\mathbf{i}+2\mathbf{j}-2\mathbf{k}$ and $2\mathbf{i}-\mathbf{j}+\mathbf{k}$ respectively. The line l has equation

$$\mathbf{r}=\mathbf{i}-\mathbf{j}+3\mathbf{k}+\mu(2\mathbf{i}-3\mathbf{j}+4\mathbf{k})$$

(a) Show that l does not intersect the line passing through A and B. **[5]**

(b) Find the position vector of the foot of the perpendicular from A to l. **[4]**

Total: 9

Examiner comment

> “Many candidates did not know how to get started – they often understood that a scalar product was required, but then did not use the correct vectors.”

> “There were also a large number of blank responses.”

Mark scheme

Question 11 - Mark Scheme

(a)

- **M1**: Carry out correct method for finding a vector equation for $\overrightarrow{AB}$

- **A1**: Obtain $[\mathbf{r} =]\mathbf{i} + 2\mathbf{j} - 2\mathbf{k} + \lambda(\mathbf{i} - 3\mathbf{j} + 3\mathbf{k})$

(OE e.g. $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + \mathbf{k} + \lambda(-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})$.)

- **M1**: Equate two pairs of components of general points on their AB and l and evaluate λ or μ

- **A1**: Obtain correct answer for λ or μ, e.g. $\lambda = -1, \mu = -2$

(Correct value from two correct component equations.)

- **A1**: Verify that all three equations are not satisfied and the lines fail to intersect ($\neq$ is sufficient justification e.g. $0 \neq -3$).

(Conclusion needs to follow correct values. Two possible cases are possible e.g. using $\mathbf{j}$ and $\mathbf{k}$ to get one parameter and then $\mathbf{i}$ to obtain the other. If two pairs of simultaneous equations and showing that the results are not the same.)

Alternatives

Case	λ	μ	Comparison	$\neq$
ij	2	1	$4 \neq 7$	-
ik	5	$\frac{5}{2}$	$-13 \neq -\frac{17}{2}$	-
jk	-1	-2	$0 \neq -3$	-

Total for (a): 5

(b)

- **B1**: Find $\overrightarrow{AP}$ for a general point P on l, e.g. $-3\mathbf{j} + 5\mathbf{k} + \mu(2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$

(Or equivalent e.g. $\overrightarrow{PA} = -2\mu\mathbf{i} + (3\mu + 3)\mathbf{j} - (4\mu + 5)\mathbf{k}$.)

- **M1**: Calculate scalar product of their $\overrightarrow{AP}$ and a direction vector for l and equate the result to zero (e.g. $4\mu + (9 + 9\mu) + (20 + 16\mu) = 0$.)

(Mo if using $\overrightarrow{OP}$.)

(Mo if using parallel line through A.)

- **A1**: Obtain $\mu = -1$

- **A1**: Obtain answer $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

(Accept coordinates in place of position vector.)

Alternative Method for Question 11(b)

- **B1**: Find $\overrightarrow{AP}$ for a general point P on l, e.g. $-3\mathbf{j}+5\mathbf{k}+\mu(2\mathbf{i}-3\mathbf{j}+4\mathbf{k})$

(Or equivalent e.g. $\overrightarrow{PA}=-2\mu\mathbf{i}+(3\mu+3)\mathbf{j}-(4\mu+5)\mathbf{k}$.)

- **M1**: Use Pythagoras and differentiate with respect to μ to obtain value of μ corresponding to minimum distance.

(No need to prove it is a minimum.)

(e.g. $\frac{d}{d\mu}\left[4\mu^2+9(\mu+1)^2+(4\mu+5)^2\right]=0$.)

- **A1**: Obtain $\mu=-1$

- **A1**: Obtain answer $-\mathbf{i}+2\mathbf{j}-\mathbf{k}$

(Accept coordinates in place of position vector.)

Total for (b): 4

Question Total: 9

Q21 May/June 2023 | Paper 3 (variant 33) | Question 3

Topics: Complex Numbers

Question 3

On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z-3-i|\leq 3$ and $|z|\geq|z-4i|$. **[4]**

Total: 4

Examiner comment

> "The most common error seen was in omitting a scale from the diagram or showing the scale on only one axis."

Mark scheme

Question 3 - Mark Scheme

Answer

- **B1**: Show a circle with centre $3+i$

(Must be some evidence of scale on both axes or centre stated as $3+i$ or (3, 1).)

- **B1**: Show a circle with radius 3 and centre not at the origin

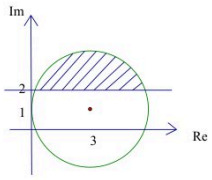
(Must be some evidence that radius = 3 or stated $r=3$.)

- **B1**: Show the line $y=2$

(Line $y=2$ can be represented by 2 or correct dashes.)

- **B1**: Shade the correct region

(Line and circle must be correct.)

3	Show a circle with centre $3+i$	**B1**	Must be some evidence of scale on both axes or centre stated as 3 + i or (3, 1).
	Show a circle with radius 3 and centre not at the origin	**B1**	Must be some evidence that radius = 3 or stated $r = 3$
	Show the line $y = 2$	**B1**	Line $y = 2$ can be represented by 2 or correct dashes.
	Shade the correct region	**B1**	Line and circle must be correct.
		**4**	Scales may be replaced by dashes on axes for all marks. Correct figure, with no scale on either axis then allow 1/3 and the B1 for correct shaded region Max 2/4. If B0 above for line but relatively correct position then B1 for correct shaded region Max 3/4. Re and Im axes interchanged but clearly labelled, allow **SCB1** for centre and radius of circle correct and **SCB1** for line and shading correct Max 2/4.

Total: 4

Guidance

Scales may be replaced by dashes on axes for all marks. Correct figure, with no scale on either axis then allow 1/3 and the **B1** for correct shaded region Max 2/4. If **B0** above for line but relatively correct position then **B1** for correct shaded region Max 3/4. Re and Im axes interchanged but clearly labelled, allow **SCB1** for centre and radius of circle correct and **SCB1** for line and shading correct Max 2/4.

Q22 May/June 2023 | Paper 3 (variant 33) | Question 6

Topics: Trigonometry

6(a) Express $3\cos x + 2\cos(x - 60^\circ)$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. State the exact value of R and give α correct to 2 decimal places. **[4]**

6(b) Hence solve the equation

$$3\cos 2\theta + 2\cos(2\theta - 60^\circ) = 2.5$$

for $0^\circ < \theta < 180^\circ$. **[4]**

Examiner comment

> "Using θ instead of 2θ or 2(θ – 23.41) instead of 2θ – 23.41 were common errors."

Mark scheme

6(a) B1 Need to see $A\cos x + B\sin x$ with A and B correct. A may be 4 or $3 + 2\cos 60^\circ$ and B may be $\sqrt{3}$ or $2\sin 60^\circ$.

B1 FT Follow through their 4 and $\sqrt{3}$. If coefficients are 3 and 2 then B0. $R = \sqrt{19}$ from $R = 4.36$ B0 but 4.36 seen after $\sqrt{19}$ ISW.

M1 Use correct trig formulae for their expansion to find α e.g. $\alpha = \tan^{-1}\frac{\sqrt{3}}{4}$ or $\cos^{-1}\frac{4}{\sqrt{19}}$ or $\sin^{-1}\frac{\sqrt{3}}{\sqrt{19}}$. If $\sin\alpha = \sqrt{3}\cos\alpha = 4$ seen then M0 A0. If $\tan\alpha = 23.41^\circ$ M0 A0 but can recover if $\alpha = 23.41^\circ$ seen later. $\alpha = \tan^{-1}\frac{2}{3}$ **M1** ($\alpha = 33.69^\circ$) but $\alpha = \tan^{-1}\frac{3}{2}$ M0

A1 Obtain $\alpha = 23.41^\circ$. Allow if x instead of α.

Total: 4

6(b) B1 FT $\cos^{-1}\left(\frac{2.5}{R}\right)$. SOI [55.0°]. Follow through their $\sqrt{19}$.

M1 Use a correct method to find a value of 2θ (not x) in the interval. Allow sign error in moving α to right side. $2\theta = \cos^{-1}\left(\frac{2.5}{R}\right) + 23.41^\circ$ or $2\theta = 360^\circ - \cos^{-1}\left(\frac{2.5}{R}\right) + 23.41^\circ$ with R substituted.

A1 Obtain one correct answer e.g. 39.2°. If working for **M1** not seen then **M1** implied by 39.2° or 164.2°. Must be at least 1 d.p.

A1 Obtain second correct answer e.g. 164.2° and no others in the interval. Must be at least 1 d.p. Ignore answers outside the given interval.

Total: 4

Q23 May/June 2023 | Paper 3 (variant 33) | Question 7

Topics: Integration

7(a) Use the substitution $u = \cos x$ to show that

$$\int_0^{\pi} \sin 2x e^{2\cos x} dx = \int_{-1}^{1} 2ue^{2u} du. \quad [4]$$

7(b) Hence find the exact value of $\int_0^{\pi} \sin 2x e^{2\cos x} dx$. **[4]**

Examiner comment

> "The most common errors were integrating e2u incorrectly to obtain 2e2u or making one or more sign errors."

> "This meant they were faced with a far more difficult integral, and had to attempt to integrate by parts twice, often making little progress."

Mark scheme

7(a) B1 $\frac{du}{dx} = -\sin x$. SOI

M1 Use double angle formula and substitute for x and dx throughout the integral. All x's must be removed, can be coefficient errors provided 2 seen in working.

A1 Obtain $\pm \int 2ue^{2u} du$. Limits may be omitted, or left as 0 and π, during the change of variable stage.

A1 Justify new limits and obtain $\int_{-1}^{1} 2ue^{2u} du$ from correct working. **AG** Must see $x = 0, u = 1$ and $x = \pi, u = -1$. Inequalities alone e.g. $0 \leq x \leq \pi$ and $1 \leq u \leq -1$ or $-1 \leq u \leq 1$ for limits are insufficient A0. If sign in expression and order of limits incorrect then A0. If negative sign is present in the integrand then this can be removed and limits introduced in correct order in a single step.

Total: 4

7(b) M1* Condone dx.

A1 Complete integration and obtain $ue^{2u} - \frac{1}{2}e^{2u}$. **OE** Allow $\left(2u\frac{1}{2}e^{2u}\right) - \frac{1}{2}e^{2u}$.

DM1 Use correct limits correctly in $cue^{2u} + de^{2u}$ having integrated twice or in $c\cos x e^{2\cos x} + de^{2\cos x}$. 1 and -1 for u, 0 and π for x e.g. $ce^2 + de^2 - (-ce^{-2} + de^{-2})$. Not decimals. Allow one sign error at most in going from $cue^{2u} + de^{2u}$ or $c\cos x e^{2\cos x} + de^{2\cos x}$ to $ce^2 + de^2 - (-ce^{-2} + de^{-2})$. [e² - ½e² - (-e^{-2} - ½e^{-2})] Complete reversal of sign by converting back to $\cos x$ and not making $x = 0$ upper limit is DM0 A0.

A1 ISW. Obtain $\frac{1}{2}e^2 + \frac{3}{2}e^{-2}$. Or equivalent 2-term expression e.g. $\frac{e^4+3}{2e^2}$ or $\frac{1}{2}\left(e^2 + \frac{3}{e^2}\right)$.

Total: 4

Q24 May/June 2023 | Paper 3 (variant 33) | Question 8

Topics: Differential Equations

8 The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = \frac{x^2 + 4}{x(y + 4)}$$

for $x > 0$. It is given that $x = 4$ when $y = 2\sqrt{3}$.

Solve the differential equation to obtain the value of x when $y = 2$. **[8]**

Examiner comment

> "Question 8 There were many complete solutions seen, however a number of candidates found this question challenging 1 and could only separate the variables correctly and integrate ."

> "A small number chose to complete the y integrand by parts, but after undertaking this correctly once, for which M1 was available, little or no progress was made in the next integration step."

Mark scheme

8 B1 Separate the variables correctly. $\frac{y+4}{y^2+4}dy = \frac{1}{x}dx$.

B1 Obtain $\ln x$.

***M1** Split the fraction and integrate to obtain $p\ln(y^2 + 4)$ or $q\tan^{-1}\frac{y}{2}$ correctly. Only following subdivision into $\frac{y}{y^2+4} + \frac{4}{y^2+4}$. If no subdivision seen then both terms $p\ln(y^2 + 4)$ and $q\tan^{-1}\frac{y}{2}$ must be present.

A1 Obtain $\frac{1}{2}\ln(y^2 + 4)$.

A1 Obtain $2\tan^{-1}\frac{y}{2}$.

DM1 Use $(4, 2\sqrt{3})$ in an expression containing at least 2 of $a\ln x, b\ln(y^2 + 4)$ and $c\tan^{-1}\frac{y}{2}$ to obtain constant of integration. Allow one sign or arithmetic error e.g. $\frac{2\pi}{3}$. May use $(4, 2\sqrt{3})$ and $(x, 2)$ as limits to find x for the final 3 marks.

Correct solution (any form) e.g. $\frac{1}{2}\ln(y^2+4)+2\tan^{-1}\frac{y}{2}=\ln x+\frac{2\pi}{3}$ or $\frac{1}{2}\ln(y^2+4)+2\tan^{-1}\frac{y}{2}=\ln x+2\tan^{-1}\sqrt{3}+\frac{1}{2}\ln 16-\ln 4$.

A1 However solution not asked for so allow $\frac{1}{2}\ln 8+2\tan^{-1}1=\ln x+2\tan^{-1}\sqrt{3}+\frac{1}{2}\ln 16-\ln 4$.

A1 ISW. Obtain $\sqrt{8}e^{-\frac{1}{6}}$ or 1.68 or more accurate or $2\sqrt{2}e^{-\frac{1}{6}}$ or $\frac{8}{\sqrt{e^{\frac{1}{3}}}}$ or $\frac{\sqrt{8}}{e^{\frac{1}{6}}}$.

Total: 8

Q25 May/June 2023 | Paper 3 (variant 33) | Question 9

Topics: none of the listed topics

9 The lines l and m have equations

$$l: \mathbf{r}=a\mathbf{i}+3\mathbf{j}+bk+\lambda(c\mathbf{i}-2\mathbf{j}+4k),$$

$$m: \mathbf{r}=\mathbf{i}+2\mathbf{j}+3k+\mu(2\mathbf{i}-3\mathbf{j}+k).$$

Relative to the origin O, the position vector of the point P is $4\mathbf{i}+7\mathbf{j}-2k$.

(a) Given that l is perpendicular to m and that P lies on l, find the values of the constants a, b and c. **[4]**

(b) The perpendicular from P meets line m at Q. The point R lies on PQ extended, with $PQ:QR=2:3$. Find the position vector of R. **[6]**

Examiner comment

> "However, the common error was for candidates to assume a point of intersection and solve by equating lines."

> "(b) Candidates found this part more difficult and many did not attempt it."

> "Common errors were using 2 instead of 3 (or 3 instead of 5), direction errors in 3 2 2 2 υυυρ υυυρ the relevant vectors, not squaring 3 when using magnitudes, or finding vector QR not OR."

Mark scheme

9(a) M1 Perform scalar product of direction vectors and set result equal to zero. $2c+6+4=0$.

M1 Use P to find the value of λ. $3-2\lambda=7\Rightarrow\lambda=-2$ [$a+\lambda c=4, b+4\lambda=-2$]. Equation for line l may contain $-\lambda$ instead of $+\lambda$ leading to $\lambda=2$ all marks available.

A1 Obtain $c=-5$ or $b=6$.

A1 $a=-6, b=6$ and $c=-5$ all correct.

Total: 4 SC1: Use P to find the value of λ **M1**. Substitute $\lambda=-2$ into point P, so $a-2c=4, b+4\lambda=-2$. Equation for line l may contain $-\lambda$ instead of $+\lambda$ leading to $\lambda=2$ all marks available. All 3 values correct **A1**. Max 2/4.

9(b) B1 Find $\overrightarrow{PQ}$ (or $\overrightarrow{QP}$) for a general point Q on m: $= \pm \begin{bmatrix} -3+2\mu \\ -5-3\mu \\ 5+\mu \end{bmatrix}$. Could be their a, b, c and λ values provided **M1 M1** gained in (a).

M1* Equate the scalar product of $\overrightarrow{PQ}$ (or $\overrightarrow{QP}$) and a direction vector for m to zero and obtain an equation in μ. Allow $\overrightarrow{PQ} = \overrightarrow{OQ} + \overrightarrow{QP}$ sign problem.

A1 $PQ^2 = (-3+2\mu)^2 + (-5-3\mu)^2 + (5+\mu)^2 = [14(\mu+1)^2 + 45]$. Min when $\mu = -1$ or by differentiation.

A1 Obtain $\overrightarrow{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ or $\overrightarrow{PQ} = -5\mathbf{i} - \mathbf{j} + 2k$. Must be labelled correctly.

DM1 Carry out a method to find the position vector of R. e.g. Use $\overrightarrow{OR} = \overrightarrow{OP} + \frac{2}{3}\overrightarrow{PQ}$ or $\overrightarrow{OR} = \overrightarrow{OQ} + \frac{3}{2}\overrightarrow{PQ}$ or $\overrightarrow{OR} = \frac{5\overrightarrow{OO} - 2\overrightarrow{OQ}}{3} = \overrightarrow{OQ} + \frac{3}{2}\overrightarrow{PQ}$ where $\overrightarrow{OR} = (x, y, z)$. $\overrightarrow{OR} = \overrightarrow{OR} - \overrightarrow{OQ}$ Solve $|\overrightarrow{OR}|^2 = \frac{9}{4}|\overrightarrow{PQ}|^2$ or $|\overrightarrow{OR}| = \frac{3}{2}|\overrightarrow{PQ}| t = 2.5$.

Alternative method for DM1: $\overrightarrow{OR} = (4, 7, -2) + t(-5, -2, 4)$ $\overrightarrow{OR} = \overrightarrow{OR} - \overrightarrow{OQ}$ Solve $|\overrightarrow{OR}|^2 = \frac{9}{4}|\overrightarrow{PQ}|^2$ or $|\overrightarrow{OR}| = \frac{3}{2}|\overrightarrow{PQ}|$ $t = 2.5$.

A1 Accept coordinates. $\begin{pmatrix} -17/2 \\ 2 \\ 8k \end{pmatrix}$. Don't accept $-\frac{17}{2}\mathbf{i} + \frac{4}{2}\mathbf{j} + \frac{16}{2}k$.

Total: 6

SC2: Equate lines, attempt to find $\mu = -1$ or $\lambda = -1$ **M1***. Substitute $\lambda = -2$ into point P, so $\overrightarrow{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ **A1**. Attempt to find $\overrightarrow{OQ}$ using the other parameter value **DM1**. $\overrightarrow{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ therefore intersect **A1**. Then use main scheme for the final **DM1 A1**.

Q26 May/June 2023 | Paper 3 (variant 33) | Question 10

Topics: Algebra

10 Let $f(x) = \frac{21 - 8x - 2x^2}{(1+2x)(3-x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Examiner comment

> "The most common errors were extracting 3–1 instead of 3–2, extracting 3 and 9 instead of 3–1 and 3–2 or not evaluating the squared term(s) correctly."

Mark scheme

10(a) B1 State or imply the form $\frac{A}{1+2x} + \frac{B}{3-x} + \frac{C}{(3-x)^2}$. Alternative form: $\frac{A}{1+2x} + \frac{Dx+E}{(3-x)^2}$.

M1 Use a correct method to find a constant.

A1 Obtain one of $A = 2, B = 2$ and $C = -3$.

A1 Obtain a second value.

A1 Obtain the third value. Do not need to substitute values back into original form.

Total: 5 If $\frac{A}{1+2x}+\frac{B}{3-x}+\frac{Cx+D}{(3-x)^2}$ Bo but **M1 A1** for A, A for B and **C** and **D**. If $C=0$ then recovers **B1** from above.

10(b) M1 Use a correct method to obtain the first two terms of one of the unsimplified expansions.
$(1+2x)^{-1}, \left(1-\frac{1}{3}x\right)^{-1}, \left(1-\frac{1}{3}x\right)^{-2}, (3-x)^{-1}, (3-x)^{-2}$.

$(1+2x)^{-1}=1+(-1)(2x)+\ldots$
$\left(1-\frac{1}{3}x\right)^{-1}=1+(-1)(-\frac{x}{3})+\ldots$
$\left(1-\frac{1}{3}x\right)^{-2}=1+(-2)(-\frac{x}{3})+((-2)(-3)/2)(-\frac{x}{3})^2+\ldots$
$(3-x)^{-1}=3^{-1}+(-1)3^{-2}(-x)\ldots$
$(3-x)^{-2}=3^{-2}+(-2)3^{-3}(-x)+\ldots$

A1 FT Follow through on their A, B, C.
$A(1+(-1)(2x)+((-1)(-2)/2)(2x)^2+\ldots)+\frac{B}{3}(1+\frac{x}{3}+(\frac{x}{3})^2+\ldots)$

A1 FT $\frac{C}{3^2}(1+(-2)(-\frac{x}{3})+((-2)(-3)/2)(-\frac{x}{3})^2+\ldots)$.

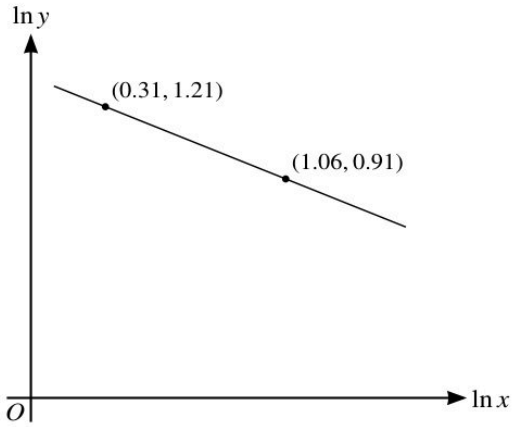
A1 FT Must be their coefficients from (a) but may be unsimplified expansions for **FT** marks. If correct, expect to see
$2(1-2x+(2x)^2)+\ldots$
$2-4x+8x^2$
or $2-x+\frac{x^2}{27}$
or $\frac{2}{3}+\frac{2x}{9}+\frac{2x^2}{27}$
or $-\frac{1}{3}\left(1+\frac{2x}{3}+(3)\left(\frac{x}{3}\right)^2+\ldots\right)$.

Obtain final answer $\frac{7}{3}-4x+\frac{215}{27}x^2$. Accept $2\frac{1}{3}-4x+7\frac{26}{27}x^2$. No **ISW**.

Q27 Feb/Mar 2022 | Paper 3 (variant 32) | Question 3

Topics: Indices, Logarithms & Exponents

3



The variables x and y satisfy the equation $x^n y^2 = C$, where n and C are constants. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points (0.31, 1.21) and (1.06, 0.91), as shown in the diagram.

Find the value of n and find the value of C correct to 2 decimal places. [5]

The variables x and y satisfy the equation $x^n y^2 = C$, where n and C are constants. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(0.31, 1.21)$ and $(1.06, 0.91)$, as shown in the diagram.

Find the value of n and find the value of C correct to 2 decimal places. [5]

Examiner comment

> "Question 3 Candidates generally found this question challenging but many were able to take logarithms of the given equation."

> "Common errors occurred when candidates either substituted lnx and lny into the given equation, for example used (0.31, 1.21), instead of the exponential values from the given coordinates."

Mark scheme

State or imply $n \ln x + 2 \ln y = \ln C$ **B1**

Substitute values of $\ln y$ and $\ln x$, or equate gradient of line to $\pm\frac{1}{2}n$, but not $\pm n$, and solve for n **M1** Using $\ln x$ and $\ln y$ values

Obtain $n = 0.8[0]$ or $0.8[00]$ or $\frac{4}{5}$ **A1**

Solve for C **M1** Using $\ln x$ and $\ln y$ values in equation of correct form, that is $\ln C$ not C. Allow $C = e^{2.668}$.

Obtain $C = 14.41$ **A1** Must be 2 d.p.

Alternative method for question 3

Obtain two correct equations in n and C by substituting x and y values in the given equation **B1**
$(2.886)^n \times (2.484)^2 = C$ and $(1.363)^n \times (3.353)^2 = C$

Solve for n **M1** Using x and y values

Obtain $n = 0.8[0]$ or $0.8[00]$ or $4/5$ **A1**

$\left(\frac{2.886}{1.363}\right)^n \times \left(\frac{2.484}{3.353}\right)^2 = 1$ leading to $n = 0.7995$

Solve for C **M1** Using x and y values

Obtain $C = 14.41$ **A1** Must be 2 d.p.

Total: 5

Q28 Feb/Mar 2022 | Paper 3 (variant 32) | Question 4

Topics: Differentiation

The parametric equations of a curve are

$$x = 1 - \cos\theta, \qquad y = \cos\theta - \frac{1}{4}\cos 2\theta.$$

Show that $\frac{dy}{dx} = -2\sin^2\left(\frac{1}{2}\theta\right)$. [5]

Examiner comment

> "Question 4 Candidates generally found this question to be challenging, a common error was incorrect chain rule dy dy dθ application."

Mark scheme

State $\frac{dx}{d\theta} = \sin\theta$ or $\frac{dy}{d\theta} = -\sin\theta + \frac{1}{2}\sin 2\theta$ **B1**

Use $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ **M1**

Obtain correct answer in any form **A1** e.g. $\frac{-\sin\theta + \frac{1}{2}\sin 2\theta}{\sin\theta}$

Use double angle correctly to obtain $\frac{dy}{dx}$ in terms of θ **M1** $\sin 2\theta = 2\sin\theta\cos\theta$

Obtain the given answer with no errors seen $-2\sin^2\left(\frac{1}{2}\theta\right)$ **A1** AG. Requires correct cancellation of ALL $\sin\theta$ terms and $\cos\theta = 1 - 2\sin^2\left(\frac{1}{2}\theta\right)$ seen.

SC For incorrect signs, consistent throughout max. B0, M1, A0, M1, A1

Total: 5

Q29 Feb/Mar 2022 | Paper 3 (variant 32) | Question 6

Topics: Complex Numbers

Find the complex numbers w which satisfy the equation $w^2 + 2iw^* = 1$ and are such that $\mathrm{Re}\, w \leq 0$. Give your answers in the form $x + iy$, where x and y are real. [6]

Examiner comment

> "Candidates who tried to work with w and w* found it more difficult to do so."

Mark scheme

Substitute and obtain a correct equation in x and y **B1** $(x+iy)^2+2i(x-iy)=1$

Use $i^2=-1$ at least once and equate real and imaginary parts **M1**

Obtain two correct equations, e.g. $x^2-y^2+2y=1$ and $2xy+2x=0$ **A1**

Solve for x or for y **M1**

Using $y=-1$, obtain answer $w=-2-i$ only **A1** A0 if $w=2-i$ as well

Using $x=0$, obtain answer $w=i$ **A1**

Total: 6

Q30 Feb/Mar 2022 | Paper 3 (variant 32) | Question 7

Topics: Numerical Solutions/Iteration

(a) By sketching a suitable pair of graphs, show that the equation $4-x^2=\sec\frac{1}{2}x$ has exactly one root in the interval $0\leq x<\pi$. [2]

(b) Verify by calculation that this root lies between 1 and 2. [2]

(c) Use the iterative formula $x_{n+1}=\sqrt{4-\sec\frac{1}{2}x_n}$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "1 2 2 1 However, it should be noted that there are many other functions that can be used at these values, 1 for example x and 4−sec1 x 2 2 (c) Many candidates used degrees to answer this question, therefore were unable to get the correct result."

Mark scheme

(a)

Sketch a relevant graph, e.g. $y=4-x^2$ **B1** Needs $(0,4)$ or marks on axis and $(2,0)$ or $(\pi,0)$

Sketch a second relevant graph, e.g. $y=\sec\frac{1}{2}x$, and justify the given statement **B1** Needs $(0,1)$ or mark on axis and $(\pi,0)$. Asymptote NOT required, but must NOT reach $x=\pi$. Sec graph must exist over at least interval $\left[0,\frac{3\pi}{4}\right]$ and quadratic graph over $[0,2.5]$.

Total: 2

(b)

Calculate the value of a relevant expression or values of a pair of relevant expressions at $x=1$ and $x=2$ **M1** Need all 4 values or the 2 values correct for M1. Angles in degrees score M0.

Complete the argument with correct calculated values **A1**

Total: 2

(c)

Use the iterative process correctly at least twice **M1**

Obtain final answer $\mathbf{1.60}$ **A1** Must be 2 d.p.

Show sufficient iterations to 4 d.p. to justify $\mathbf{1.60}$ to 2 d.p. or show there is a sign change in the interval $\mathbf{(1.595, 1.605)}$ **A1**

Total: 3

Q31 Feb/Mar 2022 | Paper 3 (variant 32) | Question 9

Topics: Differential Equations

The variables x and y satisfy the differential equation

$$(x+1)(3x+1)\frac{dy}{dx} = y,$$

and it is given that $y = 1$ when $x = 1$.

Solve the differential equation and find the exact value of y when $x = 3$, giving your answer in a simplified form. [9]

Examiner comment

> "1 Most of these candidates were successful with the integration of but struggled with the coefficient x+1 1 when integrating ."

Mark scheme

Correctly separate variables and integrate at least one side **M1** To obtain $a\ln y$ or $b\ln(x+1) + c\ln(3x+1)$

Obtain term $\ln y$ from integral of $1/y$ **B1**

State or imply the form $\frac{A}{x+1} + \frac{B}{3x+1}$ and use a correct method to find a constant **M1**

Obtain $A = -\frac{1}{2}$ and $B = \frac{3}{2}$ **A1**

Obtain terms $-\frac{1}{2}\ln(x+1) + \frac{1}{2}\ln(3x+1)$ or $-\frac{1}{2}\ln(2x+2) + \frac{1}{2}\ln(6x+2)$ or combination of these terms **A1 FT + A1 FT** The FT is on the values of A and B.

Use $x = 1$ and $y = 1$ to evaluate a constant, or expression for a constant, or as limits, in a solution containing terms $a\ln y$, $b\ln(x+1)$ and $c\ln(3x+1)$, where $abc \neq 0$ *M1 e.g. $\ln y = -\frac{1}{2}\ln(x+1) + \frac{1}{2}\ln(3x+1) - \frac{1}{2}\ln 2$

Obtain an expression for y or y^2 and substitute $x = 3$ **DM1** Do not accept decimal equivalent of $\ln$ terms

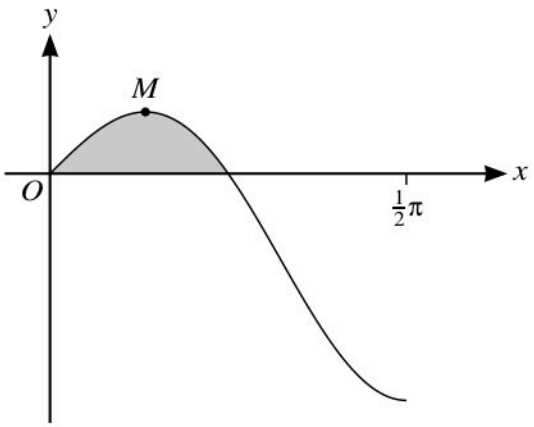
Obtain answer $y = \frac{1}{2}\sqrt{5}$ or $\sqrt{\frac{5}{4}}$ or $\sqrt{\frac{10}{8}}$ **A1** ISW. Must be simplified and exact, do not allow 1.118 or $e^{\frac{1}{2}\ln\frac{5}{4}}$.

Total: 9

Q32 Feb/Mar 2022 | Paper 3 (variant 32) | Question 11

Topics: Trigonometry, Differentiation, Integration

11



The diagram shows the curve $y = \sin x \cos 2x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M.

(a) Find the x-coordinate of M, giving your answer correct to 3 significant figures. [6]

The diagram shows the curve $y = \sin x \cos 2x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M.

(a) Find the x-coordinate of M, giving your answer correct to 3 significant figures. [6]

(b) Using the substitution $u = \cos x$, find the area of the shaded region enclosed by the curve and the x-axis in the first quadrant, giving your answer in a simplified exact form. [5]

Examiner comment

> "2sin2x was often replaced by 2sinxcosx making it very difficult to know if the double angle substitution was correct."

> "(b) Many candidates struggled to establish a correct integral in terms of u, however many good attempts were seen."

> "A common error was the inaccurate replacement of cos2x by 2cos2x – 1 accurately."

Mark scheme

(a)

Use correct product rule or chain rule **M1**

Obtain correct derivative in any form **A1** $\cos x \cos 2x - \sin x\, 2 \sin 2x$

Equate derivative to zero and use a correct double angle formula ***M1** If chain rule used then derivative set to 0 gains M1 since correct double angle formula has already been used.

Obtain an equation in one trigonometric variable **DM1** Allow following from coefficient errors in differentiation only

Obtain $6 \sin^2 x = 1$, $6 \cos^2 x = 5$ or $5 \tan^2 x = 1$ **A1** One of these 3 expressions

Obtain final answer $x = 0.421$ **A1** Must be 3 s.f.

Total: 6

(b)

State or imply $du = -\sin x\, dx$ **B1**

Using double angle formula, express integral in terms of u and du **M1** Use $\cos 2x = 2\cos^2 x - 1$

Integrate and obtain $\pm\left(u - \frac{2}{3}u^3\right)$ **A1**

Use limits $u = 1$, $u = \frac{1}{\sqrt{2}}$ in an integral of the form $au + bu^3$, where $ab \neq 0$ **M1** Require both limits substituted twice in $au + bu^3$ for M1. Do not condone decimals.

Obtain $\frac{1}{3}(\sqrt{2} - 1)$ or $\frac{1}{3}\sqrt{2} - \frac{1}{3}$ or $\frac{2}{3}\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{3}$ or simplified equivalent **A1** ISW

Total: 5

Q33 May/June 2022 | Paper 3 (variant 31) | Question 1

Topics: Indices, Logarithms & Exponents

Solve the equation $2(3^{2x-1}) = 4^{x+1}$, giving your answer correct to 2 decimal places. **[4]**

Examiner comment

> "A few candidates were successful in 32x reducing the given equation to =6, which is relatively straightforward to solve."

Mark scheme

Use law of the logarithm of a product or a quotient or a power *M1

Obtain a correct linear equation in any form **A1**

e.g. $\ln 2 + (2x - 1)\ln 3 = (x + 1)\ln 4$ or $\log_3 2 + (2x - 1)\log_3 3 = (2x + 2)\log_2 2$

Solve for x **DM1**

Allow for unsimplified expression $x = \ldots$. Allow M1 M1 for $x = 1.45$ from $6^{2x-1} = 4^{x+1}$.

Obtain answer $x = 2.21$ **A1**

The question asks for 2 dp.

Alternative method for question 1

Correct use of indices to obtain $2.25^s = 6$ or $1.5^{2x} = 6$ **M1 A1**

Correct use of logarithms to solve for x **M1**

Allow solution of $2.25^s = 6$ by trial and improvement as far as 2.2...

Obtain answer $x = 2.21$ **A1**

Need to see an intermediate step / sequence of iterations.

Total: 4

Q34 May/June 2022 | Paper 3 (variant 31) | Question 2

Topics: Algebra

(a) Expand $(2-x^2)^{-2}$ in ascending powers of x, up to and including the term in x^4, simplifying the coefficients. **[4]**

(b) State the set of values of x for which the expansion is valid. **[1]**

Examiner comment

> "The most common errors were in taking out the factor of 2 incorrectly, either not obtaining 2–2 or going on to expand(1–x2)–2 ."

> "(b) Many candidates did not attempt this question."

Mark scheme

(a) State a correct unsimplified version of the x^2 or the x^4 term of the expansion of $(2-x^2)^{-2}$ or $\left(1-\frac{1}{2}x^2\right)^{-2}$ **M1**

$$\frac{1}{4}\left(1+2\frac{x^2}{2}+\frac{-2\cdot-3}{2}\left(\frac{x^2}{2}\right)^2\ldots\right)$$

Symbolic binomial coefficients are not sufficient for the M1.

State correct first term **B1**

Accept 2^{-2}.

Obtain the next two terms **A1 A1**

$$\frac{1}{4}x^2+\frac{3}{16}x^4$$

A1 for each one correct ISW. Full marks for $\frac{1}{4}\left(1+x^2+\frac{3}{4}x^4\right)$ ISW.

SC allow M1 A1 A1 for $\frac{2}{4}$ and $1+x^2+\frac{3}{4}x^4$ SOI.

SC allow M1 A1 for $1+x^2+\frac{3}{4}x^4$

Total: 4

(b) State answer $|x|<\sqrt{2}$ **B1**

Or $-\sqrt{2}<x<\sqrt{2}$.

Total: 1

Q35 May/June 2022 | Paper 3 (variant 31) | Question 3

Topics: Trigonometry

Solve the equation $2\cot 2x + 3\cot x = 5$, for $0^\circ < x < 180^\circ$. **[6]**

Examiner comment

> "The most common errors seen were in the algebra when simplifying to obtain the quadratic 1 1 1 equation."

Mark scheme

Use correct trigonometric formula to form an equation in $\tan x$ *M1

e.g. $\frac{1-\tan^2 x}{\tan x} + \frac{3}{\tan x} = 5$

Obtain a correct linear equation in any form A1

$1 - \tan^2 x + 3 = 5\tan x$

Reduce equation to a 3-term quadratic A1

$\tan^2 x + 5\tan x - 4 = 0$, or 3-term equivalent

Solve a 3-term quadratic in $\tan x$ and obtain a value of x DM1

Obtain answer, e.g. $x = 35.1^\circ$ A1

Obtain second answer, e.g. $x = 99.9^\circ$, and no other in $(0^\circ, 180^\circ)$ A1

Ignore answers outside $(0^\circ, 180^\circ)$. Treat answers in radians $(0.612, 1.74)$ as a misread.

Alternative method for question 3

Use correct formulae for $\sin 2x$ and $\cos 2x$ to form an equation in $\sin x$ and $\cos x$ *M1

Obtain $4\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x} = 5$ A1

Reduce equation to a 3-term quadratic A1

$\tan^2 x + 5\tan x - 4 = 0$, or 3-term equivalent

Solve a 3-term quadratic in $\tan x$ and obtain a value of x DM1

Obtain answer, e.g. $x = 35.1^\circ$ A1

Obtain second answer, e.g. $x = 99.9^\circ$, and no other in $(0^\circ, 180^\circ)$ A1

Ignore answers outside $(0^\circ, 180^\circ)$. Treat answers in radians $(0.612, 1.74)$ as a misread.

Total: 6

Q36 May/June 2022 | Paper 3 (variant 31) | Question 7

Topics: Complex Numbers

The complex number u is defined by $u = \frac{\sqrt{2}-av2i}{1+2i}$, where a is a positive integer.

(a) Express u in terms of a, in the form $x + iy$, where x and y are real and exact. **[3]**

It is now given that $a = 3$.

(b) Express u in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$, giving the exact values of r and θ. **[2]**

(c) Using your answer to part **(b)**, find the two square roots of u. Give your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$, giving the exact values of r and θ. **[3]**

Examiner comment

> "(c) A few candidates followed through their answer to part (b) correctly."

Mark scheme

(a) Multiply numerator and denominator by $1 - 2i$, or equivalent **M1**

At least one multiplication completed.

Obtain correct numerator $(1 - 2a)\sqrt{2} - (2 + a)\sqrt{2}i$ **A1**

OE

Obtain final answer $\frac{1-2a}{5}\sqrt{2} - \frac{2+a}{5}\sqrt{2}i$ **A1**

OE

Alternative method for question 7(a)

Multiply $x + iy$ by $1 + 2i$ and compare real and imaginary parts **M1**

Obtain $x - 2y = \sqrt{2}$ and $2x + y = a\sqrt{2}$ **A1**

Obtain final answer $\frac{1-2a}{5}\sqrt{2} - \frac{2+a}{5}\sqrt{2}i$ **A1**

OE

Total: 3

(b) Obtain $r = 2$ **B1 FT**

Obtain $\theta = -\frac{3}{4}\pi$ **B1**

Total: 2

(c) Use correct method to find r or θ **M1**

State answer $\sqrt{2}e^{\frac{3\pi}{8}i}$ **A1 FT**

State answer $\sqrt{2}e^{\frac{5\pi}{8}i}$ **A1 FT**

Total: 3

Q37 May/June 2022 | Paper 3 (variant 31) | Question 8

Topics: Differentiation

The equation of a curve is $x^3 + y^3 + 2xy + 8 = 0$.

(a) Express $\frac{dy}{dx}$ in terms of x and y. **[4]**

The tangent to the curve at the point where $x = 0$ and the tangent at the point where $y = 0$ intersect at the acute angle α.

(b) Find the exact value of $\tan\alpha$. **[5]**

Examiner comment

> "(b) This was an unusual question, and many candidates did not know where to start."

Mark scheme

(a) State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

State or imply $2y + 2x\frac{dy}{dx}$ as derivative of $2xy$ **B1**

Complete differentiation and equate attempted derivative to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain answer $-\frac{3x^2+2y}{3y^2+2x}$ **A1**

Total: 4

(b) Find gradient at either $(0, -2)$ or $(-2, 0)$ **M1**

Obtain answers $-\frac{1}{3}$ and 3 **A1 A1**

Use $\tan(A \pm B)$ formula to find $\tan\alpha$ **M1**

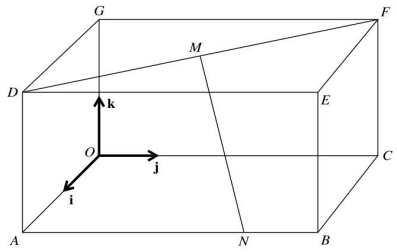
Obtain answer $\tan\alpha = \frac{4}{3}$ **A1**

Total: 5

Q38 May/June 2022 | Paper 3 (variant 31) | Question 9

Topics: none of the listed topics

9



In the diagram, $OABCDEFG$ is a cuboid in which $OA = 2$ units, $OC = 4$ units and $OG = 2$ units. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OG respectively. The point M is the midpoint of DF. The point N on AB is such that $AN = 3NB$.

(a) Express the vectors $\overrightarrow{OM}$ and $\overrightarrow{MN}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [3]

.....

.....

.....

.....

.....

.....

.....

(b) Find a vector equation for the line through M and N. [2]

In the diagram, $OABCDEFG$ is a cuboid in which $OA = 2$ units, $OC = 4$ units and $OG = 2$ units. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OG respectively. The point M is the midpoint of DF. The point N on AB is such that $AN = 3NB$.

(a) Express the vectors $\overrightarrow{OM}$ and $\overrightarrow{MN}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. **[3]**

(b) Find a vector equation for the line through M and N. **[2]**

(c) Show that the length of the perpendicular from O to the line through M and N is $\sqrt{\frac{53}{6}}$. **[4]**

Examiner comment

> "The most common error was in writing the equation without r =...."

> "(c) This part of the question was more challenging."

Mark scheme

(a) Obtain $\overrightarrow{OM} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ **B1**

Use a correct method to find $\overrightarrow{MN}$ **M1**

e.g. $\overrightarrow{MO} + \overrightarrow{OA} + \overrightarrow{AN}$ or $\overrightarrow{MO} + \overrightarrow{ON}$

Obtain $\overrightarrow{MN} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$ **A1**

Accept any notation.

Total: 3

(b) Use a correct method to form an equation for $\overrightarrow{MN}$ **M1**

Allow without $r = \ldots$

Obtain $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} + \lambda(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ **A1 FT**

OE e.g. $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \mu(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$

Must have $\mathbf{r} = \ldots$ Follow *their* answers to part 9(a).

Total: 2

(c) State $\overrightarrow{OP}$ for a general point P on MN in component form, e.g. $(2 + \lambda, 3 + \lambda, -2\lambda)$ **B1**

Equate scalar product of $\overrightarrow{OP}$ and a direction vector for MN to zero and solve for λ **M1**

Obtain $\lambda = -\frac{5}{6}$ **A1**

OE e.g. $\mu = \frac{1}{6}$

Obtain $\sqrt{\frac{53}{6}}$ correctly **A1**

AG e.g. from $\sqrt{\left(\frac{7}{6}\right)^2 + \left(\frac{13}{6}\right)^2 + \left(\frac{5}{3}\right)^2}$

Total: 4

Q39 May/June 2022 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

1 Solve the equation $\ln(e^{2x} + 3) = 2x + \ln 3$, giving your answer correct to 3 decimal places. **[4]**

Examiner comment

> "( ) The most common error was an incorrect initial statement such asln e2x +3 =lne2x +ln3."

Mark scheme

Step	Description	Mark	Notes
	Use law of the logarithm of a product, power or quotient or a law of indices (on an expression that is relevant to the question)	**M1**	e.g. $\ln(e^{2x} + 3) - \ln 3 = \ln\left(\frac{e^{2x}+3}{3}\right)$ or $e^{(2x+\ln 3)} = e^{2x}e^{\ln 3}$
	State a correct equation without logs (in any form)	**A1**	e.g. $3 + e^{2x} = 3e^{2x}$
	Carry out correct method to solve an equation of the form $e^{2x} = a$, where $a > 0$, or for solving $e^x = b$ ($b > 0$) if they have already taken the square root	**M1**	Allow for $x = \frac{1}{2}\ln\frac{1}{2}$. **M1** can be implied by correct answer.
	Obtain answer $x = 0.203$	**A1**	CAO. The question requires 3 d.p. Answer only with no working shown is 0/4.
		4	

Total: 4

Q40 May/June 2022 | Paper 3 (variant 32) | Question 3

Topics: Algebra

3 The polynomial $ax^3 + x^2 + bx + 3$ is denoted by p(x). It is given that p(x) is divisible by $(2x - 1)$ and that when p(x) is divided by $(x + 2)$ the remainder is 5.

Find the values of a and b. **[5]**

Examiner comment

> "A common mistake was to substitute x =3 into the polynomial instead of x =−2 (from 5−2=3)."

Mark scheme

Step	Description	Mark	Notes
	Substitute $x = \frac{1}{2}$, equate result to zero	**M1**	Or divide by $2x - 1$ and equate constant remainder to zero.
	Obtain a correct simplified equation	**A1**	e.g. $\frac{1}{8}a + \frac{1}{4} + \frac{1}{2}b + 3 = 0$ or $a + 4b = -26$
	Substitute $x = -2$, equate result to 5	**M1**	Or divide by $x + 2$ and equate constant remainder to 5.
	Obtain a correct simplified equation	**A1**	e.g. $-8a + 4 - 2b + 3 = 5$ or $8a + 2b = 2$
	Obtain $a = 2$ and $b = -7$	**A1**	WWW
		5	

Total: 5

Q41 May/June 2022 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry, Differentiation

4 The equation of a curve is $y = \cos^3 x\sqrt{\sin x}$. It is given that the curve has one stationary point in the interval $0 < x < \frac{1}{2}\pi$.

Find the x-coordinate of this stationary point, giving your answer correct to 3 significant figures. **[6]**

Examiner comment

> "Question 4 The differentiation in this question proved to be quite challenging."

Mark scheme

Step	Description	Mark	Notes
	Use the correct product rule and then the chain rule to differentiate either $\cos^3 x$ or $\sqrt{\sin x}$	**M1**	e.g. two terms with one part of $\frac{dy}{dx} = p\cos^2 x \sin x\sqrt{\sin x} + q\frac{\cos^3 x \cos x}{\sqrt{\sin x}}$.
	Obtain correct derivative in any form e.g. $\frac{dy}{dx} = -3\cos^2 x \sin x\sqrt{\sin x} + \frac{\cos^3 x \cos x}{2\sqrt{\sin x}}$	**A1 A1**	**A1** for each correct term substituted in the complete derivative.
	Equate their derivative to zero and obtain a horizontal equation with positive integer powers of $\sin x$ and/or $\cos x$ from an equation including $\sqrt{\sin x}$ or $\frac{1}{\sqrt{\sin x}}$ using sensible algebra.	**M1**	e.g. $-3\cos^2 x \sin^2 x + \frac{1}{2}\cos^4 x = 0$
	Use correct formula(s) to express their equation/derivative in terms of one trigonometric function	**M1**	Can be awarded before the previous **M1**. May involve more than one trigonometric term.

| | Obtain $7\cos^2 x = 6$, $7\sin^2 x = 1$, or $6\tan^2 x = 1$, or equivalent, and obtain answer $x = 0.388$ | **A1** | CAO. The question asks for 3 sf. Ignore additional answers outside $\left(0, \frac{\pi}{2}\right)$. 22.2° is Ao. |
| | | **6** | |

Total: 6

Q42 May/June 2022 | Paper 3 (variant 32) | Question 7

Topics: Differentiation

7 The equation of a curve is $x^3 + 3x^2y - y^3 = 3$.

(a) Show that $\frac{dy}{dx} = \frac{x^2+2xy}{y^2-x^2}$. **[4]**

(b) Find the coordinates of the points on the curve where the tangent is parallel to the x-axis. **[5]**

Examiner comment

> "A significant minority of candidates was not familiar with implicit differentiation and so could make no progress."

> "The most common error to overlook the factor of x in the numerator, or to reject it as not relevant."

Mark scheme

Part	Step	Description	Mark	Notes
7(a)		State or imply $6xy + 3x^2\frac{dy}{dx}$ as derivative of $3x^2y$	**B1**	Allow **B1 B1** for $\left(3x^2dx + 6xydy + 3x^2dy - 3y^2dy = [0]\right)$
		State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3	**B1**	
		Equate attempted derivative of left-hand side to zero and solve to obtain an equation with $\frac{dy}{dx}$ as subject	**M1**	Allow if zero implied by subsequent working. Allow if recover from an extra $\frac{dy}{dx} = \ldots$ at the beginning of the left-hand side.
		Obtain $\frac{dy}{dx} = \frac{x^2+2xy}{y^2-x^2}$ correctly	**A1**	AG Accept y' for $\frac{dy}{dx}$.
			4	
7(b)		Equate numerator to zero	*M1**	Must be using the given derivative.
		Obtain $x = -2y$, or equivalent	**A1**	An equation with x or y as the subject SOI.
		Use $x^3 + 3x^2y - y^3 = 3$ to obtain an equation in x or y	**DM1**	$-8y^3 + 12y^3 - y^3 = 3$ or $x^3 - \frac{3}{8}x^3 + \frac{1}{8}x^3 = 3$ or any equivalent form (do not need to evaluate powers).
		Obtain the point $(-2, 1)$ and no others from solving their cubic equation	**A1**	Allow if each component stated separately. ISW.
		State the point $(0, -\sqrt{3})$, or equivalent from correct work	**B1**	Accept $(0, -\sqrt{3})$, or $(0, -1.44)$ (-1.44225). Allow if each component stated separately. ISW.
			5	

Total: 9

Q43 May/June 2022 | Paper 3 (variant 32) | Question 9

Topics: none of the listed topics

9 The lines l and m have vector equations

$$\mathbf{r} = -\mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} - \mathbf{k}) \quad \text{and} \quad \mathbf{r} = 5\mathbf{i} + 4\mathbf{j} + 3\mathbf{k} + \mu(a\mathbf{i} + b\mathbf{j} + \mathbf{k})$$

respectively, where a and b are constants.

(a) Given that l and m intersect, show that $2b - a = 4$. **[4]**

(b) Given also that l and m are perpendicular, find the values of a and b. **[4]**

(c) When a and b have these values, find the position vector of the point of intersection of l and m. **[2]**

Examiner comment

> "Question 9 (a) Although quite simple, this task was an unfamiliar one and proved challenging to many candidates."

> "The most common errors were in the arithmetic and algebra."

Mark scheme

Part	Step	Description	Mark	Notes
9(a)		Express general point of l or m in component form, i.e. $(-1 + 2\lambda, 3 - \lambda, 4 - \lambda)$ or $(5 + a\mu, 4 + b\mu, 3 + \mu)$	**B1**	
		Equate components and eliminate either λ or μ	**M1**	e.g. $\mu = \frac{-\lambda}{1-b}, \lambda = \frac{-1-b}{1-b}, \mu = \frac{a+6}{a+2}$
		Eliminate the other parameter or obtain a second expression in the first parameters.	**M1**	λ and μ are not required to be the subject of the equations.
		Show intermediate steps to obtain $2b - a = 4$	**A1**	AG
			4	
		Alternative method for question 9(a)		
		Express general point of l or m in component form, i.e. $(-1 + 2\lambda, 3 - \lambda, 4 - \lambda)$ or $(5 + a\mu, 4 + b\mu, 3 + \mu)$	**B1**	
		Express a or b in terms of λ and μ	**M1**	$a = \frac{2\lambda-6}{\mu}, b = \frac{-1-\lambda}{\mu}, \mu = \frac{a+6}{a+2}$
		Use $\lambda = 1 - \mu$	**M1**	
		Obtain $2b - a = 4$	**A1**	AG
			4	
9(b)		Using the correct process equate the scalar product of the direction vectors to zero	***M1**	$(2\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (a\mathbf{i} + b\mathbf{j} + \mathbf{k}) = 0$ SOI.
		Obtain $2a - b - 1 = 0$	**A1**	OE e.g. $2(2b - 4) - b - 1 = 0$
		Solve simultaneous equations for a or for b	**DM1**	
		Obtain $a = 2, b = 3$	**A1**	
			4	
9(c)				

| | | | **2** | |

Total: 10

Q44 May/June 2022 | Paper 3 (variant 33) | Question 2

Topics: Trigonometry

2 Solve the equation $\cos(\theta - 60^\circ) = 3\sin\theta$, for $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Examiner comment

> "However, a common error was to show tan θ expressed as 6– 3 instead of its reciprocal, cosθ either through a simple algebraic error or mistakenly thinking that tanθ was ."

Mark scheme

Step	Description	Mark	Notes
	Use correct $\cos(A-B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$	**B1**	$\cos\theta\cos 60^\circ + \sin\theta\sin 60^\circ = 3\sin\theta$
	Use trigonometric formula and substitute values for $\cos 60$ and $\sin 60$ to obtain an equation in $\tan\theta$ (or $\cos\theta$ or $\sin\theta$)	**M1**	Allow $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$ interchanged. $\frac{1}{2} + \frac{\sqrt{3}}{2}\tan\theta = 3\tan\theta$ or $\frac{1}{4}\cos^2\theta = \left[3 - \frac{\sqrt{3}}{2}\right]\left[3 - \frac{\sqrt{3}}{2}\right](1 - \cos^2\theta)$ or $\frac{1}{4}(1-\sin^2\theta) = \left[3 - \frac{\sqrt{3}}{2}\right]\left[3 - \frac{\sqrt{3}}{2}\right]\sin^2\theta$
	Obtain $\tan\theta = \frac{1}{6-\sqrt{3}}$ or $\tan\theta = \frac{6+\sqrt{3}}{33}$ or 0.2343, $\cos\theta = \frac{3\sqrt{3}}{2\sqrt{10-3\sqrt{3}}}$ or 0.9736 or $\sin\theta = \frac{1}{2\sqrt{10-3\sqrt{3}}}$ or 0.2281	**A1**	OE
	Obtain answer, e.g. $\theta = 13.2^\circ$	**A1**	May be more accurate, allow value rounding to 13.2°. $\theta = 13.1867^\circ$.
	Obtain second answer, e.g. $\theta = 193.2^\circ$ and no others in the given interval	**A1 FT**	May be more accurate. Allow value rounding to 193.2°. **FT** is on previous value of θ, must have scored **M1**. Note if θ is negative (e.g. -13.2): $-13.2 + 180 = 166.8$ A0 but $-13.2 + 360 = 346.8$ A1 FT. Ignore answers outside the given interval. Treat answers in radians as a misread. $0.23015, 3.3717$.
			5	

Total: 5

Q45 May/June 2022 | Paper 3 (variant 33) | Question 3

Topics: Algebra, Indices, Logarithms & Exponents

(a) Show that the equation $\log_3(2x+1) = 1 + 2\log_3(x-1)$ can be written as a quadratic equation in x. **[3]**

(b) Hence solve the equation $\log_3(4y+1) = 1 + 2\log_3(2y-1)$, giving your answer correct to 2 decimal places. **[2]**

Examiner comment

> "© 2022 Cambridge International Advanced Level 9709 Mathematics June 2022 Principal Examiner Report for Teachers (b) A considerable number of candidates found this question difficult."

Mark scheme

Part	Step	Description	Mark	Notes
3(a)		Use law of logarithm of a power	**M1**	$\log_3(2x+1) = 1 + \log_3(x-1)^2$
		Use $\log_3 3 = 1$	**B1**	$\log_3(2x+1) = \log_3 3 + \log_3(x-1)$
			SC For candidates scoring M0 B0 due to combining logs before dealing with coefficient 2, and confusing coefficients, allow $\log_3(\ldots) = c$ leading to $(\ldots) = 3^c$ **B1**.
		Obtain $3x^2 - 8x + 2 = 0$ or $1.5x^2 - 4x + 1 = 0$	**A1**	OE 3 terms only and $= 0$ required.
			3	
3(b)		Solve 3-term quadratic equation from part **(a)** or restart to find y	**M1**	$\frac{4 \pm \sqrt{10}}{6}$ or $y = 1.1937\ldots$ or $y = 0.1396\ldots$ $(x = 2.3874$ or $x = 0.2792)$ May solve for x but must find $y = \frac{x}{2}$ to gain **M1**.
		Obtain answer 1.19	**A1**	CAO. 2 dp required.
			2	

Total: 5

Q46 May/June 2022 | Paper 3 (variant 33) | Question 5

Topics: Complex Numbers

5 The complex number $3 - i$ is denoted by u.

(a) Show, on an Argand diagram with origin O, the points A, B and C representing the complex numbers u, u^* and $u^* - u$ respectively.

State the type of quadrilateral formed by the points O, A, B and C. **[3]**

(b) Express $\frac{u^*}{u}$ in the form $x + iy$, where x and y are real. **[3]**

(c) By considering the argument of $\frac{u^*}{u}$, or otherwise, prove that $\tan^{-1}\left(\frac{3}{4}\right) = 2\tan^{-1}\left(\frac{1}{3}\right)$. **[2]**

Examiner comment

> "Question 5 (a) Most candidates correctly evaluated u* and u* – u, but some did not show any scale on their Argand diagram when plotting the positions of points A, B and C."

Mark scheme

Part	Step	Description	Mark	Notes
5(a)		Show u and u^* in relatively correct positions. Must have sense of scale on axes	**B1**	$u = 3 - i$, $u^* = 3 + i$ Ignore labels.
		Show $u^* - u$ in a relatively correct position. Must have sense of scale on axes	**B1**	$2i$. Scale only on Imaginary axis is sufficient for this mark.
		State that $OABC$ is a parallelogram [independent of previous marks]	**B1**	Ignore 'quadrilateral'. Allow 'trapezium' from correct work.
			3	
5(b)		Multiply their numerator and the given denominator by $3 + i$ and attempt to evaluate either	**M1**	

| Can have missing term and arithmetic errors but need $i^2 = -1$ once, seen or implied. |
		Obtain numerator $8 + 6i$ or denominator 10	**A1**	
		State final answer $\frac{4}{5} + \frac{3}{5}i$ or $0.8 + 0.6i$	**A1**	Correct answer with no working scores 0/3.
			3	
5(c)		State or imply $\arg\frac{u^*}{u} = \arg u^* - \arg u$ or $2\arg u$ or $2\arg u^*$	**M1**	
		Justify the given statement correctly	**A1**	AG $\arg\frac{u^*}{u} = \tan^{-1}\frac{3}{4}$, $\arg u^* = \tan^{-1}\frac{1}{3}$ and $\arg u = \tan^{-1}\left(-\frac{1}{3}\right)$ (or $\arg u = -\tan^{-1}\frac{1}{3}$), needed if use first expression in **M1**; or $\arg\frac{u^*}{u} = 2\tan^{-1}\frac{3}{4}$ $\arg u^* = \tan^{-1}\frac{1}{3}$, needed if use second expression in **M1**.
			2	

Total: 8

Q47 May/June 2022 | Paper 3 (variant 33) | Question 8

Topics: Differential Equations

At time t days after the start of observations, the number of insects in a population is N. The variation in the number of insects is modelled by a differential equation of the form $\frac{dN}{dt} = kN^{\frac{3}{2}}\cos 0.02t$, where k is a constant and N is a continuous variable. It is given that when $t = 0$, $N = 100$.

(a) Solve the differential equation, obtaining a relation between N, k and t. **[5]**

(b) Given also that $N = 625$ when $t = 50$, find the value of k. **[2]**

(c) Obtain an expression for N in terms of t, and find the greatest value of N predicted by this model. **[2]**

Examiner comment

> "A common error led to 0.02 in the numerator rather than the denominator."

> "(c) Again, candidates found the algebra difficult and struggled to convert to an expression for N in terms of t with their value of k substituted."

Mark scheme

8(a) Separate variables correctly. **B1**

$$\frac{dN}{N^{3/2}} = (k\cos 0.02t)dt$$

Allow without integral signs. **OE**

Obtain term $-\frac{2}{\sqrt{N}}$. **B1**

Obtain term $50\sin 0.02t$. **B1**

Use $t = 0, N = 100$ to evaluate a constant, or as limits, in a solution containing terms $\frac{a}{\sqrt{N}}$ and $b\sin 0.02t$, where $ab \neq 0$. **M1**

Obtain correct solution in any form, e.g. $-\frac{2}{\sqrt{N}} = 50k\sin 0.02t - 0.2$. **A1** ISW

e.g. $N = \frac{1}{(25k\sin 0.02t - 0.1)^2}$, $-2N^{-1/2} = \frac{k}{0.02}\sin 0.02t - \frac{1}{5}$, $\frac{2}{\sqrt{N}} = \frac{k}{50\sin 0.02t + \frac{1}{10}}$,
$50k\sin(\frac{1}{50}t) = -\frac{2}{\sqrt{N}} + \frac{1}{5}$.

Total: 5

8(b) Use the substitution $N = 625$ and $t = 50$ in expression of appropriate form to evaluate k. **M1**

Expression must contain $a + bk\sin 0.02t$, $(\sqrt{N})^{\pm n}$, where $n = -1, 1, 3$ or 5 and a and b are constants $ab \neq 0$ or $(a + bk\sin 0.02t)^{\pm n}$ and $(N)^{\pm n}$. Allow with k replaced by $\frac{1}{k}$. Error due to $k(N^{-3/2})$ when separating variables in **8(a)**. If invert term by term when 3 terms shown then M0.

Obtain $k = 0.00285[2148]$. **A1**

Total: 2

8(c) Rearrange and obtain $N = 4(0.2 - 0.142(607)\sin 0.02t)^{-2}$. **M1**
Substitution for k required.
Anything of the form $N = c(d - ek\sin 0.02t)^{-2}$, where c, d and ek are constants $cde \neq 0$ and value of k substituted. Allow with k replaced by $1/k$. Error due to $k(N^{-3/2})$ when separating variables in **8(a).**

OE ISW

e.g.

$$N = \left(-\frac{10}{0.7125\sin 0.02t - 1}\right)^2, \quad N = \frac{1}{(-0.0713\sin 0.02t + 0.1)^2}$$

$$N = \frac{100}{\left(\left(\frac{0.6}{\sin 1}\right)\sin 0.02t - 1\right)^2}, \quad N = \frac{1}{\left(-\frac{3}{-50\sin 1} \times \sin 0.02t + \frac{1}{10}\right)^2}$$

$$N = \left(-\frac{0.06}{\sin 1}\sin 0.02t + 0.1\right)^{-2}, \quad N = \left(\frac{800}{80 - 57\sin 0.02t}\right)^2$$

Do not need to substitute for $\sin(0.02t) = 1$, but must substitute for k.

Accept answers between 1209 and 1215. **A1** ISW

Substitute $\sin 0.02t = 1$ or $t = 50\sin^{-1} 1$ or 78.5 or 25π. Answer with no working (rubric) 0/2.

SC $N = \ldots$ not seen but correct numerical answer **B1** 1/2.

Total: 2

Q48 May/June 2022 | Paper 3 (variant 33) | Question 9

Topics: none of the listed topics

With respect to the origin O, the point A has position vector given by $\overrightarrow{OA} = \mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$. The line l has vector equation $\mathbf{r} = 4\mathbf{i} + \mathbf{k} + \lambda(-\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$.

(a) Find in degrees the acute angle between the directions of $\overrightarrow{OA}$ and l. **[3]**

(b) Find the position vector of the foot of the perpendicular from A to l. **[4]**

(c) Hence find the position vector of the reflection of A in l. **[2]**

Examiner comment

> "This would have avoided the very common error of setting to zero the scalar product of OP and the direction vector of l."

Mark scheme

9(a) Using the correct process find the scalar product of direction vectors of l and $\overrightarrow{OA}$. **M1**

$(1, 5, 6) \cdot (-1, 2, 3) = -1.1 + 5.2 + 6.3 = -1 + 10 + 18$

Using the correct process for the moduli, divide the scalar product by the product of the moduli and find the inverse cosine of the result. **M1**

Their scalar product $\div \sqrt{(1^2 + 5^2 + 6^2)}\sqrt{(-1)^2 + 2^2 + 3^2}$.

Angle $= \cos^{-1} \frac{27}{\sqrt{62}\sqrt{14}}$

Obtain answer 23.6°. **A1** AWRT 23.6°.

23.5889°. Radians 0.412 scores Ao ($0.4117\ldots$).

Total: 3

9(b) Taking a general point P on l, state $\overrightarrow{AP}$ (or $\overrightarrow{PA}$) in component form, e.g. $(3 - \lambda, -5 + 2\lambda, -5 + 3\lambda)$. **B1**

Note: $(4, 1, 0)$ or $(4, 1, 1)$, for $4\mathbf{i} + \mathbf{k}$ is not MR, but **M1** possible.

Either equate scalar product of $\overrightarrow{AP}$ and direction vector of l to zero and solve for λ or use Pythagoras in a relevant triangle and solve for λ. **M1**

$(3 - \lambda, -5 + 2\lambda, -5 + 3\lambda) \cdot (-1, 2, 3) = 0$

$-3 - 10 - 15 + \lambda + 4\lambda + 9\lambda = 0$

or let $\overrightarrow{OQ} = (4, 0, 1)$ so $\overrightarrow{AQ} = (3, -5, -5)$,
$\overrightarrow{QP} = (-\lambda, 2\lambda, 3\lambda)$, $\overrightarrow{AP} = (3 - \lambda, -5 + 2\lambda, -5 + 3\lambda)$ hence
$3^2 + (-5)^2 + (-5)^2 = (3 - \lambda)^2 + (-5 + 2\lambda)^2 + (-5 + 3\lambda)^2 + (-\lambda)^2 +

(2\lambda)^2 + (3\lambda)^2$

Obtain $\boldsymbol{\lambda = 2}$. **A1**

State that the position vector $\overrightarrow{OP}^*$ of the foot is $\mathbf{2i + 4j + 7k}$. **A1** OE

Condone coordinates.

Total: 4

9(c) Set up a correct method for finding the position vector of the reflection of A in l. **M1**

For all methods, allow a sign error in one component only:
$\overrightarrow{OA}' = \overrightarrow{OP}^* + (\overrightarrow{OP}^* - \overrightarrow{OA})$

their $(2, 4, 7) + ({}_t heir\ 2, 4, 7 - 1, 5, 6)$

or $\overrightarrow{OA}' = \overrightarrow{OP}^* - (\overrightarrow{OA} - \overrightarrow{OP}^*)$

their $(2, 4, 7) - (1, 5, 6 -_t heir\ 2, 4, 7)$

or $\overrightarrow{OA}' = \overrightarrow{OA} + 2(\overrightarrow{OP}^* - \overrightarrow{OA})$

$(1 + 2({}_t heir 2 - 1))$

$5 + 2({}_t heir 4 - 5)$

$(6 + 2({}_t heir 7 - 6))$

or midpoint $\overrightarrow{OP}^* = (\overrightarrow{OA} + \overrightarrow{OA}')/2$ with _their_ λ value substituted.

$1 + x$

$5 + y$

$6 + z = 2 \times_t heir\ 2, 4, 7$

Obtain answer $\mathbf{3i + 3j + 8k}$ or $3\left(\mathbf{i + j} + \frac{8}{3}\right)$. **A1** OE

Condone coordinates.

$x = 3, y = 3, z = 8$ A1.

No method shown and correct answer 2/2.
Total: 2

Q49 Oct/Nov 2022 | Paper 3 (variant 31) | Question 1

Topics: Algebra

1

(a) Sketch the graph of $y = |2x + 1|$.

[1]

(b) Solve the inequality $3x + 5 < |2x + 1|$.

[3]

Examiner comment

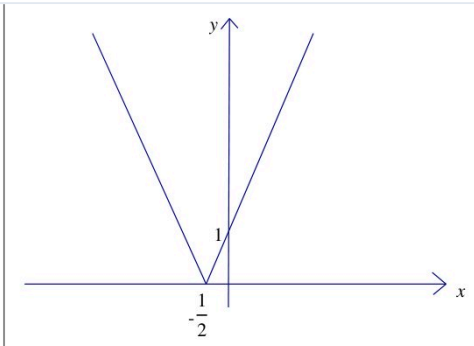
> "A few sketches did not consist of straight lines, but the most common error was to have the vertex on the vertical axis."

Mark scheme

1(a)

Show a recognisable sketch graph of $y = |2x + 1|$

B1

1(a)	Show a recognisable sketch graph of $y = \|2x+1\|$	B1	 Ignore $y = 3x + 5$ if also drawn on the sketch.
		1	

Total: 1

1(b)

Find x-coordinate of intersection with $y = 3x + 5$

M1

Obtain $x = -\frac{6}{5}$

A1

State final answer $x < -\frac{6}{5}$ only

A1 Do not condone $\leq$ for $<$ in the final answer.

Alternative method 1 for question 1(b)

Solve the linear inequality $3x + 5 < -(2x + 1)$, or corresponding equation

M1 Must solve the relevant equation.

Obtain critical value $x = -\frac{6}{5}$

A1 Ignore -4 if seen.

State final answer $x < -\frac{6}{5}$ only

A1

Alternative method 2 for question 1(b)

Solve the quadratic inequality $(3x+5)^2 < (2x+1)^2$, or corresponding equation

M1 $5x^2 + 26x + 24 < 0$

Obtain critical value $x = -\frac{6}{5}$

A1 Ignore -4 if seen.

State final answer $x < -\frac{6}{5}$ only

A1

Total: 3

Q50 Oct/Nov 2022 | Paper 3 (variant 31) | Question 3

Topics: Indices, Logarithms & Exponents

3 Solve the equation $2^{3x-1} = 5(3^{-x})$. Give your answer in the form $\frac{\ln a}{\ln b}$, where a and b are integers.

[4]

Examiner comment

> "The most common error was to rewrite 5×3−x as 15−x."

Mark scheme

3

Use law of logarithm of a product or power

M1 One correct application of a log law.

Obtain a correct linear equation in any form, e.g. $(3x-1)\ln 2 = \ln 5 - x\ln 3$

A1

Solve for x

M1 As far as $x = \ldots$ with only minor slips in processing.

Obtain answer $x = \frac{\ln 10}{\ln 24}$

A1

Alternative method for question 3

Use laws of indices to split at least one exponential term

M1 e.g. $\frac{2^{3x}}{2}$ or an arrangement with 8^x

Obtain $24^x = 10$

A1 OE

Solve for x

M1

Obtain answer $x = \frac{\ln 10}{\ln 24}$

A1

Total: 4

Q51 Oct/Nov 2022 | Paper 3 (variant 31) | Question 5

Topics: Complex Numbers

5 The complex numbers u and w are defined by $u = 2e^{\frac{i\pi}{6}}$ and $w = 3e^{\frac{5i\pi}{6}}$.

(a) Find $\frac{u^2}{w}$, giving your answer in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the exact values of r and θ.

[3]

(b) State the least positive integer n such that both $\text{Im } w^n = 0$ and $\text{Re } w^n > 0$.

[1]

Examiner comment

> "There were several blank responses, some answers that gave a negative real value, and some that appeared to be guesses."

Mark scheme

5(a)

State or imply $u^2 = 4e^{\frac{i\pi}{3}}$

B1

Obtain answer $y = \frac{4}{3}e^{\frac{i\pi}{6}}$

B1 + B1 For the modulus and the argument.

Total: 3

5(b)

State $n = 6$

B1

Total: 1

Q52 Oct/Nov 2022 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry

6

(a) Prove the identity $\cos 4\theta + 4\cos 2\theta + 3 = 8\cos^4\theta$.

[4]

(b) Hence solve the equation $\cos 4\theta + 4\cos 2\theta = 4$ for $0^\circ \leq \theta \leq 180^\circ$.

[3]

Examiner comment

> "Common errors came from slips in the algebra, in particular ( ) errors when squaring 2cos2θ−1."

Mark scheme

6(a)

Express $\cos 4\theta$ in terms of $\cos 2\theta$ and/or $\sin 2\theta$

B1

Express $\cos 2\theta$ in terms of $\cos\theta$ and/or $\sin\theta$

B1 Anywhere

Expand to obtain a correct expression in terms of $\cos\theta$

B1 e.g. $2(2\cos^2\theta - 1)^2 - 1 + 4(2\cos^2\theta - 1) + 3$

Reduce correctly to $\cos 4\theta + 4\cos 2\theta + 3 = 8\cos^4\theta$

B1 AG

Total: 4

6(b)

Use the identity and carry out method to calculate a root

M1 $8\cos^4\theta - 3 = 4$

Obtain answer, e.g. 14.7°

A1

Obtain second answer, e.g. 165.3°, and no other in the given interval

A1 FT Ignore answers outside the given interval. Treat answers in radians as a misread.

Total: 3

Q53 Oct/Nov 2022 | Paper 3 (variant 31) | Question 7

Topics: Trigonometry, Differentiation, Numerical Solutions/Iteration

7 The equation of a curve is $y = \frac{x}{\cos^2 x}$, for $0 \leq x < \frac{1}{2}\pi$. At the point where $x = a$, the tangent to the curve has gradient equal to 12.

(a) Show that $a = \cos^{-1}\left(\sqrt{\frac{3}{4}\frac{\cos a + 2a\sin a}{12}}\right)$.

[3]

(b) Verify by calculation that a lies between 0.9 and 1.

[2]

(c) Use an iterative formula based on the equation in part (a) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

[3]

Examiner comment

> "(b) There were several blank responses to this part of the question."

> "(c) There were several blank responses to this part."

Mark scheme

7(a)

Use correct product or quotient rule

M1

Obtain correct derivative in any form

A1 e.g. $\frac{dy}{dx} = \frac{\cos^2 x + 2x\sin x\cos x}{\cos^4 x}$ or $\frac{dy}{dx} = \sec^2 x + 2x\sec^2 x\tan x$

Equate derivative at $x = a$ to 12 and obtain $a = \cos^{-1}\left(\sqrt{\frac{\cos a + 2a\sin a}{12}}\right)$

A1 AG

Total: 3

7(b)

Evaluate a relevant expression or pair of expressions at $a = 0.9$ and $a = 1$

M1 Must be calculated in radians.

Complete the argument correctly with correct calculated values

A1 e.g. $\frac{\cos 0.9 = 0.622 > 0.553}{0.9 < 0.985}$, $\frac{0.0846 > 0}{1 > 0.964}$ or $\frac{-0.0358 < 0}{\text{or could be looking at values of the gradient } 8.46\&14.1}$

Total: 2

7(c)

Use the process $a_{n+1} = \cos^{-1}\left(\sqrt{\frac{\cos a_n + 2a_n \sin a_n}{12}}\right)$ correctly at least once

M1 Must be working in radians.

Obtain final answer 0.97

A1

Show sufficient iterations to 4 d.p. to justify 0.97 to 2 d.p., or show there is a sign change in the interval $(0.965, 0.975)$

A1 e.g. $0.95, 0.9743, 0.9694, 0.9704$

Total: 3

Q54 Oct/Nov 2022 | Paper 3 (variant 31) | Question 8

Topics: Differential Equations

8 In a certain chemical reaction the amount, x grams, of a substance is increasing. The differential equation satisfied by x and t, the time in seconds since the reaction began, is

$$\frac{dx}{dt} = kxe^{-0.1t},$$

where k is a positive constant. It is given that $x = 20$ at the start of the reaction.

(a) Solve the differential equation, obtaining a relation between x, t and k.

[5]

(b) Given that $x = 40$ when $t = 10$, find the value of k and find the value approached by x as t becomes large.

[3]

Examiner comment

> "(b) This part proved a challenge and there were many blank responses, often due to an incomplete response in part (a)."

Mark scheme

8(a)

Separate variables correctly

B1 $\int \frac{1}{x}dx = \int ke^{-0.1t}dt$

Obtain term $\ln x$

B1

Obtain term $-10ke^{-0.1t}$

B1 Not from $\int xe^{-0.1t}dt$

Use $x = 20, t = 0$ to evaluate a constant or as limits in a solution containing terms $a\ln x, be^{-0.1t}$ where $ab \neq 0$

M1

Obtain $\ln x = 10k(1 - e^{-0.1t}) + \ln 20$

A1 or equivalent ISW

Total: 5

8(b)

Use $x = 40, t = 10$ to find k or $10k$

M1 Available for their function of the correct structure even if they found no constant in (a).

Obtain $10k = 1.09654$

A1 or equivalent e.g. $10k = \frac{\ln 2}{1-e^{-1}}$

State that x tends to 59.9

A1 Need a number, not an expression for that value. 3 sf or better $59.87595\ldots$

Total: 3

Q55 Oct/Nov 2022 | Paper 3 (variant 31) | Question 10

Topics: Algebra

Question 10

Let $f(x) = \frac{2x^2+7x+8}{(1+x)(2+x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Examiner comment

> "(b) This part proved challenging with many candidates offering no attempt to use the binomial expansion."

> "A few candidates reached the correct conclusion but there were several errors in using the expansion and in simplifying the results."

Mark scheme

10(a) State or imply the form $\frac{A}{1+x} + \frac{B}{2+x} + \frac{C}{(2+x)^2}$ **B1**

Use a correct method to find a constant **M1**

Obtain one of $A = 3, B = -1$ and $C = -2$ **A1** (SR after B0 can score M1 A1 for one correct value)

Obtain a second value **A1**

Obtain the third value **A1** $\frac{A}{1+x} + \frac{Dx+E}{(2+x)^2}$, where $A = 3, D = -1$ and $E = -4$, is awarded **B1 M1 A1 A1 A1** as above.

Total: 5

10(b) Use a correct method to find the first two terms of the expansion of $(1+x)^{-1}, (2+x)^{-1}, \left(1+\frac{1}{2}x\right)^{-1}, (2+x)^{-2}$ or $\left(1+\frac{1}{2}x\right)^{-2}$ **M1** (For the A, D, E form of fractions, award M1 A1 FT A1 FT for the expanded partial fractions, then if $D \neq 0$, M1 for multiplying out fully, and A1 for the final answer.)

Obtain correct unsimplified expansions up to the term in x^2 of each partial fraction **A3 FT**
$3\left(1-x+x^2.....\right) - \frac{1}{2}\left(1-\frac{x}{2}+\frac{x^2}{4}.....\right) - \frac{2}{4}\left(1-x+\frac{3}{4}x^2\right)$

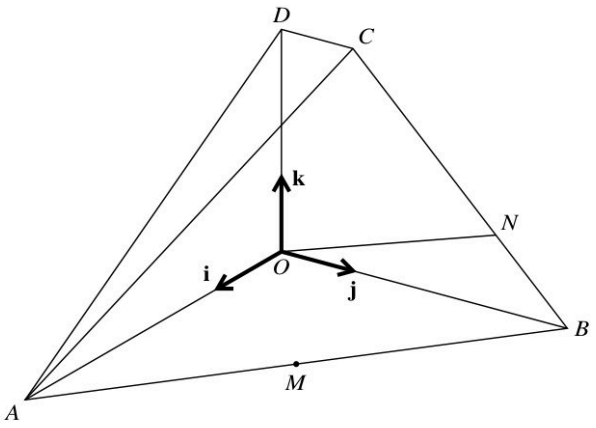
Obtain final answer $2-\frac{9}{4}x+\frac{5}{2}x^2$ **A1**

Q56 Oct/Nov 2022 | Paper 3 (variant 31) | Question 11

Topics: none of the listed topics

Question 11

11



In the diagram, $OABCD$ is a solid figure in which $OA = OB = 4$ units and $OD = 3$ units. The edge OD is vertical, DC is parallel to OB and $DC = 1$ unit. The base, OAB, is horizontal and angle $AOB = 90°$. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OD respectively. The midpoint of AB is M and the point N on BC is such that $CN = 2NB$.

(a) Express vectors $\overrightarrow{MD}$ and $\overrightarrow{ON}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [4]

In the diagram, $OABCD$ is a solid figure in which $OA = OB = 4$ units and $OD = 3$ units. The edge OD is vertical, DC is parallel to OB and $DC = 1$ unit. The base, OAB, is horizontal and angle $AOB = 90°$. Unit vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OD respectively. The midpoint of AB is M and the point N on BC is such that $CN = 2NB$.

(a) Express vectors $\overrightarrow{MD}$ and $\overrightarrow{ON}$ in terms of $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$. **[4]**

(b) Calculate the angle in degrees between the directions of $\overrightarrow{MD}$ and $\overrightarrow{ON}$. **[3]**

(c) Show that the length of the perpendicular from M to ON is $\sqrt{\frac{22}{5}}$. **[4]**

Examiner comment

> "Question 11 This question was challenging for many candidates and there was a high proportion of blank responses to parts (b) and (c)."

Mark scheme

11(a) State $\overrightarrow{OM} = 2\mathbf{i} + 2\mathbf{j}$ or equivalent **B1** (Can be implied by $\overrightarrow{MB} = -2\mathbf{i} + 2\mathbf{j}$ or $\overrightarrow{MA} = 2\mathbf{i} - 2\mathbf{j}$.)

Obtain $\overrightarrow{MD} = -2\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ **B1**

Use a correct method to find $\overrightarrow{ON}$ **M1** (e.g. $\overrightarrow{OC} + \frac{2}{3}\overrightarrow{CB}$)

Obtain answer $3\mathbf{j} + \mathbf{k}$ **A1**

Total: 4

11(b) Use the correct process for evaluating the scalar product of $\overrightarrow{MD}$ and $\overrightarrow{ON}$ **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and reach the inverse cosine of the result **M1** $\cos^{-1}\left(\frac{-6+3}{\sqrt{10}\sqrt{17}}\right)$

Obtain final answer 103.3° **A1**

Total: 3

11(c) Taking a general point P of ON to have position vector $\lambda(3\mathbf{j} + \mathbf{k})$, form an equation in λ by *either* equating the scalar product of $\overrightarrow{ON}$ and $\overrightarrow{MP}$ to zero, *or* applying Pythagoras to triangle OMP, or equating the derivative of $|\overrightarrow{MP}|$ to zero **M1** (e.g. $(-2)(0) - 2 + 3\lambda(\lambda)(1) = 0$)

Solve and obtain $\lambda = \frac{3}{5}$ **A1**

Substitute for λ and calculate $\overrightarrow{MP}$ **M1** $\overrightarrow{MP} = -2\mathbf{i} - \frac{1}{5}\mathbf{j} + \frac{3}{5}\mathbf{k}$

Obtain $\sqrt{\frac{22}{5}}$ **A1** (AG)

Alternative method for question 11(c)

Use a scalar product to find the projection OQ of OM on OM **M1**

Obtain $OQ = \frac{6}{\sqrt{10}}$ **A1**

Use Pythagoras in triangle OMQ to find MQ **M1**

Obtain $\sqrt{\frac{22}{5}}$ **A1** (AG)

Q57 Oct/Nov 2022 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Question 1

Solve the equation $2^{3x-1} = 5(3^{1-x})$. Give your answer in the form $\frac{\ln a}{\ln b}$ where a and b are integers. **[4]**

Examiner comment

> "The most common error was to rewrite 5×31−x as 151−x."

Mark scheme

1 Use law of the logarithm of a power or product **M1** (Ignoring the 3 or the 5 is not a misread.)

Obtain a correct linear equation in any form, e.g. $(3x-1)\ln 2 = \ln 5 + (1-x)\ln 3$ **A1** (Condone invisible brackets if they are used correctly later.)

Solve for x **M1** (Get as far as $x = \ldots$ Condone minor slips in the processing e.g. sign errors and losing a term that had been there, but award M0 for a fundamental error e.g. $3x\ln 2 + x\ln 3 = 3x\ln 6$ or ignoring the 3 or the 5 completely. Condone working in decimals.)

Obtain **final** answer $x = \frac{\ln 30}{\ln 24}$ **A1** (Do not ISW)

Alternative method for question 1

Use laws of indices to split at least one exponential term **M1** (e.g. $\frac{2^{3x}}{2}$ or an arrangement with 8^x and/or 3^x.)

Obtain $24^x = 30$ **A1** (Or equivalent e.g. Double exponent: use braces to clarify not for Double exponent: use braces to clarify (need two factors with the same index).)

Solve for x **M1** (Get as far as $x = \ldots$)

Obtain **final** answer $x = \frac{\ln 30}{\ln 24}$ **A1** (Do not ISW)

Q58 Oct/Nov 2022 | Paper 3 (variant 32) | Question 2

Topics: Algebra

Question 2

The polynomial $2x^3 - x^2 + a$, where a is a constant, is denoted by p(x). It is given that $(2x+3)$ is a factor of p(x).

(a) Find the value of a. **[2]**

(b) When a has this value, solve the inequality p(x) < 0. **[4]**

Examiner comment

> "Very few candidates with the correct factors demonstrated an understanding that for a negative result they required one factor to be positive and the other negative."

Mark scheme

2(a) Substitute $x = -\frac{3}{2}$ and equate result to zero **M1** (Or divide by $2x+3$ and set constant remainder equal to zero. Or state $(2x^3 - x^2 + a) = (2x+3)(x^2 + px + q)$, compare coefficients and solve for p or q .)

Obtain $a = 9$ **A1**

Total: 2

2(b) Commence division by $(2x+3)$ reaching a partial quotient x^2+kx ***M1** (The M1 is earned if inspection reaches an unknown factor: x^2+Bx+C and an equation in B and/or C, or an unknown factor Ax^2+Bx+3 and an equation in A and/or B.)

Obtain factorisation $(2x+3)(x^2-2x+3)$ **A1** (Allow if the correct quotient seen. Correct factors seen in (a) and quoted or used here scores M1 A1.)

Show that x^2-2x+3 is always positive, or $2x^3-x^2+9$ only intersects the x-axis once **DM1** (Must use their quadratic factor. SC If M0, allow B1 if state $x<-\frac{3}{2}$ and no error seen)

State **final** answer $x<-\frac{3}{2}$ from correct work **A1**

Total: 4

Q59 Oct/Nov 2022 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry

Question 4

(a) Express $4\cos x-\sin x$ in the form $R\cos(x+\alpha)$, where $R>0$ and $0^\circ<\alpha<90^\circ$. State the exact value of R and give α correct to 2 decimal places. **[3]**

(b) Hence solve the equation $4\cos 2x-\sin 2x=3$ for $0^\circ<x<180^\circ$. **[5]**

Examiner comment

> "The question 4 specifies that the angle should be given correct to 2 decimal places, but writing an answer to 1 decimal place or more than 2 decimal places were common errors."

> "The most common errors were in adjusting from an expression in x in part (a) to an equation in 2x in part (b)."

Mark scheme

4(a) State $R=\sqrt{17}$ **B1** (Allow if working from an incorrect expansion but not from decimals.)

Use correct trig formulae to find α (Correct expansion and correct expression for trig ratio for α) **M1** (NB: $\cos\alpha=4$ and $\sin\alpha=1$ scores M0A0. M0 for incorrect expansion of $\cos(x-\alpha)$. M1 for correct expression for trig ratio for α and no errors seen.)

Obtain $\alpha=14.04^\circ$ **A1** (2 d.p. required. Allow M1A1 for correct answer with no working shown. Correct answer from incorrect working (e.g. $\tan^{-1}(-\frac{1}{4})$) is awarded M0A0. $180^\circ-\tan^{-1}(-\frac{1}{4})$ is awarded M1)

Total: 3

4(b) Evaluate $\cos^{-1}\left(\frac{3}{\sqrt{17}}\right)$ to at least 1 d.p. (43.31 38...°) **B1 FT** (FT *their* R. Accept awrt 43.3° or awrt 316.7°. Can be implied by subsequent working.)

Use correct method to find a value of x in the interval **M1** (Must be working with $2x$ and *their* α.)

Obtain answer, e.g. 14.6° **A1** (Accept overspecified answers but they need to be correct. (14.6388.... and 151.3249....).)

Use a correct method to find a second answer in the interval **M1** (Must be working with $2x$, *their* α and $360^\circ -$ *their* 43.3.)

Obtain second answer in the interval, e.g. 151.3°, and no other in the interval **A1** (Ignore answers outside the given interval. Treat answers in radians (0.255... and 2.64...) as a misread.)

Total: 5

Q60 Oct/Nov 2022 | Paper 3 (variant 32) | Question 5

Topics: Complex Numbers

Question 5

(a) Solve the equation $z^2 - 6iz - 12 = 0$, giving the answers in the form $x + iy$, where x and y are real and exact. **[3]**

(b) On a sketch of an Argand diagram with origin O, show points A and B representing the roots of the equation in part (a). **[1]**

(c) Find the exact modulus and argument of each root. **[3]**

(d) Hence show that the triangle OAB is equilateral. **[1]**

Examiner comment

> "The most common error was to use 6 in place of 6i when substituting values into the quadratic formula."

Mark scheme

5(a) Use quadratic formula, or completing the square $\left((z - 3i)^2 - 3 = 0\right)$ **M1** and use $i^2 = -1$ to find a root

Obtain a root, e.g. $\sqrt{3} + 3i$ **A1** (Or exact 2 term equivalent e.g. $\frac{6i + \sqrt{12}}{2}$ ISW.)

Obtain the other root, e.g. $-\sqrt{3} + 3i$ **A1** (Or exact 2 term equivalent ISW.)

Total: 3

5(b) Show points representing the roots correctly **B1 FT** (2 roots consistent with *their* (a) and with no errors seen on the diagram. B0 if they only have one root or more than 2 roots. Must match *their* scale and $1 < \sqrt{3} < 2$ (Linear scales seen or implied. Need some indication of scale (numbers or dashes). Scales along an axis must be approximately consistent but scales may be different on the 2 axes.)

Total: 1

5(c) State modulus of either root is $2\sqrt{3}$, or simplified exact equivalent **B1 FT** (ISW if converted to decimal. Ignore modulus of second root if seen. Follow *their* root(s) not on either axis (from (a) or (b)).)

Find the argument of one of *their* roots – get as far as $\tan^{-1}(\ldots)$ **M1** (SOI but must be correct for *their* root.)

Obtain correct arguments $\frac{1}{3}\pi$ and $\frac{2}{3}\pi$, or simplified exact equivalents **A1** (Must obtain values. Allow degrees.)

Total: 3

5(d) Give a **complete justification** that the **correct** triangle is equilateral **B1** (Check *their* diagram in (b). Possible justifications: 3 equal sides, or all angles equal to $\frac{\pi}{3}$, or isosceles and an angle of $\frac{\pi}{3}$.)

Total: 1

Q61 Oct/Nov 2022 | Paper 3 (variant 32) | Question 6

Topics: none of the listed topics

Question 6

Relative to the origin O, the points A, B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}.$$

(a) Using a scalar product, find the cosine of angle BAC. **[4]**

(b) Hence find the area of triangle ABC. Give your answer in a simplified exact form. **[4]**

Examiner comment

> "The other common error υυυρ was to use an incorrect pair of vectors."

> "(b) Although the question asks for an exact answer, many candidates could not see how to obtain this and gave a decimal answer instead."

Mark scheme

6(a) State or imply $\overrightarrow{AB}$ or $\overrightarrow{AC}$ correctly in component form **B1** $(\overrightarrow{AB} = \mathbf{2i - 2j + k}, \overrightarrow{AC} = \mathbf{4i - 3k})$.

Using the correct process with relevant vectors to evaluate the scalar product $\overrightarrow{AB}.\overrightarrow{AC}$, **M1** (or $\overrightarrow{BA}.\overrightarrow{CA}$ (8 - 3 - 5).)

Using the correct process for the moduli, divide *their* scalar product by the product of *their* moduli to obtain $\cos\theta$ or θ **M1** $\left(\frac{5}{\sqrt{9}\sqrt{25}}\right)$ (Independent of the first M1.)

Obtain answer $\frac{1}{3}$ **A1** (ISW. Need to see a value for $\cos\theta$. Accept $\frac{5}{15}$ or 0.333 ($\cos^{-1}\frac{1}{3}$ alone is not sufficient))

Total: 4

6(b) Use correct method to find an **exact value** for the sine of angle BAC from *their* (a) **M1** $\left(\sqrt{1-\frac{1}{9}}\right)$ (or $\frac{1}{2}\sqrt{9}\sqrt{25} \times$ ***their*** $\sin\theta$) or $\frac{1}{2}\sqrt{9}\sqrt{25}(\cos^{-1}\frac{1}{3})$)

Obtain answer $\frac{2}{3}\sqrt{2}$, or equivalent **A1** (Only ISW)

Use correct area formula to find the area of triangle ABC with *their* versions of relevant vectors **M1** $\left(\frac{1}{2}\sqrt{9}\sqrt{25} \times \text{*their* } \sin\theta\right)$ or $\frac{1}{2} \times \text{*their* } \sqrt{24} \times \text{*their* } \frac{1}{\sqrt{75}}$) (The length they use for *their* base must be found correctly.)

Obtain answer $5\sqrt{2}$ or $\sqrt{50}$ **A1** (Only ISW)

Alternative method 1 for question 6(b)

Use correct method to find the perpendicular distance from A to BC (or B to AC or C to AB) **M1** $\left(\begin{pmatrix} 2+2\lambda \\ -2+2\lambda \\ 1-4\lambda \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix} = 0 \Rightarrow \lambda = \frac{1}{6}\right)$

Obtain $\frac{1}{6}\sqrt{75}$ **A1** $\left(\left|\frac{1}{6}\mathbf{i} - \frac{5}{6}\mathbf{j} + \frac{1}{6}\mathbf{k}\right|\right)$

Use correct area formula to find the area of triangle ABC **M1** $\left(\frac{1}{2} \times \text{*their* } \sqrt{24} \times \text{*their* } \frac{1}{\sqrt{75}}\right)$ (The length they use for *their* base must be found correctly.)

Obtain answer $5\sqrt{2}$ or $\sqrt{50}$ **A1**

Q62 Oct/Nov 2022 | Paper 3 (variant 32) | Question 7

Topics: Trigonometry, Differentiation, Integration, Differential Equations

Question 7

The variables x and θ satisfy the differential equation

$$x \sin^2\theta \frac{dx}{d\theta} = \tan^2\theta - 2\cot\theta,$$

for $0 < \theta < \frac{1}{2}\pi$ and $x > 0$. It is given that $x = 2$ when $\theta = \frac{1}{4}\pi$.

(a) Show that $\frac{d}{d\theta}(\cot^2\theta) = -\frac{2\cot\theta}{\sin^2\theta}$.

(You may assume without proof that the derivative of $\cot\theta$ with respect to θ is $-\operatorname{cosec}^2\theta$.) **[1]**

(b) Solve the differential equation and find the value of x when $\theta = \frac{1}{6}\pi$. **[7]**

Examiner comment

> "The integration of tan2θ proved to be more challenging, with relatively few sin2θ © 2022 Cambridge International Advanced Subsidiary and Advanced Level 9709 Mathematics November 2022 Principal Examiner Report for Teachers 1 candidates recognising this as sec2θ."

Mark scheme

7(a) Show sufficient working to justify the given statement **B1** (e.g. see $2\cot\theta \times -\text{cosec}^2\theta$ in the working or express in terms of $\sin\theta$ and $\cos\theta$ and use quotient rule to obtain the given result. Solution must have θ present throughout and must reach the given answer.)

Total: 1

7(b) Separate variables correctly **B1** (Check for relevant working in (a))

(Integral: $\int x dx = \int \frac{\tan^2\theta - 2\cot\theta}{\sin^2\theta} d\theta$ Condone incorrect notation e.g. missing dx. Need either the integral sign or the dx, dθ.)

Obtain term $\frac{1}{2}x^2$ **B1**

Obtain terms $\tan\theta + \cot^2\theta$ **B1** + **B1** (Alternative: $\int \frac{2\cot\theta - \cos\cot\theta}{\sin^2\theta} d\theta = -\frac{1}{\sin^2\theta}(-C)$) Condone incorrect notation e.g. missing dx. Need either the integral sign or the dx, dθ.

Form an equation for the constant of integration, or use limits $x = 2, \theta = \frac{1}{4}\pi$, in a solution with **at least two correctly obtained terms** of the form $ax^2, b\tan\theta$ and $c\cot^2\theta$, where $abc \neq 0$ **M1** (Need to have 3 terms. Constant of correct form.)

State correct solution in any form, e.g. $\frac{1}{2}x^2 = \tan\theta + \cot^2\theta - 1$ **A1** (or $\frac{1}{2}x^2 = \tan\theta + \cos ec^2\theta - 1$) If everything else is correct, allow a correct final answer to imply this A1. If the user has an error elsewhere, allow this A1 to imply that this A1 is assumed.

Substitute $\theta = \frac{1}{6}\pi$ and obtain answer $x = 2.67$ **A1** $\left(2.6748\ldots \sqrt{\frac{18+2\sqrt{3}}{3}}\right)$ If see a correctly rounded value ISW.

Q63 Oct/Nov 2022 | Paper 3 (variant 32) | Question 8

Topics: Trigonometry, Integration

The diagram shows part of the curve $y = \sin\sqrt{x}$. This part of the curve intersects the x-axis at the point where $x = a$.

(a) State the exact value of a. **[1]**

(b) Using the substitution $u = \sqrt{x}$, find the exact area of the shaded region in the first quadrant bounded by this part of the curve and the x-axis. **[7]**

Examiner comment

> "The most common errors occurred in the course of substituting for dx, with several candidates having a factor of u or 1 in their integral."

Mark scheme

(a) State $(a) = \pi^2$ **B1** Allow 32400, 180^2. Accept $(x =)\pi^2$.

(b) State or imply $dx = 2u\,du$ or equivalent **B1** e.g. $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$. Incorrect statements e.g. $du = \frac{1}{2\sqrt{x}}$ is **B0**.

Substitute for x and dx throughout the integral **M1**

Obtain $\int 2u \sin u\,du$ **A1** Allow with missing du.

Commence integration of $\int ku \sin u\,du$ by parts and reach $\mp ku\cos u \pm \int k\cos u\,du$ **M1**

Obtain integral $-ku\cos u + k\sin u$ **A1**

Substitute limits $u = 0$ and $u = \sqrt{\text{their } a}$, $a \neq 0$, a in radians or $x = 0$ and *their* a in the complete integral **DM1** $-2\pi\cos\pi + 2\sin\pi(+0 - 2\sin 0)$. Need limits stated but condone if zeros not shown in substitution.

Obtain answer 2π **A1**

Total: 7

Q64 Oct/Nov 2022 | Paper 3 (variant 32) | Question 9

Topics: Trigonometry, Numerical Solutions/Iteration

The diagram shows a semicircle with diameter AB, centre O and radius r. The shaded region is the minor segment on the chord AC and its area is one third of the area of the semicircle. The angle CAB is θ radians.

(a) Show that $\theta = \frac{1}{3}(\pi - 1.5\sin 2\theta)$. **[4]**

(b) Verify by calculation that $0.5 < \theta < 0.7$. **[2]**

(c) Use an iterative formula based on the equation in part **(a)** to determine θ correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Examiner comment

> "Some candidates found it challenging when using this to obtain the given answer; some answers contained sin(π−2θ) , and some claimed that sin(π−2θ)=sinπ−sin2θ ."

Mark scheme

(a) State or imply angle $AOC = \pi - 2\theta$ **B1** Might be seen on the printed diagram.

Use correct formulae for the area of a sector and triangle, or of a segment, and find the area of the shaded region **M1** $\frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2\sin(\pi - 2\theta)$ or $\frac{1}{2}\pi r^2 - [\frac{1}{2}r^2(2\theta) + \frac{1}{2}r^2\sin(\pi - 2\theta)]$. **M0** if subtraction the wrong way round.

Equate to $\frac{1}{6}\pi r^2$ and obtain a correct equation in any form **A1** e.g.
$\frac{1}{6}\pi r^2 = \frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2\sin(\pi - 2\theta)$.

Obtain $\theta = \frac{1}{3}(\pi - 1.5\sin 2\theta)$ correctly **A1 AG** Condone if state / imply $\sin(\pi - 2\theta) = \sin 2\theta$.

Total: 4

(b) Evaluate a relevant expression or pair of expressions at $\theta = 0.5$ and $\theta = 0.7$ **M1** Allow work on a smaller interval. Need to evaluate for both limits, with at least one correct. When using $x = f(x)$ embedded values are not sufficient e.g. $f(0.5)\ldots$ is accepted but $\frac{1}{3}(\pi - 1.5\sin 2 \times 0.5) = \ldots$ is not.

Complete the argument correctly with correct calculated values **A1** e.g. $0.5 < 0.626$, $0.7 > 0.554$ or $0.126 > 0$, $-0.146 < 0$. If using pairs then the pairing must be clear. Need to see the inequalities or an appropriate comment. Need to see values calculated to at least 2 sf.

Total: 2

(c) Use the iterative process $\theta_{n+1} = \frac{1}{3}(\pi - 1.5\sin 2\theta_n)$ correctly at least once **M1** i.e obtain one value and use that value to obtain a second value. Must be working in radians.

Obtain final answer 0.586 **A1**

Show sufficient iterations to 5 d.p. to justify 0.586 to 3 d.p., or show there is a sign change in the interval (0.5855, 0.5865). e.g. 0.5, 0.62646, 0.57225, 0.59195, 0.58416, 0.58715, 0.58599, 0.58644 or 0.6, 0.58118, 0.58833, 0.58553, 0.58661, 0.58619, 0.58636 or 0.7, 0.55447, 0.59958, 0.58133, 0.58827, 0.58556, 0.58661, 0.58620, 0.58636. Allow working to more than 5 dp, but not less. **A1**

Total: 3

Q65 Oct/Nov 2022 | Paper 3 (variant 32) | Question 10

Topics: Algebra, Indices, Logarithms & Exponents, Integration

Let $f(x) = \frac{4-x+x^2}{(1+x)(2+x^2)}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Find the exact value of $\int_0^4 f(x)\,dx$. Give your answer as a single logarithm. **[5]**

Examiner comment

> "The A second integral proved to be more challenging; many candidates ignored the numerator and assumed that the answer would be of the form ptan−1q, or they recognised that it should be a 1 logarithm but they retained x or as a coefficient."

Mark scheme

(a) State or imply the form $\frac{A}{1+x} + \frac{Bx+C}{2+x^2}$ **B1**

Use a correct method for finding a constant **M1**

Obtain one of $A = 2$, $B = -1$ and $C = 0$ **A1 SC**: A maximum of **M1A1** is available for obtaining $A = 2$ after scoring **B0**.

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(b) Integrate and obtain term $2\ln(1+x)$ **B1 FT** $A\ln(1+x)$

Integrate and obtain term of the form $k\ln(2+x^2)$ from an integral of the correct form **M1** Ignore any separate working relating to $C \neq 0$.

Obtain term $-\frac{1}{2}\ln(2+x^2)$ **A1 FT** $\frac{B}{2}\ln(2+x^2)$

Substitute limits in an integral containing terms of the form $a\ln(1+x)+b\ln(2+x^2)$, where $ab \neq 0$ **DM1** Ignore working relating to $C \neq 0$. $(2\ln 5 - 2\ln 1 - \frac{1}{4}\ln 18 + \frac{1}{4}\ln 2)$. Dependent on the first **M1**. Must be subtracting the correct way round. Must have an exact substitution.

Obtain answer $\ln\left(\frac{25}{3}\right)$ **A1 ISW**

Any exact equivalent e.g. $\ln 25\sqrt{2}/\sqrt{8}$ with no number in front of the logarithm.

Total: 5

Q66 Oct/Nov 2022 | Paper 3 (variant 33) | Question 3

Topics: Trigonometry, Differentiation, Integration

Find the exact value of $\int_0^{\pi/4} x\sec^2 x\,dx$. **[5]**

Examiner comment

> "Otherwise, the most common error was a sign error in the ln cos x term."

Mark scheme

Commence integration by parts and reach $x\tan x \pm \int \tan x.1dx$ **M1**

Use a correct method to integrate $\tan x$ **M1**

Obtain integral $x\tan x - \ln\sec x$, or equivalent **A1**

Use limits correctly, having integrated twice **DM1**

Obtain answer $\frac{1}{4}\pi - \frac{1}{2}\ln 2$, or exact equivalent **A1**

Total: 5

Q67 Oct/Nov 2022 | Paper 3 (variant 33) | Question 5

Topics: Complex Numbers

(a) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z+2| < 2$ and $\text{Im}\ z > 1$. **[4]**

(b) Find the greatest value of $\arg z$ for points in the shaded region. **[2]**

Examiner comment

> "(b) This proved challenging even for those with the required correct diagram, and many candidates made no attempt, seemingly being unable to locate the point P that would give the greatest argument."

Mark scheme

(a) Show a circle with centre -2 **B1**

Show a circle with radius 2 and centre not the origin **B1**

Show the line $y = 1$ **B1**

Shade the correct region **B1**

Total: 4

(b) Identify the correct point and carry out a correct method to find the argument **M1**

Obtain answer $\frac{11}{12}\pi$ **A1** 2.88 radians or 165°.

Total: 2

Q68 Oct/Nov 2022 | Paper 3 (variant 33) | Question 9

Topics: none of the listed topics

With respect to the origin O, the position vectors of the points A, B and C are given by

$$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix}$$

The midpoint of AC is M and the point N lies on BC, between B and C, and is such that $BN = 2NC$.

(a) Find the position vectors of M and N. **[3]**

(b) Find a vector equation for the line through M and N. **[2]**

(c) Find the position vector of the point Q where the line through M and N intersects the line through A and B. **[4]**

Examiner comment

> "Question 9 (a) Many candidates did not seem well prepared to tackle this vector question."

> "A large number of marks were lost throughout this question through carelessness and inaccuracies."

> "Common errors υυυρ υυυρ υυυρ arose through using OA–OB for vector AB, confusing the position vector of the midpoint M of υυυυρ υυυυρ υυυρ υυυρ AC, OM , with the vector AM , or misunderstanding what the statement BN =2NC meant about the position of N."

Mark scheme

(a) State $\overrightarrow{OM} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ **B1**

Use a correct method to find $\overrightarrow{ON}$ **M1**

Obtain answer $\begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$ **A1**

Total: 3

(b) Carry out a correct method to form a vector equation for MN **M1**

Obtain a correct equation in any form, e.g. $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$ **A1 OE**

Total: 2

(c) [Detailed marking scheme provided in MS, involves finding intersection of two lines through parameter equations]

Q69 Oct/Nov 2022 | Paper 3 (variant 33) | Question 10

Topics: Differential Equations

A gardener is filling an ornamental pool with water, using a hose that delivers 30 litres of water per minute. Initially the pool is empty. At time t minutes after filling begins the volume of water in the pool is V litres. The pool has a small leak and loses water at a rate of 0.01V litres per minute.

The differential equation satisfied by V and t is of the form $\frac{dV}{dt} = a - bV$.

(a) Write down the values of the constants a and b. **[1]**

(b) Solve the differential equation and find the value of t when $V = 1000$. **[6]**

(c) Obtain an expression for V in terms of t and hence state what happens to V as t becomes large. **[2]**

Examiner comment

> "(b) This question proved very challenging for many candidates who were unable to separate the variables correctly."

Mark scheme

(a) $a = 30$ and $b = 0.01$ **B1**

Total: 1

(b) Separate variables and integrate one side **M1**

Obtain terms $-100\ln(30 - 0.01V)$ and t, or equivalent **A1 FT + A1 FT** FT *their* a and b.

Evaluate a constant, or use $t = 0$, $V = 0$ as limits, in a solution containing terms $c\ln(30 - 0.01V)$ and dt where $cd \neq 0$ **M1**

Obtain solution $100\ln 30 - 100\ln(30 - 0.01V) = t$, or equivalent **A1**

Substitute $V = 1000$ and obtain answer $t = 40.5$ **A1**

Total: 6

(c) Obtain $V = 3000(1 - e^{-0.01t})$ **B1 OE**

State that V approaches 3000 **B1**

Total: 2

Q70 Feb/Mar 2021 | Paper 3 (variant 32) | Question 1

Topics: Algebra, Indices, Logarithms & Exponents

Solve the equation $\ln(x^3 - 3) = 3\ln x - \ln 3$. Give your answer correct to 3 significant figures. [3]

Examiner comment

> "However, a few candidates mistakenly gave their final answer correct to 3 decimal places instead of 3 significant figures, as demanded by the question."

Mark scheme

Use law of the logarithm of a product or power **M1**

Obtain a correct equation free of logarithms, e.g. $x^3 - 3 = \frac{x^3}{3}$ **A1**

Obtain $x = 1.65$ **A1**

Total: 3

Q71 Feb/Mar 2021 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry, Integration, Differential Equations

The variables x and y satisfy the differential equation

$$(1 - \cos x)\frac{dy}{dx} = y\sin x.$$

It is given that $y = 4$ when $x = \pi$.

(a) Solve the differential equation, obtaining an expression for y in terms of x. [6]

(b) Sketch the graph of y against x for $0 < x < 2\pi$. [1]

Examiner comment

> "(b) The majority of candidates found this question challenging."

Mark scheme

(a)

Separate variables correctly and attempt integration of at least one side **M1**

Obtain term $\ln y$ **A1**

Obtain term of the form $\pm \ln(1 - \cos x)$ **M1**

Obtain term $\ln(1 - \cos x)$ **A1**

Use $x = \pi$, $y = 4$ to evaluate a constant, or as limits, in a solution containing terms of the form $a \ln y$ and $b \ln(1 - \cos x)$ **M1**

Obtain final answer $y = 2(1 - \cos x)$ **A1** OE

Total: 6

(b)

Show a correct graph for $0 < x < 2\pi$ with the maximum at $x = \pi$ **B1 FT**

Total: 1

Q72 Feb/Mar 2021 | Paper 3 (variant 32) | Question 5

Topics: Trigonometry

(a) Express $\sqrt{7}\sin x + 2\cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. State the exact value of R and give α correct to 2 decimal places. [3]

(b) Hence solve the equation $\sqrt{7}\sin 2\theta + 2\cos 2\theta = 1$, for $0° < \theta < 180°$. [5]

Examiner comment

> "Some candidates did not introduce 2θ, □ 11□ continuing with x, and then struggled with the rest of the question."

Mark scheme

(a)

State $R = \sqrt{11}$ **B1**

Use trig formulae to find α **M1**

Obtain $\alpha = 37.09°$ **A1**

Total: 3

(b)

Evaluate $\sin^{-1}\left(\frac{1}{\sqrt{11}}\right)$ to at least 2 d.p. $(17.5484°)$ **B1 FT**

Use correct method to find a value of θ in the interval **M1**

Obtain answer, e.g. $62.7°$ **A1**

Use a correct method to obtain a second answer **M1**

Obtain second answer, e.g. $\mathbf{170.2°}$, and no other in the interval **A1**

Total: 5

Q73 Feb/Mar 2021 | Paper 3 (variant 32) | Question 6

Topics: Algebra, Indices, Logarithms & Exponents, Integration

Let $f(x) = \frac{5a}{(2x-a)(3a-x)}$, where a is a positive constant.

(a) Express $f(x)$ in partial fractions. [3]

(b) Hence show that $\int_a^{2a} f(x)\,dx = \ln 6$. [4]

Examiner comment

> "Question 6 (a) Candidates found this partial fraction question to be more challenging, likely due to the presence of a the constant a."

> "Those who multiplied out and equated coefficients, or set x = 3a and , were 2 usually successful, but those who set x to just a numerical value usually struggled to then correctly find the constants in their partial fractions."

Mark scheme

(a)

Carry out a relevant method to determine constants A and B such that $\frac{5a}{(2x-a)(3a-x)} = \frac{A}{2x-a} + \frac{B}{3a-x}$ **M1**

Obtain $A = 2$ **A1**

Obtain $B = 1$ **A1**

Total: 3

(b)

Integrate and obtain terms $\ln(2x-a)$ and $-\ln(3a-x)$ **B1 FT B1 FT**

Substitute limits correctly in a solution containing terms of the form $b\ln(2x-a)$ and $c\ln(3a-x)$, where $bc \neq 0$ **M1**

Obtain the given answer showing full and correct working **A1**

Total: 4

Q74 Feb/Mar 2021 | Paper 3 (variant 32) | Question 7

Topics: none of the listed topics

Two lines have equations

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

and

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}.$$

(a) Show that the lines are skew. [5]

(b) Find the acute angle between the directions of the two lines. [3]

Examiner comment

> "Question 7 (a) Most candidates knew how to show that the two lines did not intersect, although occasionally they had an arithmetical error present."

> "Few candidates were able to undertake this successfully, since they either used the position vectors or line vectors as opposed to the direction vectors, and related that they needed to not be equal to each other rather than not be a multiple of each other."

Mark scheme

(a)

Express general point of a line in component form, e.g. $(1 + 2s, 3 - s, 2 + 3s)$ or $(2 + t, 1 - t, 4 + 4t)$ **B1**

Equate at least two pairs of components and solve for s or for t **M1**

Obtain correct answer for s or for t **A1**

Verify that all three component equations are not satisfied **A1**

Show that the lines are not parallel and are thus skew **A1**

Total: 5

(b)

Carry out correct process for evaluating the scalar product of the direction vectors **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result **M1**

Obtain answer $19.1°$ or 0.333 radians **A1**

Total: 3

Q75 Feb/Mar 2021 | Paper 3 (variant 32) | Question 9

Topics: Indices, Logarithms & Exponents, Differentiation, Numerical Solutions/Iteration

Let $f(x) = \frac{e^{2x}+1}{e^{2x}-1}$, for $x > 0$.

(a) The equation $x = f(x)$ has one root, denoted by α. Verify by calculation that α lies between 1 and 1.5. [2]

(b) Use an iterative formula based on the equation in part (a) to determine α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(c) Find $f'(x)$. Hence find the exact value of x for which $f'(x) = -8$. [6]

Examiner comment

> "However, those candidates who found the four values of x and f(x) at x = 1 and x = 1.5 struggled with this question, as they often omitted any comparison, or, if they did undertake one, they did it between the x values and between the f(x) values instead of between x =1 and f(1) and between 1.5 and f(1.5)."

Mark scheme

(a)

Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(b)

Use the iterative formula $x_{n+1} = \frac{e^{2x_n}+1}{e^{2x_n}-1}$, or equivalent, correctly at least once **M1**

Obtain final answer 1.20 **A1**

Show sufficient iterations to 4 d.p. to justify 1.20 to 2 d.p., or show there is a sign change in $(1.195, 1.205)$ **A1**

Total: 3

(c)

Use quotient rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to -8 and obtain a quadratic in e^{2x} **M1**

Obtain $2(e^{2x})^2 - 5e^{2x} + 2 = 0$ **A1** OE

Solve a 3-term quadratic in e^{2x} for x **M1**

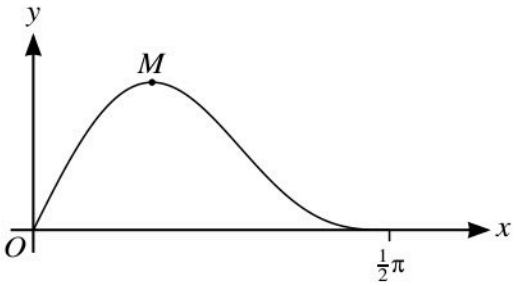
Obtain answer $x = \frac{1}{2}\ln 2$, or exact equivalent, only **A1**

Total: 6

Q76 Feb/Mar 2021 | Paper 3 (variant 32) | Question 10

Topics: Trigonometry, Differentiation, Integration

10



The diagram shows the curve $y = \sin 2x \cos^2 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M.

(a) Using the substitution $u = \sin x$, find the exact area of the region bounded by the curve and the x-axis. [5]

The diagram shows the curve $y = \sin 2x \cos^2 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M.

(a) Using the substitution $u = \sin x$, find the exact area of the region bounded by the curve and the x-axis. [5]

(b) Find the exact x-coordinate of M. [6]

Examiner comment

> "Question 10 (a) This question was well answered by many candidates, although several did not establish a correct integral in terms of u."

Mark scheme

(a)

State or imply $du = \cos x \, dx$ **B1**

Using double angle formula for $\sin 2x$ and Pythagoras, express integral in terms of u and du **M1**

Obtain integral $\int (2u - 2u^3) \, du$ **A1** OE

Use limits $u = 0$ and $u = 1$ in an integral of the form $au^2 + bu^4$, where $ab \neq 0$ **M1**

Obtain answer $\frac{1}{2}$ **A1**

Total: 5

(b)

Use product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero and use a double angle formula **M1**

Obtain an equation in one trig variable **DM1**

Obtain $4\sin^2 x = 1$, $4\cos^2 x = 3$ or $3\tan^2 x = 1$ **A1**

Obtain answer $x = \frac{1}{6}\pi$ **A1**

Total: 6

Q77 May/June 2021 | Paper 3 (variant 31) | Question 1

Topics: Algebra

1 Solve the inequality $2|3x - 1| < |x + 1|$. **[4]**

Examiner comment

> "For those candidates who attempted to remove the modulus signs by squaring both sides of the inequality, the most common error was not squaring the 2."

> "For candidates attempting to form two linear inequalities, the most common errors were sign errors and slips in the algebra."

Mark scheme

1

B1 State or imply non-modular inequality $2^2(3x - 1)^2 < (x + 1)^2$, or corresponding quadratic equation, or pair of linear equations

M1 Form and solve a 3-term quadratic, or solve two linear equations for x: e.g. $35x^2 - 26x + 3 = 0$

A1 Obtain critical values $x = \frac{3}{5}$ and $x = \frac{1}{7}$

Allow 0.143 or better

A1 State final answer $\frac{1}{7} < x < \frac{3}{5}$

Exact values required. Accept $x > \frac{1}{7}$ and $x < \frac{3}{5}$

Do not condone $<$ for $<$ in the final answer.
Fractions need not be in lowest terms.

Alternative method for Question 1

B1 Obtain critical value $x = \frac{3}{5}$ from a graphical method, or by solving a linear equation or linear inequality

B2 Obtain critical value $x = \frac{1}{7}$ similarly

Allow 0.143 or better

B1 State final answer $\frac{1}{7} < x < \frac{3}{5}$

OE. Exact values required. Accept $x > \frac{1}{7}$ and $x < \frac{3}{5}$

Do not condone $<$ for $<$ in the final answer.
Fractions need not be in lowest terms.

Total: 4

Q78 May/June 2021 | Paper 3 (variant 31) | Question 3

Topics: Algebra, Trigonometry

3 (a) Given that $\cos(x - 30^\circ) = 2\sin(x + 30^\circ)$, show that $\tan x = \frac{2-\sqrt{3}}{1-2\sqrt{3}}$. **[4]**

(b) Hence solve the equation

$$\cos(x - 30^\circ) = 2\sin(x + 30^\circ),$$

for $0^\circ < x < 360^\circ$. **[2]**

Examiner comment

> "There were some sign errors in quoting the formulae, but the most common error was to use an incorrect formula such as cos(x−30°)=cosx−cos30°."

Mark scheme

3(a)

M1 Use correct trig expansions and obtain an equation in $\sin x$ and $\cos x$

B1 FT Use correct exact trig ratios for 30° in their expansion: e.g. $\cos x(\frac{\sqrt{3}}{2} - 1) = \sin x(\sqrt{3} - \frac{1}{2})$

DM1 Obtain an equation in $\tan x$

Allow if their error in line 1 was a sign error

A1 AG: Obtain $\tan x = \frac{2-\sqrt{3}}{1-2\sqrt{3}}$ from correct working

Total: 4

3(b)

B1 Obtain answer in the given interval, e.g. 173.8°

Accept $174^\circ, 354^\circ$ or better

B1 Obtain a second answer and no other in the given interval, e.g. 353.8°

Ignore answers outside the given interval.
Treat answers in radians $(3.03 and 6.17)$ as a misread.

Total: 2

Q79 May/June 2021 | Paper 3 (variant 31) | Question 4

Topics: Trigonometry, Integration

4 (a) Prove that

$$\frac{1-\cos 2\theta}{1+\cos 2\theta} \equiv \tan^2\theta.$$

[2]

(b) Hence find the exact value of

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1-\cos 2\theta}{1+\cos 2\theta}\, d\theta.$$

[4]

Examiner comment

> "(b) Many candidates found this definite integration question challenging."

Mark scheme

4(a)

M1 Use correct double angle formula or t-substitution twice

A1 AG: Obtain $\frac{1-\cos 2\theta}{1+\cos 2\theta} = \tan^2\theta$ from correct working

Total: 2

4(b)

(No marking scheme provided for part b in the image)

Total:

Q80 May/June 2021 | Paper 3 (variant 31) | Question 5

Topics: Algebra, Complex Numbers

(a) Solve the equation $z^2 - 2piz - q = 0$, where p and q are real constants. [2]

In an Argand diagram with origin O, the roots of this equation are represented by the distinct points A and B.

(b) Given that A and B lie on the imaginary axis, find a relation between p and q. [2]

(c) Given instead that triangle OAB is equilateral, express q in terms of p. [3]

Examiner comment

> "Candidates who did this often made slips in simplifying (−2pi)2+4q, common errors being 4p2+4q and −4p+4q."

Mark scheme

(a) Use quadratic formula and $i^2 = -1$ **M1**

Obtain answers $pi + \sqrt{q - p^2}$ and $pi - \sqrt{q - p^2}$ **A1**

Accept $\frac{2pi \pm \sqrt{-4p^2 + 4q}}{2}$ and ISW

Total: 2

(b) State or imply that the discriminant must be negative **M1**

State condition $q < p^2$ **A1**

Total: 2

(c) Carry out a correct method for finding a relation, e.g. use the fact that the argument of one of the roots is $(\pm)60°$ **M1**

State a correct relation in any form, e.g. $\frac{p}{\sqrt{q - p^2}} = (\pm)\sqrt{3}$ **A1**

Simplify to $q = \frac{4}{3}p^2$ **A1**

Alternative method for Question 5(c)

Carry out a correct method for finding a relation, e.g. use the fact that the sides have equal length **M1**

State a correct relation in any form, e.g. $4(q - p^2) = p^2 + q - p^2$ **A1**

Simplify to $q = \frac{4}{3}p^2$ **A1**

Total: 3

Q81 May/June 2021 | Paper 3 (variant 32) | Question 1

Topics: Algebra

(1) Solve the inequality $|2x - 1| < 3|x + 1|$. [4]

Examiner comment

> "The most common error at this point was to forget to square the 3."

Mark scheme

(1) State or imply non-modular inequality $(2x - 1)^2 < 3^2(x + 1)^2$, or corresponding quadratic equation **B1**

e.g. $5x^2 + 22x + 8 = 0$ Allow recovery from 'invisible brackets' on RHS

Form and solve a 3-term quadratic in x **M1**

Obtain critical values $x = -4$ and $x = -\frac{2}{5}$ **A1**

State final answer $x < -4, x > -\frac{2}{5}$ **A1**

Do not condone $<$ for $<$, or $>$ for $>$ in the final answer. Allow 'or' but not 'and'.

$-\frac{2}{5} < x < -4$ scores A0.

Accept equivalent forms using brackets e.g. $x \in (-\infty, -4) \cup (-0.4, \infty)$

Alternative method for Question 1

Obtain critical value $x = -4$ from a graphical method, or by solving a linear equation or linear inequality **B1**

Obtain critical value $x = -\frac{2}{5}$ similarly **B2**

State final answer $x < -4, x > -\frac{2}{5}$ **B1**

Do not condone $<$ for $<$, or $>$ for $>$ in the final answer. Allow 'or' but not 'and'.

$-\frac{2}{5} < x < -4$ scores A0.

Accept equivalent forms using brackets e.g. $x \in (-\infty, -4) \cup (-0.4, \infty)$

Total: 4

Q82 May/June 2021 | Paper 3 (variant 32) | Question 2

Topics: Complex Numbers

(2) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z + 1 - i| \leq 1$ and $\arg(z - 1) \leq \frac{3}{4}\pi$. [4]

Examiner comment

> "The half-line proved to be more challenging, with several candidates thinking that the line needed to pass through the centre of the circle."

Mark scheme

(2) Show a circle with centre $-1 + i$. **B1**

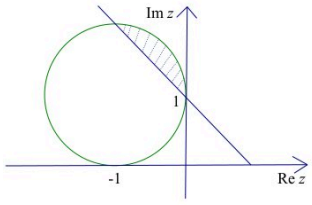
Need some indication of scale or a correct label. Could just be mark(s) on the axes

Show a circle with radius 1 and centre not at the origin (or relevant part thereof). **B1**

Show correct half line from 1 (or relevant part thereof). **B1**

Shade the correct region on a correct diagram. **B1**

2	Show a circle with centre – 1 + i.	**B1**	Need some indication of scale or a correct label. Could just be mark(s) on the axes
	Show a circle with radius 1 and centre not at the origin (or relevant part thereof).	**B1**	
	Show correct half line from 1(or relevant part thereof) .	**B1**	

	Shade the correct region on a correct diagram.	B1	
		4	N.B. If they have very different scales on *their* 2 axes the diagram must match *their* scale - the 'circle' should be an ellipse. Allow freehand diagrams with clear correct intention.

Question	Answer	Marks	Guidance

N.B. If they have very different scales on their 2 axes the diagram must match their scale - the 'circle' should be an ellipse.

Allow freehand diagrams with clear correct intention.

Total: 4

Q83 May/June 2021 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry, Differentiation, Integration

(4) Using integration by parts, find the exact value of $\int_0^1 \tan^{-1}\left(\frac{x}{2}\right)dx$. [5]

Examiner comment

> "Question 4 This was a challenging question for many, with candidates not knowing how to use the given formulae or x x mistaking tan−1 for cot ."

Mark scheme

(4) Commence integration and reach $ax\tan^{-1}\frac{x}{2} + b\int \frac{1}{c+x^2}dx$ **M1** (OE. Denominator might be $1+\frac{x^2}{4}$ or $2+\frac{x^2}{2}$.)

Obtain $x\tan^{-1}\left(\frac{1}{2}x\right) - \int x.\frac{1}{4+x^2}dx$ **A1** (OE)

Complete integration and obtain $x\tan^{-1}\left(\frac{1}{2}x\right) - \ln\left(4+x^2\right)$ **A1**

OE e.g. with $\ln\left|1+\frac{x^2}{4}\right|$

Substitute limits correctly in an expression of the form $px\tan^{-1}x + q\ln(c+x^2)$ **DM1**

$2\tan^{-1}1 - \ln 8 + \ln 4$ OE

Obtain final answer $\frac{1}{2}\pi - \ln 2$ **A1**

OE exact answer. Needs a value for $\tan^{-1}1$ and a single log term

Alternative method for Question 4

Use the substitution $\theta = \tan^{-1}\frac{x}{2}$ to obtain $\lambda\int 2\theta\sec^2\theta d\theta$ and reach **M1**

$p\theta\tan\theta + q\int\tan\theta d\theta$

Obtain $2\theta\tan\theta - 2\int\tan\theta d\theta$ **A1**

Complete integration and obtain $2\theta\tan\theta + 2\ln(\cos\theta)$ **A1**

Substitute correct limits correctly in an expression of the form $r\theta\tan\theta + s\ln(\cos\theta)$ **DM1**

Limits should be $\frac{\pi}{4}$ and 0. Limits must be in radians.

Obtain final answer $\frac{1}{2}\pi - \ln 2$ **A1**

OE exact answer. Need values for trig. functions and a single log term.

Total: 5

Q84 May/June 2021 | Paper 3 (variant 33) | Question 3

Topics: Algebra, Indices, Logarithms & Exponents, Differentiation

(3) The parametric equations of a curve are

$$x = t + \ln(t+2), \quad y = (t-1)e^{-2t},$$

where $t > -2$.

(a) Express $\frac{dy}{dx}$ in terms of t, simplifying your answer. [5]

(b) Find the exact y-coordinate of the stationary point of the curve. [2]

Examiner comment

> "Question 3 (a) Many candidates did not simplify their answer, for example leaving the exponential terms within a bracket, hence could not be awarded the final accuracy mark."

> "A common error involved differentiating the t term within the equation for x as 0, not 1."

Mark scheme

(3a) State $\frac{dx}{dt} = 1 + \frac{1}{t+2}$ **B1**

Use product rule **M1**

Obtain $\frac{dy}{dt} = e^{-2t} - 2(t-1)e^{-2t}$ **A1**

Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ **M1**

Obtain correct answer in any simplified form, e.g. $\frac{(3-2t)(t+2)}{t+3}e^{-2t}$ **A1**

Total: 5

(3b) Equate derivative to zero and solve for t **M1**

Obtain $t = \frac{3}{2}$ and obtain answer $y = \frac{1}{2}e^{-3}$, or exact equivalent **A1**

Total: 2

Q85 May/June 2021 | Paper 3 (variant 33) | Question 6

Topics: Indices, Logarithms & Exponents, Trigonometry, Numerical Solutions/Iteration

(6) (a) By sketching a suitable pair of graphs, show that the equation $\cot \frac{1}{2}x = 1 + e^{-x}$ has exactly one root in the interval $0 < x \leq \pi$. [2]

(b) Verify by calculation that this root lies between 1 and 1.5. [2]

(c) Use the iterative formula $x_{n+1} = 2\tan^{-1}\left(\frac{1}{1+e^{-x_n}}\right)$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "Question 6 (a) Many candidates' attempts to sketch both graphs showed they did not quite understand the properties of each type of graph and its transformations."

Mark scheme

(6a) Sketch a relevant graph, e.g. $y = \cot \frac{1}{2}x$ **B1**

Sketch a second relevant graph, e.g. $y = 1 + e^{-x}$, and justify the given statement **B1**

Total: 2

(6b) Calculate values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(6c) Use the iterative formula correctly at least once **M1**

Obtain final answer 1.34 **A1**

Show sufficient iterations to 4 d.p. to justify 1.34 to 2 d.p. or show there is a sign change in the interval (1.335, 1.345) **A1**

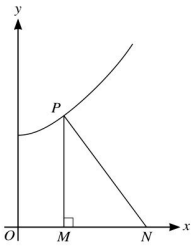
Total: 3

Q86 May/June 2021 | Paper 3 (variant 33) | Question 7

Topics: Trigonometry, Differentiation, Integration, Differential Equations

(7)

7



For the curve shown in the diagram, the normal to the curve at the point P with coordinates (x, y) meets the x-axis at N. The point M is the foot of the perpendicular from P to the x-axis.

The curve is such that for all values of x in the interval $0 \leqslant x < \frac{1}{2}\pi$, the area of triangle PMN is equal to $\tan x$.

(a) **(i)** Show that $\frac{MN}{y} = \frac{dy}{dx}$. [1]

.....

.....

.....

.....

.....

(ii) Hence show that x and y satisfy the differential equation $\frac{1}{2}y^2\frac{dy}{dx} = \tan x$. [2]

For the curve shown in the diagram, the normal to the curve at the point $\boldsymbol{P}$ with coordinates $\boldsymbol{(x, y)}$ meets the x-axis at $\boldsymbol{N}$. The point $\boldsymbol{M}$ is the foot of the perpendicular from $\boldsymbol{P}$ to the x-axis.

The curve is such that for all values of $\boldsymbol{x}$ in the interval $\boldsymbol{0 \leq x < \frac{1}{2}\pi}$, the area of triangle $\boldsymbol{PMN}$ is equal to $\boldsymbol{\tan x}$.

(a) (i) Show that $\boldsymbol{\frac{MN}{y} = \frac{dy}{dx}}$. [1]

(ii) Hence show that $\boldsymbol{x}$ and $\boldsymbol{y}$ satisfy the differential equation $\boldsymbol{\frac{1}{2}y^2\frac{dy}{dx} = \tan x}$. [2]

(b) Given that $\boldsymbol{y = 1}$ when $\boldsymbol{x = 0}$, solve this differential equation to find the equation of the curve, expressing $\boldsymbol{y}$ in terms of $\boldsymbol{x}$. [6]

Examiner comment

> "Question 7 (a) (i) Very few candidates were able to justify the statement, with most simply writing the given expression."

Mark scheme

(7a)(i) Justify the given statement $\boldsymbol{\frac{MN}{y} = \frac{dy}{dx}}$ **B1**

Total: 1

(7a)(ii) Express the area of $\boldsymbol{PMN}$ in terms of $\boldsymbol{y}$ and $\boldsymbol{\frac{dy}{dx}}$ and equate to $\boldsymbol{\tan x}$ **M1**

Obtain the given equation correctly **A1**

Total: 2

(7b) Separate variables and integrate at least one side **M1**

Obtain term $\frac{1}{6}y^3$ **A1**

Obtain term of the form $\pm \ln \cos x$ **M1**

Evaluate a constant or use $x = 0$ and $y = 1$ in a solution containing terms ay^3 and $\pm \ln \cos x$, or equivalent **M1**

Obtain correct answer in any form, e.g. $\frac{1}{6}y^3 = -\ln \cos x + \frac{1}{6}$ **A1**

Obtain final answer $y = \sqrt[3]{(1 - 6 \ln \cos x)}$ **A1**

OE

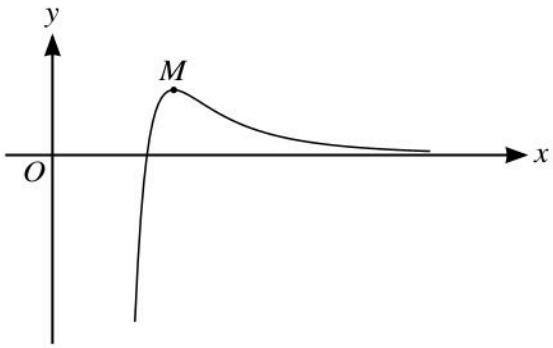
Total: 6

Q87 May/June 2021 | Paper 3 (variant 33) | Question 8

Topics: Differentiation, Integration

(8)

8



The diagram shows the curve $y = \dfrac{\ln x}{x^4}$ and its maximum point M.

(a) Find the exact coordinates of M. [4]

The diagram shows the curve $y = \frac{\ln x}{x^4}$ and its maximum point M.

(a) Find the exact coordinates of M. [4]

(b) By using integration by parts, show that for all $a > 1$, $\int_1^a \frac{\ln x}{x^4} dx < \frac{1}{6}$. [6]

Examiner comment

> "4 (b) Many candidates demonstrated excellent use of integration by parts although the conclusion proved challenging for most candidates."

Mark scheme

(8a) Use quotient or product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero and solve for x **M1**

Obtain $x = \sqrt[4]{e}$ and $y = \frac{1}{4e}$, or exact equivalents **A1**

Total: 4

(8b) Commence integration and reach $ax^{-3}\ln x + b\int x^{-3}.\frac{1}{x}dx$ **M1** (OE. Denominator might be $1 + \frac{x^2}{4}$ or $2 + \frac{x^2}{2}$.)

Obtain $-\frac{1}{3}x^{-3}\ln x + \frac{1}{3}\int x^{-3}.\frac{1}{x}dx$ **A1**

Complete integration and obtain $-\frac{1}{3}x^{-3}\ln x - \frac{1}{9}x^{-3}$ **A1**

Substitute limits correctly, having integrated twice **DM1**

Obtain answer $\frac{1}{9} - \frac{1}{3}a^{-3}\ln a - \frac{1}{9}a^{-3}$ **A1**

Justify the given statement **A1**

Total: 6

Q88 May/June 2021 | Paper 3 (variant 33) | Question 9

Topics: none of the listed topics

Question 9

The quadrilateral $ABCD$ is a trapezium in which AB and DC are parallel. With respect to the origin O, the position vectors of A, B and C are given by $\overrightarrow{OA} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\overrightarrow{OB} = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\overrightarrow{OC} = 2\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$.

(a) Given that $\overrightarrow{DC} = 3\overrightarrow{AB}$, find the position vector of D. [3]

(b) State a vector equation for the line through A and B. [1]

(c) Find the distance between the parallel sides and hence find the area of the trapezium. [5]

Examiner comment

> "(b) Many candidates did not give their answer in the correct form as a vector equation as required by the question."

Mark scheme

9(a)

B1 OE State or imply $\overrightarrow{AB} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

M1 Carry out a correct method to find $\overrightarrow{OD}$

A1 OE Obtain answer $-4\mathbf{i} - \mathbf{j} + 3\mathbf{k}$

Total: 3

9(b)

B1FT State $\mathbf{r} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$

OE. The FT is on $\overrightarrow{AB}$.

Total: 1

9(c)

M1 For a general point P on AB, state $\overrightarrow{CP}$ or $\overrightarrow{DP}$ in component form, e.g. $\overrightarrow{CP} = (3 - 2\lambda, -\lambda, -6 + 2\lambda)$

DM1 Equate a relevant scalar product to zero or equate derivative of $|\overrightarrow{CP}|$ to zero or use Pythagoras in a relevant triangle and solve for λ

A1 Obtain $\lambda = 2$

A1 Show the perpendicular is of length 3

A1 Carry out a correct method to find the area of $ABCD$ and obtain the answer 18

Alternative method for Question 9(c)

M1 Use a scalar product to find the projection CN (or DN) of BC (or AD) on CD

A1 Obtain $CN = 3$ (or $DN = 3$)

DM1 Use Pythagoras to obtain BN (or AN)

Total: 5

Q89 Oct/Nov 2020 | Paper 3 (variant 31) | Question 1

Topics: Algebra

Solve the inequality $2 - 5x > 2|x - 3|$. [4]

Examiner comment

> "A few candidates chose to sketch a graph and reached a correct inequality as the sketch enabled them to visualise it."

Mark scheme

Make a recognisable sketch graph of $y = 2|x - 3|$ and the line $y = 2 - 5x$

B1 Need to see correct V at $x = 3$, roughly symmetrical, $x = 3$ stated, domain at least $(-2, 5)$.

Find x-coordinate of intersection with $y = 2 - 5x$

M1 Find point of intersection with $y = 2|x - 3|$ or solve $2 - 5x$ with $2(x - 3)$ or $-2(x - 3)$

Obtain $x = -\frac{4}{3}$

A1

State final answer $x < -\frac{4}{3}$

A1 Do not accept $x < -1.33$ [Do not condone $<$ for $<$ in the final answer.]

Alternative method for question 1

State or imply non-modular inequality/equality $(2-5x)^2 >, =, +, 2^2(x-3)^2$, or corresponding quadratic equation, or pair of linear equations

$(2-5x) >, =, +2(x-3)$

B1 Two correct linear equations only

Make reasonable attempt at solving a 3-term quadratic, or solve one linear equation, or linear inequality for x

M1 $21x^2 + 4x - 32 = (3x+4)(7x-8) = 0$ or $2-5x$ or $(-2-5x)$ with $2(x-3)$ or $-2(x-3)$

Obtain critical value $x = -\frac{4}{3}$

A1

State final answer $x < -\frac{4}{3}$

A1 Do not accept $x < -1.33$ [Do not condone $<$ for $<$ in the final answer.]

Total: 4

Q90 Oct/Nov 2020 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Numerical Solutions/Iteration

(a) By sketching a suitable pair of graphs, show that the equation $\operatorname{cosec} x = 1 + e^{-\frac{x}{2}}$ has exactly two roots in the interval $0 < x < \pi$. **[2]**

(b) The sequence of values given by the iterative formula

$$x_{n+1} = \pi - \sin^{-1}\left(\frac{1}{e^{-\frac{1}{2}x_n} + 1}\right),$$

with initial value $x_1 = 2$, converges to one of these roots.

Use the formula to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Examiner comment

> "Many candidates were unable to sketch the graph ofy =cosec x."

> "Common mistakes in sketching the exponential graph were drawing a straight line, not including y(0)2= and not indicating the asymptote at y =1 even for x > π."

Mark scheme

(5(a)) Sketch a relevant graph, e.g. $y = \csc x$ | **B1** | $\csc x$, U shaped, roughly symmetrical about $x = \frac{\pi}{2}$, $y(\frac{\pi}{2}) = 1$ and domain at least $(\frac{\pi}{6}, \frac{5\pi}{6})$ |

Sketch a second relevant graph, e.g. $y = 1 + e^{-\frac{x}{2}}$, and justify the given statement | **B1** | Exponential graph needs $y(0) = 2$, negative gradient, always increasing, and $y(\pi) > 1$. Needs to mark intersections with dots, crosses, or say roots at points of intersection, or equivalent |
| | | **2** |

(5(b)) | Use the iterative formula correctly at least twice | **M1** | $2, 2.3217, 2.2760, 2.2824\ldots$ Need to see 2 iterations and following value inserted correctly |
	Obtain final answer 2.28	**A1**	Must be supported by iterations
	Show sufficient iterations to at least 4 d.p. to justify 2.28 to 2 d.p., or show there is a sign change in the interval $(2.275, 2.285)$	**A1**	
		3

Q91 Oct/Nov 2020 | Paper 3 (variant 31) | Question 7

Topics: Algebra, Complex Numbers

(a) Verify that $-1 + \sqrt{5}i$ is a root of the equation $2x^3 + x^2 + 6x - 18 = 0$. **[3]**

(b) Find the other roots of this equation. **[4]**

Examiner comment

> "A fairly common mistake was to write 2x3 + x2+ 6x – 18 = (x – (–1 – √5i)) (x – (–1 + √5i)) (x – a) and to find a."

Mark scheme

(7(a)) | Substitute $-1 + \sqrt{5}i$ in the equation and attempt expansions of x^2 and x^3 | **M1** | All working must be seen. Allow M1 if small errors in $1 - 2\sqrt{5}i - 5$ or $1 - \sqrt{5}i - \sqrt{5}i - 5$ and $4 - 2\sqrt{5}i + 10$ or $4 - 4\sqrt{5}i + 2\sqrt{5}i + 10$ |
	Use $i^2 = -1$ correctly at least once	**M1**	$1 - 5$ or $4 + 10$ seen
	Complete the verification correctly	**A1**	$2(14 - 2\sqrt{5}i) + (-4 - 2\sqrt{5}i) + 6(-1 + \sqrt{5}i) - 18 = 0$
		3

(7(b)) | State second root $-1 - \sqrt{5}i$ | **B1** | |
	Carry out a complete method for finding a quadratic factor with zeros $-1 + \sqrt{5}i$ and $-1 - \sqrt{5}i$	**M1**	
	Obtain $x^2 + 2x + 6$	**A1**	
	Obtain root $x = \frac{3}{2}$	**A1**	OE

Alternative method for question 7(b)

State second root $-1 - \sqrt{5}i$	**B1**	
	$(x + 1 - \sqrt{5}i)(x + 1 + \sqrt{5}i)(2x + a) = 2x^3 + x^2 + 6x - 18$	**M1**	
	$(1 - \sqrt{5}i)(1 + \sqrt{5}i)a = -18$	**A1**	
	$6a = -18, a = -3$ leading to $x = \frac{3}{2}$	**A1**	OE
		4

Q92 Oct/Nov 2020 | Paper 3 (variant 31) | Question 9

Topics: Algebra

Let $f(x) = \frac{8+5x+12x^2}{(1-x)(2+3x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Examiner comment

> "The most common error was to multiply throughout by an expression involving (2 + 3x)3."

> "(b) This question proved more challenging for some candidates."

Mark scheme

(9(a)) | State or imply the form $\frac{A}{1-x} + \frac{B}{2+3x} + \frac{C}{(2+3x)^2}$ | **B1** | |

| | Use a correct method for finding a coefficient | **M1** | |

| | Obtain one of $A = 1, B = -1, C = 6$ | **A1** | |

| | Obtain a second value | **A1** | |

| | Obtain the third value | **A1** | In the form $\frac{A}{1-x} + \frac{Dx+E}{(2+3x)^2}$, where $A = 1, D = -3$ and $E = 4$ can score B1 M1 A1 A1 A1 as above. |

| | | **5** |

(9(b)) | Use a correct method to find the first two terms of the expansion of $(1-x)^{-1}, (2+3x)^{-1}, (1+\frac{3}{2}x)^{-1}, (2+3x)^{-2}$ or $(1+\frac{3}{2}x)^{-2}$ | **M1** | Symbolic coefficients are not sufficient for the M1. $A\frac{1+(-1)(-x)+(-1)(-2)(-x)^2}{2\ldots}$, $A = 1$ |

| | | $\frac{B}{2}\left[\frac{1+(-1)(\frac{3x}{2})+(-1)(-2)(\frac{3x}{2})^2}{2\ldots}\right]$, $B = 1$ |

| | | $\frac{C}{4}\left[\frac{1+(-2)(\frac{3x}{2})+(-2)(-3)(\frac{3x}{2})^2}{2\ldots}\right]$, $C = 6$ |

| | Obtain correct un-simplified expansions up to the term in of each partial fraction | **A1 FT** + | $(1+x+x^2) + (-\frac{1}{2} + (\frac{3}{4})x - (\frac{9}{8})x^2)$ |

| | | **A1 FT** + | $(\frac{6}{4} - (\frac{18}{4})x + (\frac{81}{8})x^2)$ [The FT is on A, B, C] |

| | | **A1 FT** | $(1 - \frac{1}{2} + \frac{4}{1}) + (1 + \frac{3}{4} - \frac{18}{4})x + (1 - \frac{8}{8} + \frac{81}{8})x^2$ |

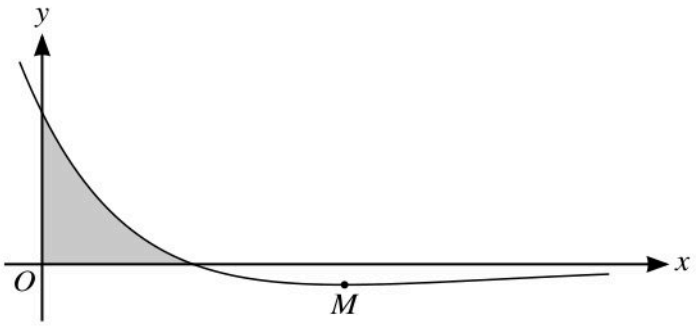
| | Obtain final answer $2 - \frac{11}{4}x + 10x^2$, or equivalent | **A1** | Allow unsimplified fractions. $(Dx+E)(1+(-2)(\frac{3x}{2}) + (-2)(-3)(\frac{3x}{2})^2)$ where $D = -3, E = 4$ can score B1 M1 A1 A1 A1 as above. The FT is on A, D, E. |

| | | **5** |

Q93 Oct/Nov 2020 | Paper 3 (variant 31) | Question 10

Topics: Differentiation, Integration

10



The diagram shows the curve $y = (2 - x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. [5]

(a) Find the exact coordinates of M. **[5]**

(b) Find the area of the shaded region bounded by the curve and the axes. Give your answer in terms of e. **[5]**

Examiner comment

> "Question 10 (a) This should have been a straightforward application of the product rule for differentiation, but many –x candidates struggled because they were unable to differentiate e2 ."

> "Candidates often found the integration of e2 difficult, with similar errors to (a) seen."

Mark scheme

(10(a)) | Use correct product or quotient rule | *M1 | $\frac{dy}{dx} = (-\frac{1}{2})(2-x)e^{-\frac{1}{2}x} - e^{-\frac{1}{2}x}$ **M1** requires at least one of derivatives correct |

| | Obtain correct derivative in any form | **A1** | |

| | Equate derivative to zero and solve for x | **DM1** | |

| | Obtain $x = 4$ | **A1** | ISW |

| | Obtain $y = -2e^{-2}$, or exact equivalent | **A1** | |

| | | **5** |

(10(b)) | Commence integration and reach | *M1 | Condone omission of dx.
$a(2-x)e^{-\frac{1}{2}x} + b\int e^{-\frac{1}{2}x}dx$ |

| | Obtain $-2(2-x)e^{-\frac{1}{2}x} - 2\int e^{-\frac{1}{2}x}dx$ | **A1** | OE |

| | Complete integration and obtain $2xe^{-\frac{1}{2}x}$ | **A1** | OE |

| | Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice | **DM1** | Ignore omission of zeros and allow max of 1 error |

| | Obtain answer $4e^{-1}$, or exact equivalent | **A1** | ISW |

Alternative method for question 10(b)

	$\frac{d}{dx}(2xe^{-\frac{1}{2}x})$	*M1 A1	
	: $2xe^{-\frac{1}{2}x}$	A1	
	Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice	DM1	Ignore omission of zeros and allow max of 1 error
	Obtain answer $4e^{-1}$, or exact equivalent	A1	ISW
		5

Q94 Oct/Nov 2020 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Solve the equation

$$\ln(1 + e^{-3x}) = 2.$$

Give the answer correct to 3 decimal places. **[3]**

Examiner comment

> "A few candidates who used the correct method did not give the final answer to the required accuracy."

Mark scheme

1	State that $1 + e^{-3x} = e^2$	B1	With no errors seen to that point
	Use correct method to solve an equation of the form $e^{-3x} = a$, where $a > 0$, for x or equivalent	M1	$(e^{-3x} = 6.389\ldots)$ Evidence of method must be seen.
	Obtain answer $x = -0.618$ only	A1	Must be 3 decimal places
		3

Alternative method for question 1

State that $1 + e^{-3x} = e^2$	B1	
Rearrange to obtain an expression for e^x and solve an equation of the form $e^x = a$, where $a > 0$, or equivalent	M1	$e^x = 3\sqrt[3]{\frac{1}{e^2 - 1}}$
Obtain answer $x = -0.618$ only	A1	Must be 3 decimal places
	3

Q95 Oct/Nov 2020 | Paper 3 (variant 32) | Question 2

Topics: Algebra

(a) Expand $\sqrt[4]{1 + 6x}$ in ascending powers of x, up to and including the term in x^3, simplifying the coefficients. **[4]**

(b) State the set of values of x for which the expansion is valid. **[1]**

Examiner comment

> "There were a number of slips in the arithmetic to simplify the coefficients, but the most common errors were using powers of x rather than 6x or using an incorrect index for the expansion."

Mark scheme

(2(a)) | State a correct unsimplified version of the x or x^2 or x^3 term | **M1** | For the given expression |
	State correct first two terms $1 + 2x$	**A1**	
	Obtain the next two terms $-4x^2 + \frac{40}{3}x^3$	**A1** + **A1**	One mark for each correct term. ISW Accept $13\frac{1}{3}$. The question asks for simplified coefficients, so candidates should cancel fractions.
		4

(2(b)) | State answer $|x| < \frac{1}{6}$ | **B1** | OE. Strict inequality |
| | | **1** |

Q96 Oct/Nov 2020 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry

(a) Show that the equation $\tan(\theta + 60^\circ) = 2\cot\theta$ can be written in the form

$$\tan^2\theta + 3\sqrt{3}\tan\theta - 2 = 0. \quad [3]$$

(b) Hence solve the equation $\tan(\theta + 60^\circ) = 2\cot\theta$, for $0^\circ < \theta < 180^\circ$. **[3]**

Examiner comment

> "A common error was to rewrite tan(θ+60°) as cos(θ+60°) 1 tanθ+tan60°."

> "The most common errors were due to premature approximation of decimal values or giving incorrect over-specified answers."

Mark scheme

(4(a)) | Use correct $\tan(A + B)$ formula and obtain an equation in $\tan\theta$ | **M1** | e.g. $\frac{\tan\theta + \tan 60^\circ}{1 - \tan\theta\tan 60^\circ} = \frac{2}{\tan\theta}$ |
	Use $\tan 60^\circ = \sqrt{3}$ and obtain a correct horizontal equation in any form	**A1**	e.g. $\tan\theta(\tan\theta + \sqrt{3}) = 2(1 - \sqrt{3}\tan\theta)$
	Reduce to $\tan^2\theta + 3\sqrt{3}\tan\theta - 2 = 0$ correctly	**A1**	AG
		3

(4(b)) | Solve the given quadratic to obtain a value for θ | **M1** | $\left(\tan\theta = \frac{-3\sqrt{3} \pm \sqrt{35}}{2} = 0.3599, -5.556\right)$ |
	Obtain one correct answer e.g. $\theta = 19.8^\circ$	**A1**	Accept 1d.p. or better. If over-specified must be correct. $19.797\ldots, 100.2029\ldots$
	Obtain second correct answer $\theta = 100.2^\circ$ and no others in the given interval	**A1**	Ignore answers outside the given interval.
		3

Q97 Oct/Nov 2020 | Paper 3 (variant 32) | Question 6

Topics: Complex Numbers

The complex number u is defined by

$$u = \frac{7+i}{1-i}.$$

(a) Express u in the form $x + iy$, where x and y are real. **[3]**

(b) Show on a sketch of an Argand diagram the points A, B and C representing u, $7+i$ and $1-i$ respectively. **[2]**

(c) By considering the arguments of $7+i$ and $1-i$, show that

$$\tan^{-1}\left(\frac{4}{3}\right) = \tan^{-1}\left(\frac{1}{7}\right) + \frac{1}{4}\pi. \quad [3]$$

Examiner comment

> "There were several blank responses."

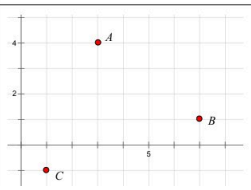
Mark scheme

(6(a)) | Multiply numerator and denominator by $1+i$, or equivalent | **M1** | Must multiply out |
| | Obtain numerator $6+8i$ or denominator 2 | **A1** | |
| | Obtain final answer $u = 3+4i$ | **A1** | |

Alternative method for question 6(a)

Multiply out $(1-i)(x+iy) = 7+i$ and compare real and imaginary parts	**M1**	
	Obtain $x+y=7$ or $y-x=1$	**A1**	
	Obtain final answer $u = 3+4i$	**A1**	
		3	

(6(b)) | Show the point A representing u in a relatively correct position | **B1 FT** | The FT is on $xy \neq 0$. |
| | Show the other two points B and C in relatively correct positions: approximately equal distance above / below real axis | **B1** |

6(a)	Multiply numerator and denominator by 1 + i, or equivalent	**M1**	Must multiply out
	Obtain numerator 6 + 8i or denominator 2	**A1**	
	Obtain final answer $u = 3 + 4i$	**A1**	
	Alternative method for question 6(a)		
	Multiply out $(1-i)(x+iy) = 7+i$ and compare real and imaginary parts	**M1**	
	Obtain $x+y=7$ or $y-x=1$	**A1**	
	Obtain final answer $u = 3 + 4i$	**A1**	
		3	
6(b)	Show the point A representing u in a relatively correct position	**B1 FT**	The FT is on $xy \neq 0$.
	Show the other two points B and C in relatively correct positions: approximately equal distance above / below real axis	**B1**	

			Take the position of A as a guide to 'scale' if axes not marked
		2	

|
|---|
| \| \| \| **2** \| |

(6(c)) | State or imply $\arg(1-i) = -\frac{1}{4}\pi$ | **B1** | ArgC |
	Substitute exact arguments in $\arg(7+i) - \arg(1-i) = \arg u$	**M1**	Must see a statement about the relationship between the Args e.g. $\arg A = \arg B - \arg C$ or equivalent exact method
	Obtain $\tan^{-1}\left(\frac{4}{3}\right) = \tan^{-1}\left(\frac{1}{7}\right) + \frac{1}{4}\pi$ correctly	**A1**	Obtain given answer correctly from *their* $u = k(3+4i)$
			3

Q98 Oct/Nov 2020 | Paper 3 (variant 32) | Question 7

Topics: Differential Equations

(a) Solve the differential equation and obtain an expression for x in terms of t.

Total: 7

(b) State what happens to the value of x when t tends to infinity.

Total: 1

Question: The variables x and t satisfy the differential equation

$$e^{3t}\frac{dx}{dt} = \cos^2 2x$$

for $t \geq 0$. It is given that $x = 0$ when $t = 0$.

Examiner comment

> "Common errors at this point were sign errors and putting the 3 in the wrong 3 place."

> "The integration of cos–22x proved to be much more difficult."

> "Common errors in attempts to complete the correct integration centred on the 2."

Mark scheme

7(a) Correct separation of variables **B1**

$$\int \sec^2 2x\, dx = \int e^{-3t}\, dt$$

Needs correct structure

Obtain term $-\frac{1}{3}e^{-3t}$ **B1**

Obtain term of the form $k\tan 2x$ **M1** From correct working

Obtain term $\frac{1}{2}\tan 2x$ **A1**

Use $x = 0, t = 0$ to evaluate a constant, or as limits in a solution containing terms of the form $a\tan 2x$ and be^{-3t}, where $ab \neq 0$ **M1**

Obtain correct solution in any form **A1**

e.g. $\frac{1}{2}\tan 2x = -\frac{1}{3}e^{-3t} + \frac{1}{3}$

Obtain final answer $x = \frac{1}{2}\tan^{-1}\left(\frac{2}{3}(1 - e^{-3t})\right)$ **A1**

Total: 7

7(b) State that x approaches $\frac{1}{2}\tan^{-1}\left(\frac{2}{3}\right)$ **B1 FT**

Correct value. Accept $x \to 0.294$

The FT is dependent on letting $e^{-3t} \to 0$ in a solution containing e^{-3t}.

Total: 1

Q99 Oct/Nov 2020 | Paper 3 (variant 32) | Question 9

Topics: Algebra, Integration

Question 9: Let $f(x) = \frac{lx+18}{(3x+2)(x^2+4)}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence find the exact value of $\int_0^2 f(x)\,dx$. **[6]**

Examiner comment

> "Apart from slips in the arithmetic, common errors were to use the © 2020 Cambridge International Advanced Level 9709 Mathematics November 2020 Principal Examiner Report for Teachers A B form + or to rewrite x2+4 as (x+2)2 or as (x−2)(x+2) ."

> "The integration of proved to be much more difficult, with only a minority of x2+4 candidates realising that this needed to be rewritten as two separate terms."

Mark scheme

9(a) State or imply the form

$$\frac{A}{3x+2} + \frac{Bx+C}{x^2+4}$$

B1

Use a correct method for finding a constant **M1**

Obtain one of $A = 3, B = -1, C = 3$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

9(b) Integrate and obtain $\ln(3x+2)\ldots$ **B1 FT** The FT is on A

State a term of the form $k\ln(x^2+4)$. **M1** From $\int \frac{\lambda x}{x^2+4}\,dx$

$\ldots -\frac{1}{2}\ln(x^2+4)\ldots$ **A1 FT** The FT is on B

$\ldots +\frac{3}{2}\tan^{-1}\frac{x}{2}$ **B1 FT** The FT is on C

Substitute limits correctly in an integral with at least two terms of the form $a\ln(3x+2), b\ln(x^2+4)$ and $c\tan^{-1}\left(\frac{x}{2}\right)$, and subtract in correct order **M1**

Using terms that have been obtained correctly from completed integrals

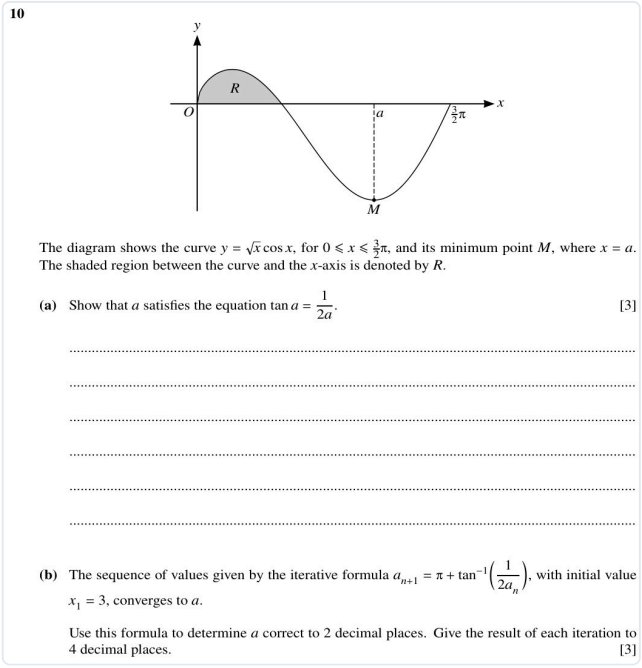
Obtain answer $\frac{3}{2}\ln 2+\frac{3}{8}\pi$, or exact 2-term equivalent **A1**

Total: 6

Q100 Oct/Nov 2020 | Paper 3 (variant 32) | Question 10

Topics: Trigonometry, Differentiation, Integration, Numerical Solutions/Iteration

Question 10:



10

The diagram shows the curve $y = \sqrt{x}\cos x$, for $0 \leqslant x \leqslant \frac{3}{2}\pi$, and its minimum point M, where $x = a$. The shaded region between the curve and the x-axis is denoted by R.

(a) Show that a satisfies the equation $\tan a = \frac{1}{2a}$. [3]

.....

.....

.....

.....

.....

.....

(b) The sequence of values given by the iterative formula $a_{n+1} = \pi + \tan^{-1}\left(\frac{1}{2a_n}\right)$, with initial value $x_1 = 3$, converges to a.

Use this formula to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

The diagram shows the curve $y = \sqrt{x}\cos x$, for $0 < x \leq \frac{3\pi}{2}$, and its minimum point M, where $x = a$. The shaded region between the curve and the x-axis is denoted by R.

(a) Show that a satisfies the equation $\tan a = \frac{1}{2a}$. **[3]**

(b) The sequence of values given by the iterative formula $a_{n+1} = \pi + \tan^{-1}\left(\frac{1}{2a_n}\right)$, with initial value $x_1 = 3$, converges to a.

Use this formula to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

(c) Find the volume of the solid obtained when the region R is rotated completely about the x-axis. Give your answer in terms of π. **[6]**

Examiner comment

> "Some candidates made no attempt to rearrange their answer to the given form."

> "The most common errors were to work in degrees, or to use an incorrect formula."

Mark scheme

10(a) Use correct product rule **M1**

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\cos x - \sqrt{x}\sin x$$

Accept in a or in x

Obtain correct derivative in any form **A1**

Equate derivative to zero and obtain $\tan a = \frac{1}{2a}$ **A1**

Obtain given answer from correct working. The question says 'show that ...' so there should be an intermediate step e.g. $\cos x = 2x \sin x$.

Allow $\tan x = \frac{1}{2x}$

Total: 3

10(b) Use the iterative process correctly at least once (get one value and go on to use it in a second use of the formula) **M1**

Must be working in radians. Degrees gives 1, 12.6039, 5.4133, ... M0

Obtain final answer 3.29 **A1** Clear conclusion

Show sufficient iterations to at least 4 d.p. to justify 3.29, or show there is a sign change in the interval (3.285, 3.295) **A1**

3, 3.3067, 3.2917, 3.2923

Allow more than 4d.p. Condone truncation.

Total: 3

10(c) State or imply the indefinite integral for the volume is $\pi \int (\sqrt{x} \cos x)^2 \, dx$ **B1**

[If π omitted, or 2π or $\frac{1}{2}\pi$ used, give B0 and follow through. 4/6 available]

Use correct $\cos 2A$ formula, commence integration by parts and reach
$x(ax + b \sin 2x) + \int ax + b \sin 2x \, dx$ *M1

Alternative: $\frac{x^2}{4} + \frac{x}{4} \sin 2x - \int \frac{1}{4} \sin 2x \, dx$

Obtain $x\left(\frac{1}{2}x + \frac{1}{4}\sin 2x\right) - \int \frac{1}{2}x + \frac{1}{4}\sin 2x \, dx$ or equivalent **A1**

Complete integration and obtain $\frac{1}{4}x^2 + \frac{1}{4}x \sin 2x + \frac{1}{8}\cos 2x$ **A1** OE

Substitute limits $x = 0$ and $x = \frac{1}{2}\pi$, having integrated twice **DM1**

$$\pi\left[\frac{\pi^2}{8} + 0 - \frac{1}{4} - 0 - 0 - \frac{1}{4}\right]$$

Obtain answer $\frac{1}{16}\pi(\pi^2 - 4)$, or exact equivalent **A1** CAO

Total: 6

Q101 Oct/Nov 2020 | Paper 3 (variant 33) | Question 1

Topics: Algebra

Question 1: Solve the inequality $2 - 5x > 2|x - 3|$. **[4]**

Examiner comment

> "A few candidates chose to sketch a graph and reached a correct inequality as the sketch enabled them to visualise it."

Mark scheme

1 Make a recognisable sketch graph of $y = 2|x - 3|$ and the line $y = 2 - 5x$ **B1**

Need to see correct V at $x = 3$, roughly symmetrical, $x = 3$ stated, domain at least $(-2, 5)$.

Find x-coordinate of intersection with $y = 2 - 5x$ **M1**

Find point of intersection with $y = 2|x - 3|$ or solve $2 - 5x$ with $2(x - 3)$ or $-2(x - 3)$

Obtain $x = -\frac{4}{3}$ **A1**

State final answer $x < -\frac{4}{3}$ **A1**

Do not accept $x < -1.33$

[Do not condone $<$ for $<$ in the final answer.]

Alternative method for question 1

State or imply non-modular inequality/equality $(2 - 5x)^2 >, =, >, =, 2^2(x - 3)^2$ or corresponding quadratic equation, or pair of linear equations: $(2 - 5x) >, =, +2(x - 3)$ **B1**

Two correct linear equations only

Make reasonable attempt at solving a 3-term quadratic, or solve one linear equation, or linear inequality for x **M1**

$21x^2 + 4x - 32 = (3x + 4)(7x - 8) = 0$ or $2 - 5x$ or $-2(5x) - 2(x - 3)$

Obtain critical value $x = -\frac{4}{3}$ **A1**

State final answer $x < -\frac{4}{3}$ **A1**

Do not accept $x < -1.33$

[Do not condone $<$ for $<$ in the final answer.]

Total: 4

Q102 Oct/Nov 2020 | Paper 3 (variant 33) | Question 5

Topics: Trigonometry, Numerical Solutions/Iteration

(a) By sketching a suitable pair of graphs, show that the equation $\operatorname{cosec} x = 1 + e^{-\frac{x}{2}}$ has exactly two roots in the interval $0 < x < \pi$. **[2]**

(b) The sequence of values given by the iterative formula

$$x_{n+1} = \pi - \sin^{-1}\left(\frac{1}{e^{-\frac{1}{2}x_n} + 1}\right)$$

with initial value $x_1 = 2$, converges to one of these roots.

Use the formula to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Examiner comment

> "Many candidates were unable to sketch the graph ofy =cosec x."

> "Common mistakes in sketching the exponential graph were drawing a straight line, not including y(0)2= and not indicating the asymptote at y =1 even for x > π."

Mark scheme

5(a) Sketch a relevant graph, e.g. $y = \operatorname{cosec} x$ **B1**

$\operatorname{cosec} x$, U shaped, roughly symmetrical about $x = \frac{\pi}{2}$, $y\left(\frac{\pi}{2}\right) = 1$ and domain at least $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$.

Sketch a second relevant graph, e.g. $y = 1 + e^{-\frac{x}{2}}$, and justify the given statement **B1**

Exponential graph needs $y(0) = 2$, negative gradient, always increasing, and $y(\pi) > 1$

Needs to mark intersections with dots, crosses, or say roots at points of intersection, or equivalent

Total: 2

5(b) Use the iterative formula correctly at least twice **M1**

2, 2.3217, 2.2760, 2.2824...

Need to see 2 iterations and following value inserted correctly

Obtain final answer 2.28 **A1** Must be supported by iterations

Show sufficient iterations to at least 4 d.p. to justify 2.28 to 2 d.p., or show there is a sign change in the interval (2.275, 2.285) **A1**

Total: 3

Q103 Oct/Nov 2020 | Paper 3 (variant 33) | Question 7

Topics: Algebra, Complex Numbers

(a) Verify that $-1+\sqrt{5}i$ is a root of the equation $2x^3+x^2+6x-18=0$. **[3]**

(b) Find the other roots of this equation. **[4]**

Examiner comment

> "A fairly common mistake was to write 2x3 + x2+ 6x – 18 = (x – (–1 – √5i)) (x – (–1 + √5i)) (x – a) and to find a."

Mark scheme

7(a) Substitute $-1+\sqrt{5}i$ in the equation and attempt expansions of x^2 and x^3 **M1**

All working must be seen. Allow M1 if small errors in $1-2\sqrt{5}i-5$ or $1-\sqrt{5}i-\sqrt{5}i-5$ and $4-2\sqrt{5}i+10$ or $4-4\sqrt{5}i+2\sqrt{5}i+10$

Use $i^2=-1$ correctly at least once **M1**

$1-5$ or $4+10$ seen

Complete the verification correctly **A1**

$$2(14-2\sqrt{5}i)+(-4-2\sqrt{5}i)+6(-1+\sqrt{5}i)-18=0$$

Total: 3

7(b) State second root $-1-\sqrt{5}i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $-1+\sqrt{5}i$ and $-1-\sqrt{5}i$ **M1**

Obtain x^2+2x+6 **A1**

Obtain root $x=\frac{3}{2}$ **A1** OE

Alternative method for question 7(b)

State second root $-1-\sqrt{5}i$ **B1**

POR $(-1+\sqrt{5}i)(-1-\sqrt{5}i)a=9$ or **SOR** $(-1+\sqrt{5}i)+(-1-\sqrt{5}i)+a=-\frac{1}{2}$ **M1 A1**

Obtain root $x=\frac{3}{2}$ **A1** OE

Alternative method for question 7(b)

State second root $-1-\sqrt{5}i$ **B1**

$$(x+1-\sqrt{5}i)(x+1+\sqrt{5}i)(2x+a)=2x^3+x^2+6x-18$$

M1 A1

Obtain root $x=\frac{3}{2}$ **A1** OE

Total: 4

Q104 Oct/Nov 2020 | Paper 3 (variant 33) | Question 9

Topics: Algebra

Question 9: Let $f(x) = \frac{8+5x+12x^2}{(1-x)(2+3x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Examiner comment

> "The most common error was to multiply throughout by an expression involving (2 + 3x)3."

> "(b) This question proved more challenging for some candidates."

Mark scheme

9(a) State or imply the form

$$\frac{A}{1-x} + \frac{B}{2+3x} + \frac{C}{(2+3x)^2}$$

B1

Use a correct method for finding a coefficient **M1**

Obtain one of $A = 1, B = -1, C = 6$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

In the form $\frac{A}{1-x} + \frac{Dx+E}{(2+3x)^2}$, where $A = 1, D = -3$ and $E = 4$ can score **B1 M1 A1 A1 A1** as above.

Total: 5

9(b) Use a correct method to find the first two terms of the expansion of $(1-x)^{-1}, (2+3x)^{-1}$, or $\left(1+\frac{3}{2}x\right)^{-1}, (2+3x)^{-2}$ or $\left(1+\frac{3}{2}x\right)^{-2}$ **M1**

Symbolic coefficients are not sufficient for the M1:

$$\left[\frac{1+(-1)(-x)+(-1)(-2)(-x)^2}{2\ldots}\right] \quad A = 1$$

$$\left[\frac{1+(-1)\left(\frac{3x}{2}\right)+(-1)(-2)\left(\frac{3x}{2}\right)^2}{2\ldots}\right] \quad B = 1$$

$$\left[\frac{1+(-2)\left(\frac{3x}{2}\right)+(-2)(-3)\left(\frac{3x}{2}\right)^2}{4}\right] \quad C = 6$$

Obtain correct un-simplified expansions up to the term in x^2 of each partial fraction **A1 FT** plus

$$(1+x+x^2) + \left(-\frac{1}{2} + \frac{3}{4}x - \frac{9}{8}x^2\right)$$

A1 FT plus

$$\left(6-\frac{18}{4}x+\frac{81}{8}x^2\right)$$

[The FT is on A, B, C]

Obtain final answer $2-\frac{11}{4}x+10x^2$, or equivalent **A1**

Allow unsimplified fractions: $(Dx+E)\frac{1+(-2)\left(\frac{3x}{2}\right)+(-2)(-3)\left(\frac{3x}{2}\right)^2}{4}$ $D=-3, E=4$

The FT is on A, D, E.

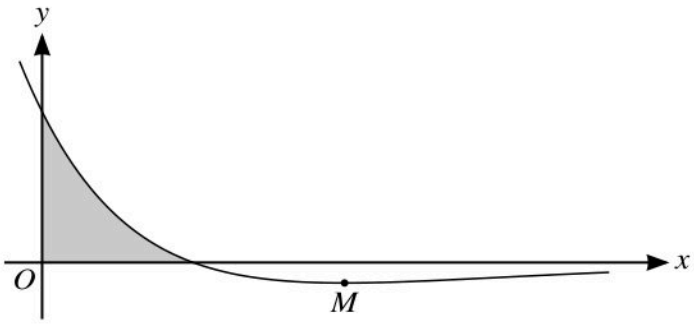
Total: 5

Q105 Oct/Nov 2020 | Paper 3 (variant 33) | Question 10

Topics: Differentiation, Integration

Question 10:

10



The diagram shows the curve $y = (2-x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. [5]

The diagram shows the curve $y=(2-x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. **[5]**

(b) Find the area of the shaded region bounded by the curve and the axes. Give your answer in terms of e. **[5]**

Examiner comment

> "Question 10 (a) This should have been a straightforward application of the product rule for differentiation, but many –x candidates struggled because they were unable to differentiate e2 ."

> "Candidates often found the integration of e2 difficult, with similar errors to (a) seen."

Mark scheme

10(a) Use correct product or quotient rule *M1

$$\frac{dy}{dx} = \left(-\frac{1}{2}\right)(2-x)e^{-\frac{1}{2}x} - e^{-\frac{1}{2}x}$$

M1 requires at least one of derivatives correct

Obtain correct derivative in any form **A1**

Equate derivative to zero and solve for x **DM1**

Obtain $x = 4$ **A1** ISW

Obtain $y = -2e^{-2}$, or exact equivalent **A1**

Total: 5

10(b) Commence integration and reach

$$a(2-x)e^{-\frac{1}{2}x} + b\int e^{-\frac{1}{2}x}\,dx$$

***M1**

Condone omission of dx

$$-2(2-x)e^{-\frac{1}{2}x} - 4e^{-\frac{1}{2}x} \quad \text{or} \quad 2xe^{-\frac{1}{2}x}$$

Obtain $-2(2-x)e^{-\frac{1}{2}x} - 2\int e^{-\frac{1}{2}x}\,dx$ **A1** OE

Complete integration and obtain $2xe^{-\frac{1}{2}x}$ **A1** OE

Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice **DM1**

Ignore omission of zeros and allow max of 1 error

Obtain answer $4e^{-1}$, or exact equivalent **A1** ISW

Alternative method for question 10(b)

$$\frac{d}{dx}\left(2xe^{-\frac{1}{2}x}\right) = 2e^{-\frac{1}{2}x} - xe^{-\frac{1}{2}x}$$

***M1 A1**

$$\therefore 2xe^{-\frac{1}{2}x}$$

Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice **DM1**

Ignore omission of zeros and allow max of 1 error

Obtain answer $4e^{-1}$, or exact equivalent **A1** ISW

Total: 5

Q106 Feb/Mar 2019 | Paper 3 (variant 32) | Question 2

Topics: Numerical Solutions/Iteration

The sequence of values given by the iterative formula

$$x_{n+1} = \frac{2x_n^6 + 12x_n}{3x_n^5 + 8},$$

with initial value $x_1 = 2$, converges to α.

(i) Use the formula to calculate α correct to 4 decimal places. Give the result of each iteration to 6 decimal places. [3]

(ii) State an equation satisfied by α and hence find the exact value of α. [2]

Examiner comment

> "Answer: 1.3195 (ii) Again most candidates gained both marks, although a few candidates did not use consistent symbols throughout their equations, whilst others gave a decimal answer rather than an exact answer."

Mark scheme

(i)

Use the iterative formula correctly at least once **M1**

Obtain answer **1.3195 A1**

Show sufficient iterations to 6 d.p. to justify **1.3195** to 4 d.p., or show there is a sign change in $(1.31945, 1.31955)$ **A1**

Total: 3

(ii)

State $x = \frac{2x^6+12x}{3x^5+8}$, or equivalent **B1**

State answer $\sqrt[5]{4}$, or exact equivalent **B1**

Total: 2

Q107 Feb/Mar 2019 | Paper 3 (variant 32) | Question 3

Topics: Trigonometry

(i) Given that $\sin(\theta + 45°) + 2\cos(\theta + 60°) = 3\cos\theta$, find the exact value of $\tan\theta$ in a form involving surds. You need not simplify your answer. [4]

(ii) Hence solve the equation $\sin(\theta + 45°) + 2\cos(\theta + 60°) = 3\cos\theta$ for $0° < \theta < 360°$. [2]

Examiner comment

> "Question 3 (i) Most candidates were able to correctly expand sin(θ + 45°), but cos(θ + 60°) proved more difficult."

> "Answer: tan θ = (2√2 – 1)/(1 – √6) or equivalent (ii) Many candidates did not gain full marks due to incorrectly opting for (360° – θ*) instead of (180° + θ*) as their second answer."

Mark scheme

(i)

Use trig formulae and obtain an equation in $\sin\theta$ and $\cos\theta$ **M1**

Obtain a correct equation in any form **A1**

Substitute exact trig ratios and obtain an expression for $\tan\theta$ **M1**

Obtain answer $\tan\theta = \frac{2\sqrt{2}-1}{1-\sqrt{6}}$, or equivalent **A1**

Total: 4

(ii)

State answer, e.g. $\theta = 128.4°$ **B1**

State second answer, e.g. $\theta = 308.4°$ **B1 ft**

Total: 2

Q108 Feb/Mar 2019 | Paper 3 (variant 32) | Question 5

Topics: Differentiation

The variables x and y satisfy the relation $\sin y = \tan x$, where $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$. Show that

$$\frac{dy}{dx} = \frac{1}{\cos x\sqrt{\cos 2x}}.$$

[5]

Examiner comment

> "Question 5 Nearly all candidates found this to be a difficult question."

> "Unless they realised that either implicit differentiation or the derivative of sin–1 (tan x) involving the chain rule was required, no progress was possible."

> "However whilst many candidates did reach this stage, the conversion to the given answer as a function of x proved too difficult for most."

Mark scheme

State $\cos y\frac{dy}{dx}$ as derivative of $\sin y$ **B1**

State correct derivative in terms of x and y, e.g. $\frac{\sec^2 x}{\cos y}$ **B1**

State correct derivative in terms of x, e.g. $\frac{\sec^2 x}{\sqrt{1-\tan^2 x}}$ **B1**

Use double angle formula **M1**

Obtain the given answer correctly **A1**

Total: 5

Q109 May/June 2019 | Paper 3 (variant 31) | Question 1

Topics: Integration

Use the trapezium rule with 3 intervals to estimate the value of

$$\int_0^3 |2^x - 4|\, dx.$$

[3]

Examiner comment

> "The most common error was to ignore the modulus signs and use ordinates –3, –2, 0 and 4."

Mark scheme

State or imply the ordinates

$$3,\ 2,\ 0,\ 4$$

B1

Use the trapezium rule correctly with $h = 1$ and the four ordinates **M1**

Obtain

$$5.5$$

A1

Total: 3

Q110 May/June 2019 | Paper 3 (variant 31) | Question 2

Topics: Algebra, Indices, Logarithms & Exponents

Showing all necessary working, solve the equation

$$\ln(2x - 3) = 2\ln x - \ln(x - 1).$$

Give your answer correct to 2 decimal places. [4]

Examiner comment

> "The most common errors in the working occurred when candidates started with $-(-1)=$ before dealing with the 2."

Mark scheme

Use a logarithm law for a product, quotient or power **M1**

Obtain a correct equation free of logarithms, for example

$$(2x-3)(x-1)=x^2$$

or

$$x^2-5x+3=0$$

A1

Solve the three-term quadratic, obtaining at least one root **M1**

Obtain the valid answer only:

$$x=4.30$$

A1

Total: 4

Q111 May/June 2019 | Paper 3 (variant 31) | Question 3

Topics: Differentiation

Find the gradient of the curve

$$x^3+3xy^2-y^3=1$$

at the point with coordinates $(1, 3)$. [4]

Examiner comment

> "The most common errors were to introduce an additional term =⋯ at the start of their work or to not differentiate the 1 on the right hand side of the equation."

Mark scheme

State or imply

$$3y^2+6xy\frac{dy}{dx}$$

as the derivative of $3xy^2$ **B1**

State or imply

$$3y^2\frac{dy}{dx}$$

as the derivative of y^3 **B1**

Equate the derivative of the left hand side to zero, substitute $(1, 3)$ and find the gradient **M1**

Obtain

$$\frac{10}{3}$$

A1

Total: 4

Q112 May/June 2019 | Paper 3 (variant 31) | Question 4

Topics: Trigonometry

By first expressing the equation

$$\cot\theta - \cot(\theta + 45°) = 3$$

as a quadratic equation in $\tan\theta$, solve the equation for

$$0° < \theta < 180°.$$

[6]

Examiner comment

> "Question 4 Most candidates were aware that θ = , but some found dealing with (θ + 45°) challenging."

Mark scheme

Use a correct trigonometric formula to obtain an equation in $\tan\theta$ **M1**

Obtain a correct equation in any form, for example

$$\frac{1}{\tan\theta} - \frac{1 - \tan\theta}{1 + \tan\theta} = 3$$

A1

Reduce to

$$2\tan^2\theta + 3\tan\theta - 1 = 0$$

A1

Solve the three-term quadratic and find a value of θ **M1**

Obtain

$$\theta = 15.7°$$

A1

Obtain

$$\theta = 119.3°$$

with no other values in the range. **A1**

Total: 6

Q113 May/June 2019 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry, Integration

(i) By first expanding $\sin(2x + x)$, show that

$$\sin 3x \equiv 3\sin x - 4\sin^3 x.$$

[4]

(ii) Hence, showing all necessary working, find the exact value of

$$\int_0^{\pi/3} \sin^3 x \, dx.$$

[4]

Examiner comment

> "(ii) Most candidates did not use the result from part (i) to tackle this integral."

Mark scheme

(i)

State the correct expansion of

$$\sin(2x + x)$$

B1

Use trigonometric formulae and Pythagoras to express $\sin 3x$ in terms of $\sin x$ **M1**

Obtain a correct expression in any form **A1**

Obtain

$$\sin 3x \equiv 3\sin x - 4\sin^3 x$$

A1

Total: 4

(ii)

Use the identity, integrate, and obtain the terms

$$-\frac{1}{4}\cos x$$

and

$$\frac{1}{12}\cos 3x$$

B1 B1

Use the limits correctly in an integral of the form

$$a\cos x + b\cos 3x,$$

where $ab \neq 0$ **M1**

Obtain the exact value

$$\frac{5}{24}$$

A1

Total: 4

Q114 May/June 2019 | Paper 3 (variant 32) | Question 1

Topics: Algebra

Find the coefficient of x^3 in the expansion of

$$(3-x)(1+3x)^{\frac{1}{3}}$$

in ascending powers of x. [4]

Examiner comment

> "The most common error in expanding (1+3) was to use in place of 3."

Mark scheme

State the unsimplified term in x^2, or its coefficient, in the expansion of $(1+3x)^{1/3}$ **B1**

State the unsimplified term in x^3, or its coefficient, in the expansion of $(1+3x)^{1/3}$ **B1**

Multiply by $(3-x)$ to give the two terms contributing to x^3 **M1**

Obtain the coefficient

$$6$$

A1

Total: 4

Q115 May/June 2019 | Paper 3 (variant 32) | Question 2

Topics: Algebra, Indices, Logarithms & Exponents

Showing all necessary working, solve the equation

$$9^x = 3^x + 12.$$

Give your answer correct to 2 decimal places. [4]

Examiner comment

> "Another common error was to use 9 =3×3."

Mark scheme

State or imply, with $u = 3^x$,

$$u^2 - u - 12 = 0$$

B1

Solve for u or 3^x and obtain the positive root

$$u = 4$$

B1

Use a correct method to solve

$$3^x = 4$$

M1

Obtain

$$x = 1.26$$

only. **A1**

Total: 4

Q116 May/June 2019 | Paper 3 (variant 32) | Question 5

Topics: Algebra, Complex Numbers

Throughout this question the use of a calculator is not permitted.

It is given that the complex number $-1 + \sqrt{3}\,i$ is a root of the equation

$$kx^3 + 5x^2 + 10x + 4 = 0,$$

where k is a real constant.

(i) Write down another root of the equation. [1]

(ii) Find the value of k and the third root of the equation. [6]

Examiner comment

> "However, the common error was to combine the quadratic factor with a linear factor + , in which case no further progress was possible."

Mark scheme

(i)

Since the coefficients are real, another root is

$$-1 - \sqrt{3}\,i$$

B1

Total: 1

(ii)

Substitute $x = -1 + \sqrt{3}\,i$ in the equation and attempt expansions of x^2 and x^3 **M1**

Use $i^2 = -1$ correctly at least once **M1**

Obtain

$$k = 2$$

A1

Carry out a complete method for finding the quadratic factor with roots $-1 + \sqrt{3}\,i$ and $-1 - \sqrt{3}\,i$ **M1**

Obtain the quadratic factor

$$x^2 + 2x + 4$$

A1

Obtain the third root

$$x = -\frac{1}{2}$$

A1

Total: 6

Q117 May/June 2019 | Paper 3 (variant 32) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

In the diagram, A is the mid-point of the semicircle with centre O and radius r. A circular arc with centre A meets the semicircle at B and C. The angle OAB is equal to x radians. The area of the shaded region bounded by AB, AC and the arc with centre A is equal to half the area of the semicircle.

(i) Use triangle OAB to show that

$$AB = 2r\cos x.$$

[1]

(ii) Hence show that

$$x = \cos^{-1}\sqrt{\frac{\pi}{16x}}.$$

[2]

(iii) Verify by calculation that x lies between 1 and 1.5. [2]

(iv) Use an iterative formula based on the equation in part (ii) to determine x correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Examiner comment

> "The most common errors occurred in the area of the sector, with many candidates using an angle of rather than2."

Mark scheme

(i)

Use correct trigonometry in triangle OAB to obtain

$$AB = 2r\cos x$$

B1

Total: 1

(ii)

Use the correct method for finding the area of the sector and the semicircle and form an equation in x **M1**

Obtain

$$x = \cos^{-1}\sqrt{\frac{\pi}{16x}}$$

correctly. **A1**

Total: 2

(iii)

Calculate values of a relevant expression at $x = 1$ and $x = 1.5$, working in radians **M1**

Use correct values and complete the sign-change or bounding argument correctly **A1**

Total: 2

(iv)

Use the iteration

$$x_{n+1} = \cos^{-1}\sqrt{\frac{\pi}{16x_n}}$$

correctly at least twice, working in radians **M1**

Obtain

$$x = 1.144$$

A1

Show sufficient iterations to at least 5 decimal places to justify the answer to 3 decimal places **A1**

Total: 3

Q118 May/June 2019 | Paper 3 (variant 32) | Question 9

Topics: none of the listed topics

The points A and B have position vectors

$$\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

and

$$3\mathbf{i} + \mathbf{j} + \mathbf{k}$$

respectively. The line l has equation

$$\mathbf{r} = 2\mathbf{i} + \mathbf{j} + \mathbf{k} + \mu(\mathbf{i} + \mathbf{j} + 2\mathbf{k}).$$

(i) Show that l does not intersect the line passing through A and B. [5]

(ii) The plane m is perpendicular to AB and passes through the mid-point of AB. The plane m intersects the line l at the point P. Find the equation of m and the position vector of P. [5]

Examiner comment

> "There was one very common error: candidates who reached a correct equation 3=2 often concluded that = ."

Mark scheme

(i)

Find a vector equation for the line AB **M1**

Obtain, for example,

$$\mathbf{r} = (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) + \lambda(2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

A1

Equate two pairs of components of general points on this line and on l, and solve for λ or μ **M1**

Obtain a correct value, for example $\lambda = 0$ and $\mu = -1$ from one pair of component equations **A1**

Verify that all three component equations are not satisfied, so the lines do not intersect **A1**

Total: 5

(ii)

State or imply that the midpoint of AB has position vector

$$2\mathbf{i} + \frac{3}{2}\mathbf{j}$$

B1

Use the normal direction parallel to AB and substitute the midpoint to find the constant in the plane equation **M1**

Obtain the plane equation

$$4x - 2y + 4z = 5$$

(or equivalent). **A1**

Substitute the components of l in the plane equation and solve for μ **M1**

Obtain

$$\mu = -\frac{1}{2}$$

and hence the position vector of P:

$$\frac{3}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$$

A1

Total: 5

Q119 May/June 2019 | Paper 3 (variant 33) | Question 1

Topics: Indices, Logarithms & Exponents

Use logarithms to solve the equation

$$5^{3-2x} = 4(7^x),$$

giving your answer correct to 3 decimal places. [4]

Examiner comment

> "Question 1 Many fully correct answers were seen, and only a few candidates were unable to present their final answer to the accuracy required."

Mark scheme

Use the logarithm law for a product or quotient **M1**

Use the logarithm law for powers twice **M1**

Obtain a correct linear equation in x, for example

$$(3-2x)\ln 5 = \ln 4 + x\ln 7$$

A1

Obtain

$$x = 0.666$$

A1

Total: 4

Q120 May/June 2019 | Paper 3 (variant 33) | Question 4

Topics: Indices, Logarithms & Exponents, Differentiation

The equation of a curve is

$$y = \frac{1+e^{-x}}{1-e^{-x}}, \qquad x > 0.$$

(i) Show that $\frac{dy}{dx}$ is always negative. [3]

(ii) The gradient of the curve is equal to -1 when $x = a$.

Show that a satisfies the equation

$$e^{2a} - 4e^{a} + 1 = 0.$$

Hence find the exact value of a. [4]

Examiner comment

> "Many candidates with a correct expression were unable to make any progress, for example inverting terms individually to change the equation from one in e–a to one in ea."

> "Although few candidates obtained the equation correctly, most realised it was a quadratic in ea and were able to solve it."

> "Most candidates did not reject one of the roots, and many did not give the final answer in exact form."

Mark scheme

(i)

Use the quotient rule or product rule **M1**

Obtain the correct derivative in any form **A1**

Reduce to

$$-\frac{2e^{-x}}{(1-e^{-x})^2}$$

or an equivalent expression, and explain why this is always negative for $x > 0$. **A1**

Total: 3

(ii)

Equate the derivative to -1 and obtain the given equation

$$e^{2a} - 4e^{a} + 1 = 0$$

B1

State or imply the quadratic

$$u^2 - 4u + 1 = 0,$$

where $u = e^a$. **B1**

Solve for a **M1**

Obtain

$$a = \ln(2 + \sqrt{3})$$

with no other value. **A1**

Total: 4

Q121 Oct/Nov 2019 | Paper 3 (variant 31) | Question 1

Topics: Indices, Logarithms & Exponents

1 Given that $\ln(1 + e^{2y}) = x$, express y in terms of x. **[3]**

Examiner comment

> "The next step of rearranging to make y the subject of the equation proved more challenging for candidates."

Mark scheme

State $1 + e^{2y} = e^x$ **B1**

Make y the subject **M1** Rearrange to $e^{2y} = \ldots$ and use logs

Obtain answer $y = \frac{1}{2}\ln(e^x - 1)$ **A1** OE

Total: 3

Q122 Oct/Nov 2019 | Paper 3 (variant 31) | Question 4

Topics: Differential Equations

4 The number of insects in a population t weeks after the start of observations is denoted by N. The population is decreasing at a rate proportional to $Ne^{-0.02t}$. The variables N and t are treated as continuous, and it is given that when $t = 0$, $N = 1000$ and $\frac{dN}{dt} = -10$.

(i) Show that N and t satisfy the differential equation

$$\frac{dN}{dt} = -0.01e^{-0.02t}N.$$

[1]

(ii) Solve the differential equation and find the value of t when $N = 800$. **[6]**

(iii) State what happens to the value of N as t becomes large. **[1]**

Examiner comment

> "(iii) Very few correct answers were seen."

Mark scheme

(i) State $\frac{dN}{dt} = ke^{-0.02t}N$ and show $k = -0.01$ **B1** OE ($-10 = k \times 1 \times 1000$)

Total: 1

(ii) Separate variables correctly and integrate at least one side **B1** $\int \frac{1}{N}dN = \int -0.01e^{-0.02t}dt$

Obtain term $\ln N$ **B1** OE

Obtain term $0.5e^{-0.02t}$ **B1** OE

Use $N = 1000, t = 0$ to evaluate a constant, or as limits, in a solution with terms $a \ln N$ and $be^{-0.02t}$, where $ab \neq 0$ **M1**

Obtain correct solution in any form
e.g. $\ln N - \ln 1000 = 0.5(e^{-0.02t} - 1)$ **A1** $\ln 1000 - \frac{1}{2} = 6.41$

Substitute $N = 800$ and obtain $t = 29.6$ **A1**

Total: 6

(iii) State that N approaches $\frac{1000}{\sqrt{e}}$ **B1** Accept 606 or 607 or 606.5

Total: 1

Q123 Oct/Nov 2019 | Paper 3 (variant 31) | Question 7

Topics: none of the listed topics

7 Two lines l and m have equations $\mathbf{r} = a\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \lambda(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k} + \mu(2\mathbf{i} - \mathbf{j} + \mathbf{k})$ respectively, where a is a constant. It is given that the lines intersect.

(i) Find the value of a. **[4]**

(ii) When a has this value, find the equation of the plane containing l and m. **[5]**

Examiner comment

> "Common errors were often due to slips with signs when forming equations or in the arithmetic."

> "The most common errors were sign errors and arithmetic errors in completing the vector product."

Mark scheme

(i) Express general point of l or m in component form

e.g. $(a + \lambda, 2 - 2\lambda, 3 + 3\lambda)$ or $(2 + 2\mu, 1 - \mu, 2 + \mu)$ **B1**

Equate at least two pairs of corresponding components and solve for λ or for μ **M1**

Obtain **either** $\lambda = -2$ or $\mu = -5$ **A1**

or $\lambda = \frac{1}{3}a$ or $\mu = \frac{2}{3}a - 1$ **A1**

or $\lambda = \frac{1}{5}(a - 4)$ or $\mu = \frac{1}{5}(3a - 7)$ **A1**

Obtain $a = -6$ **A1**

Total: 4

(ii) Use scalar product to obtain a relevant equation in a, b and c, e.g. $a - 2b + 3c = 0$ **B1**

Obtain a second equation, e.g. $2a - b + c = 0$ and solve for one ratio **M1**

Obtain $a : b : c = 1 : 5 : 3$ **A1** OE

Substitute a relevant point and values of a, b, c in general equation and find d **M1**

Obtain correct answer $x + 5y + 3z = 13$ **A1FT** OE. The FT is on a from part (i), if used

Alternative method for question 7(ii)

Attempt to calculate vector product of relevant vectors, **M1** e.g. $(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}).(2\mathbf{i} - \mathbf{j} + \mathbf{k})$

Obtain two correct components **A1**

Obtain correct answer, e.g. $\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ **A1**

Substitute a relevant point and find d **M1**

Obtain correct answer $x + 5y + 3z = 13$ **A1FT** OE. The FT is on a from part (i), if used

Total: 5

Q124 Oct/Nov 2019 | Paper 3 (variant 31) | Question 8

Topics: Algebra, Integration

8 Let $f(x) = \frac{x^2+x+6}{x^2(x+2)}$.

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence, showing full working, show that the exact value of $\int_1^4 f(x)\,dx$ is $\frac{9}{4}$. **[5]**

Examiner comment

> "The most common error, apart from in arithmetic, was to omit either the term with denominator x, or the term with denominator x2."

Mark scheme

(i) State or imply the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$ **B1**

Use a correct method for finding a constant **M1**

Obtain one of $A = -1$, $B = 3$, $C = 2$ **A1**

Obtain a second value **A1**

Obtain the third value **A1** Allow in the form $\frac{4x+B}{x^2} + \frac{C}{x+2}$

Total: 5

(ii) Integrate and obtain terms $\ln x - \frac{3}{x} + 2\ln(x+2)$ **B1FT** + The FT is on A, B, C; or on A, D, E.

Substitute limits correctly in an integral with terms $a\ln x$, $\frac{b}{x}$ and $c\ln(x+2)$, where $abc \neq 0$ **M1**
$-\ln 4 - \frac{3}{4} + 2\ln 6 (+\ln 1) + 3 - 2\ln 3$

Obtain $\frac{9}{4}$ following full and exact working **A1** AG - work to combine or simplify logs is required

Total: 5

Q125 Oct/Nov 2019 | Paper 3 (variant 32) | Question 2

Topics: Algebra, Differentiation

2 The curve with equation $y = \frac{e^{-x}}{1-x^2}$ has a stationary point in the interval $-1 < x < 1$. Find $\frac{dy}{dx}$ and hence find the x-coordinate of this stationary point, giving the answer correct to 3 decimal places. **[5]**

3 The polynomial $x^4 + 3x^3 + ax + b$, where a and b are constants, is denoted by $p(x)$. When $p(x)$ is divided by $x^2 + x - 1$ the remainder is $2x + 3$. Find the values of a and b. **[5]**

Examiner comment

> "A few candidates used the product rule, but the derivative of (1 – x2)–1 proved challenging."

Mark scheme

2

Use correct quotient rule or correct product rule **M1**

Obtain correct derivative in any form **A1** $\frac{dy}{dx} = \frac{-2e^{-x}(1-x^2)+2xe^{-x}}{(1-x^2)^2}$

Equate derivative to zero and obtain a 3 term quadratic in x **M1** From correct work only
e.g. $2x^2 + 2x - 2 = 0$ or $x^2 + x - 1$ **A1**

Solve and obtain $x = 0.618$ only **A1** From correct work only

Total: 5

3

Divide $x^4 + 3x^3 + ax + b$ by $x^2 + x - 1$ to obtain remainder in the form $cx + d$ or divide and use remainder theorem directly **M1** [3 marks]

Quotient of form $x^2 + 2x + k$ and remainder $cx + d$ where c and d depend on a and k **A1**

Form two equations, e.g. $c = a + 2k + 2 = 2$ and $d = -k + b - 2 = 3$ [needs 3 marks] **M1**

Obtain both a and b **A1A1**

Total: 5

Q126 Oct/Nov 2019 | Paper 3 (variant 32) | Question 3

Topics: Indices, Logarithms & Exponents

1 Solve the equation $5\ln(4 - 3^x) = 6$. Show all necessary working and give the answer correct to 3 decimal places. **[3]**

Examiner comment

> "Question 3 This question appeared to be particularly challenging for candidates."

> "Having reached a linear remainder, many candidates did not go on to set their remainder identically equal to 2x+3, and to solve for a and b accordingly."

> "A common error was for candidates to treat2x+3 as a factor."

Mark scheme

Question 3

Commence division and reach partial quotient $x^2 + kx$ | M1

Obtain correct quotient $x^2 + 2x - 1$ | A1

Set their linear remainder equal to $2x + 3$ and solve for a or for b | M1 | Remainder $= (a + 3)x + (b - 1)$

Obtain answer $a = -1$ | A1

Obtain answer $b = 4$ | A1

Alternative method for question 3

State $x^4 + 3x^3 + ax + b = (x^2 + x - 1)(x^2 + Ax + B) + 2x + 3$ and form and solve two equations in A and B | **M1** | e.g. $3 = 1 + A$ and $0 = -1 + A + B$

Obtain $A = 2, B = -1$ | **A1**

Form and solve equations for a or b | **M1** | e.g. $a = B - A + 2, b = -B + 3$

Obtain answer $a = -1$ | **A1**

Obtain answer $b = 4$ | **A1**

Total: 5

Q127 Oct/Nov 2019 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry

(i) Express $(\sqrt{6})\sin x + \cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. State the exact value of R and give α correct to 3 decimal places. **[3]**

(ii) Hence solve the equation $(\sqrt{6})\sin 2\theta + \cos 2\theta = 2$, for $0^\circ < \theta < 180^\circ$. **[4]**

Examiner comment

> "(ii) Most candidates found this question challenging."

> "Many candidates did not work with enough accuracy in answering this question, given that part (i) required an answer to 3 decimal places, candidates should have identified that at least 3 decimal places was needed for this part."

Mark scheme

4(i) | **State $R = \sqrt{7}$** | **B1**

Use correct trig formulae to find a | **M1** | e.g. $\tan\alpha = \frac{1}{\sqrt{6}}, \sin\alpha = \frac{1}{\sqrt{7}}$, or $\cos\alpha = \frac{\sqrt{6}}{\sqrt{7}}$

Obtain $\alpha = 22.208^\circ$ | **A1** | ISW

4(ii) | **Evaluate $\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)$ to at least 1 d.p.** | **B1FT** | 49.107° to 3 d.p. **B1 can be implied by correct answer(s) later. The FT is on their R**

SC: allow B1 for a correct alternative equation e.g. $3\tan^2\theta - 2\sqrt{6}\tan\theta + 1 = 0$

Use correct method to find a value of θ in the interval | **M1** | **Must get to θ**

Obtain answer, e.g. 13.4° | **A1** | Accept correct over-specified answers. $13.449\ldots, 54.3425\ldots$

Obtain second answer, e.g. 54.3° and no extras in the given interval | **A1** | Ignore answers outside the given interval.

Total: 4

Q128 Oct/Nov 2019 | Paper 3 (variant 32) | Question 6

Topics: Differential Equations

6 The variables x and θ satisfy the differential equation

$$\sin\frac{1}{2}\theta\frac{dx}{d\theta} = (x+2)\cos\frac{1}{2}\theta$$

for $0 < \theta < \pi$. It is given that $x = 1$ when $\theta = \frac{1}{3}\pi$. Solve the differential equation and obtain an expression for x in terms of $\cos\theta$. **[8]**

Examiner comment

> "Most candidates went on to obtain kln sin1 θ but 2 only a minority had k = 2.Many candidates found difficulty in the correct simplification and removal of the 1 1 logarithms."

> "Very few fully correct solutions were seen as the majority of candidates did not attempt the final step to express the answer in terms of cosθ."

Mark scheme

6 | Separate variables correctly to obtain $\int \frac{1}{x+2}dx = \int \cot\frac{1}{2}\theta d\theta$ | B1 | Or equivalent integrands. Integral signs SOI

Obtain term $\ln(x+2)$ | B1 | Modulus signs not needed.

Obtain term of the form $k\ln\sin\frac{1}{2}\theta$ | M1

Obtain term $2\ln\sin\frac{1}{2}\theta$ | A1

Use $x = 1, \theta = \frac{1}{3}\pi$ to evaluate a constant, or as limits, in an expression containing $p\ln(x+2)$ and $q\ln\left(\sin\frac{1}{2}\theta\right)$ | M1 | Reach $C =$ an expression or a decimal value

Obtain correct solution in any form e.g. $\ln(x+2) = 2\ln\sin\frac{1}{2}\theta + \ln 12$ | **A1** | $\ln 12 = 2.4849\ldots$ Accept constant to at least 3 s.f. Accept with $\ln 3 - 2\ln\frac{1}{2}$

Remove logarithms and use correct double angle formula | M1 | Need correct algebraic process. $\left(\frac{x+2}{12} = \frac{1-\cos\theta}{2}\right)$

Obtain answer $x = 4 - 6\cos\theta$ | A1

Total: 8

Q129 Oct/Nov 2019 | Paper 3 (variant 32) | Question 7

Topics: Complex Numbers

(a) Find the complex number z satisfying the equation

$$z + \frac{iz}{z^*} - 2 = 0,$$

where z^* denotes the complex conjugate of z. Give your answer in the form $x + iy$, where x and y are real. **[5]**

(b) (i) On a single Argand diagram sketch the loci given by the equations $|z - 2i| = 2$ and $\text{Im } z = 3$, where Im z denotes the imaginary part of z. **[2]**

(ii) In the first quadrant the two loci intersect at the point P. Find the exact argument of the complex number represented by P. **[2]**

Examiner comment

> "Question 7 (a) Many candidates found this question challenging, often candidates would confuse their working involving z or z* rather than x and y."

> "Many candidates found drawing the line Imz=3 challenging, with several candidates drawing a second circle."

Mark scheme

7(a) | Substitute and obtain a correct horizontal equation in x and y in any form | B1 |

$zz^* + iz - 2z^* = 0 \Rightarrow x^2 + y^2 + ix - y - 2x + 2iy = 0$ Allow if still includes brackets and/or i^2

Use $i^2 = -1$ and equate real and imaginary parts to zero OE | M1 | For their horizontal equation

Obtain two correct equations e.g. $x^2 + y^2 - y - 2x = 0$ and $x + 2y = 0$ **| A1 |** Allow $ix + 2iy = 0$

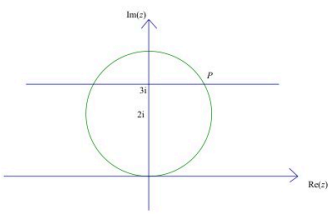
Solve for x or for y | DM1

Obtain answer $\frac{6}{5} - \frac{3}{5}i$ and no other | A1 | OE, condone $\frac{1}{5}(6 - 3i)$

Total: 5

7(b)(i) | Show a circle with centre $2i$ and radius 2 | B1

Show horizontal line $y = 3$ in first and second quadrant | B1 |

Question	Answer	Marks	Guidance
7(b)(i)	Show a circle with centre 2i and radius 2	B1	
	Show horizontal line y = 3 – in first and second quadrant	B1	
			SC: For clearly labelled axes not in the conventional directions, allow B1 for a fully 'correct' diagram.
		2	

7(b)(ii)	Carry out a complete method for finding the argument. (Not by measuring the sketch)	**M1**	$\left(z=\sqrt{3}+3\mathrm{i}\right)$ Must show working if using 1.7 in place of $\sqrt{3}$.
	Obtain answer $\frac{1}{3}\pi$ (or 60°)	**A1**	**SC:** Allow B2 for 60° with no working
		2	

SC: For clearly labelled axes not in the conventional directions, allow B1 for a fully 'correct' diagram.

Total: 2

7(b)(ii) | Carry out a complete method for finding the argument. (Not by measuring the sketch) | M1 | $(z = \sqrt{3} + 3i)$ Must show working if using 1.7 in place of $\sqrt{3}$.

Obtain answer $\frac{1}{3}\pi$ (or 60°) | A1 | SC: Allow B2 for 60° with no working

Total: 2

Q130 Oct/Nov 2019 | Paper 3 (variant 32) | Question 8

Topics: Algebra, Integration

8 Let

$$f(x) = \frac{2x^2 + x + 8}{(2x-1)(x^2+2)}.$$

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence, showing full working, find

$$\int_1^5 f(x)\,dx,$$

giving the answer in the form $\ln c$, where c is an integer. **[5]**

Examiner comment

> "In x2 +2 x2 +2 this question many candidates did not include brackets, which was a common error seen."

Mark scheme

8(i) | State or imply the form $\frac{4}{2x-1} + \frac{Bx+C}{x^2+2}$ | B1

Use a correct method for finding a constant | M1

Obtain one of $A = 4, B = -1, C = 0$ | A1

Obtain a second value | A1

Obtain the third value | A1

Total: 5

8(ii) | Integrate and obtain term $2\ln(2x-1)$ | B1FT | The FT is on A. $\frac{1}{2}4\ln(2x-1)$

Integrate and obtain term of the form $k\ln(x^2+2)$ | M1 | From $\frac{nx}{x^2+2}$

Obtain term $-\frac{1}{2}\ln(x^2+2)$ | A1FT | The FT is on B

Substitute limits correctly in an integral of the form $a\ln(2x-1)+b\ln(x^2+2)$, where $ab \neq 0$ | DM1 | $2\ln 9(-2\ln 1)-\frac{1}{2}\ln 27+\frac{1}{2}\ln 3$

Obtain answer $\ln 27$ after full and correct exact working | **A1** | ISW

Total: 5

Q131 Oct/Nov 2019 | Paper 3 (variant 33) | Question 2

Topics: Algebra

2 The polynomial $6x^3+ax^2+bx-2$, where a and b are constants, is denoted by **p**(x). It is given that $(2x+1)$ is a factor of **p**(x) and that when **p**(x) is divided by $(x+2)$ the remainder is -24. Find the values of a and b. **[5]**

Examiner comment

> "Candidates doing so usually gained full marks, although a few candidates incorrectly used the remainder + 24 instead of – 24."

Mark scheme

2 | Substitute $x=-\frac{1}{2}$, equate result to zero and obtain a correct equation, e.g. $-\frac{6}{8}+\frac{1}{4}a-\frac{1}{2}b-2=0$ | B1

Substitute $x=-2$ and equate result to -24 | M1

Obtain a correct equation, e.g. $-48+4a-2b-2=-24$ | A1

Solve for a or for b | DM1

Obtain $a=5$ and $b=-3$ | A1

Total: 5

Q132 Oct/Nov 2019 | Paper 3 (variant 33) | Question 4

Topics: Trigonometry

(i) By first expanding $\tan(2x+x)$, show that the equation $\tan 3x = 3\cot x$ can be written in the form $\tan^4 x - 12\tan^2 x + 3 = 0$. **[4]**

(ii) Hence solve the equation $\tan 3x = 3\cot x$ for $0^\circ < x < 90^\circ$. **[3]**

Examiner comment

> "A common error was to find inverse tan directly from t2 instead of first obtaining t."

Mark scheme

4(i) | Use $\tan(A+B)$ formula to express the LHS in terms of $\tan 2x$ and $\tan x$ | M1

Using the $\tan 2A$ formula, express the entire equation in terms of $\tan x$ | M1

Obtain a correct equation in $\tan x$ in any form | **A1**

Obtain the given form correctly | **A1** | AG

Total: 4

4(ii) | **Use correct method to solve the given equation for x** | **M1**

Obtain answer, e.g. $x = 26.8^\circ$ | **A1**

Obtain second answer, e.g. $x = 73.7^\circ$ and no other | **A1** | Ignore answers outside the given interval

Total: 3

Q133 Oct/Nov 2019 | Paper 3 (variant 33) | Question 5

Topics: Indices, Logarithms & Exponents, Numerical Solutions/Iteration

(i) By sketching a suitable pair of graphs, show that the equation $\ln(x+2) = 4e^{-x}$ has exactly one real root. **[2]**

(ii) Show by calculation that this root lies between $x = 1$ and $x = 1.5$. **[2]**

(iii) Use the iterative formula $x_{n+1} = \ln\left(\frac{4}{f(x_n)+2}\right)$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Examiner comment

> "Question 5 (i) Candidates appeared to find this question quite challenging."

Mark scheme

5(i) | **Sketch a relevant graph, e.g. $y = \ln(x+2)$** | **B1**

Sketch a second relevant graph, e.g. $y = 4e^{-x}$, and justify the given statement | **B1** | Consideration of behaviour for $x < 0$ is needed for the second B1

Total: 2

5(ii) | **Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$** | **M1**

Complete the argument correctly with correct calculated values | **A1**

Total: 2

5(iii) | **Set up and use the iterative formula or similar. Must interpret the notation correctly.** | **M1**

Give results of at least 2 iterations to 4 d.p. | **M1**

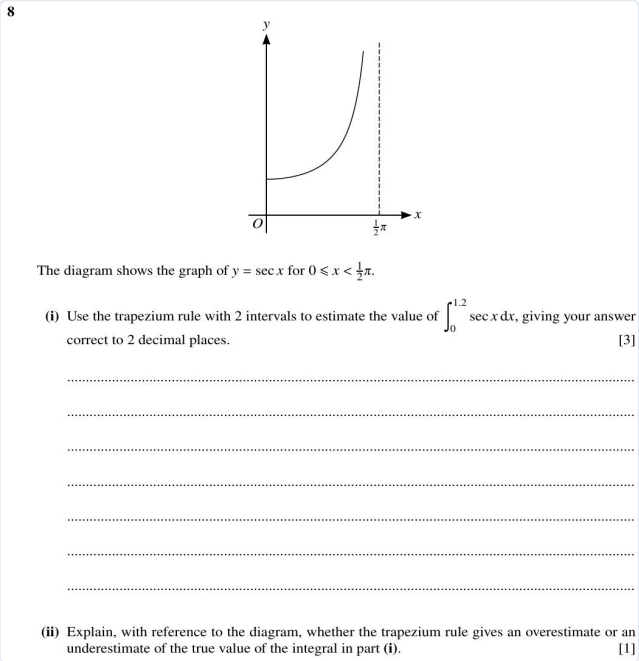
Obtain correct root to 2 d.p. | **A1**

Total: 3

Q134 Oct/Nov 2019 | Paper 3 (variant 33) | Question 8

Topics: Trigonometry, Differentiation, Integration

8 The diagram shows the graph of $y = \sec x$ for $0 \leq x < \frac{1}{2}\pi$.



8

The diagram shows the graph of $y = \sec x$ for $0 \leqslant x < \frac{1}{2}\pi$.

(i) Use the trapezium rule with 2 intervals to estimate the value of $\int_0^{1.2} \sec x \, dx$, giving your answer correct to 2 decimal places. [3]

(ii) Explain, with reference to the diagram, whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part (i). [1]

(i) Use the trapezium rule with 2 intervals to estimate the value of $\int_0^{1.2} \sec x \, dx$, giving your answer correct to 2 decimal places. **[3]**

(ii) Explain, with reference to the diagram, whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part **(i)**. **[1]**

(iii) P is the point on the part of the curve $y = \sec x$ for $0 \leq x < \frac{1}{2}\pi$ at which the gradient is 2. By first differentiating $\frac{1}{\cos x}$, find the x-coordinate of P, giving your answer correct to 3 decimal places. **[6]**

Examiner comment

> "Question 8 (i) Many candidates found this trapezium rule question challenging."

Mark scheme

(i) B1 State or imply ordinates $1, 1.2116\ldots, 2.7597\ldots$

M1 Use correct formula, or equivalent, with $h = 0.6$

A1 Obtain answer 1.85

Total: 3

(ii) B1 Explain why the rule gives an overestimate

Total: 1

(iii) M1 Differentiate using quotient or chain rule

A1 Obtain correct derivative in terms of $\sin x$ and $\cos x$

M1 Equate derivative to 2, use Pythagoras and obtain an equation in $\sin x$

A1 Obtain $2\sin^2 x + \sin x - 2 = 0$ OE

M1 Solve a 3-term quadratic for x

A1 Obtain answer $x = 0.896$ only

Total: 6

Q135 Oct/Nov 2019 | Paper 3 (variant 33) | Question 9

Topics: Algebra, Integration, Differential Equations

9 The variables x and t satisfy the differential equation $5\frac{dx}{dt} = (20 - x)(40 - x)$. It is given that $x = 10$ when $t = 0$.

(i) Using partial fractions, solve the differential equation, obtaining an expression for x in terms of t. **[9]**

(ii) State what happens to the value of x when t becomes large. **[1]**

Examiner comment

> "The most common error occurred after finding the value (40–x) of the arbitrary constant when candidates attempted to exponentiate."

Mark scheme

(i) B1 Separate variables correctly and integrate one side

B1 Obtain term $0.2t$, or equivalent

M1 Carry out a relevant method to obtain A and B such that $\frac{1}{(20-x)(40-x)} \equiv \frac{A}{20-x} + \frac{B}{40-x}$ OE

A1 Obtain $A = \frac{1}{20}$ and $B = -\frac{1}{20}$

A1FT+A1FT Integrate and obtain terms $-\frac{1}{20}\ln(20 - x) + \frac{1}{20}\ln(40 - x)$ OE. The FT is on A and B + **A1FT**

DM1 Use $x = 10, t = 0$ to evaluate a constant, or as limits

A1 Obtain correct answer in any form

A1 Obtain final answer $x = \frac{60e^{4t} - 40}{3e^{4t} - 1}$ OE

Total: 9

(ii) B1 State that x approaches 20

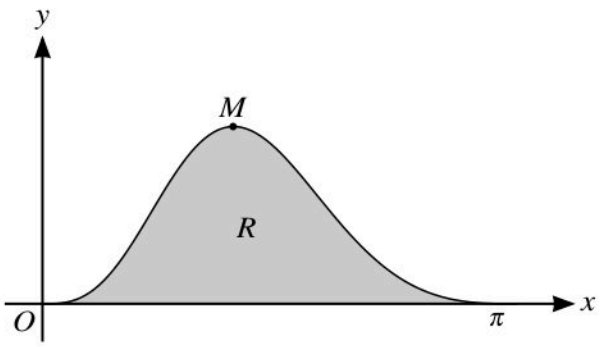
Total: 1

Q136 Oct/Nov 2019 | Paper 3 (variant 33) | Question 10

Topics: Trigonometry, Differentiation, Integration

10

10



The diagram shows the graph of $y = e^{\cos x} \sin^3 x$ for $0 \leqslant x \leqslant \pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M. Show all necessary working and give your answer correct to 2 decimal places. [5]

The diagram shows the graph of $y = e^{\cos x} \sin^3 x$ for $0 \leq x \leq \pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M. Show all necessary working and give your answer correct to 2 decimal places. **[5]**

(ii) By first using the substitution $u = \cos x$, find the exact value of the area of R. **[7]**

Examiner comment

> "(ii) Many candidates did not do the requested substitution and attempted integration by parts instead; candidates who used this method often made little progress."

Mark scheme

(i) M1 Use product rule and chain rule at least once

A1 Obtain correct derivative in any form

M1 Equate derivative to zero, use Pythagoras and obtain an equation in $\cos x$

A1 Obtain $\cos^2 x + 3\cos x - 1 = 0$, or 3-term equivalent

A1 Obtain answer $x = 1.26$

Total: 5

(ii) M1 Using $du = \pm \sin x \, dx$ express integrand in terms of u and du

A1 Obtain integrand $e^u(u^2 - 1)$ OE

M1 Commence integration by parts and reach $ae^u(u^2 - 1) + b\int ue^u \, du$

A1 Obtain $e^u(u^2 - 1) - 2\int ue^u \, du$ OE

A1 Complete integration, obtaining $e^u(u^2 - 2u + 1)$ OE

DM1 Substitute limits $u = 1$ and $u = -1$ (or $x = 0$ and $x = \pi$), having integrated completely

A1 Obtain answer $\frac{4}{e}$, or exact equivalent

Total: 7

Q137 Feb/Mar 2018 | Paper 3 (variant 32) | Question 3

Topics: Trigonometry, Integration

(i) Using the expansions of $\cos(3x + x)$ and $\cos(3x - x)$, show that

$$\frac{1}{2}(\cos 4x + \cos 2x) \equiv \cos 3x \cos x.$$

[3]

(ii) Hence show that

$$\int_{-\frac{1}{6}\pi}^{\frac{1}{6}\pi} \cos 3x \cos x \, dx = \frac{3}{8}\sqrt{3}.$$

[3]

Examiner comment

> "Although limits were usually correctly substituted, many candidates did not show sufficient working to convincingly obtain the given answer."

Mark scheme

(i)

State correct expansion of $\cos(3x + x)$ or $\cos(3x - x)$ **B1**

Substitute in $\frac{1}{2}(\cos 4x + \cos 2x)$ **M1**

Obtain the given identity correctly AG **A1**

Total: 3

(ii)

Obtain integral $\frac{1}{8}\sin 4x + \frac{1}{4}\sin 2x$ **B1**

Substitute limits correctly **M1**

Obtain the given answer following full, correct and exact working AG **A1**

Total: 3

Q138 Feb/Mar 2018 | Paper 3 (variant 32) | Question 4

Topics: Indices, Logarithms & Exponents

The variables x and y satisfy the equation $y^n = Ax^3$, where n and A are constants. It is given that $y = 2.58$ when $x = 1.20$, and $y = 9.49$ when $x = 2.51$.

(i) Explain why the graph of $\ln y$ against $\ln x$ is a straight line. [2]

(ii) Find the values of n and A, giving your answers correct to 2 decimal places. [4]

Examiner comment

> "Numerous other such basic ln errors were very prevalent with the result that few candidates had even a correct method for n and for A, with even fewer obtaining the correct answers."

Mark scheme

(i)

State or imply $n \ln y = \ln A + 3 \ln x$ **B1**

State that the graph of $\ln y$ against $\ln x$ has an equation which is linear in $\ln y$ and $\ln x$, or has equation of the form $nY = \ln A + 3X$, where $Y = \ln y$ and $X = \ln x$, and is thus a straight line **B1**

Total: 2

(ii)

Substitute x- and y-values in $n \ln y = \ln A + 3 \ln x$ or in the given equation and solve for one of the constants **M1**

Obtain a correct constant, e.g. $n = 1.70$ **A1**

Solve for a second constant **M1**

Obtain the other constant, e.g. $A = 2.90$ **A1**

Total: 4

Q139 Feb/Mar 2018 | Paper 3 (variant 32) | Question 5

Topics: Trigonometry, Differentiation

The parametric equations of a curve are

$$x = 2t + \sin 2t, \qquad y = 1 - 2\cos 2t,$$

for $-\frac{1}{2}\pi < t < \frac{1}{2}\pi$.

(i) Show that $\frac{dy}{dx} = 2\tan t$. [5]

(ii) Hence find the x-coordinate of the point on the curve at which the gradient of the normal is 2. Give your answer correct to 3 significant figures. [2]

Examiner comment

> "☐ 1☐ (ii) Few candidates scored both marks, and many candidates who reached tant =☐– ☐ were unable ☐ 4☐ ☐ 1☐ to move to t =tant–1 ☐– ☐, the usual error being the omission of the minus sign."

Mark scheme

(i)

State correct derivative of x or y with respect to t **B1**

Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ **M1**

Obtain $\frac{dy}{dx} = \frac{4\sin 2t}{2+2\cos 2t}$, or equivalent **A1**

Use double angle formulae throughout **M1**

Obtain the given answer correctly AG **A1**

Total: 5

(ii)

State or imply $t = \tan^{-1}\left(-\frac{1}{4}\right)$ **B1**

Obtain answer $x = -0.961$ **B1**

Total: 2

Q140 Feb/Mar 2018 | Paper 3 (variant 32) | Question 6

Topics: Trigonometry, Differentiation, Integration, Differential Equations

The variables x and θ satisfy the differential equation

$$x\cos^2\theta\frac{dx}{d\theta} = 2\tan\theta + 1,$$

for $0 \leq \theta < \frac{1}{2}\pi$ and $x > 0$. It is given that $x = 1$ when $\theta = \frac{1}{4}\pi$.

(i) Show that $\frac{d}{d\theta}(\tan^2\theta) = \frac{2\tan\theta}{\cos^2\theta}$. [1]

(ii) Solve the differential equation and calculate the value of x when $\theta = \frac{1}{3}\pi$, giving your answer correct to 3 significant figures. [7]

Examiner comment

> "Question 6 (i) Some candidates seemed to believe that this relationship was to be found within the differential equation, hence made no progress."

Mark scheme

(i)

Show sufficient working to justify the given statement AG **B1**

Total: 1

(ii)

Separate variables correctly and attempt integration of at least one side **B1**

Obtain term $\frac{1}{2}x^2$ **B1**

Obtain terms $\tan^2\theta$ and $\tan\theta$ **B1+B1**

Evaluate a constant, or use limits $x = 1, \theta = \frac{1}{4}\pi$, in a solution with two terms of the form ax^2 and $b\tan\theta$, where $ab \neq 0$ **M1**

State correct answer in any form, e.g. $\frac{1}{2}x^2 = \tan^2\theta + \tan\theta - \frac{3}{2}$ **A1**

Substitute $\theta = \frac{1}{3}\pi$ and obtain $x = 2.54$ **A1**

Total: 7

Q141 Feb/Mar 2018 | Paper 3 (variant 32) | Question 7

Topics: Indices, Logarithms & Exponents, Numerical Solutions/Iteration

(i) By sketching suitable graphs, show that the equation $e^{2x} = 6 + e^{-x}$ has exactly one real root. [2]

(ii) Verify by calculation that this root lies between 0.5 and 1. [2]

(iii) Show that if a sequence of values given by the iterative formula

$$x_{n+1} = \frac{1}{3}\ln(1 + 6e^{x_n})$$

converges, then it converges to the root of the equation in part (i). [2]

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Examiner comment

> "© 2018 Cambridge International Advanced Level 9709 Mathematics March 2018 Principal Examiner Report for Teachers (iii) Another part that candidates found very difficult."

Mark scheme

(i)

Sketch a relevant graph, e.g. $y = e^{2x}$ **B1**

Sketch a second relevant graph, e.g. $y = 6 + e^{-x}$, and justify the given statement **B1**

Total: 2

(ii)

Calculate the value of a relevant expression or values of a pair of relevant expressions at $x = 0.5$ and $x = 1$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

State a suitable equation, e.g. $x = \frac{1}{3}\ln(1 + 6e^x)$ **B1**

Rearrange this as $e^{2x} = 6 + e^{-x}$, or commence working vice versa **B1**

Total: 2

(iv)

Use the iterative formula correctly at least once **M1**

Obtain final answer 0.928 **A1**

Show sufficient iterations to 5 d.p. to justify 0.928 to 3 d.p., or show there is a sign change in the interval $(0.9275, 0.9285)$ **A1**

Total: 3

Q142 Feb/Mar 2018 | Paper 3 (variant 32) | Question 9

Topics: Algebra, Complex Numbers

The complex number $1 + 2i$ is denoted by u.

(i) It is given that u is a root of the equation $2x^3 - x^2 + 4x + k = 0$, where k is a constant.

(a) Showing all working and without using a calculator, find the value of k. [3]

(b) Showing all working and without using a calculator, find the other two roots of this equation. [4]

(ii) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - u| = 1$. Determine the least value of $\arg z$ for points on this locus. Give your answer in radians correct to 2 decimal places. [4]

Examiner comment

> "(b) The majority of candidates realised that one of the other roots was 1 – 2i, but were then unclear how to establish the quadratic term, or the correct quadratic term but struggled with the long division."

Mark scheme

(i)(a)

Substitute $x = 1 + 2i$ in the equation and attempt expansions of x^2 and x^3 **M1**

Use $i^2 = -1$ correctly at least once and solve for k **M1**

Obtain answer $k = 15$ **A1**

Total: 3

(i)(b)

State answer $1 - 2i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $1+2i$ and $1-2i$ **M1**

Obtain x^2-2x+5 **A1**

Obtain root $-\frac{3}{2}$, or equivalent, via division or inspection **A1**

Total: 4

(ii)

Show a circle with centre $1+2i$ **B1**

Show a circle with radius 1 **B1**

Carry out a complete method for calculating the least value of $\arg z$ **M1**

Obtain answer 0.64 **A1**

Total: 4

Q143 May/June 2018 | Paper 3 (variant 31) | Question 1

Topics: Indices, Logarithms & Exponents

Showing all necessary working, solve the equation

$$\ln(x^4-4)=4\ln x-\ln 4,$$

giving your answer correct to 2 decimal places. [4]

Examiner comment

> "Many candidates made no progress with the question at all because they started with the incorrect statement ln(x4 −4) =lnx4 −ln4."

Mark scheme

Use a law for the logarithm of a product, quotient or power **M1**

Obtain a correct equation free of logarithms, e.g.

$$4(x^4-4)=x^4$$

A1

Solve for x **M1**

Obtain answer

$$x=1.52$$

only **A1**

Total: 4

Q144 May/June 2018 | Paper 3 (variant 31) | Question 2

Topics: Trigonometry

(i) Given that

$$\sin(x - 60°) = 3\cos(x - 45°),$$

find the exact value of $\tan x$. [4]

(ii) Hence solve the equation

$$\sin(x - 60°) = 3\cos(x - 45°),$$

for $0° < x < 360°$. [2]

Examiner comment

> "The most common errors were to use tanx = , or a completely false start sinx with expansions such as sin(x–60°)=sinx–sin60°."

Mark scheme

(i)

Use trigonometric formulae and obtain an equation in $\sin x$ and $\cos x$ **M1**

Obtain a correct equation in any form **A1**

Substitute exact trigonometric ratios and obtain an expression for $\tan x$ **M1**

Obtain

$$\tan x = \frac{\sqrt{3} + 3\sqrt{2}}{1 - 3\sqrt{2}},$$

or equivalent **A1**

Total: 4

(ii)

State answer

$$x = 118.5°$$

B1

State second answer

$$x = 298.5°$$

B1

Total: 2

Q145 May/June 2018 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Integration

Let

$$I = \int_{1/4}^{3/4} \sqrt{\frac{x}{1-x}}\, dx.$$

(i) Using the substitution $x = \cos^2\theta$, show that

$$I = \int_{\pi/6}^{\pi/3} 2\cos^2\theta\, d\theta.$$

[4]

(ii) Hence find the exact value of I. [4]

Examiner comment

> "Many candidates did not seem to have a confident working knowledge of complex numbers or of vectors."

Mark scheme

(i)

State or imply

$$dx = -2\cos\theta\sin\theta\, d\theta,$$

or equivalent **B1**

Substitute for x and dx, and use Pythagoras **M1**

Obtain integrand $\pm 2\cos^2\theta$ **A1**

Justify the change of limits and obtain the given answer correctly **A1**

Total: 4

(ii)

Obtain an indefinite integral of the form

$$a\theta + b\sin 2\theta$$

M1

Obtain

$$\theta + \frac{1}{2}\sin 2\theta$$

A1

Use the correct limits correctly **M1**

Obtain

$$\frac{\pi}{6}$$

with no errors seen **A1**

Total: 4

Q146 May/June 2018 | Paper 3 (variant 31) | Question 6

Topics: Integration, Differential Equations

In a certain chemical reaction the amount, x grams, of a substance is decreasing. The differential equation relating x and t, the time in seconds since the reaction started, is

$$\frac{dx}{dt} = -kx\sqrt{t},$$

where k is a positive constant. It is given that $x = 100$ at the start of the reaction.

(i) Solve the differential equation, obtaining a relation between x, t and k. [5]

(ii) Given that $t = 25$ when $x = 80$, find the value of t when $x = 40$. [3]

Examiner comment

> "After a correct start, the most common error was to confuse k with a constant of integration."

Mark scheme

(i)

Separate variables correctly and integrate at least one side **B1**

Obtain term $\ln x$ **B1**

Obtain term

$$-\frac{2}{3}kt^{3/2},$$

or equivalent **B1**

Evaluate a constant, or use limits $x = 100$ and $t = 0$, in a solution containing terms $a \ln x$ and $bt^{3/2}$ **M1**

Obtain correct solution in any form, e.g.

$$\ln x = -\frac{2}{3}kt^{3/2} + \ln 100$$

A1

Total: 5

(ii)

Substitute $x = 80$ and $t = 25$ to form an equation in k **M1**

Substitute $x = 40$ and eliminate k **M1**

Obtain

$$t = 64.1$$

A1

Total: 3

Q147 May/June 2018 | Paper 3 (variant 31) | Question 7

Topics: Algebra, Complex Numbers

(i) Showing all working and without using a calculator, solve the equation

$$z^2 + 2\sqrt{6}\,z + 8 = 0,$$

giving your answers in the form $x + iy$, where x and y are real and exact. [3]

(ii) Sketch an Argand diagram showing the points representing the roots. [1]

(iii) The points representing the roots are A and B, and O is the origin. Find angle AOB. [3]

(iv) Prove that triangle AOB is equilateral. [1]

Examiner comment

> "8i was a common error, and several candidates did not split 8 as 4×2."

> "(ii) There were many blank responses to this question implying candidates were not familiar with drawing an argand diagram."

Mark scheme

(i)

Use the quadratic formula, completing the square, or the substitution $z = x + iy$ to find a root, using $i^2 = -1$ **M1**

Obtain one root, e.g.

$$-\sqrt{6} - \sqrt{2}\,i$$

A1

Obtain the other root, e.g.

$$-\sqrt{6} + \sqrt{2}\,i$$

A1

Total: 3

(ii)

Represent both roots in relatively correct positions on an Argand diagram **B1**

Total: 1

(iii)

State or imply a correct relevant length or angle, e.g. OA, OB, AB, or the angle between OA or OB and the real axis **B1**

Carry out a complete method for finding angle AOB **M1**

Obtain

Math input error

A1

Total: 3

(iv)

Give a complete justification that triangle AOB is equilateral **B1**

Total: 1

Q148 May/June 2018 | Paper 3 (variant 31) | Question 8

Topics: Indices, Logarithms & Exponents, Integration, Numerical Solutions/Iteration

The positive constant a is such that

$$\int_0^a xe^{-x/2}\,dx = 2.$$

(i) Show that a satisfies the equation

$$a = 2\ln(a+2).$$

[5]

(ii) Verify by calculation that a lies between 3 and 3.5. [2]

(iii) Use an iteration based on the equation in part (i) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "The most common error 1 was in division by ."

> "Most candidates used the limits correctly, although a minority did not appear 2 to consider the lower limit."

> "A large number of incorrect algebraic statements were seen."

Mark scheme

(i)

Integrate by parts and reach an expression of the form

$$lxe^{-x/2} + m\int e^{-x/2}\,dx$$

M1

Obtain

$$-2xe^{-x/2} + 2\int e^{-x/2}\,dx$$

A1

Complete the integration and obtain

$$-2xe^{-x/2} - 4e^{-x/2},$$

or equivalent **A1**

Having integrated twice, use limits and equate the result to **2 M1**

Obtain the given equation

$$a = 2\ln(a+2)$$

correctly **A1**

Total: 5

(ii)

Calculate values of a relevant expression at $a = 3$ and $a = 3.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use the iterative formula

$$a_{n+1} = 2\ln(a_n + 2)$$

correctly at least once **M1**

Obtain final answer **3.36 A1**

Show sufficient iterations to 4 d.p. to justify **3.36** to 2 d.p., or show there is a sign change in the interval $(3.355, 3.365)$ **A1**

Total: 3

Q149 May/June 2018 | Paper 3 (variant 31) | Question 10

Topics: none of the listed topics

The point P has position vector

$$3\mathbf{i} - 2\mathbf{j} + \mathbf{k}.$$

The line l has equation

$$\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}).$$

(i) Find the length of the perpendicular from P to l, giving your answer correct to 3 significant figures. [5]

(ii) Find the equation of the plane containing l and P, giving your answer in the form $ax + by + cz = d$. [5]

Examiner comment

> "For both of these approaches, the most common error was to use the direction of the line, rather than the position vector of a point, to evaluate d."

Mark scheme

(i)

Find $\overrightarrow{PQ}$ for a general point Q on l, e.g.

$$(1 + \lambda)\mathbf{i} + (4 + 2\lambda)\mathbf{j} + (4 + 3\lambda)\mathbf{k}$$

B1

Calculate the scalar product of $\overrightarrow{PQ}$ and a direction vector for l and equate to zero **M1**

Solve and obtain, for example, $\lambda = -\frac{3}{2}$ **A1**

Carry out a method to calculate PQ **M1**

Obtain answer **1.22 A1**

Total: 5

(ii)

Use scalar product to obtain a relevant equation in a, b and c, e.g.

$$a + 2b + 3c = 0$$

B1

Obtain a second relevant equation, e.g. using $\overrightarrow{PA}$,

$$a + 4b + 4c = 0,$$

and solve for one ratio **M1**

Obtain

$$a : b : c = 4 : 1 : -2$$

A1

Substitute a relevant point and values of a, b, c in the general equation and find d **M1**

Obtain

$$4x + y - 2z = 8$$

A1

Total: 5

Q150 May/June 2018 | Paper 3 (variant 32) | Question 1

Topics: Algebra, Indices, Logarithms & Exponents

Showing all necessary working, solve the equation

$$3|2^x - 1| = 2^x,$$

giving your answers correct to 3 significant figures. [4]

Examiner comment

> "The most common errors centred in attempts to square the original equation."

Mark scheme

Either state or imply a non-modular equation, or pair of equations,

$$3(2^x - 1) = \pm 2^x$$

M1

Obtain

$$2^x = \frac{3}{2} \quad \text{and} \quad 2^x = \frac{3}{4},$$

or equivalent **A1**

Use a correct method for solving an equation of the form $2^x = a$, where $a > 0$ **M1**

Obtain final answers

$$x = 0.585 \quad \text{and} \quad x = -0.415$$

only **A1**

Total: 4

Q151 May/June 2018 | Paper 3 (variant 32) | Question 2

Topics: Trigonometry

Showing all necessary working, solve the equation

$$\cot\theta + \cot(\theta + 45°) = 2,$$

for $0° < \theta < 180°$. [5]

Examiner comment

> "Another common error was to 2 cos(θ+45°) claim that cot(θ+45°)= ."

Mark scheme

Use the correct $\tan(A \pm B)$ formula and obtain an equation in $\tan\theta$ **M1**

Obtain a correct equation in any form **A1**

Reduce to

$$3\tan^2\theta = 1,$$

or equivalent **A1**

Obtain answer

$$\theta = 30°$$

A1

Obtain answer

$$\theta = 150°$$

and no others in the range **A1**

Total: 5

Q152 May/June 2018 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry, Integration

(i) Show that

$$\frac{2\sin x - \sin 2x}{1 - \cos 2x} \equiv \frac{\sin x}{1 + \cos x}.$$

[4]

(ii) Hence, showing all necessary working, find

$$\int_{\pi/3}^{\pi/2} \frac{2\sin x - \sin 2x}{1 - \cos 2x}\, dx,$$

giving your answer in the form $\ln k$. [4]

Examiner comment

> "The most common error was to omit the "–"in the integration, but the limits were usually applied correctly."

Mark scheme

(i)

Use correct double-angle formulae and express the left hand side in terms of $\cos x$ and $\sin x$ **M1**

Obtain a correct expression **A1**

Complete a method to get the correct denominator, e.g. by factorising to remove a factor of $1 - \cos x$ **M1**

Obtain the given right hand side correctly **A1**

Total: 4

(ii)

State an integral of the form

$$a\ln(1 + \cos x)$$

M1

Obtain integral

$$-\ln(1 + \cos x)$$

A1

Substitute correct limits in the correct order **M1**

Obtain

$$\ln\left(\frac{3}{2}\right)$$

A1

Total: 4

Q153 May/June 2018 | Paper 3 (variant 32) | Question 5

Topics: Algebra, Differentiation

The equation of a curve is

$$x^2(x+3y)-y^3=3.$$

(i) Show that

$$\frac{dy}{dx}=\frac{x^2+2xy}{y^2-x^2}.$$

[4]

(ii) Hence find the exact coordinates of the two points on the curve at which the gradient of the normal is 1. [4]

Examiner comment

> "A disappointing feature of the work this time was the number of candidates who reached x3 =1 and concluded that x = ±1."

Mark scheme

(i)

State or imply $3y^2\frac{dy}{dx}$ as the derivative of y^3 **B1**

State or imply $6xy+x^2\frac{dy}{dx}$ as part of the derivative of $3x^2y$ **B1**

Equate the derivative of the left hand side to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate the tangent gradient to -1 and solve for y **M1**

Use $y = -2x$ or $y = 0$ to obtain an equation in x or y **M1**

Obtain point

$$(1, -2)$$

A1

Obtain point

$$(\sqrt[3]{3}, 0)$$

B1

Total: 4

Q154 May/June 2018 | Paper 3 (variant 32) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

The diagram shows a triangle ABC in which $AB = AC = a$ and angle $BAC = \theta$ radians. Semicircles are drawn outside the triangle with AB and AC as diameters. A circular arc with centre A joins B and C. The area of the shaded segment is equal to the sum of the areas of the semicircles.

(i) Show that

$$\theta = \frac{1}{2}\pi + \sin\theta.$$

[3]

(ii) Verify by calculation that θ lies between 2.2 and 2.4. [2]

(iii) Use an iterative formula based on the equation in part (i) to determine θ correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "The most common error was use of the incorrect radius for the semi-circles."

Mark scheme

(i)

Use a correct method for finding the area of a segment and the area of a semicircle, and form an equation in θ **M1**

State a correct equation in any form **A1**

Obtain

$$\theta = \frac{1}{2}\pi + \sin\theta$$

A1

Total: 3

(ii)

Calculate values of a relevant expression at $\theta = 2.2$ and $\theta = 2.4$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use

$$\theta_{n+1} = \frac{1}{2}\pi + \sin\theta_n$$

correctly at least once **M1**

Obtain final answer 2.31 **A1**

Show sufficient iterations to 4 d.p. to justify 2.31 to 2 d.p., or show there is a sign change in the interval $(2.305, 2.315)$ **A1**

Total: 3

Q155 May/June 2018 | Paper 3 (variant 32) | Question 7

Topics: Complex Numbers

Throughout this question the use of a calculator is not permitted.

The complex numbers $-3\sqrt{3} + i$ and $\sqrt{3} + 2i$ are denoted by u and v respectively.

(i) Find, in the form $x + iy$, where x and y are real and exact, the complex numbers uv and $\frac{u}{v}$. [5]

(ii) On a sketch of an Argand diagram with origin O, show the points A and B representing the complex numbers u and v respectively. Prove that angle $AOB = \frac{2}{3}\pi$. [3]

Examiner comment

> "Very few candidates provided a correct demonstration that ∠AOB = π."

Mark scheme

(i)

Substitute in uv, expand the product and use $i^2 = -1$ **M1**

Obtain

$$uv = -11 - 5\sqrt{3}\,i$$

A1

Substitute in $\frac{u}{v}$ and multiply numerator and denominator by the conjugate of v, or equivalent **M1**

Obtain numerator $-7 + 7\sqrt{3}\,i$ or denominator **7 A1**

Obtain final answer

$$\frac{u}{v} = -1 + \sqrt{3}\,i$$

A1

Total: 5

(ii)

Show the points A and B representing u and v in relatively correct positions **B1**

Carry out a complete method for finding angle AOB, e.g. calculate Math input error **M1**

Prove

Math input error

A1

Total: 3

Q156 May/June 2018 | Paper 3 (variant 33) | Question 1

Topics: Algebra

Expand

$$\frac{4}{\sqrt{4 - 3x}}$$

in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. [4]

Examiner comment

> "However, there were a considerable number of candidates for whom it proved to be difficult or impossible."

Mark scheme

Obtain a correct unsimplified version of the x or x^2 term of the expansion of

$$(4-3x)^{-1/2}$$

or

$$2\left(1-\frac{3x}{4}\right)^{-1/2}$$

M1

State correct first term 2 **B1**

Obtain the next two terms

$$\frac{3}{4}x+\frac{27}{64}x^2$$

A2

Total: 4

Q157 May/June 2018 | Paper 3 (variant 33) | Question 5

Topics: Trigonometry

(i) By first expanding $(\cos^2 x+\sin^2 x)^3$, or otherwise, show that

$$\cos^6 x+\sin^6 x=1-\frac{3}{4}\sin^2 2x.$$

[4]

(ii) Hence solve the equation

$$\cos^6 x+\sin^6 x=\frac{2}{3},$$

for $0°<x<180°$. [4]

Examiner comment

> "Question 5 (i) Candidates found this question extremely difficult."

Mark scheme

(i)

Attempt the cubic expansion and equate to 1 **M1**

Obtain a correct equation **A1**

Use Pythagoras and a double-angle formula in the expansion **M1**

Obtain the given result correctly **A1**

Total: 4

(ii)

Use the identity and carry out a method for finding a root **M1**

Obtain answer $20.9°$ **A1**

Obtain a second answer, e.g. $69.1°$ **A1**

Obtain the remaining answers $110.9°$ and $159.1°$, and no others in the given interval **A1**

Total: 4

Q158 May/June 2018 | Paper 3 (variant 33) | Question 6

Topics: Algebra, Integration, Differential Equations

(i) Express

$$\frac{1}{4-y^2}$$

in partial fractions. [2]

(ii) The variables x and y satisfy the differential equation

$$x\frac{dy}{dx} = 4 - y^2,$$

and $y = 1$ when $x = 1$. Solve the differential equation, obtaining an expression for y in terms of x. [6]

Examiner comment

> "Many candidates failed to realise how to extract an expression for y in terms of x from their correct □(2+y)□ (2+y) ln□ □ or expressions."

Mark scheme

(i)

Carry out a relevant method to find A and B such that

$$\frac{1}{4-y^2} \equiv \frac{A}{2+y} + \frac{B}{2-y}$$

M1

Obtain

$$A = B = \frac{1}{4}$$

A1

Total: 2

(ii)

Separate variables correctly and integrate at least one side to obtain one of the terms $a \ln x$, $b \ln(2+y)$ or $c \ln(2-y)$ **M1**

Obtain term $\ln x$ **B1**

Integrate and obtain terms

$$\frac{1}{4}\ln(2+y) - \frac{1}{4}\ln(2-y)$$

A1

Use $x = 1$, $y = 1$ to evaluate a constant, or use limits **M1**

Obtain a correct solution in any form, e.g.

$$\ln x = \frac{1}{4}\ln(2+y) - \frac{1}{4}\ln(2-y) - \frac{1}{4}\ln 3$$

A1

Rearrange to obtain

$$y = \frac{2(3x^4 - 1)}{3x^4 + 1},$$

or equivalent **A1**

Total: 6

Q159 May/June 2018 | Paper 3 (variant 33) | Question 8

Topics: Algebra, Differentiation

The equation of a curve is

$$2x^3 - y^3 - 3xy^2 = 2a^3,$$

where a is a non-zero constant.

(i) Show that

$$\frac{dy}{dx} = \frac{2x^2 - y^2}{y^2 + 2xy}.$$

[4]

(ii) Find the coordinates of the two points on the curve at which the tangent is parallel to the y-axis. [5]

Examiner comment

> "(ii) This question, along with Question 5(i) and Question 10(ii), proved very challenging for the majority of the candidates."

Mark scheme

(i)

State or imply $3y^2 \frac{dy}{dx}$ as the derivative of y^3 **B1**

State or imply $3y^2 + 6xy\frac{dy}{dx}$ as the derivative of $3xy^2$ **B1**

Equate the derivative of the left hand side to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate the denominator to zero and solve for y **M1**

Obtain $y = 0$ and $x = a$ **A1**

Obtain $y = -2x$ and substitute in the curve equation to find x or y **M1**

Obtain $x = -a$ **A1**

Obtain $y = 2a$ **A1**

Total: 5

Q160 May/June 2018 | Paper 3 (variant 33) | Question 9

Topics: Complex Numbers

(a) Find the complex number z satisfying the equation

$$3z - iz^* = 1 + 5i,$$

where z^* denotes the complex conjugate of z. [4]

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z which satisfy both the inequalities

$$|z| \leq 3$$

and

$$\operatorname{Im} z \geq 2,$$

where $\operatorname{Im} z$ denotes the imaginary part of z. Calculate the greatest value of Math input error for points in this region. Give your answer in radians correct to 2 decimal places. [5]

Examiner comment

> "For some reason the trigonometry to calculate the greatest value of arg proved difficult."

Mark scheme

(a)

Substitute $z = x + iy$ and obtain a correct equation in x and y **B1**

Use $i^2 = -1$ and equate real and imaginary parts **M1**

Obtain two correct equations, e.g.

$$3x - y = 1, \qquad 3y - x = 5$$

A1

Solve and obtain

$$z = 1 + 2i$$

A1

Total: 4

(b)

Show a circle with radius 3 **B1**

Show the line $y = 2$ extending in both quadrants **B1**

Shade the correct region **B1**

Carry out a complete method for finding the greatest value of Math input error **M1**

Obtain answer **2.41** **A1**

Total: 5

Q161 May/June 2018 | Paper 3 (variant 33) | Question 10

Topics: none of the listed topics

The points A and B have position vectors $2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$ and $4\mathbf{i} + \mathbf{j} + \mathbf{k}$ respectively. The line l has equation

$$\mathbf{r} = 4\mathbf{i} + 6\mathbf{j} + t(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}).$$

(i) Show that l does not intersect the line passing through A and B. [5]

The point P, with parameter t, lies on l and is such that angle PAB is equal to $120°$.

(ii) Show that

$$3t^2 + 8t + 4 = 0.$$

Hence find the position vector of P. [6]

Examiner comment

> "(ii) All candidates found this question difficult."

Mark scheme

(i)

Carry out a correct method for finding a vector equation for AB **M1**

Obtain

$$\mathbf{r} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} - 2\mathbf{k}),$$

or equivalent **A1**

Equate pairs of components for AB and l and solve for λ or t **M1**

Obtain a correct value for λ or t **A1**

Verify that all three component equations are not satisfied **A1**

Total: 5

(ii)

State or imply a direction vector for AP has components

$$(2+t,\ 5+2t,\ -3-2t)$$

B1

State or imply that $\cos 120°$ equals the scalar product of $\overrightarrow{AP}$ and $\overrightarrow{AB}$ divided by the product of their moduli **M1**

Carry out the correct processes for finding the scalar product and product of moduli in terms of t **M1**

Obtain the given equation

$$3t^2+8t+4=0$$

A1

Solve the quadratic and use a root to find a position vector for P **M1**

Obtain position vector

$$2\mathbf{i}+2\mathbf{j}+4\mathbf{k}$$

from $t=-2$, having rejected $t=-\frac{2}{3}$ **A1**

Total: 6

Q162 Oct/Nov 2018 | Paper 3 (variant 31) | Question 1

Topics: Algebra

1 Find the set of values of x satisfying the inequality $2|2x-a|<|x+3a|$, where a is a positive constant. **[4]**

Examiner comment

> "Question 1 This question proved to be challenging for candidates."

> "Many candidates failed to square the 2 outside the modulus sign or poor algebra prevented them from arriving at the correct equation."

Mark scheme

EITHER: State or imply non-modular inequality **[B1]**

$2^2(2x-a)^2<(x+3a)^2$, or corresponding quadratic equation, or pair of linear equations

$2(2x-a)=\pm(x+3a)$

Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x **[M1]**

Obtain critical values $x = \frac{5}{3}a$ and $x = -\frac{1}{5}a$ **[A1]**

State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$ **[A1]**

OR: Obtain critical value $x = \frac{5}{3}a$ from a graphical method, or by inspection, or **[B1]**

by solving a linear equation or an inequality

Obtain critical value $x = -\frac{1}{5}a$ similarly **[B2]**

State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$ **[B1]**

[Do not condone $\leq$ for $<$ in the final answer.]

Total: 4

Q163 Oct/Nov 2018 | Paper 3 (variant 31) | Question 2

Topics: Indices, Logarithms & Exponents

2 Showing all necessary working, solve the equation $\frac{2e^x + e^{-x}}{e^x - e^{-x}} = 4$, giving your answer correct to 2 decimal places.

Total: 4

Examiner comment

> "Question 2 This question was usually well done, however there were still many candidates who did not reach an b equation of the form ae2x = b, aex = be–x or aex = with the correct values of both a and b."

> "Most candidates knew they then needed to take lns but some did not show this step despite the additional instruction to show all necessary working."

Mark scheme

2 Rearrange the equation in the form $ae^{2x} = b$ or $ae^x = be^{-x}$ — **M1**

Obtain correct equation in either form with $a = 2$ and $b = 5$ — **A1**

Use correct method to solve for x — **M1**

Obtain answer $x = 0.46$ — **A1**

Total: 4

Q164 Oct/Nov 2018 | Paper 3 (variant 31) | Question 3

Topics: Algebra, Numerical Solutions/Iteration

(i) By sketching a suitable pair of graphs, show that the equation $x^3 = 3 - x$ has exactly one real root.

Total: 2

(ii) Show that if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{2x_n^3 + 3}{3x_n^2 + 1}$$

converges, then it converges to the root of the equation in part (i).

Total: 2

(iii) Use this iterative formula to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Total: 3

Examiner comment

> "Question 3 (i) This part proved challenging more many candidate, for various reasons."

Mark scheme

3(i) Sketch a relevant graph, e.g. $y = x^3$ — **B1**

Sketch a second relevant graph, e.g. $y = 3 - x$, and justify the given statement — **B1** (Consideration of behaviour for $x < 0$ is needed for the second **B1**)

Total: 2

3(ii) State or imply the equation $x = \frac{2x^3+3}{3x^2+1}$ — **B1**

Rearrange this in the form $x^3 = 3 - x$, or commence work *vice versa* — **B1**

Total: 2

Q165 Oct/Nov 2018 | Paper 3 (variant 31) | Question 5

Topics: Differential Equations

5 The coordinates (x, y) of a general point on a curve satisfy the differential equation

$$x\frac{dy}{dx} = (2 - x^2)y.$$

The curve passes through the point $(1, 1)$. Find the equation of the curve, obtaining an expression for y in terms of x.

Total: 7

Examiner comment

> "The integral in x proved more difficult, since instead of considering and x, some candidates x resorted to

> integrating by parts and increased the possibility of sign errors."

Mark scheme

5 Separate variables correctly and integrate at least one side — **B1**

Obtain term $\ln y$ — **B1**

Obtain terms $2\ln x - \frac{1}{2}x^2$ — **B1+B1**

Use $x = 1, y = 1$ to evaluate a constant, or as limits — **M1**

Obtain correct solution in any form, e.g. $\ln y = 2\ln x - \frac{1}{2}x^2 + \frac{1}{2}$ — **A1**

Rearrange as $y = x^2 \exp\left(\frac{1}{2} - \frac{1}{2}x^2\right)$, or equivalent — **A1**

Total: 7

Q166 Oct/Nov 2018 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry

(i) Show that the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$ can be expressed in the form $R\sin(x - \alpha) = \sqrt{2}$, where $R > 0$ and $0° < \alpha < 90°$.

Total: 4

(ii) Hence solve the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$, for $0° < x < 180°$.

Total: 4

Examiner comment

> "Question 6 (i) Many candidates found it difficult to obtain the given expression, taking far more steps than were required."

Mark scheme

6(i) Rearrange in the form $\sqrt{3}\sin x - \cos x = \sqrt{2}$ — **B1**

State $R = 2$ — **B1**

Use trig formulae to obtain a — **M1**

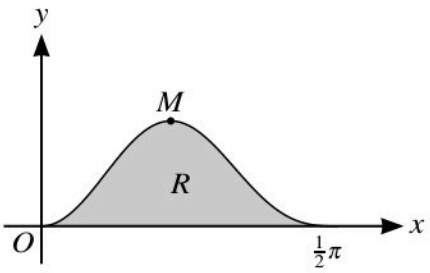
Obtain $a = 30°$ with no errors seen — **A1**

Total: 4

Q167 Oct/Nov 2018 | Paper 3 (variant 31) | Question 7

Topics: Trigonometry, Differentiation, Integration

7



The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. [5]

The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places.

Total: 5

(ii) Using the substitution $u = \sin x$ and showing all necessary working, find the exact area of R.

Total: 4

Examiner comment

> "(ii) Nearly all candidates struggled with this question despite being given the substitution required."

> "Common errors were poor substitution of du = cos x dx, inability to express cos2 x correctly in terms of u, substituting π limits 0 and for u or substituting limits 0 and 1 for x."

Mark scheme

7(i) Use product rule — **M1***

Obtain correct derivative in any form — **A1**

Equate derivative to zero and obtain an equation in a single trig function — **depM1***

Obtain a correct equation, e.g. $3\tan^2 x = 2$ — **A1**

Obtain answer $x = 0.685$ — **A1**

Total: 5

7(ii) Use the given substitution and reach $a\int(u^2 - u^4)du$ — **M1**

Obtain correct integral with $a = 5$ and limits 0 and 1 — **A1**

Use correct limits in an integral of the form $a\left(\frac{1}{3}u^3 - \frac{1}{5}u^5\right)$ — **M1**

Obtain answer $\frac{2}{3}$ — **A1**

Total: 4

Q168 Oct/Nov 2018 | Paper 3 (variant 31) | Question 8

Topics: Complex Numbers

(a) Showing all necessary working, express the complex number $\frac{2+3i}{1-2i}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the values of r and θ correct to 3 significant figures.

Total: 5

(b) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - 3 + 2i| = 1$. Find the least value of $|z|$ for points on this locus, giving your answer in an exact form.

Total: 4

Examiner comment

> "Few candidates managed to obtain the correct argument, the usual answer being θ = −1.05, something that candidates arrived at from their calculators without any reference to the position of the complex number −4 7 + i in the Argand diagram."

Mark scheme

8(i) EITHER: Multiply numerator and denominator by $1 + 2i$, or equivalent, or equate to $x + iy$, obtain two equations in x and y and solve for x or y — **M1**

Obtain quotient $-\frac{4}{5} + \frac{7}{5}i$, or equivalent — **A1**

Use correct method to find either r or θ — **M1**

Obtain $r = 1.61$ — **A1**

Obtain $\theta = 2.09$ — **A1**

OR: Find modulus or argument of $2 + 3i$ or of $1 - 2i$ — **B1**

Use correct method to find r — **M1**

Obtain $r = 1.61$ — **A1**

Use correct method to find θ — **M1**

Obtain $\theta = 2.09$ — **A1**

Total: 5

8(ii) Show a circle with centre $3 - 2i$ — **B1**

Show a circle with radius 1 — **B1ft** (Centre not at the origin)

Carry out a correct method for finding the least value of $|z|$ — **M1**

Obtain answer $\sqrt{13} - 1$ — **A1**

Total: 4

Q169 Oct/Nov 2018 | Paper 3 (variant 31) | Question 10

Topics: none of the listed topics

10 The planes m and n have equations $3x + y - 2z = 10$ and $x - 2y + 2z = 5$ respectively. The line l has equation $\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.

(i) Show that l is parallel to m.

Total: 3

(ii) Calculate the acute angle between the planes m and n.

Total: 3

(iii) A point P lies on the line l. The perpendicular distance of P from the plane n is equal to 2. Find the position vectors of the two possible positions of P.

Total: 4

Examiner comment

> "(iii) Those candidates who knew the formula for the distance of a point from a plane usually quickly scored full marks, whilst most others struggled."

Mark scheme

10(i) EITHER: Expand scalar product of a normal to m and a direction vector of l — **M1**

Verify scalar product is zero — **A1**

Verify that one point of l does not lie in the plane — **A1**

OR: Substitute coordinates of a general point of l in the equation of the plane m — **M1**

Obtain correct equation in λ in any form — **A1**

Verify that the equation is not satisfied for any value of λ — **A1**

Total: 3

10(ii) Use correct method to evaluate a scalar product of normal vectors to m and n — **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result — **M1**

Obtain answer 74.5° or 1.30 radians — **A1**

Total: 3

10(iii) EITHER: Using the components of a general point P of l form an equation in λ by equating the perpendicular distance from n to 2 — **M1**

Obtain a correct modular or non-modular equation in any form — **A1**

Solve for λ and obtain a position vector for P, e.g. $7\mathbf{i} + 5\mathbf{j} + 7\mathbf{j}$ from $\lambda = 3$ — **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ from $\lambda = -1$ — **A1**

OR: Take a point Q on l, e.g. $(5, 3, 3)$ and form an equation in λ by equating the length of the projection of QP onto a normal to plane n to 2 — **M1**

Total: 4

Q170 Oct/Nov 2018 | Paper 3 (variant 32) | Question 1

Topics: Algebra

1 Solve the inequality $3|2x - 1| > |x + 4|$.

Total: 4

Examiner comment

> "The most common errors were in squaring the brackets and in simplifying to obtain a quadratic equation/inequality."

> "1 7 The incorrect answer – > x > was a common error."

Mark scheme

1 State or imply non-modular inequality $3^2(2x - 1)^2 > (x + 4)^2$, or corresponding quadratic equation, or pair of linear equations/inequalities $3(2x - 1) = \pm(x + 4)$ — **B1**

Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x — **M1**

Obtain critical values $x = \frac{7}{5}$ and $x = -\frac{1}{7}$ — **A1** (Accept **1.4** and **−0.143** or better for penultimate **A1** mark)

State final answer $x > \frac{7}{5}, x < -\frac{1}{7}$ — **A1** ('*and*' is **A0**, $\frac{7}{5} < x < -\frac{1}{7}$ is **A0**. Must be exact values. Must be strict inequalities in final answer)

Alternative

Obtain critical value $x = \frac{7}{5}$ from a graphical method or by inspection, or by solving a linear equation or an inequality — **B1** (or by inspection, or by solving a linear equation or an inequality)

Obtain critical value $x = -\frac{1}{7}$ similarly — **B2**

State final answer $x > \frac{7}{5}$ or $x < -\frac{1}{7}$ or equivalent — **B1** ([Do not condone $>$ for $<$, or $<$ for $<$.])

Total: 4

Q171 Oct/Nov 2018 | Paper 3 (variant 32) | Question 4

Topics: Indices, Logarithms & Exponents

Question 4

Showing all necessary working, solve the equation

$$\frac{e^x + e^{-x}}{e^x + 1} = 4,$$

giving your answer correct to 3 decimal places.

Marks: [5]

Examiner comment

> "© 2018 Cambridge International Advanced Level 9709 Mathematics November 2018 Principal Examiner Report for Teachers A significant minority of candidates made no progress at all because they attempted to apply incorrect rules of logarithms to the original fraction, or to ex +e−x = 4ex +4, in order to remove the exponentials."

Mark scheme

Question 4

B1 Substitute and obtain 3-term quadratic $3u^2 + 4u - 1 = 0$, or equivalent. e.g. $3(e^x)^2 + 4e^x - 1 = 0$

M1 Solve a 3-term quadratic for u

A1 Obtain root $\left(\sqrt{7} - 2\right)/3$, or decimal in $[0.21, 0.22]$

M1 Use correct method for finding x from a positive value of e^x. Must see some indication of method: use of $x = \ln u$

A1 Obtain answer $x = -1.536$ only. **CAO**. Must be 3 dp

Total: 5

Q172 Oct/Nov 2018 | Paper 3 (variant 32) | Question 6

Topics: Differential Equations

Question 6

A certain curve is such that its gradient at a general point with coordinates (x, y) is proportional to $\frac{y^2}{x}$. The curve passes through the points with coordinates $(1, 1)$ and $(e, 2)$. By setting up and solving a differential equation, find the equation of the curve, expressing y in terms of x.

Marks: [8]

Examiner comment

> "1 1 The majority of candidates completed ∫ dx correctly, but ∫ dy proved to be more difficult."

Mark scheme

Question 6

B1 State equation $\frac{dy}{dx} = k\frac{y^2}{x}$, or equivalent

B1 Separate variables correctly and integrate at least one side: $\int \frac{k}{x}dx = \int \frac{1}{y^2}dy$. Allow with incorrect value substituted for k

B1 + B1 Obtain terms $-\frac{1}{y}$ and $k\ln x$

M1 Use given coordinates correctly to find k and/or a constant of integration C in an equation containing terms $\frac{c}{x}$, $b\ln x$ and C

A1 + A1 Obtain $k = \frac{1}{2}$ and $c = -1$, or equivalent. $\frac{1}{2}\ln x = 1 - \frac{1}{y}$. A0 for fortuitous answers.

A1 Obtain answer $y = \frac{-1}{-1+\ln x}$ or equivalent, and **ISW**. $y = \frac{-1}{-1+\ln x}$

Alternative schemes:

B1 SC: MR of the fraction. $\frac{dy}{dx} = k\frac{y^2}{x^2}$. Separate variables and integrate: $-\frac{1}{y} = -k\left(\frac{1}{x} + C\right)$

B1 Separate variables and integrate

B1 + B1 Substitute to find k and/or c: $k = \frac{e}{2(e-1)}$, $c = \frac{2-e}{2(e-1)}$

M1 Substitute to find k and/or c

A1 + A1 Answer

A0 A0 for fortuitous answers.

Total: 8

Q173 Oct/Nov 2018 | Paper 3 (variant 32) | Question 7

Topics: Trigonometry, Differentiation, Numerical Solutions/Iteration

Question 7

A curve has equation $y = \frac{3\cos x}{2+\sin x}$, for $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$.

(i) Find the exact coordinates of the stationary point of the curve.

Marks: [6]

(ii) Verify by calculation that a lies between 2.9 and 3.1.

Marks: [2]

(iii) Use an iterative formula based on the equation in part (i) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

Marks: [3]

Examiner comment

> "The common errors were to attempt to use integration by parts, or to integrate the numerator and denominator separately."

> "A common error in the final stages of the solution was the incorrect statement ln(2+sinx)=ln2+ln(sinx)."

Mark scheme

Question 7(i)

M1 Use correct quotient or product rule

A1 Obtain correct derivative in any form: $\frac{dy}{dx} = \frac{-3\sin x(2+\sin x) - 3\cos x \cos x}{(2+\sin x)^2}$

Condone invisible brackets if recovery implied later.

M1 Equate numerator to zero

M1 Use $\cos^2 x + \sin^2 x = 1$ and solve for $\sin x$: $-6\sin x - 3 = 0 \Rightarrow \sin x = \ldots$

A1 + A1 Obtain coordinates $x = -\frac{\pi}{6}$ and $y = \sqrt{3}$ **ISW**. From correct working. No others in range

SR: A candidate who only states the numerator of the derivative, but justifies this, can have full marks. Otherwise they score **M0 A0 M1 A0 A0**

Total: 6

Question 7(ii)

M1* State indefinite integral of the form $k\ln(2 + \sin x)$

A1 Substitute limits correctly, equate result to 1 and obtain: $3\ln(2 + \sin a) - 3\ln 2 = 1$

M1(dep*) Use correct method to solve for a. Allow for a correct method to solve an incorrect equation, so long as that equation has a solution.

$1 + \frac{1}{3}\sin a = e^{a/2} \Rightarrow a = \sin^{-1}\left[2\left(e^{a/2} - 1\right)\right]$

Can be implied by 52.3 degrees

A1 Obtain answer $a = 0.913$ or better. Ignore additional solutions. Must be in radians.

Total: 4

Q174 Oct/Nov 2018 | Paper 3 (variant 32) | Question 9

Topics: Differentiation, Integration

Question 9(a)(i)

[Question part not shown in image - mark scheme only provided]

Question 9(a)(ii)

[Question part not shown in image - mark scheme only provided]

Question 9(b)

[Question part not shown in image - mark scheme only provided]

Examiner comment

> "Many candidates did not attempt to find the greatest value of argz."

Mark scheme

Question 9(a)(i)

M1 Multiply numerator and denominator by $1 + 2i$, or equivalent

Requires at least one of $2 + 10i + 12i^2$ and $1 - 4i^2$ together with use of $i^2 = -1$. Can be implied by $\frac{-10+10i}{5}$

A1 Obtain quotient $-2 + 2i$

Alternative:

M1 Equate to $x + iy$, obtain two equations in x and y and solve for x or for y: $x + 2y = 2$, $y - 2x = 6$

A1 Obtain quotient $-2 + 2i$

Total: 2

Question 9(a)(ii)

M1 Use correct method to find either r or θ

A1ft Obtain $r = 2\sqrt{2}$, or exact equivalent

A1ft Obtain $\theta = \frac{3}{4}\pi$ from exact work. ft on $k(-1 + i)$ for $k > 0$. Do not **ISW**

Total: 3

Question 9(b)

B1 Show a circle with centre $3i$

B1ft Show a circle with radius 1. Follow through their centre provided not at the origin. For clearly unequal scales, should be an ellipse

B1 All correct with even scales and shade the correct region

9(a)(i)	Multiply numerator and denominator by 1 + 2i, or equivalent	**M1**	Requires at least one of $2+10i+12i^2$ and $1-4i^2$ together with use of $i^2=-1$. Can be implied by $\frac{-10+10i}{5}$
	Obtain quotient – 2 + 2i	**A1**	
	Alternative		
	Equate to $x + iy$, obtain two equations in x and y and solve for x or for y	**M1**	$x+2y=2, \quad y-2x=6$
	Obtain quotient – 2 + 2i	**A1**	
		2	
9(a)(ii)	Use correct method to find either r or θ	**M1**	If only finding θ, need to be looking for θ in the correct quadrant
	Obtain $r=2\sqrt{2}$, or exact equivalent	**A1ft**	ft their $x + iy$
	Obtain $\theta=\frac{3}{4}\pi$ from exact work	**A1ft**	ft on $k(-1 + i)$ for $k > 0$ Do not ISW
		3	

M1 Carry out a correct method for calculating greatest value of $\arg z$

e.g. $\arg z = \frac{\pi}{2} + \sin^{-1}\frac{1}{3}$

A1 Obtain answer 1.91

Total: 5

Q175 Oct/Nov 2018 | Paper 3 (variant 32) | Question 10

Topics: none of the listed topics

Question 10(i)

[Question part not shown in provided images - mark scheme indicates vector work]

Question 10(ii)

Calculate the acute angle between l and p.

Marks: [4]

Question 10(iii)

Find the equation of the line which lies in p and intersects l at right angles.

Marks: [4]

Examiner comment

> "Question 10 (i) Although there have been questions in the past using the form r.n= a.n, many candidates did not recognise this use of (r−a).n=0."

Mark scheme

Question 10(i)

M1* Substitute for $\mathbf{r}$ and expand the scalar product to obtain an equation in λ: e.g. $3(5+\lambda)+(-3-2\lambda)+(-1+\lambda)=5\ (2\lambda=5-11)$. Must attempt to deal with $\mathbf{i}+2\mathbf{j}$

M1(dep*) Solve a linear equation for λ

A1 Obtain $\lambda = -3$ and position vector $\mathbf{r}_A = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ for A. Accept coordinates

Total: 3

Question 10(ii)

B1 State or imply a normal vector of p is $3\mathbf{i} + \mathbf{j} + \mathbf{k}$, or equivalent

M1 Use correct method to evaluate a scalar product of relevant vectors: e.g. $(\mathbf{i} - 2\mathbf{j} + \mathbf{k}).(3\mathbf{i} + \mathbf{j} + \mathbf{k})$

M1 Using the correct process for calculating the moduli, divide the scalar product by the product of the moduli and evaluate the inverse sine or cosine of the result: $\cos\theta = \frac{2}{\sqrt{6}\sqrt{11}}$. Second **M1** available if working with the wrong vectors

A1 Obtain answer 14.3° or 0.249 radians. Or better

Total: 4

Question 10(iii)

B1 Taking the direction vector of the line to be $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, state a relevant equation in a, b, c, e.g. $3a + b + c = 0$

M1 State a second relevant equation, e.g. $a - 2b + c = 0$, and solve for one ratio, e.g. $a : b$

A1 Obtain $a : b : c = 3 : -2 : -7$, or equivalent

A1ft State answer $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} + \mu(3\mathbf{i} - 2\mathbf{j} - 7\mathbf{k})$. Or equivalent. The f.t. is on $\mathbf{r}_A$. Requires "$\mathbf{r} = \ldots$ "

Alternative:

M1 Attempt to calculate the vector product of relevant vectors, e.g. $(3\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

A1 Obtain two correct components of the product

A1 Obtain correct product, e.g. $3\mathbf{i} - 2\mathbf{j} - 7\mathbf{k}$

A1ft State answer $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} + \mu(3\mathbf{i} - 2\mathbf{j} - 7\mathbf{k})$. Or equivalent. The f.t. is on $\mathbf{r}_A$. Requires "$\mathbf{r} = \ldots$ "

Total: 4

Q176 Oct/Nov 2018 | Paper 3 (variant 33) | Question 1

Topics: Algebra

Question 1

Find the set of values of x satisfying the inequality $2|2x - a| < |x + 3a|$, where a is a positive constant.

Marks: [4]

Examiner comment

> "Question 1 This question proved to be challenging for candidates."

> "Many candidates failed to square the 2 outside the modulus sign or poor algebra prevented them from arriving at the correct equation."

Mark scheme

Question 1

B1 EITHER: State or imply non-modular inequality:

$$2^2(2x - a)^2 < (x + 3a)^2,$$

or corresponding quadratic equation, or pair of linear equations:

$$2(2x - a) = \pm(x + 3a)$$

M1 Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x

A1 Obtain critical values $x = \frac{5}{3}a$ and $x = -\frac{1}{5}a$

A1 State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$

B1 OR: Obtain critical value $x = \frac{5}{3}a$ from a graphical method, or by inspection, or by solving a linear equation or an inequality

B2 Obtain critical value $x = -\frac{1}{5}a$ similarly

B1 State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$

[Do not condone $\leq$ for $<$ in the final answer.]

Total: 4

Q177 Oct/Nov 2018 | Paper 3 (variant 33) | Question 2

Topics: Indices, Logarithms & Exponents

Question 2

Showing all necessary working, solve the equation

$$\frac{2e^{x}+e^{-x}}{e^{x}-e^{-x}}=4,$$

giving your answer correct to 2 decimal places.

Marks: [4]

Examiner comment

> "Question 2 This question was usually well done, however there were still many candidates who did not reach an b equation of the form ae2x = b, aex = be−x or aex = with the correct values of both a and b."

> "Most candidates knew they then needed to take lns but some did not show this step despite the additional instruction to show all necessary working."

Mark scheme

Question 2

M1 Rearrange the equation in the form $ae^{2x}=b$ or $ae^{x}=be^{-x}$

A1 Obtain correct equation in either form with $a=2$ and $b=5$

M1 Use correct method to solve for x

A1 Obtain answer $x=0.46$

Total: 4

Q178 Oct/Nov 2018 | Paper 3 (variant 33) | Question 3

Topics: Numerical Solutions/Iteration

Question 3(i)

By sketching a suitable pair of graphs, show that the equation $x^3 = 3 - x$ has exactly one real root.

Marks: [2]

Question 3(ii)

Show that if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{2x_n^3 + 3}{3x_n^2 + 1}$$

converges, then it converges to the root of the equation in part (i).

Marks: [2]

Question 3(iii)

Use this iterative formula to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Marks: [3]

Examiner comment

> "Question 3 (i) This part proved challenging more many candidate, for various reasons."

Mark scheme

Question 3(i)

B1 Sketch a relevant graph, e.g. $y = x^3$

B1 Sketch a second relevant graph, e.g. $y = 3 - x$, and justify the given statement. Consideration of behaviour for $x < 0$ is needed for the second B1

Total: 2

Question 3(ii)

B1 State or imply the equation $x = \left(2x^3 + 3\right) / \left(3x^2 + 1\right)$

B1 Rearrange this in the form $x^3 = 3 - x$, or commence work *vice versa*

Total: 2

Q179 Oct/Nov 2018 | Paper 3 (variant 33) | Question 5

Topics: Differential Equations

5 The coordinates (x, y) of a general point on a curve satisfy the differential equation

$$x\frac{dy}{dx} = (2 - x^2)y$$

The curve passes through the point $(1, 1)$. Find the equation of the curve, obtaining an expression for y in terms of x. **[7]**

Examiner comment

> "The integral in x proved more difficult, since instead of considering and x, some candidates x resorted to integrating by parts and increased the possibility of sign errors."

Mark scheme

5 Separate variables correctly and integrate at least one side | **B1**

Obtain term ln y | **B1**

Obtain terms $2\ln x - \frac{1}{2}x^2$ | **B1+B1**

Use $x = 1, y = 1$ to evaluate a constant, or as limits | **M1**

Obtain correct solution in any form, e.g. | **A1**
$\ln y = 2\ln x - \frac{1}{2}x^2 + \frac{1}{2}$

Rearrange as $y = x^2 \exp\left(\frac{1}{2} - \frac{1}{2}x^2\right)$, or equivalent | **A1**

Total: 7

Q180 Oct/Nov 2018 | Paper 3 (variant 33) | Question 6

Topics: Trigonometry

(i) Show that the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$ can be expressed in the form $R\sin(x - \alpha) = \sqrt{2}$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. **[4]**

(ii) Hence solve the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$, for $0^\circ < x < 180^\circ$. **[4]**

Examiner comment

> "Question 6 (i) Many candidates found it difficult to obtain the given expression, taking far more steps than were required."

Mark scheme

6(i) Rearrange in the form $\sqrt{3}\sin x - \cos x = \sqrt{2}$ | **B1**

State $R = 2$ | **B1**

Use trig formulae to obtain α | **M1**

Obtain $\alpha = 30^\circ$ with no errors seen | **A1**

Total: 4

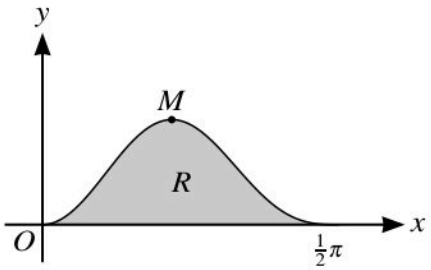
6(ii) [Part (ii) marking scheme not clearly visible in MS image]

Q181 Oct/Nov 2018 | Paper 3 (variant 33) | Question 7

Topics: Trigonometry, Differentiation, Integration

7

7



The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. [5]

The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. **[5]**

(ii) Using the substitution $u = \sin x$ and showing all necessary working, find the exact area of R. **[4]**

Examiner comment

> "(ii) Nearly all candidates struggled with this question despite being given the substitution required."

> "Common errors were poor substitution of du = cos x dx, inability to express cos2 x correctly in terms of u, substituting π limits 0 and for u or substituting limits 0 and 1 for x."

Mark scheme

7(i) Use product rule | **M1***

Obtain correct derivative in any form | **A1**

Equate derivative to zero and obtain an equation in a single trig function | **depM1***

Obtain a correct equation, e.g. $3\tan^2 x = 2$ | **A1**

Obtain answer $x = 0.685$ | **A1**

Total: 5

7(ii) Use the given substitution and reach $a\int(u^2 - u^4)du$ | **M1**

Obtain correct integral with $a = 5$ and limits 0 and 1 | **A1**

Use correct limits in an integral of the form $a\left(\frac{1}{3}u^3 - \frac{1}{5}u^5\right)$ | **M1**

Obtain answer $\frac{2}{3}$ | **A1**

Total: 4

Q182 Oct/Nov 2018 | Paper 3 (variant 33) | Question 8

Topics: Complex Numbers

8(a) Showing all necessary working, express the complex number $\frac{2+3i}{1-2i}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the values of r and θ correct to 3 significant figures. **[5]**

8(b) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - 3 + 2i| = 1$. Find the least value of $|z|$ for points on this locus, giving your answer in an exact form. **[4]**

Examiner comment

> "Few candidates managed to obtain the correct argument, the usual answer being θ = −1.05, something that candidates arrived at from their calculators without any reference to the position of the complex number −4 7 + i in the Argand diagram."

Mark scheme

8(i) EITHER: Multiply numerator and denominator by $1 + 2i$, or equivalent, or equate to $x + iy$, obtain two equations in x and y and solve for x or for y | **M1**

Obtain quotient $-\frac{4}{5} + \frac{7}{5}i$, or equivalent | **A1**

Use correct method to find either r or θ | **M1**

Obtain $r = 1.61$ | **A1**

Obtain $\theta = 2.09$ | **A1**

OR: Find modulus or argument of $2 + 3i$ or of $1 - 2i$ | **B1**

Use correct method to find r | **M1**

Obtain $r = 1.61$ | **A1**

Use correct method to find θ | **M1**

Obtain $\theta = 2.09$ | **A1**

Total: 5

8(ii) Show a circle with centre $3 - 2i$ | **B1**

Show a circle with radius 1 | **B1ft** (Centre not at the origin)

Carry out a correct method for finding the least value of $|z|$ | **M1**

Obtain answer $\sqrt{13} - 1$ | **A1**

Total: 4

Q183 Oct/Nov 2018 | Paper 3 (variant 33) | Question 10

Topics: none of the listed topics

10 The planes m and n have equations $3x + y - 2z = 10$ and $x - 2y + 2z = 5$ respectively. The line l has equation $\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.

(i) Show that l is parallel to m. **[3]**

(ii) Calculate the acute angle between the planes m and n. **[3]**

(iii) A point P lies on the line l. The perpendicular distance of P from the plane n is equal to 2. Find the position vectors of the two possible positions of P. **[4]**

Examiner comment

> "(iii) Those candidates who knew the formula for the distance of a point from a plane usually quickly scored full marks, whilst most others struggled."

Mark scheme

10(i) EITHER: Expand scalar product of a normal to m and a direction vector of l | **M1**

Verify scalar product is zero | **A1**

Verify that one point of l does not lie in the plane | **A1**

OR: Substitute coordinates of a general point of l in the equation of the plane m | **M1**

Obtain correct equation in λ in any form | **A1**

Verify that the equation is not satisfied for any value of λ | **A1**

Total: 3

10(ii) Use correct method to evaluate a scalar product of normal vectors to m and n | **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result | **M1**

Obtain answer 74.5° or 1.30 radians | **A1**

Total: 3

10(iii) EITHER: Using the components of a general point P of l form an equation in λ by equating the perpendicular distance from n to 2 | **M1**

Obtain a correct modular or non-modular equation in any form | **A1**

Solve for λ and obtain a position vector for P, e.g. $7\mathbf{i} + 5\mathbf{j} + 7\mathbf{j}$ from $\lambda = 3$ | **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ from $\lambda = -1$ | **A1**

OR: Take a point Q on l, e.g. $(5, 3, 3)$ and form an equation in λ by equating the length of the projection of QP onto a normal to plane n to 2 | **M1**

Obtain a correct modular or non-modular equation in any form | **A1**

Obtain a correct modular or non-modular equation in any form | **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ from $\lambda=-1$ | **A1**

Total: 4

Q184 Feb/Mar 2017 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Solve the equation $\ln(1+2^x)=2$, giving your answer correct to 3 decimal places. [3]

Examiner comment

> "Question 1 Many candidates produced a completely correct solution, however, some either failed to give their answer to 3 decimal places or rounded it incorrectly."

Mark scheme

Remove logarithm and obtain $1+2^x=e^2$ **B1**

Use correct method to solve an equation of the form $2^x=a$, where $a>0$ **M1**

Obtain answer $x=2.676$ **A1**

Total: 3

Q185 Feb/Mar 2017 | Paper 3 (variant 32) | Question 4

Topics: Trigonometry

(i) Express $8\cos\theta-15\sin\theta$ in the form $R\cos(\theta+\alpha)$, where $R>0$ and $0°<\alpha<90°$, stating the exact value of R and giving the value of α correct to 2 decimal places. [3]

(ii) Hence solve the equation

$$8\cos 2x-15\sin 2x=4,$$

for $0°<x<180°$. [4]

Examiner comment

> "The common errors seen were the negative sign within the expression for α, so α = tan−1(−15/8), or a muddled tan−1, so α = tan−1(8/15) or the failure to give the value of α to 2 decimal places."

> "Answer: R = 17, α = 61.93° (ii) Many candidates successfully found the solution x = 7.2°, however, few candidates were able to find the second solution."

Mark scheme

(i)

State $R=17$ **B1**

Use trig formula to find α **M1**

Obtain $\alpha = 61.93°$ with no errors seen **A1**

Total: 3

(ii)

Evaluate $\cos^{-1}(4/17)$ to at least 1 d.p. ($76.39°$ to 2 d.p.) **B1**

Use a correct method to find a value of x in the interval $0° < x < 180°$ **M1**

Obtain answer, e.g. $x = 7.2°$ **A1**

Obtain second answer, e.g. $x = 110.8°$ and no others **A1**

Total: 4

Q186 Feb/Mar 2017 | Paper 3 (variant 32) | Question 5

Topics: Trigonometry, Differentiation

The curve with equation $y = e^{-ax}\tan x$, where a is a positive constant, has only one point in the interval $0 < x < \frac{1}{2}\pi$ at which the tangent is parallel to the x-axis. Find the value of a and state the exact value of the x-coordinate of this point. [7]

Examiner comment

> "Question 5 Most candidates were unable to progress far with this question, although usually the correct product rule was applied apart from some weak attempts at the differentiation of e–ax."

Mark scheme

Use product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero, use Pythagoras and obtain a quadratic equation in $\tan x$ **M1**

Obtain $\tan^2 x - a\tan x + 1 = 0$, or equivalent **A1**

Use the condition for a quadratic to have only one root **M1**

Obtain answer $a = 2$ **A1**

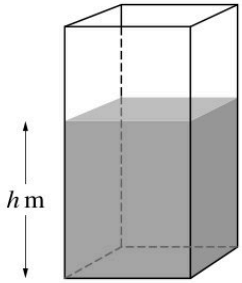
Obtain answer $x = \frac{1}{4}\pi$ **A1**

Total: 7

Q187 Feb/Mar 2017 | Paper 3 (variant 32) | Question 7

Topics: Integration, Differential Equations

7



A water tank has vertical sides and a horizontal rectangular base, as shown in the diagram. The area of the base is 2 m^2. At time $t = 0$ the tank is empty and water begins to flow into it at a rate of 1 m^3 per hour. At the same time water begins to flow out from the base at a rate of $0.2\sqrt{h}$ m^3 per hour, where h m is the depth of water in the tank at time t hours.

(i) Form a differential equation satisfied by h and t, and show that the time T hours taken for the depth of water to reach 4 m is given by

$$T = \int_0^4 \frac{10}{5 - \sqrt{h}}\,\mathrm{d}h.$$ [3]

A water tank has vertical sides and a horizontal rectangular base, as shown in the diagram. The area of the base is $\mathbf{2\ m^2}$. At time $t = 0$ the tank is empty and water begins to flow into it at a rate of $\mathbf{1\ m^3}$ per hour. At the same time water begins to flow out from the base at a rate of $\mathbf{0.2\sqrt{h}\ m^3}$ per hour, where h m is the depth of water in the tank at time t hours.

(i) Form a differential equation satisfied by h and t, and show that the time T hours taken for the depth of water to reach $\mathbf{4}$ m is given by

$$T = \int_0^4 \frac{10}{5 - \sqrt{h}}\,dh.$$

[3]

(ii) Using the substitution $u = 5 - \sqrt{h}$, find the value of T. [6]

Examiner comment

> "Unfortunately a common error was the omission of the constant 10, despite the integration producing a lnu and a u term."

Mark scheme

(i)

State or imply $\frac{dV}{dt} = 2\frac{dh}{dt}$ **B1**

State or imply $\frac{dV}{dt} = 1 - 0.2\sqrt{h}$ **B1**

Obtain the given answer correctly **B1**

Total: 3

(ii)

State or imply $\frac{du}{dh} = -\frac{1}{2\sqrt{h}}$, or equivalent **B1**

Substitute for h and dh throughout **M1**

Obtain $T = \int_3^5 \frac{20(5-u)}{u}\, du$, or equivalent **A1**

Integrate and obtain terms $100 \ln u - 20u$, or equivalent **A1**

Substitute limits $u = 3$ and $u = 5$ correctly **M1**

Obtain answer 11.1, with no errors seen **A1**

Total: 6

Q188 Feb/Mar 2017 | Paper 3 (variant 32) | Question 8

Topics: Algebra, Complex Numbers

Throughout this question the use of a calculator is not permitted.

The polynomial $z^4 + 3z^2 + 6z + 10$ is denoted by $p(z)$. The complex number $-1 + i$ is denoted by u.

(i) Showing all your working, verify that u is a root of the equation $p(z) = 0$. [3]

(ii) Find the other three roots of the equation $p(z) = 0$. [7]

Examiner comment

> "(ii) Candidates found this section difficult, since many failed to use the fact that another root of p(z) was the conjugate of u."

Mark scheme

(i)

Substitute $z = -1 + i$ and attempt expansions of the z^2 and z^4 terms **M1**

Use $i^2 = -1$ at least once **M1**

Complete the verification correctly **A1**

Total: 3

(ii)

State second root $z = -1 - i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $-1 + i$ and $-1 - i$ **M1**

Obtain $z^2 + 2z + 2$, or equivalent **A1**

Attempt division of $p(z)$ by $z^2 + 2z + 2$ and reach a partial quotient $z^2 + kz$ **M1**

Obtain quadratic factor $z^2 - 2z + 5$ **A1**

Solve 3-term quadratic and use $i^2 = -1$ **M1**

Obtain roots $1+2i$ and $1-2i$ **A1**

Total: 7

Q189 May/June 2017 | Paper 3 (variant 31) | Question 2

Topics: Algebra

Expand

$$(1+6x)^{-1/3}$$

in ascending powers of x, up to and including the term in x^3, simplifying the coefficients. [4]

Examiner comment

> "The most common error was to omit the 6 in (6x)2 and (6x)3 ."

Mark scheme

State a correct unsimplified version of the x, x^2 or x^3 term in the expansion of $(1+6x)^{-1/3}$ **M1**

State correct first two terms

$$1-2x$$

A1

Obtain term

$$8x^2$$

A1

Obtain term

$$-\frac{112}{3}x^3$$

in the final answer **A1**

Total: 4

Q190 May/June 2017 | Paper 3 (variant 31) | Question 3

Topics: Indices, Logarithms & Exponents, Integration

It is given that

$$x = \ln(1-y) - \ln y,$$

where $0 < y < 1$.

(i) Show that

$$y = \frac{e^{-x}}{1+e^{-x}}.$$

[2]

(ii) Hence show that

$$\int_0^1 y\,dx = \ln\left(\frac{2e}{e+1}\right).$$

[4]

Examiner comment

> "After correct substitution of limits, some candidates struggled to combine the two terms correctly."

Mark scheme

(i)

Remove logarithms correctly and obtain

$$e^x = \frac{1-y}{y}$$

B1

Obtain the given answer

$$y = \frac{e^{-x}}{1+e^{-x}}$$

following full working **B1**

Total: 2

(ii)

State an integral of the form

$$k\ln(1+e^{-x})$$

where $k = \pm 1$ **M1**

State the correct integral

$$-\ln(1+e^{-x})$$

A1

Use limits correctly **M1**

Obtain the given answer

$$\ln\left(\frac{2e}{e+1}\right)$$

following full working **A1**

Total: 4

Q191 May/June 2017 | Paper 3 (variant 31) | Question 4

Topics: Trigonometry, Differentiation

The parametric equations of a curve are

$$x = \ln\cos\theta,$$

$$y = 3\theta - \tan\theta,$$

where $0 \leq \theta < \frac{1}{2}\pi$.

(i) Express $\frac{dy}{dx}$ in terms of $\tan\theta$. [5]

(ii) Find the exact y-coordinate of the point on the curve at which the gradient of the normal is equal to 1. [3]

Examiner comment

> "dy (ii) Many candidates did not connect the gradient of the normal being 1 with =−1 and therefore dx made no progress here."

Mark scheme

(i)

Use the chain rule to differentiate x, obtaining

$$\frac{dx}{d\theta} = -\tan\theta$$

M1

State

$$\frac{dy}{d\theta} = 3 - \sec^2\theta$$

B1

Use

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

M1

Obtain a correct expression in any form, e.g.

$$\frac{3 - \sec^2\theta}{-\tan\theta}$$

A1

Obtain

$$\frac{dy}{dx} = \frac{\tan^2\theta - 2}{\tan\theta}$$

A1

Total: 5

(ii)

Equate the tangent gradient to -1 and obtain an equation in $\tan\theta$ **M1**

Solve the quadratic

$$\tan^2\theta + \tan\theta - 2 = 0$$

M1

Obtain $\theta = \frac{\pi}{4}$ and hence

$$y = \frac{3\pi}{4} - 1$$

A1

Total: 3

Q192 May/June 2017 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Numerical Solutions/Iteration

The diagram shows a semicircle with centre O, radius r and diameter AB. The point P on its circumference is such that the area of the minor segment on AP is equal to half the area of the minor segment on BP. The angle AOP is x radians.

(i) Show that x satisfies the equation

$$x = \frac{1}{3}(\pi + \sin x).$$

[3]

(ii) Verify by calculation that x lies between 1 and 1.5. [2]

(iii) Use an iterative formula based on the equation in part (i) to determine x correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Examiner comment

> "© 2017 Cambridge International Advanced Level 9709 Mathematics June 2017 Principal Examiner Report for Teachers (iii) Again, many candidates did not attempt this question, possibly because they simply did not continue after not completing the first part."

Mark scheme

(i)

Use the correct sector formula at least once and form an equation in r and x **M1**

Obtain a correct equation in any form, e.g.

$$x - \sin x = \frac{1}{2}(\pi - x - \sin x)$$

A1

Rearrange to obtain

$$x = \frac{1}{3}(\pi + \sin x)$$

A1

Total: 3

(ii)

Calculate values of a relevant expression at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use the iterative formula, for example

$$x_{n+1} = \frac{1}{3}(\pi + \sin x_n),$$

correctly at least once **M1**

Obtain final answer $\mathbf{1.374}$ **A1**

Show sufficient iterations to 5 d.p. to justify $\mathbf{1.374}$ to 3 d.p., or show there is a sign change in the interval $(1.3745, 1.3755)$ **A1**

Total: 3

Q193 May/June 2017 | Paper 3 (variant 31) | Question 6

Topics: none of the listed topics

The plane with equation

$$2x + 2y - z = 5$$

is denoted by m. Relative to the origin O, the points A and B have coordinates $(3, 4, 0)$ and $(-1, 0, 2)$ respectively.

(i) Show that the plane m bisects AB at right angles. [5]

A second plane p is parallel to m and nearer to O. The perpendicular distance between the planes is 1.

(ii) Find the equation of p, giving your answer in the form $ax + by + cz = d$. [3]

Examiner comment

> "(ii) This proved to be one of the most difficult questions on the paper."

> "Many candidates did not attempt to use the fact that the equation r.n = a.n gives you the distance of the plane from the origin if n is the unit normal vector."

> "A few candidates reached the correct answer, sometimes via quite difficult routes."

> "Many candidates did not appear to recognise that the equation of a parallel plane would be of the form 2x+2y −z =d Answer: (ii) 2x+2y −z =2"

Mark scheme

(i)

State or obtain coordinates $(1, 2, 1)$ for the midpoint of AB **B1**

Verify that this midpoint lies on m **B1**

State or imply a correct normal vector to the plane, e.g.

$$2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

B1

State or imply a direction vector for AB, e.g.

$$-4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$$

B1

Confirm that m is perpendicular to AB **B1**

Total: 5

(ii)

State or imply that the perpendicular distance of m from the origin is $\frac{5}{3}$, or an unsimplified equivalent **B1**

State or imply that p has an equation of the form

$$2x + 2y - z = k$$

B1

Obtain

$$2x + 2y - z = 2$$

B1

Total: 3

Q194 May/June 2017 | Paper 3 (variant 31) | Question 7

Topics: Complex Numbers

Throughout this question the use of a calculator is not permitted.

The complex numbers u and w are defined by

$$u = -1 + 7i \quad \text{and} \quad w = 3 + 4i.$$

(i) Showing all your working, find in the form $x + iy$, where x and y are real, the complex numbers $u - 2w$ and $\frac{u}{w}$. [4]

In an Argand diagram with origin O, the points A, B and C represent the complex numbers u, w and $u - 2w$ respectively.

(ii) Prove that angle $AOB = \frac{1}{4}\pi$. [2]

(iii) State fully the geometrical relation between the line segments OB and CA. [2]

Examiner comment

> "Some marks were lost needlessly through arithmetic errors and poor use of brackets."

> "Some candidates lost marks because they used the trigonometry of a right angled triangle without demonstrating that they had a right angle triangle."

Mark scheme

(i)

State

$$u - 2w = -7 - i$$

B1

Multiply numerator and denominator of $\frac{u}{w}$ by $3 - 4i$, or equivalent **M1**

Simplify the numerator to $25 + 25i$ or the denominator to 25 **A1**

Obtain

$$\frac{u}{w} = 1 + i$$

A1

Total: 4

(ii)

Find the argument of $\frac{u}{w}$ **M1**

Since Math input error, obtain the given answer **A1**

Total: 2

(iii)

State that OB and CA are parallel **B1**

State that $CA = 2OB$, or equivalent **B1**

Total: 2

Q195 May/June 2017 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Solve the equation

$$\ln(x^2 + 1) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. [3]

Examiner comment

> "The false statement ln(x2 + 1)=lnx2 +ln1 was a common error."

Mark scheme

Use a law of logarithms for a power or a quotient **M1**

Remove logarithms and obtain a correct equation in x, e.g.

$$x^2 + 1 = ex^2$$

A1

Obtain answer $x = 0.763$ and no other **A1**

Total: 3

Q196 May/June 2017 | Paper 3 (variant 32) | Question 2

Topics: Algebra

Solve the inequality

$$|x - 3| < 3x - 4.$$

[4]

Examiner comment

> "Common errors included slips in the arithmetic, errors in squaring 2 brackets, or squaring only the modular side of the inequality."

Mark scheme

Either state or imply the non-modular inequality

$$(x - 3)^2 < (3x - 4)^2,$$

or the corresponding equation **B1**

Make a reasonable attempt at solving a three-term quadratic **M1**

Obtain critical value

$$x = \frac{7}{4}$$

A1

State final answer

$$x > \frac{7}{4}$$

only **A1**

Total: 4

Q197 May/June 2017 | Paper 3 (variant 32) | Question 3

Topics: Trigonometry

(i) Express the equation

$$\cot\theta - 2\tan\theta = \sin 2\theta$$

in the form

$$a\cos^4\theta + b\cos^2\theta + c = 0,$$

where a, b and c are constants to be determined. [3]

(ii) Hence solve the equation

$$\cot\theta - 2\tan\theta = \sin 2\theta$$

for $90° < \theta < 180°$. [2]

Examiner comment

> "The most common errors were slips in the arithmetic and algebra, which prevented several candidates from reaching a correct final answer."

> "Many candidates considered only the positive square root and consequently did not find any solution in the given range."

Mark scheme

(i)

Use correct formulae to express the equation in terms of $\cos\theta$ and $\sin\theta$ **M1**

Use Pythagoras and express the equation in terms of $\cos\theta$ only **M1**

Obtain a correct three-term equation, e.g.

$$2\cos^4\theta + \cos^2\theta - 2 = 0$$

A1

Total: 3

(ii)

Solve a three-term quadratic in $\cos^2\theta$ for $\cos\theta$ **M1**

Obtain answer

$$\theta = 152.1°$$

only **A1**

Total: 2

Q198 May/June 2017 | Paper 3 (variant 32) | Question 4

Topics: Differentiation

The parametric equations of a curve are

$$x = t^2 + 1,$$

$$y = 4t + \ln(2t - 1).$$

(i) Express $\frac{dy}{dx}$ in terms of t. [3]

(ii) Find the equation of the normal to the curve at the point where $t = 1$. Give your answer in the form $ax + by + c = 0$. [3]

Examiner comment

> "The most common error dx d 1 was to obtain ln(2t−1)= ."

Mark scheme

(i)

State

$$\frac{dy}{dt} = 4 + \frac{2}{2t - 1}$$

B1

Use

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

M1

Obtain

$$\frac{dy}{dx} = \frac{8t - 2}{2t(2t - 1)},$$

or equivalent **A1**

Total: 3

(ii)

Use a correct method to find the gradient of the normal at $t = 1$ **M1**

Use a correct method to form an equation for the normal at $t = 1$ **M1**

Obtain

$$x + 3y - 14 = 0$$

A1

Total: 3

Q199 May/June 2017 | Paper 3 (variant 32) | Question 5

Topics: Indices, Logarithms & Exponents, Integration, Differential Equations

In a certain chemical process a substance A reacts with and reduces a substance B. The masses of A and B at time t after the start of the process are x and y respectively. It is given that

$$\frac{dy}{dt} = -0.2xy$$

and

$$x = \frac{10}{(1+t)^2}.$$

At the beginning of the process $y = 100$.

(i) Form a differential equation in y and t, and solve this differential equation. [6]

(ii) Find the exact value approached by the mass of B as t becomes large. State what happens to the mass of A as t becomes large. [2]

Examiner comment

> "Several candidates never formed a differential equation in y and t and made no progress at all."

Mark scheme

(i)

State

$$\frac{dy}{dt} = -\frac{2y}{(1+t)^2}$$

B1

Separate variables correctly and attempt integration of one side **M1**

Obtain term $\ln y$, or equivalent **A1**

Obtain term

$$\frac{2}{1+t}$$

A1

Use $y = 100$ and $t = 0$ to evaluate a constant, or use limits, in an expression containing terms of the form $a \ln y$ and $\frac{b}{1+t}$ **M1**

Obtain a correct solution in any form, e.g.

$$\ln y = \frac{2}{1+t} - 2 + \ln 100$$

A1

Total: 6

(ii)

State that the mass of B approaches

$$\frac{100}{e^2}$$

B1

State or imply that the mass of A tends to zero **B1**

Total: 2

Q200 May/June 2017 | Paper 3 (variant 32) | Question 7

Topics: Trigonometry, Differentiation, Integration

(i) Prove that if

$$y = \frac{1}{\cos\theta}$$

then

$$\frac{dy}{d\theta} = \sec\theta\tan\theta.$$

[2]

(ii) Prove the identity

$$\frac{1+\sin\theta}{1-\sin\theta} \equiv 2\sec^2\theta + 2\sec\theta\tan\theta - 1.$$

[3]

(iii) Hence find the exact value of

$$\int_0^{\frac{1}{4}\pi} \frac{1+\sin\theta}{1-\sin\theta}\, d\theta.$$

[4]

Examiner comment

> "The most common error here was that candidates could not integrate secθtanθ – they often offered a complicated expression involving logarithms."

Mark scheme

(i)

Use the quotient rule or chain rule **M1**

Obtain the given answer correctly **A1**

Total: 2

(ii)

Multiply numerator and denominator of the left hand side by $1+\sin\theta$ **M1**

Use Pythagoras and express the result in terms of $\sec\theta$ and $\tan\theta$ **M1**

Complete the proof **A1**

Total: 3

(iii)

Use the identity and obtain the integral

$$\int (2\tan\theta + 2\sec\theta - 1)\, d\theta$$

B2

Use the correct limits in an integral containing terms $a\tan\theta$ and $b\sec\theta$ **M1**

Obtain

$$2\sqrt{2} - \frac{\pi}{4}$$

A1

Total: 4

Q201 May/June 2017 | Paper 3 (variant 32) | Question 8

Topics: Algebra

Let

$$f(x) = \frac{5x^2 - 7x + 4}{(3x+2)(x^2+5)}.$$

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

Examiner comment

> "The most common error was to mishandle the extracted factors of 2 and 5."

Mark scheme

(i)

State or imply the form

$$\frac{A}{3x+2} + \frac{Bx+C}{x^2+5}$$

B1

Use a relevant method to determine a constant **M1**

Obtain one of $A = 2$, $B = 1$, $C = -3$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Use a correct method to find the first two terms of the expansion of $(3x+2)^{-1}$ or $(x^2+5)^{-1}$ **M1**

Obtain correct unsimplified expansions up to the term in x^2 of each partial fraction **A2**

Multiply out up to the term in x^2 by $Bx+C$, where $BCe0$ **M1**

Obtain final answer

$$\frac{2}{5}-\frac{13}{10}x+\frac{237}{100}x^2$$

A1

Total: 5

Q202 May/June 2017 | Paper 3 (variant 32) | Question 9

Topics: none of the listed topics

Relative to the origin O, the point A has position vector

$$\overrightarrow{OA}=\mathbf{i}+2\mathbf{j}+4\mathbf{k}.$$

The line l has equation

$$\mathbf{r}=9\mathbf{i}-\mathbf{j}+8\mathbf{k}+\lambda(3\mathbf{i}-\mathbf{j}+2\mathbf{k}).$$

(i) Find the position vector of the foot of the perpendicular from A to l. Hence find the position vector of the reflection of A in l. [5]

(ii) Find the equation of the plane through the origin which contains l. Give your answer in the form $ax+by+cz=d$. [3]

(iii) Find the exact value of the perpendicular distance of A from this plane. [3]

Examiner comment

> "Question 9 (i) Although very straight forward, this question was not similar to anything that the candidates had seen on recent papers, and many of them struggled to find a correct approach."

> "Many candidates made no attempt to find the position vector of the reflection of A."

> "The most common error in applying the formula for distance of a point from a plane was to use the modulus of i+2j+4k in the denominator rather than the modulus of their normal vector."

Mark scheme

(i)

Find $\overrightarrow{AP}$ for a general point P on l with parameter λ, e.g.

$$(8+3\lambda)\mathbf{i}+(-3-\lambda)\mathbf{j}+(4+2\lambda)\mathbf{k}$$

B1

Equate the scalar product of $\overrightarrow{AP}$ and the direction vector of l to zero and solve for λ **M1**

Obtain $\lambda=-\frac{5}{2}$ and foot of perpendicular

$$\frac{3}{2}\mathbf{i}+\frac{3}{2}\mathbf{j}+3\mathbf{k}$$

A1

Carry out a complete method for finding the position vector of the reflection of A in l **M1**

Obtain

$$2\mathbf{i}+\mathbf{j}+2\mathbf{k}$$

A1

Total: 5

(ii)

Use a scalar product to obtain an equation in a, b and c, e.g.

$$3a-b+2c=0$$

B1

Form a second relevant equation, e.g.

$$9a-b+8c=0,$$

and solve for one ratio **M1**

Obtain $a:b:c=1:1:-1$ and state plane equation

$$x+y-z=0$$

A1

Total: 3

(iii)

Apply the perpendicular distance formula using $A(1, 2, 4)$ and plane $x + y - z = 0$ **M1**

Obtain a correct expression, e.g.

$$\frac{|1 + 2 - 4|}{\sqrt{1^2 + 1^2 + (-1)^2}}$$

A1

Obtain

$$\frac{1}{\sqrt{3}}$$

A1

Total: 3

Q203 May/June 2017 | Paper 3 (variant 32) | Question 10

Topics: Trigonometry, Differentiation, Integration, Numerical Solutions/Iteration

The diagram shows the curve

$$y = x^2 \cos 2x$$

for $0 \leq x \leq \frac{1}{4}\pi$. The curve has a maximum point at M where $x = p$.

(i) Show that p satisfies the equation

$$p = \frac{1}{2}\tan^{-1}\left(\frac{1}{p}\right).$$

[3]

(ii) Use the iterative formula

$$p_{n+1} = \frac{1}{2}\tan^{-1}\left(\frac{1}{p_n}\right)$$

to determine the value of p correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(iii) Find, showing all necessary working, the exact area of the region bounded by the curve and the x-axis. [5]

Examiner comment

> "The most common errors were to 1 see 2 in place of , or sign errors."

Mark scheme

(i)

Use the product rule correctly **M1**

Obtain a correct derivative in any form, e.g.

$$y' = 2x\cos 2x - 2x^2 \sin 2x$$

A1

Equate to zero and derive the given equation **A1**

Total: 3

(ii)

Use the iterative formula correctly at least once, e.g.

$$0.5 \to 0.55357 \to 0.53261 \to 0.54070 \to 0.53755$$

M1

Obtain final answer 0.54 **A1**

Show sufficient iterations to 4 d.p. to justify 0.54 to 2 d.p., or show there is a sign change in the interval $(0.535, 0.545)$ **A1**

Total: 3

(iii)

Integrate by parts and reach an expression of the form

$$ax^2 \sin 2x + b \int x \sin 2x \, dx$$

M1

Obtain

$$\frac{1}{2}x^2 \sin 2x - \frac{1}{2} \int x \sin 2x \, dx$$

A1

Complete the integration and obtain

$$\frac{1}{2}x^2 \sin 2x + \frac{1}{2}x \cos 2x - \frac{1}{4}\sin 2x,$$

or equivalent **A1**

Substitute limits $x = 0$ and $x = \frac{1}{4}\pi$, having integrated twice **M1**

Obtain

$$\frac{\pi^2 - 8}{32}$$

A1

Total: 5

Q204 May/June 2017 | Paper 3 (variant 33) | Question 3

Topics: Indices, Logarithms & Exponents

Using the substitution $u = e^x$, solve the equation

$$4e^{-x} = 3e^x + 4.$$

Give your answer correct to 3 significant figures. [4]

Examiner comment

> "It was disappointing to see some candidates going straight from their quadratic equation to the solution, hence risking losing the method mark should any error occur."

Mark scheme

Rearrange as

$$3u^2 + 4u - 4 = 0,$$

or

$$3e^{2x} + 4e^x - 4 = 0,$$

or equivalent **B1**

Solve a three-term quadratic for e^x or for u **M1**

Obtain

$$e^x = \frac{2}{3}$$

A1

Obtain answer $x = -0.405$ and no other **A1**

Total: 4

Q205 May/June 2017 | Paper 3 (variant 33) | Question 5

Topics: Indices, Logarithms & Exponents, Trigonometry, Differentiation

A curve has equation

$$y = \frac{2}{3}\ln(1 + 3\cos^2 x)$$

for $0 \leq x \leq \frac{1}{2}\pi$.

(i) Express $\frac{dy}{dx}$ in terms of $\tan x$. [4]

(ii) Hence find the x-coordinate of the point on the curve where the gradient is -1. Give your answer correct to 3 significant figures. [2]

Examiner comment

> "Question 5 (i) Although many candidates could differentiate the ln function correctly using the chain rule, either by finding the derivative of 1 + 3cos2x or by converting the expression into one involving cos 2x, many found it challenging to progress to an expression in tan x as the question demanded."

> "© 2017 Cambridge International Advanced Level 9709 Syllabus June 2017 Principal Examiner Report for Teachers (ii) Few candidates had a suitable expression from (i) to enable them to proceed to solving a 3-term quadratic in this part, although it was quite possible to solve from the initial correct derivative in terms of sin x and cos x."

Mark scheme

(i)

Use the chain rule **M1**

Obtain a correct derivative in any form **A1**

Use correct trigonometry to express the derivative in terms of $\tan x$ **M1**

Obtain

$$\frac{dy}{dx} = \frac{-4\tan x}{4 + \tan^2 x},$$

or equivalent **A1**

Total: 4

(ii)

Equate derivative to -1 and solve a three-term quadratic for $\tan x$ **M1**

Obtain answer $x = 1.11$ and no other in the given interval **A1**

Total: 2

Q206 May/June 2017 | Paper 3 (variant 33) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

The equation

$$\cot x = 1 - x$$

has one root in the interval $0 < x < \pi$, denoted by ω.

(i) Show by calculation that ω is greater than 2.5. [2]

(ii) Show that, if a sequence of values in the interval $0 < x < \pi$ given by the iterative formula

$$x_{n+1} = \pi + \tan^{-1}\left(\frac{1}{1 - x_n}\right)$$

converges, then it converges to ω. [2]

(iii) Use this iterative formula to determine ω correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Examiner comment

> "In fact a considerable number of candidates made no attempt at this section or at (ii)."

Mark scheme

(i)

Calculate the value of a relevant expression at $x = 2.5$ and at another relevant value, e.g. $x = 3$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(ii)

State a suitable equation, e.g.

$$\tan^{-1}\left(\frac{1}{1 - x}\right) = x - \pi$$

B1

Rearrange this as

$$\cot x = 1 - x,$$

or commence working vice versa **B1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer $\mathbf{2.576}$ only **A1**

Show sufficient iterations to 5 d.p. to justify $\mathbf{2.576}$ to 3 d.p., or show there is a sign change in the interval $\mathbf{(2.5755, 2.5765)}$ **A1**

Total: 3

Q207 May/June 2017 | Paper 3 (variant 33) | Question 7

Topics: Differentiation, Integration

The diagram shows a sketch of the curve

$$y = \frac{e^{x/2}}{x}$$

for $x > 0$, and its minimum point M.

(i) Find the x-coordinate of M. [4]

(ii) Use the trapezium rule with two intervals to estimate the value of

$$\int_1^3 \frac{e^{x/2}}{x}\, dx,$$

giving your answer correct to 2 decimal places. [3]

(iii) The estimate found in part (ii) is denoted by E. Explain, without further calculation, whether another estimate found using the trapezium rule with four intervals would be greater than E or less than E. [1]

Examiner comment

> "(iii) Few candidates were able to explain whether the estimate with two intervals would be greater than or less than the estimate with four intervals."

Mark scheme

(i)

Use the correct quotient rule or product rule **M1**

Obtain a correct derivative in any form **A1**

Equate derivative to zero and solve for x **M1**

Obtain $x = 2$ **A1**

Total: 4

(ii)

State or imply ordinates $1.6487\ldots$, $1.3591\ldots$, $1.4938\ldots$ **B1**

Use the correct trapezium rule formula, or equivalent, with $h = 1$ and three ordinates **M1**

Obtain answer 2.93 only **A1**

Total: 3

(iii)

Explain why the estimate with four intervals would be less than E **B1**

Total: 1

Q208 May/June 2017 | Paper 3 (variant 33) | Question 8

Topics: Indices, Logarithms & Exponents, Integration, Differential Equations

In a certain chemical reaction, a compound A is formed from a compound B. The masses of A and B at time t after the start of the reaction are x and y respectively and the sum of the masses is equal to 50 throughout the reaction. At any time the rate of increase of the mass of A is proportional to the mass of B at that time.

(i) Explain why

$$\frac{dx}{dt} = k(50 - x),$$

where k is a constant. [1]

It is given that $x = 0$ when $t = 0$, and $x = 25$ when $t = 10$.

(ii) Solve the differential equation in part (i) and express x in terms of t. [8]

Examiner comment

> "Only a few candidates were unable to separate the variables correctly."

Mark scheme

(i)

Justify the given differential equation **B1**

Total: 1

(ii)

Separate variables correctly and attempt to integrate one side **B1**

Obtain term kt, or equivalent **B1**

Obtain term $-\ln(50 - x)$, or equivalent **B1**

Evaluate a constant, or use $x = 0$, $t = 0$ as limits in a solution containing terms $a\ln(50 - x)$ and bt **M1**

Obtain solution

$$-\ln(50-x) = kt - \ln 50,$$

or equivalent **A1**

Use $x = 25$, $t = 10$ to determine k **M1**

Obtain correct solution in any form, e.g.

$$\ln 50 - \ln(50-x) = \frac{t}{10}\ln 2$$

A1

Obtain

$$x = 50\left(1 - e^{-0.0693t}\right),$$

or equivalently

$$x = 50\left(1 - 2^{-t/10}\right)$$

A1

Total: 8

Q209 May/June 2017 | Paper 3 (variant 33) | Question 9

Topics: Algebra, Indices, Logarithms & Exponents, Integration

Let

$$f(x) = \frac{3x^2 - 4}{x^2(3x+2)}.$$

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence show that

$$\int_1^2 f(x)\,dx = \ln\left(\frac{25}{8}\right) - 1.$$

[5]

Examiner comment

> “A few candidates incorrectly introduced a constant term within their partial fraction, but fortunately most were able to recover by finding that the value of this constant was zero.”

Mark scheme

(i)

State or imply the form

$$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{3x+2}$$

B1

Use a relevant method to determine a constant **M1**

Obtain one of the values $A=3$, $B=-2$, $C=-6$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Integrate and obtain terms

$$3\ln x+\frac{2}{x}-2\ln(3x+2)$$

B3

Use limits correctly, having integrated all the partial fractions, in a solution containing terms $a\ln x+b\ln(3x+2)+\frac{c}{x}$ **M1**

Obtain the given answer following full and exact working **A1**

Total: 5

Q210 Oct/Nov 2017 | Paper 3 (variant 31) | Question 2

Topics: Indices, Logarithms & Exponents

2 Two variable quantities x and y are believed to satisfy an equation of the form $y = C(a^x)$, where C and a are constants. An experiment produced four pairs of values of x and y. The table below gives the corresponding values of x and $\ln y$.

x	0.9	1.6	2.4	3.2
$\ln y$	1.7	1.9	2.3	2.6

By plotting $\ln y$ against x for these four pairs of values and drawing a suitable straight line, estimate the values of C and a. Give your answers correct to 2 significant figures. **[5]**

Examiner comment

> "Question 2 This was another question that many candidates found difficult."

Mark scheme

Question	Answer	Marks

Q211 Oct/Nov 2017 | Paper 3 (variant 31) | Question 4

Topics: Trigonometry

(i) Prove the identity $\tan(45^\circ + x) + \tan(45^\circ - x) \equiv 2\sec 2x$. **[4]**

(ii) Hence sketch the graph of $y = \tan(45^\circ + x) + \tan(45^\circ - x)$ for $0^\circ \leq x \leq 90^\circ$. **[3]**

Examiner comment

> "A large proportion of candidates found it difficult to progress beyond the first three marks since they did not recognise that sec2x could be expressed as 1/cos2x."

Mark scheme

4(i) Use correct $\tan(A \pm B)$ formula and express the LHS in terms of $\tan x$ **M1**

Using $\tan 45^\circ = 1$ express LHS as a single fraction **A1**

Use Pythagoras or correct double angle formula **M1**

Obtain given answer **A1**

Total: 4

4(ii) Show correct sketch for one branch **B1**

Both branches correct and nothing else seen in the interval **B1**

Show asymptote at $x = 45^\circ$ **B1**

Total: 3

Q212 Oct/Nov 2017 | Paper 3 (variant 31) | Question 6

Topics: Indices, Logarithms & Exponents, Trigonometry, Integration, Differential Equations

6 The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = 4\cos^2 y \tan x,$$

for $0 < x < \frac{1}{2}\pi$, and $x = 0$ when $y = \frac{1}{4}\pi$. Solve this differential equation and find the value of x when $y = \frac{3}{8}\pi$. **[8]**

Examiner comment

> "A common error was failing to write 1/cos2y as sec2y, or not realising it integrated to tany."

Mark scheme

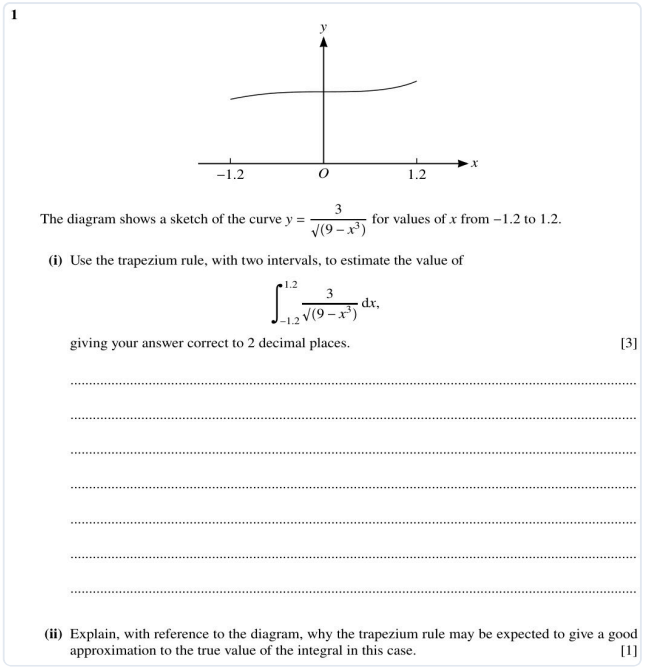
Question Answer Marks

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Q213 Oct/Nov 2017 | Paper 3 (variant 32) | Question 1

Topics: Integration

1



The diagram shows a sketch of the curve $y = \frac{3}{\sqrt{9-x^3}}$ for values of x from -1.2 to 1.2.

(i) Use the trapezium rule, with two intervals, to estimate the value of

$$\int_{-1.2}^{1.2} \frac{3}{\sqrt{9-x^3}}\,dx,$$

giving your answer correct to 2 decimal places. **[3]**

(ii) Explain, with reference to the diagram, why the trapezium rule may be expected to give a good approximation to the true value of the integral in this case. **[1]**

Examiner comment

> "A few candidates lost marks through not working to sufficient accuracy, or incorrect rounding of their final answer."

Mark scheme

1(i) State or imply ordinates $0.915929\ldots, 1, 1.112485\ldots$ **B1**

Use correct formula, or equivalent, with $h = 1.2$ and three ordinates **M1**

Obtain answer 2.42 only **A1**

Total: 3

1(ii) Justify the given statement **B1**

Total: 1

Q214 Oct/Nov 2017 | Paper 3 (variant 32) | Question 2

Topics: Indices, Logarithms & Exponents

2 Showing all necessary working, solve the equation $2\log_3 x = 3 + \log_2(x+1)$, giving your answer correct to 3 significant figures. **[5]**

Examiner comment

> "A common error at the very end of the solution was the acceptance of a negative value of x as a possible solution."

Mark scheme

2 Use law for the logarithm of a power or a quotient on the given equation **M1**

Use $\log_3 8 = 3$ or $2^3 = 8$ **M1**

Obtain $x^2 - 8x - 8 = 0$, or horizontal equivalent **A1**

Solve a 3-term quadratic equation **M1**

Obtain final answer $x = 8.90$ only **A1**

Total: 5

Q215 Oct/Nov 2017 | Paper 3 (variant 32) | Question 6

Topics: Algebra, Differentiation

6 The equation of a curve is $x^3y - 3xy^3 = 2a^4$, where a is a non-zero constant.

(i) Show that $\frac{dy}{dx} = \frac{3x^2y - 3y^3}{9xy^2 - x^3}$. **[4]**

(ii) Hence show that there are only two points on the curve at which the tangent is parallel to the x-axis and find the coordinates of these points. **[4]**

Examiner comment

> "The most common errors were sign errors and transcription errors in rearranging the terms."

Mark scheme

EITHER: Use scalar product to obtain an equation in a, b and c, e.g. $a + b + 3c = 0$ **B1**

Form a second relevant equation, e.g. $2a - 2b + c = 0$, and solve for one ratio, e.g. $a : b$ **M1**

Obtain final answer $a : b : c = 7 : 5 : -4$ **A1**

Use coordinates of A and values of a, b and c in general equation and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR1: Calculate the vector product of relevant vectors, e.g. $(\mathbf{i} + 3\mathbf{k}) \times (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **(*M1**

Obtain two correct components **A1**

Obtain correct answer, e.g. $7\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR2: Using relevant vectors, form a two-parameter equation for the plane **(*M1**

State a correct equation, e.g. $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + \mu(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **A1 FT**

State 3 correct equations in x, y, z, λ and μ **A1 FT**

Eliminate λ and μ **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR3: Use the direction vector of the line of intersection of the two planes as normal vector to the plane **(*M1**

Two correct components **A1**

Three correct components **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

Total: 7

Q216 Oct/Nov 2017 | Paper 3 (variant 32) | Question 8

Topics: Algebra

8 Let $f(x) = \frac{8x+9x+8}{(1-x)(2x+3)^2}$.

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Examiner comment

> "Some candidates lost marks through inaccurate algebraic manipulation or numerical errors, even though they were using a correct method."

> "The expansions of –1 –2 □ 2 □ □ 2 □ □1+ x□ and □1+ x□ were often correct, but the candidates often did not deal with the factors □ 3 □ □ 3 □ 1 1 1 of and correctly: the most common errors were to multiply by 3 and by 9, or to use in place 3 9 3 1 of ."

Mark scheme

EITHER: Use scalar product to obtain an equation in a, b and c, e.g. $a + b + 3c = 0$ **B1**

Form a second relevant equation, e.g. $2a - 2b + c = 0$, and solve for one ratio, e.g. $a : b$ **M1**

Obtain final answer $a : b : c = 7 : 5 : -4$ **A1**

Use coordinates of A and values of a, b and c in general equation and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR1: Calculate the vector product of relevant vectors, e.g. $(\mathbf{i} + 3\mathbf{k}) \times (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **(*M1**

Obtain two correct components **A1**

Obtain correct answer, e.g. $7\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR2: Using relevant vectors, form a two-parameter equation for the plane **(*M1**

State a correct equation, e.g. $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + \mu(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **A1 FT**

State 3 correct equations in x, y, z, λ and μ **A1 FT**

Eliminate λ and μ **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR3: Use the direction vector of the line of intersection of the two planes as normal vector to the plane **(*M1**

Two correct components **A1**

Three correct components **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

Total: 7

Q217 Oct/Nov 2017 | Paper 3 (variant 32) | Question 9

Topics: Indices, Logarithms & Exponents, Integration, Numerical Solutions/Iteration

Question 9

It is given that $\int_1^a x^2 \ln x \, dx = 2$, where $a > 1$.

(i) Show that $a^{3/2} = \frac{7+2a^3}{3\ln a}$. **[5]**

(ii) Show by calculation that a lies between 2 and 4. **[2]**

(iii) Use the iterative formula

$$a_{n+1} = \left(\frac{7 + 2a_n^3}{3 \ln a_n}\right)^{3/4}$$

to determine a correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Examiner comment

> "There were many incorrect answers, partly because some candidates struggled to key this function correctly on their calculators (often not using brackets when they needed to), and partly because candidates were not considering an appropriate function."

Mark scheme

9(iii)

M1 Use the iterative formula correctly at least once

A1 Obtain final answer 3.031

A1 Show sufficient iterations to 5 d.p. to justify 3.031 to 3 d.p., or show there is a sign change in the interval (3.0305, 3.0315)

Total: 3

Q218 Oct/Nov 2017 | Paper 3 (variant 33) | Question 4

Topics: Trigonometry

(i) Prove the identity $\tan(45^\circ + x) + \tan(45^\circ - x) \equiv 2\sec 2x$. **[4]**

(ii) Hence sketch the graph of $y = \tan(45^\circ + x) + \tan(45^\circ - x)$ for $0^\circ \leq x \leq 90^\circ$. **[3]**

Examiner comment

> "A large proportion of candidates found it difficult to progress beyond the first 3 marks since they did not recognise that sec2x could be expressed as 1/cos2x."

Mark scheme

4(i)

M1 Use correct $\tan(A \pm B)$ formula and express the LHS in terms of $\tan x$

A1 Using $\tan 45^\circ = 1$ express LHS as a single fraction

M1 Use Pythagoras or correct double angle formula

A1 Obtain given answer

Total: 4

4(ii)

B1 Show correct sketch for one branch

B1 Both branches correct and nothing else seen in the interval

B1 Show asymptote at $x = 45^\circ$

Total: 3

Q219 Oct/Nov 2017 | Paper 3 (variant 33) | Question 6

Topics: Trigonometry, Differential Equations

Question 6

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = 4\cos^2 y \tan x,$$

for $0 \leq x < \frac{1}{2}\pi$, and $x = 0$ when $y = \frac{1}{4}\pi$. Solve this differential equation and find the value of x when $y = \frac{1}{3}\pi$. **[8]**

Examiner comment

> "A common error was failing to write 1/cos2y as sec2y, or not realising it integrated to tany."

Mark scheme

Question	Answer	Marks
	Obtain answer $2x-2y+z=9$, or equivalent	A1)
	*OR*3: Using the relevant point and relevant vectors, form a determinant equation for the plane	(M1
	State a correct equation, e.g. $\begin{vmatrix} x-3 & y+2 & z+1 \\ -1 & 1 & 4 \\ 2 & 1 & -2 \end{vmatrix} = 0$	A1
	Attempt to expand the determinant	M1
	Obtain two correct cofactors	A1
	Obtain answer $-2x+2y-z=-9$, or equivalent	A1)
		5

Q220 Feb/Mar 2016 | Paper 3 (variant 32) | Question 4

Topics: Algebra

The polynomial $4x^3 + ax + 2$, where a is a constant, is denoted by $p(x)$. It is given that $(2x+1)$ is a factor of $p(x)$.

(i) Find the value of a. [2]

(ii) When a has this value,

(a) factorise $p(x)$, [2]

(b) solve the inequality $p(x) > 0$, justifying your answer. [3]

Examiner comment

> "(b) Most candidates obtained the correct answer, but often failed to justify this and so lost marks."

Mark scheme

(i)

Substitute $x = -\frac{1}{2}$ and equate to zero, or divide by $(2x+1)$ and equate constant remainder to zero **M1**

$a = 3$ **A1**

Total: 2

(ii)(a)

Commence division by $(2x+1)$ reaching a partial quotient of $2x^2 + kx$ **M1**

$(2x+1)(2x^2 - x + 2)$ **A1**

Total: 2

(ii)(b)

State or imply critical value $x = -\frac{1}{2}$ **B1**

Show that $2x^2 - x + 2$ is always positive, or that the gradient of $4x^3 + 3x + 2$ is always positive **B1***

Justify final answer $x > -\frac{1}{2}$ **B1(dep*)**

Total: 3

Q221 Feb/Mar 2016 | Paper 3 (variant 32) | Question 8

Topics: none of the listed topics

The line l has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$. The plane p has equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} = 6$.

(i) Show that l is parallel to p. [3]

(ii) A line m lies in the plane p and is perpendicular to l. The line m passes through the point with coordinates $(5, 3, 1)$. Find a vector equation for m. [6]

Examiner comment

> "(ii) Few candidates made much progress with this part, incorrectly using either i + 2j –k or 5i + 3j + k in trying to establish the direction of the line m."

Mark scheme

(i)

EITHER: Substitute for $\mathbf{r}$ in the given equation of p and expand scalar product **M1**

Obtain equation in λ in any correct form **A1**

Verify this is not satisfied for any value of λ **A1**

OR: Expand scalar product of the normal to p and the direction vector of l **M1**

Verify scalar product is zero **A1**

Verify that one point of l does not lie in the plane **A1**

Total: 3

(ii)

Calling the unknown direction vector $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, state equation $2a + b + 3c = 0$ **B1**

State equation $2a - b - c = 0$ **B1**

Solve for one ratio, e.g. $a : b : c = 1 : 4 : -2$, or equivalent **M1 A1**

Form line equation with relevant vectors **M1**

$\mathbf{r} = 5\mathbf{i} + 3\mathbf{j} + \mathbf{k} + \mu(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$, or equivalent **A1**

Total: 6

Q222 Feb/Mar 2016 | Paper 3 (variant 32) | Question 9

Topics: Algebra, Integration

Let $f(x) = \frac{3x^3+6x-8}{x(x^2+2)}$.

(i) Express $f(x)$ in the form $A + \frac{B}{x} + \frac{Cx+D}{x^2+2}$. [5]

(ii) Show that $\int_1^2 f(x)\,dx = 3 - \ln 4$. [5]

Examiner comment

> "C+Dx (ii) Unless candidates had found the constant D in (i) to be zero their integration of the term x2 +2 proved difficult."

Mark scheme

(i)

State or obtain $A = 3$ **B1**

Use a relevant method to find a constant **M1**

Obtain one of $B = -4$, $C = 4$ and $D = 0$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Integrate and obtain $3x - 4\ln x$ **B1**

Integrate and obtain term of the form $k\ln(x^2+2)$ **M1**

Obtain term $2\ln(x^2+2)$ **A1**

Substitute limits in an integral of the form $ax + b\ln x + c\ln(x^2+2)$, where $abc \neq 0$ **M1**

Obtain given answer $3 - \ln 4$ after full and correct working **A1**

Total: 5

Q223 May/June 2016 | Paper 3 (variant 31) | Question 6

Topics: Indices, Logarithms & Exponents, Numerical Solutions/Iteration

(i) By sketching a suitable pair of graphs, show that the equation

$$5e^{-x} = \sqrt{x}$$

has one root. [2]

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \frac{1}{2}\ln\left(\frac{25}{x_n}\right)$$

converges, then it converges to the root of the equation in part (i). [2]

(iii) Use this iterative formula, with initial value $x_1 = 1$, to calculate the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "2 ☐ x ☐ Many candidates did not appear to understand what they needed to do, or did not give sufficient explanation to support the given answer."

Mark scheme

(i)

Make recognisable sketch of a relevant graph **B1**

Sketch the other relevant graph and justify the given statement **B1**

Total: 2

(ii)

State $x = \frac{1}{2}\ln(25/x)$ **B1**

Rearrange this in the form $5e^{-x} = \sqrt{x}$ **B1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer **1.43 A1**

Show sufficient iterations to 4 d.p. to justify **1.43** to 2 d.p., or show there is a sign change in the interval $(1.425, 1.435)$ **A1**

Total: 3

Q224 May/June 2016 | Paper 3 (variant 31) | Question 7

Topics: Algebra, Differentiation

The equation of a curve is $x^3 - 3x^2y + y^3 = 3$.

(i) Show that $\frac{dy}{dx} = \frac{x^2-2xy}{x^2-y^2}$. [4]

(ii) Find the coordinates of the points on the curve where the tangent is parallel to the x-axis. [5]

Examiner comment

> "(ii) This part of the question proved to be challenging for many candidates It required an dy understanding that for the tangent to be parallel to the x-axis, they needed =0and therefore dx x2 −2xy =0."

Mark scheme

(i)

State or imply $6xy + 3x^2\frac{dy}{dx}$ as derivative of $3x^2y$ **B1**

State $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

Equate attempted derivative of the LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate numerator to zero **M1***

Obtain $x = 2y$, or equivalent **A1**

Obtain an equation in x or y **DM1***

Obtain the point $(-2, -1)$ **A1**

State the point $(0, 1.44)$ **B1**

Total: 5

Q225 May/June 2016 | Paper 3 (variant 31) | Question 9

Topics: none of the listed topics

With respect to the origin O, the points A, B, C, D have position vectors given by

$$\overrightarrow{OA} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k},$$

$$\overrightarrow{OB} = 2\mathbf{i} + \mathbf{j} - \mathbf{k},$$

$$\overrightarrow{OC} = 2\mathbf{i} + 4\mathbf{j} + \mathbf{k},$$

$$\overrightarrow{OD} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k}.$$

(i) Find the equation of the plane containing A, B and C, giving your answer in the form $ax + by + cz = d$. [6]

(ii) The line through D parallel to OA meets the plane with equation $x + 2y - z = 7$ at the point P. Find the position vector of P and show that the length of DP is $2\sqrt{14}$. [5]

Examiner comment

> "The most common error was to include the position vector of D in the working, when it is not given that D lies on the plane."

Mark scheme

(i)

Obtain a vector parallel to the plane, e.g. $\overrightarrow{AB} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$ **B1**

Use scalar product to obtain an equation in a, b, c, e.g. $a - 2b - 3c = 0$, $a + b - c = 0$, or $3b + 2c = 0$ **M1**

State two correct equations **A1**

Solve to obtain ratio $a : b : c$ **M1**

Obtain $a : b : c = 5 : -2 : 3$ **A1**

Obtain equation $5x - 2y + 3z = 5$, or equivalent **A1**

Total: 6

(ii)

Correctly form an equation for the line through D parallel to OA **M1**

Obtain a correct equation e.g. $\mathbf{r} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ **A1**

Substitute components in the equation of the plane and solve for λ **M1**

Obtain $\lambda = 2$ and position vector $-\mathbf{i} + 7\mathbf{j} + 6\mathbf{k}$ for P **A1**

Obtain the given answer correctly **A1**

Total: 5

Q226 May/June 2016 | Paper 3 (variant 31) | Question 10

Topics: Complex Numbers

(a) Showing all your working and without the use of a calculator, find the square roots of the complex number $7 - (6\sqrt{2})i$. Give your answers in the form $x + iy$, where x and y are real and exact. [5]

(b)(i) On an Argand diagram, sketch the loci of points representing complex numbers w and z such that $|w - 1 - 2i| = 1$ and $\arg(z - 1) = \frac{3}{4}\pi$. [4]

(b)(ii) Calculate the least value of $|w - z|$ for points on these loci. [2]

Examiner comment

> "The half line for argz−1 = proved to be more challenging."

Mark scheme

(a)

Square $x + iy$ and equate real and imaginary parts to 7 and $-6\sqrt{2}$ respectively **M1**

Obtain equations $x^2 - y^2 = 7$ and $2xy = -6\sqrt{2}$ **A1**

Eliminate one variable and find an equation in the other **M1**

Obtain $x^4 - 7x^2 - 18 = 0$ or $y^4 + 7y^2 - 18 = 0$, or 3-term equivalent **A1**

Obtain answers $\pm(3 - \sqrt{2}i)$ **A1**

Total: 5

(b)(i)

Show point representing $1 + 2i$ **B1**

Show circle with radius 1 and centre $1 + 2i$ **B1**

Show a half-line from the point representing 1 **B1**

Show line making the correct angle with the real axis **B1**

Total: 4

(b)(ii)

State or imply the relevance of the perpendicular from $1 + 2i$ to the line **M1**

Obtain answer $\sqrt{2} - 1$ or 0.414 **A1**

Total: 2

Q227 May/June 2016 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

Use logarithms to solve the equation

$$4^{3x-1} = 3(5^x),$$

giving your answer correct to 3 decimal places. [4]

Examiner comment

> "On the left the omission of brackets in (3x–1)ln4 was a common error."

Mark scheme

Use a law of logarithms for a product, power or quotient **M1**

Obtain a correct linear equation, e.g.

$$(3x - 1)\ln 4 = \ln 3 + x \ln 5$$

A1

Solve a linear equation for x **DM1**

Obtain answer $x = 0.975$ **A1**

Total: 4

Q228 May/June 2016 | Paper 3 (variant 32) | Question 9

Topics: none of the listed topics

The points A, B and C have position vectors, relative to the origin O, given by

$$\overrightarrow{OA} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k},$$

$$\overrightarrow{OB} = 4\mathbf{j} + \mathbf{k}$$

and

$$\overrightarrow{OC} = 2\mathbf{i} + 5\mathbf{j} - \mathbf{k}.$$

A fourth point D is such that the quadrilateral $ABCD$ is a parallelogram.

(i) Find the position vector of D and verify that the parallelogram is a rhombus. [5]

(ii) The plane p is parallel to OA and the line BC lies in p. Find the equation of p, giving your answer in the form $ax + by + cz = d$. [5]

Examiner comment

> "A common error was to equate CD to AB rather than BA."

> "A common error was to think that the equality of a pair of parallel sides was sufficient."

Mark scheme

(i)

Either state or imply $\overrightarrow{AB}$ or $\overrightarrow{BC}$ in component form, or state the position vector of the midpoint of AC **B1**

Use a correct method for finding the position vector of D **M1**

Obtain answer $\mathbf{3i} + \mathbf{3j} + \mathbf{k}$, or equivalent **A1**

Either use the correct process for the moduli and compare lengths of a pair of adjacent sides, e.g. AB and BC **M1**

Show that $ABCD$ has a pair of adjacent sides that are equal **A1**

Alternatively, calculate $\overrightarrow{AC} \cdot \overrightarrow{BD}$, or equivalent, and show that the diagonals are perpendicular.

Total: 5

(ii)

Either state

$$a + 2b + 3c = 0$$

or

$$2a + b - 2c = 0$$

B1

Obtain two relevant equations and solve for one ratio, e.g. $a : b$ **M1**

Obtain $a : b : c = -7 : 8 : -3$, or equivalent **A1**

Substitute coordinates of a relevant point in $-7x + 8y - 3z = d$ and evaluate **M1**

Obtain answer

$$-7x + 8y - 3z = 29,$$

or equivalent **A1**

Total: 5

Q229 May/June 2016 | Paper 3 (variant 32) | Question 10

Topics: Complex Numbers

(a) Showing all necessary working, solve the equation

$$iz^2 + 2z - 3i = 0,$$

giving your answers in the form $x + iy$, where x and y are real and exact. [5]

(b)

(i) On a sketch of an Argand diagram, show the locus representing complex numbers satisfying the equation

$$|z| = |z - 4 - 3i|.$$

[2]

(ii) Find the complex number represented by the point on the locus where $|z|$ is least. Find the modulus and argument of this complex number, giving the argument correct to 2 decimal places. [3]

Examiner comment

> “Those with unequal scales on the axes found this difficult.”

> “A common error was to take the locus to be a circle with centre at the above point (ii) This was well answered by those with a correct diagram.”

Mark scheme

(a)

Either use the quadratic formula to solve for z **M1**

Use $i^2 = -1$ **M1**

Obtain a correct answer in any form, simplified as far as

$$\frac{-2 \pm i\sqrt{8}}{2i}$$

A1

Multiply numerator and denominator by i, or equivalent **M1**

Obtain final answers $\sqrt{2} + i$ and $-\sqrt{2} + i$ **A1**

Alternatively, substitute $x + iy$ and equate real and imaginary parts to zero, obtaining

$$2xy + 2x = 0$$

and

$$x^2 - y^2 + 2y - 3 = 0,$$

or equivalent, then solve for x and y.

Total: 5

(b)(i)

Either show the point representing $4 + 3i$ in relatively correct position **B1**

Show the perpendicular bisector of the line segment joining this point to the origin **B1**

Alternatively, obtain the correct Cartesian equation of the locus in any form, e.g.

$$8x + 6y = 25,$$

and show this line.

Total: 2

(b)(ii)

State or imply the relevant point is represented by $2 + 1.5i$, or is at $(2, 1.5)$ **B1**

Obtain modulus 2.5 **B1**

Obtain argument 0.64, or $36.9°$ **B1**

Total: 3

Q230 May/June 2016 | Paper 3 (variant 33) | Question 5

Topics: Trigonometry, Differential Equations

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = e^{-2y} \tan^2 x,$$

for $0 \leq x < \frac{1}{2}\pi$, and it is given that $y = 0$ when $x = 0$. Solve the differential equation and calculate the value of y when $x = \frac{1}{4}\pi$. [8]

Examiner comment

> "Another common error in the x term was a muddle of signs."

Mark scheme

Separate variables and make a reasonable attempt at integration of either integral **M1**

Obtain term $\frac{1}{2}e^{2y}$ **B1**

Use Pythagoras **M1**

Obtain terms $\tan x - x$ **A1**

Evaluate a constant or use $x = 0$, $y = 0$ as limits in a solution containing terms $ae^{\pm 2y}$ and $b \tan x$, where $abe0$ **M1**

Obtain correct solution in any form, e.g.

$$\frac{1}{2}e^{2y} = \tan x - x + \frac{1}{2}$$

A1

Set $x = \frac{1}{4}\pi$ and use a correct method to solve an equation of the form $e^{\pm 2y} = a$, or $e^{\pm y} = a$, where $a > 0$ **M1**

Obtain answer $y = 0.179$ **A1**

Total: 8

Q231 Oct/Nov 2016 | Paper 3 (variant 31) | Question 1

Topics: Indices, Logarithms & Exponents

(1) Solve the equation $\frac{3^x+2}{3^x-2} = 8$, giving your answer correct to 3 decimal places.

Total: 3 marks

Examiner comment

> "The most common errors involved incorrect statements such as 8×3x =24x and 24x −3x =21x."

Mark scheme

Solve for 3^x and obtain $3^x = \frac{18}{7}$

B1 Use correct method for solving an equation of the form $3^x = a$, where $a > 0$

M1 Obtain answer $x = 0.860$ 3 d.p. only

A1

Total: 3 marks

Q232 Oct/Nov 2016 | Paper 3 (variant 31) | Question 2

Topics: Algebra

(2) Expand $(2 - x)(1 + 2x)^{-2}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients.

Total: 4 marks

Examiner comment

> "Many candidates lost marks in this question due to errors in their algebra and arithmetic when multiplying their expansion by (2−x)."

Mark scheme

(2)

B1 State correct unsimplified first two terms of the expansion of $(1 + 2x)^{-2}$, e.g. $1 + (-\frac{2}{1})(2x)$

B1 State correct unsimplified term in x^2, e.g. $(-\frac{3}{2})(-\frac{1}{2} - 1)(2x)^2/2!$

M1 Obtain sufficient terms of the product of $(2 - x)$ and the expansion up to the term in x^2

A1 Obtain **final** answer $2 - 7x + 18x^2$ (Do not ISW)

Total: 4 marks

Q233 Oct/Nov 2016 | Paper 3 (variant 31) | Question 4

Topics: Algebra, Differentiation

(4) The equation of a curve is $xy(x - 6y) = 9a^3$, where a is a non-zero constant. Show that there is only one point on the curve at which the tangent is parallel to the x-axis, and find the coordinates of this point.

Total: 7 marks

Examiner comment

> "Question 4 Many candidates found this question challenging."

Mark scheme

EITHER:

B1 State $2xy + x^2\frac{dy}{dx}$, or equivalent, as derivative of x^2y

B1 State $6y^2 + 12xy\frac{dy}{dx}$, or equivalent, as derivative of $6xy^2$

B1 Differentiating LHS using correct product rule, state term $xy(1 - 6\frac{dy}{dx})$, or equivalent

B1 State term $(y + x\frac{dy}{dx})(x - 6y)$, or equivalent

M1* Equate attempted derivative of LHS to zero and set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

A1 Explicitly reject $y = 0$ as a possibility $py^2 - qxy = 0$

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a, -a)$

OR:

B1 Rearrange to $y = \frac{9a^3}{x(x-6y)}$ and use correct quotient rule to obtain $-\frac{9a^3}{x^2(x-6y)^2}$....

B1 State term $(x - 6y) + x(1 - 6y')$, or equivalent

B1 Justify division by $x(x - 6y)$

M1* Set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

A1 Obtain an equation in x or y

DM1 Obtain answer $(-3a, -a)$

Total: 7 marks

Q234 Oct/Nov 2016 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Integration

(i) Prove the identity $\tan 2\theta - \tan\theta \equiv \tan\theta \sec 2\theta$.

(ii) Hence show that $\int_0^{\frac{\pi}{6}} \tan\theta \sec 2\theta \, d\theta = \frac{1}{2}\ln\frac{2}{5}$.

Total: 8 marks

Examiner comment

> "The most common errors involved losing the denominator when combining the two terms on the left hand side, and sign errors in removing brackets (which were often omitted)."

> "(ii) Many candidates did not see the link between the two parts of this question."

> "The lower limit of the integral often appeared to have been ignored, and several candidates struggled to deal with the logarithms correctly."

Mark scheme

(i)

EITHER: Use $\tan 2A$ formula to express LHS in terms of $\tan\theta$

M1 Express as a single fraction in any correct form

A1 Use Pythagoras or $\cos 2A$ formula

M1 Obtain the given result correctly

A1

OR: Express LHS in terms of $\sin 2\theta, \cos 2\theta, \sin\theta$ and $\cos\theta$

M1 Express as a single fraction in any correct form

A1 Use Pythagoras or $\cos 2A$ formula or $\sin(A-B)$ formula

M1 Obtain the given result correctly

A1

Total: 4 marks

(ii)

M1* Integrate and obtain a term of the form $a\ln(\cos 2\theta)$ or $b\ln(\cos\theta)$ (or secant equivalents)

A1 Obtain integral $-\frac{1}{2}\ln(\cos 2\theta) + \ln(\cos\theta)$, or equivalent

DM1 Substitute limits correctly (expect to see use of **both** limits)

A1 Obtain the given answer following full and correct working

Total: 4 marks

Q235 Oct/Nov 2016 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

(i) By sketching a suitable pair of graphs, show that the equation

$$\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$$

has one root in the interval $0 < x \leq \pi$.

(ii) Show by calculation that this root lies between 1.4 and 1.6.

(iii) Show that, if a sequence of values in the interval $0 < x < \pi$ given by the iterative formula

$$x_{n+1} = 2\sin^{-1}\left(\frac{3}{x_n + 3}\right)$$

converges, then it converges to the root of the equation in part (i).

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Total: 9 marks

Examiner comment

> "There were several candidates who clearly did not understand the notation sin−1(θ), the most common error being to 1 1 interpret it as ."

> "sinθ sin There were several candidates who struggled with the algebra, particularly among those who 3 3 3 tackled the problem in reverse – the error = + was common."

Mark scheme

(i)

B1 Make recognizable sketch of a relevant graph

B1 Sketch the other relevant graph and justify the given statement

Total: 2 marks

(ii)

M1 Use calculations to consider the value of a relevant expression at $x = 1.4$ and $x = 1.6$, or the values of relevant expressions at $x = 1.4$ and $x = 1.6$

A1 Complete the argument correctly with correct calculated values

Total: 2 marks

(iii)

B1 State $x = 2\sin^{-1}\left(\frac{3}{x+3}\right)$

B1 Rearrange this in the form $\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$

B1 If working in reverse, need $\sin\frac{x}{2} = \left(\frac{3}{x+3}\right)$ for first **B1**

Total: 2 marks

(iv)

M1 Use the iterative formula correctly at least once

A1 Obtain final answer 1.471

A1 Show sufficient iterations to 5 d.p. to justify 1.471 to 3 d.p., or show there is a sign change in the interval (1.4705, 1.4715)

Total: 3 marks

Q236 Oct/Nov 2016 | Paper 3 (variant 31) | Question 7

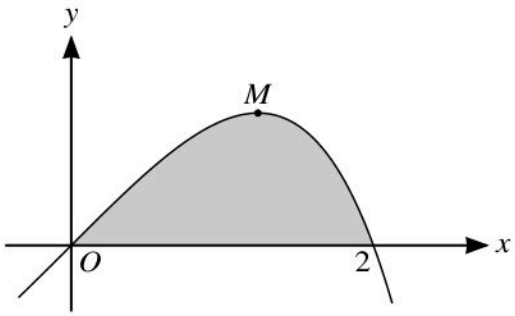
Topics: Differentiation, Integration

(i) Find the exact x-coordinate of M.

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis.

The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

7



The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

(i) Find the exact x-coordinate of M. [4]

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis. [5]

Total: 9 marks

Examiner comment

> "There were some sign errors at this stage, but the most common error was incorrect 1 1 1 1 x x x 1 – x differentiation of e2 , obtaining either e2 , 2e2 or occasionally e 2 ."

Mark scheme

(i)

M1 Use the correct product rule

A1 Obtain correct derivative in any form, e.g. $(2-2x)e^{\frac{1}{2}x}+\frac{1}{2}(2x-x^2)e^{\frac{1}{2}x}$

M1 Equate derivative to zero and solve for x

A1 Obtain $x=\sqrt{5}-1$ only

Total: 4 marks

(ii)

M1* Integrate by parts and reach $a(2x-x^2)e^{\frac{1}{2}x}+b\int(2-2x)e^{\frac{1}{2}x}\,dx$

A1 Obtain $2e^{\frac{1}{2}x}(2x-x^2)-2\int(2-2x)e^{\frac{1}{2}x}\,dx$, or equivalent

A1 Complete the integration correctly, obtaining $(12x-2x^2-24)e^{\frac{1}{2}x}$, or equivalent

DM1 Use limits $x=0, x=2$ correctly having integrated by parts twice

A1 Obtain answer $24-8e$, or exact simplified equivalent

Total: 5 marks

Q237 Oct/Nov 2016 | Paper 3 (variant 31) | Question 8

Topics: none of the listed topics

(i) Show that the planes are perpendicular.

(ii) Find a vector equation for the line of intersection of the two planes.

Two planes have equations $3x+y-z=2$ and $x-y+2z=3$.

Total: 9 marks

Examiner comment

> "A common error was to have the normal vector of the first plane as 3i+j−2k ."

Mark scheme

(i)

B1 State or imply a correct normal vector to either plane, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ or $\mathbf{i}-\mathbf{j}+2\mathbf{k}$

M1 Use correct method to calculate their scalar product

A1 Show value is zero and planes are perpendicular

Total: 3 marks

(ii)

EITHER: Carry out a complete strategy for finding a point on l the line of intersection

M1 Obtain such a point, e.g. $(0,7,5), (1,0,1), (5/4,-7/4,0)$

A1 State two equations for a direction vector $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ for l, e.g. $3a + b - c = 0$ and $a - b + 2c = 0$

B1 Solve for one ratio, e.g. $a : b$

M1 Obtain $a : b : c = 1 : -7 : -4$, or equivalent

A1 State a correct answer, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$

EITHER2: Obtain a second point on l, e.g. $(1, 0, 1)$

B1 Subtract vectors and obtain a direction vector for l

M1 Obtain $\mathbf{i} + \mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$

A1 State a correct answer, e.g. $\mathbf{r} = \mathbf{i} + \mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$

OR2: Attempt to find the vector product of the two normal vectors

M1 Obtain two correct components of the product

A1 Obtain $\mathbf{i} - 7\mathbf{j} - 4\mathbf{k}$, or equivalent

A1 State a correct answer, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$

OR1: Express one variable in terms of a second variable

M1 Obtain a correct simplified expression, e.g. $y = 7 - 7x$

M1 Express the same variable in terms of the second

A1 Obtain a correct simplified expression e.g. $z = 5 - 4x$

M1 Form a vector equation for the line

A1 Obtain a correct equation, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$

OR2: Express one variable in terms of a second variable

M1 Obtain a correct simplified expression, e.g. $z = 5 - 4x$

M1 Express the same variable in terms of the third

A1 Obtain a correct simplified expression e.g. $z = (7 + 4y)/7$

M1 Form a vector equation for the line

A1 Obtain a correct equation, e.g. $\mathbf{r} = \frac{7}{4}\mathbf{i} + \frac{7}{4}\mathbf{k} + \lambda(- \frac{7}{4}\mathbf{i} + \frac{7}{4}\mathbf{j} + \mathbf{k})$

Total: 6 marks

Q238 Oct/Nov 2016 | Paper 3 (variant 31) | Question 9

Topics: Complex Numbers

9 Throughout this question the use of a calculator is not permitted.

(a) Solve the equation $(1+2i)w^2 + 4w - (1-2i) = 0$, giving your answers in the form $x + iy$, where x and y are real. **Total: 5**

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers satisfying the inequalities $|z - 1 - i| \leq 2$ and $-\frac{1}{4}\pi \leq \arg z \leq \frac{1}{4}\pi$. **Total: 5**

Examiner comment

> "The second inequality describes two half lines which should start at the origin; it was a common error to see them starting at the centre of the circle."

Mark scheme

(a) M1 EITHER: Use quadratic formula to solve for w

M1 Use $i^2 = -1$

A1 Obtain one of the answers $w = \frac{1}{2i+1}$ and $w = -\frac{5}{2i+1}$

M1 Multiply numerator and denominator of an answer by $-2i + 1$, or equivalent

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$

OR1: **M1** Multiply the equation by $1 - 2i$

M1 Use $i^2 = -1$

A1 Obtain $5w^2 + 4w(1-2i) - (1-2i)^2 = 0$, or equivalent

M1 Use quadratic formula or factorise to solve for w

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$

OR2: **M1** Substitute $w = x + iy$ and form equations for real and imaginary parts

M1 Use $i^2 = -1$

A1 Obtain $(x^2 - y^2) - 4xy + 4x - 1 = 0$ and $2(x^2 - y^2) + 2xy + 4y + 2 = 0$ o.e.

M1 Form equation in x only or y only and solve

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **[5]**

(b) B1 Show a circle with centre $1 + i$

B1 Show a circle with radius 2

B1 Show half-line $\arg z = \frac{1}{4}\pi$

B1 Show half-line $\arg z = -\frac{1}{4}\pi$

B1 Shade the correct region **[5]**

Q239 Oct/Nov 2016 | Paper 3 (variant 31) | Question 10

Topics: Differential Equations

10 A large field of area $4\ \text{km}^2$ is becoming infected with a soil disease. At time t years the area infected is $x\ \text{km}^2$ and the rate of growth of the infected area is given by the differential equation

$$\frac{dx}{dt} = kx(4 - x),$$

where k is a positive constant. It is given that when $t = 0$, $x = 0.4$ and that when $t = 2$, $x = 2$.

(i) Solve the differential equation and show that $k = \frac{1}{4}\ln 3$. **Total: 9**

(ii) Find the value of t when 90% of the area of the field is infected. **Total: 2**

Examiner comment

> "Many candidates did not realise that ∫ dx requires x(4−x) the use of partial fractions."

Mark scheme

(i) M1 Separate variables correctly and integrate at least one side

A1 Integrate and obtain term kt, or equivalent

M1** Carry out a relevant method to obtain A and B such that

$$\frac{1}{x(4 - x)} = \frac{A}{x} + \frac{B}{4 - x},$$

or equivalent

A1 Obtain $A = B = \frac{1}{4}$, or equivalent

A1** Integrate and obtain terms $\frac{1}{4}\ln x - \frac{1}{4}\ln(4 - x)$, or equivalent

DM1 EITHER: Use a pair of limits in an expression containing $p\ln x$, $q\ln(4 - x)$ and rt and evaluate a constant. Obtain correct answer in any form, e.g. $\ln x - \ln(4 - x) = 4kt - \ln 9$, or

$$\ln\left(\frac{x}{4 - x}\right) = 4kt - 8k$$

DM1 Use a second pair of limits and determine k. Obtain the given exact answer correctly

A1

OR: **M1**** Use both pairs of limits in a definite integral. Obtain the given exact answer correctly

A1 Substitute k and either pair of limits in an expression containing $p\ln x$, $q\ln(4 - x)$ and rt and evaluate a constant. Obtain

$$\ln\frac{x}{4 - x} = t\ln 3 - \ln 9,$$

or equivalent **[9]**

(ii) M1 Substitute $x = 3.6$ and solve for t

A1 Obtain answer $t = 4$ **[2]**

Q240 Oct/Nov 2016 | Paper 3 (variant 32) | Question 1

Topics: Indices, Logarithms & Exponents

1 Solve the equation

$$\frac{3^x + 2}{3^x - 2} = 8,$$

giving your answer correct to 3 decimal places. **Total: 3**

Examiner comment

> "The most common errors involved incorrect statements such as 8×3x =24x and 24x −3x =21x."

Mark scheme

B1 Solve for 3^x and obtain $3^x = \frac{18}{7}$

M1 Use correct method for solving an equation of the form $3^x = a$, where $a > 0$

A1 Obtain answer $x = 0.860$ 3 d.p. only **[3]**

Q241 Oct/Nov 2016 | Paper 3 (variant 32) | Question 2

Topics: Algebra

2 Expand $(2 - x)(1 + 2x)^{-2}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. **Total: 4**

Examiner comment

> "Many candidates lost marks in this question due to errors in their algebra and arithmetic when multiplying their expansion by (2−x)."

Mark scheme

B1 State correct unsimplified first two terms of the expansion of $(1 + 2x)^{-2}$, e.g. $1 + (-2)(2x)$

B1 State correct unsimplified term in x^2, e.g. $\left(-\frac{3}{2}\right)\left(-\frac{3}{2} - 1\right)(2x)^2/2!$

M1 Obtain sufficient terms of the product of $(2 - x)$ and the expansion up to the term in x^2

A1 Obtain **final** answer $2 - 7x + 18x^2$. Do not ISW **[4]**

Q242 Oct/Nov 2016 | Paper 3 (variant 32) | Question 4

Topics: Differentiation

4 The equation of a curve is $xy(x - 6y) = 9a^3$, where a is a non-zero constant. Show that there is only one point on the curve at which the tangent is parallel to the x-axis, and find the coordinates of this point. **Total: 7**

Examiner comment

> "Question 4 Many candidates found this question challenging."

Mark scheme

B1 EITHER: EITHER: State $2xy + x^2\frac{dy}{dx}$, or equivalent, as derivative of x^2y

B1 State $6y^2 + 12xy\frac{dy}{dx}$, or equivalent, as derivative of $6xy^2$

OR: Differentiating LHS using correct product rule, state term $xy\left(1-6\frac{dy}{dx}\right)$, or equivalent

B1 State term $\left(y+x\frac{dy}{dx}\right)(x-6y)$, or equivalent

B1

M1** Equate attempted derivative of LHS to zero and set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2-2xy=0$ (from correct work only)

A1 Explicitly reject $y=0$ as a possibility $py^2-qxy=0$

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a,-a)$

OR: Rearrange to $y=\frac{9a^3}{x(x-6y)}$ and use correct quotient rule to obtain $-\frac{9a^3}{x^2(x-6y)^2}\cdots$

B1 State term $(x-6y)+x(1-6y')$, or equivalent

B1 Justify division by $x(x-6y)$

M1** Set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2-2xy=0$ (from correct work only)

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a,-a)$ **[7]**

Q243 Oct/Nov 2016 | Paper 3 (variant 32) | Question 5

Topics: Trigonometry, Integration

(i) Prove the identity $\tan 2\theta-\tan\theta\equiv\tan\theta\sec 2\theta$. **Total: 4**

(ii) Hence show that

$$\int_0^{\frac{1}{6}\pi}\tan\theta\sec 2\theta\,d\theta=\frac{1}{2}\ln\frac{3}{2}.$$

Total: 4

Examiner comment

> "The most common errors involved losing the denominator when combining the two terms on the left hand side, and sign errors in removing brackets (which were often omitted)."

> "(ii) Many candidates did not see the link between the two parts of this question."

> "The lower limit of the integral often appeared to have been ignored, and several candidates struggled to deal with the logarithms correctly."

Mark scheme

(i) M1 EITHER: Use $\tan 2A$ formula to express LHS in terms of $\tan\theta$

A1 Express as a single fraction in any correct form

M1 Use Pythagoras or $\cos 2A$ formula

A1 Obtain the given result correctly

OR: Express LHS in terms of $\sin 2\theta$, $\cos 2\theta$, $\sin\theta$ and $\cos\theta$

A1 Express as a single fraction in any correct form

M1 Use Pythagoras or $\cos 2A$ formula or $\sin(4-B)$ formula

A1 Obtain the given result correctly **[4]**

(ii) M1** Integrate and obtain a term of the form $a\ln(\cos 2\theta)$ or $b\ln(\cos\theta)$ (or secant equivalents)

A1 Obtain integral $-\frac{1}{2}\ln(\cos 2\theta) + \ln(\cos\theta)$, or equivalent

DM1 Substitute limits correctly (expect to see use of **both** limits)

A1 Obtain the given answer following full and correct working **[4]**

Q244 Oct/Nov 2016 | Paper 3 (variant 32) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration

(i) By sketching a suitable pair of graphs, show that the equation

$$\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$$

has one root in the interval $0 < x \leq \pi$. **Total: 2**

(ii) Show by calculation that this root lies between 1.4 and 1.6. **Total: 2**

(iii) Show that, if a sequence of values in the interval $0 < x \leq \pi$ given by the iterative formula

$$x_{n+1} = 2\sin^{-1}\left(\frac{3}{x_n + 3}\right)$$

converges, then it converges to the root of the equation in part (i). **Total: 2**

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **Total: 3**

Examiner comment

> "There were several candidates who clearly did not understand the notation sin−1(θ), the most common error being to 1 1 interpret it as ."

> "sinθ sin There were several candidates who struggled with the algebra, particularly among those who 3 3 3 tackled the problem in reverse – the error = + was common."

Mark scheme

(i) B1 Make recognizable sketch of a relevant graph

B1 Sketch the other relevant graph and justify the given statement **[2]**

(ii) M1 Use calculations to consider the value of a relevant expression at $x = 1.4$ and $x = 1.6$, or the values of relevant expressions at $x = 1.4$ and $x = 1.6$

A1 Complete the argument correctly with correct calculated values **[2]**

(iii) B1 State $x = 2\sin^{-1}\left(\frac{3}{x+3}\right)$

B1 Rearrange this in the form $\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$ **[2]**

If working in reverse, need $\sin\frac{x}{2} = \left(\frac{3}{x+3}\right)$ for first **B1**

(iv) M1 Use the iterative formula correctly at least once

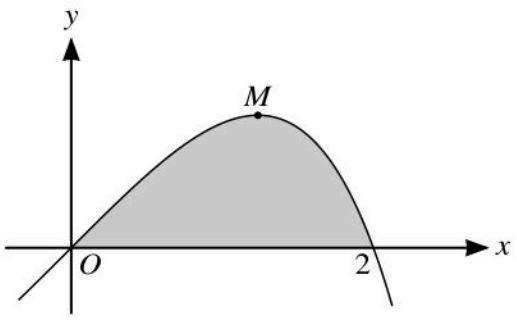
A1 Obtain final answer 1.471

A1 Show sufficient iterations to 5 d.p. to justify 1.471 to 3 d.p., or show there is a sign change in the interval (1.4705, 1.4715) **[3]**

Q245 Oct/Nov 2016 | Paper 3 (variant 32) | Question 7

Topics: Differentiation, Integration

7



The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

(i) Find the exact x-coordinate of M. [4]

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis. [5]

The diagram shows part of the curve $y = (2x - x^2)e^{\frac{x}{2}}$ and its maximum point M.

(i) Find the exact x-coordinate of M. **Total: 4**

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis.

Total: 5

Examiner comment

> "There were some sign errors at this stage, but the most common error was incorrect 1 1 1 1 x x x 1 – x differentiation of e2 , obtaining either e2 , 2e2 or occasionally e 2 ."

Mark scheme

(i) M1 Use the correct product rule

A1 Obtain correct derivative in any form, e.g. $(2 - 2x)e^{\frac{x}{2}} + \frac{1}{2}(2x - x^2)e^{\frac{x}{2}}$

M1 Equate derivative to zero and solve for x

A1 Obtain $x = \sqrt{5} - 1$ only **[4]**

(ii) M1** Integrate by parts and reach $a(2x - x^2)e^{\frac{x}{2}} + b\int(2 - 2x)e^{\frac{x}{2}}dx$

A1 Obtain $2e^{\frac{x}{2}}(2x - x^2) - 2\int(2 - 2x)e^{\frac{x}{2}}dx$, or equivalent

A1 Complete the integration correctly, obtaining $(12x - 2x^2 - 24)e^{\frac{x}{2}}$, or equivalent

DM1 Use limits $x = 0, x = 2$ correctly having integrated by parts twice

A1 Obtain answer $24 - 8e$, or **exact** simplified equivalent **[5]**

Q246 Oct/Nov 2016 | Paper 3 (variant 32) | Question 8

Topics: none of the listed topics

8 Two planes have equations $3x + y - z = 2$ and $x - y + 2z = 3$.

(i) Show that the planes are perpendicular. [3]

(ii) Find a vector equation for the line of intersection of the two planes. [6]

Examiner comment

> "A common error was to have the normal vector of the first plane as 3i+j−2k ."

Mark scheme

8 (i) State or imply a correct normal vector to either plane, e.g. $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ or $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ **M1**

Use correct method to calculate their scalar product **M1**

Show value is zero and planes are perpendicular **A1** [3]

(ii) EITHER: Carry out a complete strategy for finding a point on l the line of intersection **M1**

Obtain such a point, e.g. $(0, 7, 5)$, $(1, 0, 1)$, $(5/4, -7/4, 0)$ **A1**

EITHER: State two equations for a direction vector $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ for l, e.g. $3a + b - c = 0$ and $a - b + 2c = 0$ **B1**

Solve for one ratio, e.g. $a : b$ **M1**

Obtain $a : b : c = 1 : -7 : -4$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$ *A1^$

OR1: Obtain a second point on l, e.g. $(1, 0, 1)$ **B1**

Subtract vectors and obtain a direction vector for l **M1**

Obtain $\mathbf{i} - 7\mathbf{j} + 4\mathbf{k}$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r} = \mathbf{i} + \mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} + 4\mathbf{k})$ *A1^$

OR2: Attempt to find the vector product of the two normal vectors **M1**

Obtain two correct components of the product **A1**

Obtain $\mathbf{i} - 7\mathbf{j} - 4\mathbf{k}$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$ *A1^$

OR1: Express one variable in terms of a second variable **M1**

Obtain a correct simplified expression, e.g. $y = 7 - 7x$ **A1**

Express the third variable in terms of the second **M1**

Obtain a correct simplified expression, e.g. $z = 5 - 4x$ **A1**

Form a vector equation for the line **M1**

Obtain a correct equation, e.g. $\mathbf{r} = 7\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} - 7\mathbf{j} - 4\mathbf{k})$ *A1^$

OR2: Express one variable in terms of a second variable **M1**

Obtain a correct simplified expression, e.g. $z = 5 - 4x$ **A1**

Express the same variable in terms of the third **M1**

Obtain a correct simplified expression e.g. $z = (7 + 4y)/7$ **A1**

Form a vector equation for the line **M1**

Obtain a correct equation, e.g. $\mathbf{r} = \frac{1}{4}\mathbf{i} - \frac{7}{4}\mathbf{j} + \lambda(\frac{11}{4}\mathbf{i} + \frac{7}{4}\mathbf{j} + \mathbf{k})** A1* [6]

Q247 Oct/Nov 2016 | Paper 3 (variant 32) | Question 9

Topics: Complex Numbers

9 Throughout this question the use of a calculator is not permitted.

(a) Solve the equation $(1 + 2i)w^2 + 4w - (1 - 2i) = 0$, giving your answers in the form $x + iy$, where x and y are real. [5]

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers satisfying the inequalities $|z - 1 - i| \leq 2$ and $-\frac{1}{4}\pi \leq \arg z \leq \frac{1}{4}\pi$. [5]

Examiner comment

> "The second inequality describes two half lines which should start at the origin; it was a common error to see them starting at the centre of the circle."

Mark scheme

9 (a) EITHER: Use quadratic formula to solve for w **M1**

Use $i^2 = -1$ **M1**

Obtain one of the answers $w = \frac{1}{2i+1}$ and $w = -\frac{5}{2i+1}$ **A1**

Multiply numerator and denominator of an answer by $-2i + 1$, or equivalent **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1**

OR1: Multiply the equation by $1 - 2i$ **M1**

Use $i^2 = -1$ **M1**

Obtain $5w^2 + 4w(1 - 2i) - (1 - 2i)^2 = 0$, or equivalent **A1**

Use quadratic formula or factorise to solve for w **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1**

OR2: Substitute $w = x + iy$ and form equations for real and imaginary parts **M1**

Use $i^2 = -1$ **M1**

Obtain $(x^2 - y^2) - 4xy + 4x - 1 = 0$ and $2(x^2 - y^2) + 2xy + 4y + 2 = 0$ o.e. **A1**

Form equation in x only or y only and solve **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1** [5]

(b) Show a circle with centre $1 + i$ **B1**

Show a circle with radius 2 **B1**

Show half-line arg $z = \frac{1}{4}\pi$ **B1**

Show half-line arg $z = -\frac{1}{4}\pi$ **B1**

Shade the correct region **B1** [5]

Q248 Oct/Nov 2016 | Paper 3 (variant 32) | Question 10

Topics: Differential Equations

10 A large field of area $4\ \text{km}^2$ is becoming infected with a soil disease. At time t years the area infected is $x\ \text{km}^2$ and the rate of growth of the infected area is given by the differential equation

$$\frac{dx}{dt} = kx(4 - x),$$

where k is a positive constant. It is given that when $t = 0$, $x = 0.4$ and that when $t = 2$, $x = 2$.

(i) Solve the differential equation and show that $k = \frac{1}{4}\ln 3$. [9]

(ii) Find the value of t when 90% of the area of the field is infected. [2]

Examiner comment

> "Many candidates did not realise that ∫ dx requires x(4−x) the use of partial fractions."

Mark scheme

10 (i) Separate variables correctly and integrate at least one side **M1**

Integrate and obtain term kt, or equivalent **A1**

Carry out a relevant method to obtain A and B such that

$$\frac{1}{x(4 - x)} = \frac{A}{x} + \frac{B}{4 - x},$$

or equivalent *M1$

Obtain $A = B = \frac{1}{4}$, or equivalent **A1**

Integrate and obtain terms $\frac{1}{4}\ln x - \frac{1}{4}\ln(4-x)$, or equivalent *A1^$

EITHER: Use a pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant **DM1**

Obtain correct answer in any form, e.g. $\ln x - \ln(4-x) = 4kt - \ln 9$, or

$$\ln\left(\frac{x}{4-x}\right) = 4kt - 8k$$

A1

Use a second pair of limits and determine k **DM1**

Obtain the given exact answer correctly **A1**

OR: Use both pairs of limits in a definite integral *M1$

Obtain the given exact answer correctly **A1**

Substitute k and either pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant **DM1**

Obtain $\ln\frac{x}{4-x} = t\ln 3 - \ln 9$, or equivalent **A1** [9]

(ii) Substitute $x = 3.6$ and solve for t **M1**

Obtain answer $t = 4$ **A1** [2]

Q249 Oct/Nov 2016 | Paper 3 (variant 33) | Question 1

Topics: Indices, Logarithms & Exponents

1 It is given that $z = \ln(y+2) - \ln(y+1)$. Express y in terms of z. [3]

Examiner comment

> "Most candidates quickly produced the correct solution, although a number struggled with the algebra at the end."

Mark scheme

1 Use law of the logarithm of a quotient **M1**

Remove logarithms and obtain a correct equation, e.g. $e^z = \frac{y+2}{y+1}$ **A1**

Obtain answer $y = \frac{2-e^z}{e^z-1}$, or equivalent **A1** [3]

Q250 Oct/Nov 2016 | Paper 3 (variant 33) | Question 4

Topics: Algebra

4 The polynomial $4x^4 + ax^2 + 11x + b$, where a and b are constants, is denoted by $p(x)$. It is given that $p(x)$ is divisible by $x^2 - x + 2$.

(i) Find the values of a and b. [5]

(ii) When a and b have these values, find the real roots of the equation $p(x) = 0$. [2]

Examiner comment

> "Unfortunately many candidates who opted for the long division approach became confused as to whether they were subtracting or adding rows, so few candidates finished with the correct remainder."

Mark scheme

4 (i) Commence division by $x^2 - x + 2$ and reach a partial quotient $4x^2 + kx$ **M1**

Obtain quotient $4x^2 + 4x + a - 4$ or $4x^2 + 4x + b/2$ **A1**

Equate x or constant term to zero and solve for a or b **M1**

Obtain $a = 1$ **A1**

Obtain $b = -6$ **A1** [5]

(ii) Show that $x^2 - x + 2 = 0$ has no real roots **B1**

Obtain roots $\frac{1}{2}$ and $-\frac{3}{2}$ from $4x^2 + 4x - 3 = 0$ **B1** [2]

Q251 Oct/Nov 2016 | Paper 3 (variant 33) | Question 8

Topics: Algebra

8 Let $f(x) = \frac{3x^2+x+6}{(x+2)(x^2+4)}$.

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

Examiner comment

> "A few candidates made arithmetical errors, while others started from an incorrect form, opting for C/(x2 + 4) instead of (Bx + C) /(x2 + 4) as one of their partial fractions."

Mark scheme

8 (i) State or imply the form

$$\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$$

B1

Use a correct method to determine a constant **M1**

Obtain one of $A = 2$, $B = 1$, $C = -1$ **A1**

Obtain a second value **A1**

Obtain a third value **A1** [5]

Q252 Oct/Nov 2016 | Paper 3 (variant 33) | Question 10

Topics: none of the listed topics

10 The line l has vector equation $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} + \mathbf{k})$.

(i) Find the position vectors of the two points on the line whose distance from the origin is $\sqrt{(10)}$. [5]

(ii) The plane p has equation $ax + y + z = 5$, where a is a constant. The acute angle between the line l and the plane p is equal to $\sin^{-1}\left(\frac{2}{3}\right)$. Find the possible values of a. [5]

Examiner comment

> "After achieving the correct answer it was disappointing to see some candidates for some unknown reason decide to reject their negative solution."

Mark scheme

10 (i) Express general point of l in component form e.g. $(1 + 2\lambda, 2 - \lambda, 1 + \lambda)$ **B1**

Using the correct process for the modulus form an equation in λ *M1$

Reduce the equation to a quadratic, e.g. $6\lambda^2 + 2\lambda - 4 = 0$ **A1**

Solve for λ (usual requirements for solution of a quadratic) **DM1**

Obtain final answers $-\mathbf{i} + 3\mathbf{j}$ and $\frac{5}{3}\mathbf{i} + \frac{5}{3}\mathbf{j} + \frac{5}{3}\mathbf{k}$ **A1** [5]

(ii) Using the correct process, find the scalar product of a direction vector for l and a normal for p **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and equate the result to $\frac{2}{3}$ **M1**

State a correct equation in any form, e.g.

$$\frac{2a - 1 + 1}{\sqrt{(a^2 + 1 + 1).\sqrt{2^2 + (-1)^2 + 1}}} = \pm\frac{2}{3}$$

A1

Solve for a^2 **M1**

Obtain answer $a = \pm 2$ **A1** [5]

Q253 May/June 2015 | Paper 3 (variant 31) | Question 2

Topics: Numerical Solutions/Iteration

Use the trapezium rule with three intervals to find an approximation to

$$\int_0^3 |3^x - 10|\, dx.$$

[4]

Examiner comment

> "Question 2 Many candidates did not understand the meaning of the function 'modulus' and it was very common to see ordinates with the values –9 and –7."

Mark scheme

Attempt calculation of at least 3 ordinates **M1**

Obtain $\mathbf{9, 7, 1, 17}$ **A1**

Use trapezium rule with $h = 1$ **M1**

Obtain $\frac{1}{2}(9 + 14 + 2 + 17)$ or equivalent and hence **21** **A1**

Total: 4

Q254 May/June 2015 | Paper 3 (variant 31) | Question 3

Topics: Algebra

Show that, for small values of x^2,

$$(1 - 2x^2)^{-2} - (1 + 6x^2)^{\frac{2}{3}} \approx kx^4,$$

where the value of the constant k is to be determined. [6]

Examiner comment

> "Question 3 1 Many candidates decided to express the first term in the expression as and to then expand this (1–2x2)2 denominator, hence making no progress."

Mark scheme

Obtain correct unsimplified version of x^2 or x^4 term in $(1 - 2x^2)^{-2}$ **M1**

Obtain $1 + 4x^2$ **A1**

Obtain $+12x^4$ **A1**

Obtain correct unsimplified version of x^2 or x^4 term in $(1 + 6x^2)^{\frac{2}{3}}$ **M1**

Obtain $1 + 4x^2 - 4x^4$ **A1**

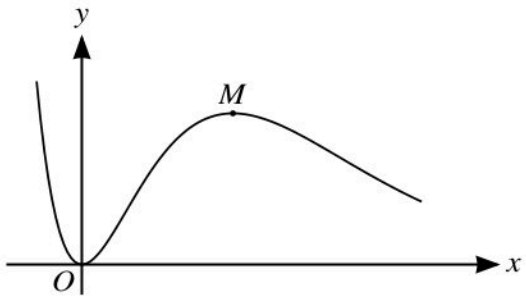
Combine expansions to obtain $k = 16$ with no error seen **A1**

Total: 6

Q255 May/June 2015 | Paper 3 (variant 31) | Question 9

Topics: Differentiation, Integration

9



The diagram shows the curve $y = x^2e^{2-x}$ and its maximum point M.

(i) Show that the x-coordinate of M is 2. [3]

(ii) Find the exact value of $\int_0^2 x^2e^{2-x}\,dx$. [6]

The diagram shows the curve $y = x^2e^{2-x}$ and its maximum point M.

(i) Show that the x-coordinate of M is 2. [3]

(ii) Find the exact value of $\int_0^2 x^2e^{2-x}\,dx$. [6]

Examiner comment

> "In addition with a more difficult example, following exactly what the candidate is doing may not be so easy."

Mark scheme

(i)

Use product rule to find first derivative **M1**

Obtain $2xe^{2-x} - x^2e^{2-x}$ **A1**

Confirm $x = 2$ at M **A1**

Total: 3

(ii)

Attempt integration by parts and reach $\pm x^2e^{2-x} \pm \int 2xe^{2-x}\,dx$ **M1***

Obtain $-x^2e^{2-x} + \int 2xe^{2-x}\,dx$ **A1**

Attempt integration by parts and reach $-x^2e^{2-x} \pm 2xe^{2-x} \pm 2e^{2-x}$ **M1***

Obtain $-x^2e^{2-x} - 2xe^{2-x} - 2e^{2-x}$ **A1**

Use limits 0 and 2 having integrated twice **M1 dep***

Obtain $2e^2 - 10$ **A1**

Total: 6

Q256 Oct/Nov 2015 | Paper 3 (variant 31) | Question 4

Topics: Numerical Solutions/Iteration

The equation $x^3 - x^2 - 6 = 0$ has one real root, denoted by α.

(i) Find by calculation the pair of consecutive integers between which α lies. **[2]**

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \sqrt{x_n + \frac{6}{x_n}}$$

converges, then it converges to α. **[2]**

(iii) Use this iterative formula to determine α correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Examiner comment

> "Common errors included truncation to 2.218 and giving the answer 2.22 to 3 s.f."

Mark scheme

(i) Evaluate, or consider the sign of, $x^3 - x^2 - 6$ for two integer values of x, or equivalent **M1** Obtain the pair $x = 2$ and $x = 3$, with no errors seen **A1 [2]**

(ii) State a suitable equation, e.g. $x = \sqrt{(x + (6/x))}$ **B1**

Rearrange this as $x^3 - x^2 - 6 = 0$, or work vice versa **B1 [2]**

(iii) Use the iterative formula correctly at least once **M1** Obtain final answer 2.219 **A1** Show sufficient iterates to 5 d.p. to justify 2.219 to 3 d.p., or show there is a sign change in the interval (2.2185, 2.2195) **A1 [3]**

Q257 Oct/Nov 2015 | Paper 3 (variant 31) | Question 6

Topics: Algebra

The polynomial $8x^3 + ax^2 + bx - 1$, where a and b are constants, is denoted by **p(x)**. It is given that $(x + 1)$ is a factor of **p(x)** and that when **p(x)** is divided by $(2x + 1)$ the remainder is 1.

(i) Find the values of a and b. **[5]**

(ii) When a and b have these values, factorise **p(x)** completely. **[3]**

Examiner comment

> "A few candidates with correct working did not factorise the quotient 8x2−2x−1."

Mark scheme

(i) Substitute $x = -1$, equate to zero and simplify at least as far as $-8 + a - b - 1 = 0$ **B1**

Substitute $x = -\frac{1}{2}$ and equate the result to 1 **M1**

Obtain a correct equation in any form, e.g. $-1 + \frac{1}{4}a - \frac{1}{2}b - 1 = 1$ **A1**

Solve for a or for b **M1** Obtain $a = 6$ and $b = -3$ **A1 [5]**

(ii) Commence division by $(x + 1)$ reaching a partial quotient $8x^2 + kx$ **M1**

Obtain quadratic factor $8x^2 - 2x - 1$ **A1** Obtain factorisation $(x + 1)(4x + 1)(2x - 1)$ **A1 [3]**

[The M1 is earned if inspection reaches an unknown factor $8x^2 + Bx + C$ and an equation in B and/or C, or an unknown factor $Ax^2 + Bx - 1$ and an equation in A and/or B.]

[If linear factors are found by the factor theorem, give B1B1 for $(2x - 1)$ and $(4x + 1)$, and B1 for the complete factorisation.]

Q258 Oct/Nov 2015 | Paper 3 (variant 32) | Question 4

Topics: Numerical Solutions/Iteration

The equation $x^3 - x^2 - 6 = 0$ has one real root, denoted by α.

(i) Find by calculation the pair of consecutive integers between which α lies. **[2]**

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \sqrt{x_n + \frac{6}{x_n}}$$

converges, then it converges to α. **[2]**

(iii) Use this iterative formula to determine α correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Examiner comment

> "Common errors included truncation to 2.218 and giving the answer 2.22 to 3 s.f."

Mark scheme

(i) Evaluate, or consider the sign of, $x^3 - x^2 - 6$ for two integer values of x, or equivalent **M1** Obtain the pair $x = 2$ and $x = 3$, with no errors seen **A1 [2]**

(ii) State a suitable equation, e.g. $x = \sqrt{(x + (6/x))}$ **B1**

Rearrange this as $x^3 - x^2 - 6 = 0$, or work vice versa **B1 [2]**

(iii) Use the iterative formula correctly at least once **M1** Obtain final answer 2.219 **A1** Show sufficient iterates to 5 d.p. to justify 2.219 to 3 d.p., or show there is a sign change in the interval (2.2185, 2.2195) **A1 [3]**

Q259 Oct/Nov 2015 | Paper 3 (variant 32) | Question 6

Topics: Algebra

The polynomial $8x^3 + ax^2 + bx - 1$, where a and b are constants, is denoted by $\mathbf{p(x)}$. It is given that $(x + 1)$ is a factor of $\mathbf{p(x)}$ and that when $\mathbf{p(x)}$ is divided by $(2x + 1)$ the remainder is 1.

(i) Find the values of a and b. **[5]**

(ii) When a and b have these values, factorise $\mathbf{p(x)}$ completely. **[3]**

Examiner comment

> "A few candidates with correct working did not factorise the quotient 8x2−2x−1."

Mark scheme

(i) Substitute $x = -1$, equate to zero and simplify at least as far as $-8 + a - b - 1 = 0$ **B1**

Substitute $x = -\frac{1}{2}$ and equate the result to 1 **M1**

Obtain a correct equation in any form, e.g. $-1 + \frac{1}{4}a - \frac{1}{2}b - 1 = 1$ **A1**

Solve for a or for b **M1** Obtain $a = 6$ and $b = -3$ **A1 [5]**

(ii) Commence division by $(x + 1)$ reaching a partial quotient $8x^2 + kx$ **M1**

Obtain quadratic factor $8x^2 - 2x - 1$ **A1** Obtain factorisation $(x + 1)(4x + 1)(2x - 1)$ **A1 [3]**

[The M1 is earned if inspection reaches an unknown factor $8x^2 + Bx + C$ and an equation in B and/or C, or an unknown factor $Ax^2 + Bx - 1$ and an equation in A and/or B.]

[If linear factors are found by the factor theorem, give B1B1 for $(2x - 1)$ and $(4x + 1)$, and B1 for the complete factorisation.]

Q260 Oct/Nov 2015 | Paper 3 (variant 33) | Question 2

Topics: Algebra

2 Given that $\sqrt[3]{(1 + 9x)} \approx 1 + 3x + ax^2 + bx^3$ for small values of x, find the values of the coefficients a and b. **[3]**

Examiner comment

> "The most common error in the expansion was to use powers of x rather than of 9x."

Mark scheme

2 Either State correct unsimplified x^2 or x^3 term **M1**

Obtain $a = -9$ **A1**

Obtain $b = 45$ **A1**

Or Use chain rule to differentiate twice to obtain form $k(1 + 9x)^{-\frac{5}{3}}$ **M1**

Obtain $f''(x) = -18(1 + 9x)^{-\frac{5}{3}}$ and hence $a = -9$ **A1**

Obtain $f'''(x) = 270(1+9x)^{-\frac{8}{3}}$ and hence $b = 45$ **A1 [3]**

Q261 Oct/Nov 2015 | Paper 3 (variant 33) | Question 6

Topics: Trigonometry

6 The angles A and B are such that

$$\sin(A+45^\circ) = (2\sqrt{2})\cos A \quad \text{and} \quad 4\sec^2 B + 5 = 12\tan B.$$

Without using a calculator, find the exact value of $\tan(A-B)$. **[8]**

Examiner comment

> "Candidates who did not recognise sinA that they could use sec2B =1+tan2B often made no progress after forming an equation in cosB and ( ) sinB."

Mark scheme

6 State or imply $\sin A \times \cos 45 + \cos A \times \sin 45 = 2\sqrt{2}\cos A$ **B1**

Divide by $\cos A$ to find value of $\tan A$ **M1**

Obtain $\tan A = 3$ **A1**

Use identity $\sec^2 B = 1 + \tan^2 B$ **B1**

Solve three-term quadratic equation and find $\tan B$ **M1**

Obtain $\tan B = \frac{3}{2}$ only **A1**

Substitute **numerical values** in $\frac{\tan A - \tan B}{1+\tan A \tan B}$ **M1**

Obtain $\frac{3}{11}$ **A1 [8]**

Q262 Oct/Nov 2015 | Paper 3 (variant 33) | Question 7

Topics: Algebra, Integration

(i) Show that $(x+1)$ is a factor of $4x^3 - x^2 - 11x - 6$. **[2]**

(ii) Find $\int \frac{4x^2+9x-1}{4x^3-x^2-11x-6}\,dx$. **[8]**

Examiner comment

> "Many candidates did not use the most efficient route to finding the values of the constants - several worked through the solution of three 3 equations in three unknowns rather than making the substitutions x =−1, x =2, and x =– ."

> "3 Many candidates who started with roots –1, 2 and – did not show where these values came from 4 ☐ 3☐ and they went on to use the incorrect factorised form (x+1)(x–2) ☐x+ ☐ so they gained very ☐ 4☐ little credit."

> "A small number of candidates made no progress at all with the integral, not recognising the potential use of partial fractions."

Mark scheme

(i) Either Substitute $x = -1$ and evaluate **M1**

Obtain **0** and conclude $x + 1$ is a factor **A1**

Or Divide by $x + 1$ and obtain a constant remainder **M1**

Obtain remainder $= 0$ and conclude $x + 1$ is a factor **A1 [2]**

(ii) Attempt division, or equivalent, at least as far as quotient $4x^2 + kx$ **M1**

Obtain complete quotient $4x^2 - 5x - 6$ **A1**

State form $\frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{4x+3}$ **A1**

Use relevant method for finding at least one constant **M1**

Obtain one of $A = -2, B = 1, C = 8$ **A1**

Obtain all three values **A1**

Integrate to obtain three terms each involving natural logarithm of linear form **M1**

Obtain $-2\ln(x+1) + \ln(x-2) + 2\ln(4x+3)$, condoning no use of modulus signs and absence of $+c$ **A1 [8]**

Q263 Oct/Nov 2015 | Paper 3 (variant 33) | Question 9

Topics: Complex Numbers

(a) It is given that $(1 + 3i)w = 2 + 4i$. Showing all necessary working, prove that the exact value of $|w^2|$ is **2** and find $\arg(w^2)$ correct to **3** significant figures. **[6]**

(b) On a single Argand diagram sketch the loci $|z| = 5$ and $|z - 5| = |z|$. Hence determine the complex numbers represented by points common to both loci, giving each answer in the form $re^{i\theta}$. **[4]**

Examiner comment

> "The straight line proved to be more difficult - the most common errors were to draw it as a second circle, or to draw a horizontal straight line."

Mark scheme

(a) Either Find w using conjugate of $1 + 3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Square $x + iy$ form to find w^2 **M1**

Obtain $w^2 = \frac{48-14i}{25}$ and confirm modulus is **2** **A1**

Use correct process for finding argument of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 1 Find w using conjugate of $1+3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Find modulus of w and hence of w^2 **M1**

Confirm modulus is 2 **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 2 Square both sides to obtain $(-8+6i)w^2 = -12+16i$ **B1**

Find w^2 using relevant conjugate **M1**

Use correct process for finding modulus of w^2 **M1**

Confirm modulus is 2 **A1**

Use correct process for finding argument of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 3 Find modulus of LHS and RHS **M1**

Find argument of LHS and RHS **M1**

Obtain $\sqrt{10}e^{i.2491}w = \sqrt{20}e^{i.1.1071}$ or equivalent **A1**

Obtain $w = \sqrt{2}e^{-0.1419i}$ or equivalent **A1**

Use correct process for finding w^2 **M1**

Obtain 2 and -0.284 radians or -16.3° **A1**

Or 4 Find moduli of $2+4i$ and $1+3i$ **M1**

Obtain $\sqrt{20}$ and $\sqrt{10}$ **A1**

Obtain $|w^2| = 2$ correctly **A1**

Find $\arg(2+4i)$ and $\arg(1+3i)$ **M1**

Use correct process for $\arg(w^2)$ **A1**

Obtain -0.284 radians or -16.3° **A1**

Or 5 Let $w = a+ib$, form and solve simultaneous equations in a and b **M1**

$a = \frac{7}{5}$ and $b = -\frac{1}{5}$ **A1**

Find modulus of w and hence of w^2 **M1**

Confirm modulus is 2 **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 6 Find w using conjugate of $1+3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Use $|w^2| = w\bar{w}$ **M1**

Confirm modulus is 2 **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1** **[6]**

(b) Draw circle with centre the origin and radius 5 **B1**

Draw straight line parallel to imaginary axis in correct position **B1**

Use relevant trigonometry on a correct diagram to find argument(s) **M1**

Obtain $5e^{\pm\frac{\pi}{3}}$ or equivalents in required form **A1** **[4]**

Q264 Oct/Nov 2015 | Paper 3 (variant 33) | Question 10

Topics: Differential Equations

10 Naturalists are managing a wildlife reserve to increase the number of plants of a rare species. The number of plants at time t years is denoted by N, where N is treated as a continuous variable.

(i) It is given that the rate of increase of N with respect to t is proportional to $(N-150)$. Write down a differential equation relating N, t and a constant of proportionality. **[1]**

(ii) Initially, when $t=0$, the number of plants was 650. It was noted that, at a time when there were 900 plants, the number of plants was increasing at a rate of 60 per year. Express N in terms of t. **[7]**

(iii) The naturalists had a target of increasing the number of plants from 650 to 2000 within 15 years. Will this target be met? **[2]**

Examiner comment

> "The common errors were to omit the constant of proportionality, to use t as the constant, or to include an additional factor of t."

> "The most common error was in finding the value for k, usually because candidates overlooked the information that 60 was the rate of change of the number of plants, not a value for N or for t."

Mark scheme

(i) State $\frac{dN}{dt} = k(N - 150)$ **B1 [1]**

(ii) Substitute $\frac{dN}{dt} = 60$ and $N = 900$ to find value of k **M1**

Obtain $k = 0.08$ **A1**

Separate variables and obtain general solution involving $\ln(N - 150)$ **M1***

Obtain $\ln(N - 150) = 0.08t + c$ (following their k) or $\ln(N - 150) = kt + c$ **A1***

Substitute $t = 0$ and $N = 650$ to find c **dep M1***

Obtain $\ln(N - 150) = 0.08t + \ln 500$ or equivalent **A1**

Obtain $N = 500e^{0.08t} + 150$ **A1 [7]**

(iii) Either Substitute $t = 15$ to find N or solve for t with $N = 2000$ **M1**

Obtain Either $N = 1810$ or $t = 16.4$ and conclude target not met **A1 [2]**

Q265 May/June 2014 | Paper 3 (variant 31) | Question 1

Topics: Trigonometry

(i) Simplify $\sin 2\alpha \sec \alpha$. [2]

(ii) Given that $3\cos 2\beta + 7\cos \beta = 0$, find the exact value of $\cos \beta$. [3]

Examiner comment

> "Question 1 (i) Many candidates seemed unable to express both sin2α and secα in terms of sinα and cosα."

> "(ii) Here, many candidates did not use the double angle formula, and of those that did, many subsequently made errors."

Mark scheme

(i)

State $\sin 2\alpha = 2\sin \alpha \cos \alpha$ and $\sec \alpha = \frac{1}{\cos \alpha}$ **B1**

Obtain $2\sin \alpha$ **B1**

Total: 2

(ii)

Use $\cos 2\beta = 2\cos^2 \beta - 1$ or equivalent to produce correct equation in $\cos \beta$ **B1**

Solve three-term quadratic equation for $\cos \beta$ **M1**

Obtain $\cos \beta = \frac{1}{3}$ only **A1**

Total: 3

Q266 May/June 2014 | Paper 3 (variant 31) | Question 4

Topics: Differential Equations

The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{6ye^{3x}}{2 + e^{3x}}.$$

Given that $y = 36$ when $x = 0$, find an expression for y in terms of x. [6]

Examiner comment

> "Question 4 Few candidates were successful at this question since the separation of variables was nearly always not 6e3x undertaken, and even when it was the integration of proved too difficult."

Mark scheme

Separate variables correctly and recognisable attempt at integration of at least one side **M1**

Obtain $\ln y$, or equivalent **B1**

Obtain $\ln(2 + e^{3x})^k$ **B1**

Use $y(0) = 36$ to find constant in $\ln y = A\ln(2 + e^{3x}) + c$ or equivalent **M1***

Obtain equation correctly without logarithms from $\ln y = A\ln(2 + e^{3x}) + c$ ***M1**

Obtain $y = 4(2 + e^{3x})^2$ **A1**

Total: 6

Q267 May/June 2014 | Paper 3 (variant 31) | Question 6

Topics: Algebra, Indices, Logarithms & Exponents

It is given that $2\ln(4x - 5) + \ln(x + 1) = 3\ln 3$.

(i) Show that $16x^3 - 24x^2 - 15x - 2 = 0$. [3]

(ii) By first using the factor theorem, factorise $16x^3 - 24x^2 - 15x - 2$ completely. [4]

(iii) Hence solve the equation $2\ln(4x - 5) + \ln(x + 1) = 3\ln 3$. [1]

Examiner comment

> "Question 6 (i) Few candidates were able to use correctly both the product and power laws of logarithms."

> "1 (iii) Most candidates did not realise that only x = 2 was a possible solution, with x = being rejected."

Mark scheme

(i)

Use law for the logarithm for a product or quotient or exponentiation and for a power **M1**

Obtain $(4x - 5)^2(x + 1) = 27$ **B1**

Obtain given equation correctly $16x^3 - 24x^2 - 15x - 2 = 0$ **A1**

Total: 3

(ii)

Obtain $x = 2$ is root or $(x - 2)$ is a factor, or likewise with $x = -\frac{1}{4}$ **B1**

Divide by $(x - 2)$ to reach a quotient of the form $16x^2 + kx$ **M1**

Obtain quotient $16x^2 + 8x + 1$ **A1**

Obtain $(x - 2)(4x + 1)^2$ or $(x - 2)$, $(4x + 1)$, $(4x + 1)$ **A1**

Total: 4

(iii)

State $x = 2$ only **A1**

Total: 1

Q268 May/June 2014 | Paper 3 (variant 31) | Question 8

Topics: Trigonometry, Numerical Solutions/Iteration

(i) By sketching each of the graphs $y = \operatorname{cosec} x$ and $y = x(\pi - x)$ for $0 < x < \pi$, show that the equation

$$\operatorname{cosec} x = x(\pi - x)$$

has exactly two real roots in the interval $0 < x < \pi$. [3]

(ii) Show that the equation $\operatorname{cosec} x = x(\pi - x)$ can be written in the form

$$x = \frac{1 + x^2 \sin x}{\pi \sin x}.$$

[2]

(iii) The two real roots of the equation $\operatorname{cosec} x = x(\pi - x)$ in the interval $0 < x < \pi$ are denoted by α and β, where $\alpha < \beta$.

(a) Use the iterative formula

$$x_{n+1} = \frac{1 + x_n^2 \sin x_n}{\pi \sin x_n}$$

to find α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(b) Deduce the value of β correct to 2 decimal places. [1]

Examiner comment

> "Question 8 (i) The sketching of y =cosecx proved difficult, although better success was forthcoming for y = x(π−x)."

Mark scheme

(i)

Sketch $y = \operatorname{cosec} x$ for at least $0 < x < \pi$ **B1**

Sketch $y = x(\pi - x)$ for at least $0 < x < \pi$ **B1**

Justify statement concerning two roots, with evidence of 1 and $\frac{1}{4}\pi^2$ for y-values on graph via scales **B1**

Total: 3

(ii)

Use $\operatorname{cosec} x = \frac{1}{\sin x}$ and commence rearrangement **M1**

Obtain given equation correctly, showing sufficient detail **A1**

Total: 2

(iii)(a)

Use the iterative formula correctly at least once **M1**

Obtain final answer 0.66 **A1**

Show sufficient iterations to 4 decimal places to justify answer or show a sign change in the interval $(0.655, 0.665)$ **A1**

Total: 3

(iii)(b)

Obtain 2.48 **B1**

Total: 1

Q269 Oct/Nov 2014 | Paper 3 (variant 31) | Question 2

Topics: Trigonometry, Integration

(i) Use the trapezium rule with 3 intervals to estimate the value of

$$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \operatorname{cosec} x \, dx,$$

giving your answer correct to 2 decimal places. **[3]**

(ii) Using a sketch of the graph of $y = \operatorname{cosec} x$, explain whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part **(i)**. **[2]**

Examiner comment

> "Question 2 In general this was poorly answered."

> "Common errors seen in the attempts to apply the trapezium rule included the use of abscissae (x-coordinates)

> instead of ordinates (y-coordinates); incorrect weightings of the ordinates; 3 or 5 ordinates instead of 4; an incorrect value for the interval width; and ordinates calculated at unequal intervals."

Mark scheme

(i) State or imply ordinates $\mathbf{2, 1.1547 \ldots, 1, 1.1547 \ldots}$ **B1**

Use correct formula, or equivalent, with $h = \frac{1}{6}\pi$ and four ordinates **M1**

Obtain answer **1.95 A1 [3]**

(ii) Make recognisable sketch of $y = \operatorname{cosec} x$ for the given interval **B1**

Justify a statement that the estimate will be an overestimate **B1 [2]**

Q270 Oct/Nov 2014 | Paper 3 (variant 31) | Question 4

Topics: Differentiation

The parametric equations of a curve are

$$x = \frac{1}{\cos^3 t}, \quad y = \tan^3 t,$$

where $0 \leq t < \frac{1}{2}\pi$.

(i) Show that $\frac{dy}{dx} = \sin t$. **[4]**

(ii) Hence show that the equation of the tangent to the curve at the point with parameter t is $y = x \sin t - \tan t$. **[3]**

Examiner comment

> "However, finding the derivatives of x and y with respect to dx t proved difficult for some candidates."

Mark scheme

(i) Use chain rule correctly at least once **M1**

Obtain either $\frac{dx}{dt} = \frac{3 \sin t}{\cos^4 t}$ or $\frac{dy}{dt} = 3 \tan^2 \sec^2 t$, or equivalent **A1**

Use $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ **M1**

Obtain the given answer **A1 [4]**

(ii) State a correct equation for the tangent in any form **B1**

Use Pythagoras **M1**

Obtain the given answer **A1 [3]**

Q271 Oct/Nov 2014 | Paper 3 (variant 31) | Question 5

Topics: Complex Numbers

Throughout this question the use of a calculator is not permitted.

The complex numbers w and z satisfy the relation

$$w = \frac{z+i}{iz+2}.$$

(i) Given that $z = 1 + i$, find w, giving your answer in the form $x + iy$, where x and y are real. **[4]**

(ii) Given instead that $w = z$ and the real part of z is negative, find z, giving your answer in the form $x + iy$, where x and y are real. **[4]**

Examiner comment

> "A common error here was to take -3 to be ± 3i rather than ±i 3."

Mark scheme

(i) Substitute $z = 1 + i$ and obtain $w = \frac{2+i}{1+i}$ **B1**

EITHER: Multiply numerator and denominator by the conjugate of the denominator, or equivalent **M1**

Simplify numerator to $3 + i$ or denominator to 2 **A1**

Obtain final answer $\frac{3}{2} + \frac{1}{2}i$, or equivalent **A1**

OR: Obtain two equations in x and y, and solve for x or for y **M1**

Obtain $x = \frac{3}{2}$ or $y = \frac{1}{2}$, or equivalent **A1**

Obtain final answer $\frac{3}{2} + \frac{1}{2}i$, or equivalent **A1 [4]**

Q272 Oct/Nov 2014 | Paper 3 (variant 31) | Question 6

Topics: Indices, Logarithms & Exponents, Integration, Numerical Solutions/Iteration

It is given that $\int_1^a \ln(2x)\,dx = 1$, where $a > 1$.

(i) Show that $a = \frac{1}{2}\exp\left(1 + \frac{\ln 2}{a}\right)$, where $\exp(x)$ denotes e^x. **[6]**

(ii) Use the iterative formula

$$a_{n+1} = \frac{1}{2}\exp\left(1 + \frac{\ln 2}{a_n}\right)$$

to determine the value of a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Examiner comment

> "Question 6 (i) Though some could make no progress at all, the majority did try to integrate by parts."

Mark scheme

(i) Integrate and reach $bx \ln 2x - c\,|x\frac{dx}{x}$, or equivalent **M1***

Obtain $x \ln 2x - \int x.\ \frac{dx}{x}$, or equivalent **A1**

Obtain integral $x \ln 2x - x$, or equivalent **A1**

Substitute limits correctly and equate to **1**, having integrated twice **M1(dep*)**

Obtain a correct equation in any form, e.g. $a \ln 2a - a + 1 - \ln 2 = 1$ **A1**

Obtain the given answer **A1 [6]**

(ii) Use the iterative formula correctly at least once **M1**

Obtain final answer **1.94 A1**

Show sufficient iterations to 4 d.p. to justify **1.94** to 2d.p. or show that there is a sign change in the interval $(1.935, 1.945)$. **A1 [3]**

Q273 Oct/Nov 2014 | Paper 3 (variant 33) | Question 2

Topics: Differentiation

A curve is defined for $0 < \theta < \frac{1}{2}\pi$ by the parametric equations

$$x = \tan\theta, \quad y = 2\cos^2\theta \sin\theta.$$

Show that $\frac{dy}{dx} = 6\cos^3\theta - 4\cos^3\theta$. **[5]**

Examiner comment

> "Some candidates focused on the cos2θ and appeared to forget the sinθ, and the incorrect result ( ) d dθcos2θ=sin2θ was a common error."

Mark scheme

Use correct product rule or correct chain rule to differentiate y **M1**

Use $\frac{dy}{dx} = \frac{d\theta/dx}{d\theta}$ **M1***

Obtain $\frac{-4\cos\theta\sin^2\theta + 2\cos^3\theta}{\sec^2\theta}$ or equivalent **A1**

Express $\frac{dy}{dx}$ in terms of $\cos\theta$ **DM1***

Confirm given answer $6\cos^5\theta - 4\cos^3\theta$ legitimately **A1 [5]**

Q274 Oct/Nov 2014 | Paper 3 (variant 33) | Question 5

Topics: Trigonometry, Complex Numbers

The complex numbers w and z are defined by $w = 5 + 3i$ and $z = 4 + i$.

(i) Express $\frac{iw}{z}$ in the form $x + iy$, showing all your working and giving the exact values of x and y. **[3]**

(ii) Find wz and hence, by considering arguments, show that

$$\tan^{-1}\left(\frac{3}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \frac{1}{4}\pi. \quad [4]$$

Examiner comment

> "Many candidates lost marks in part (ii) because they simply wrote down the given answer without explaining that they were using the general result arg (wz) = arg w + arg z and they did not explain what tan−13 and 5 tan−11 represented or where the 1 π comes from."

Mark scheme

(i) State or imply $iw = -3 + 5i$ **B1**

Carry out multiplication by $\frac{4-i}{4-i}$ **M1**

Obtain final answer $-\frac{7}{17} + \frac{23}{17}i$ or equivalent **A1 [3]**

(ii) Multiply w by z to obtain $17 + 17i$ **B1**

State $\arg w = \tan^{-1}\frac{3}{5}$ or $\arg z = \tan^{-1}\frac{1}{4}$ **B1**

State $\arg wz = \arg w + \arg z$ **M1**

Confirm given result $\tan^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4} = \frac{1}{4}\pi$ legitimately **A1 [4]**

Q275 Oct/Nov 2014 | Paper 3 (variant 33) | Question 6

Topics: Algebra, Integration

It is given that $I = \int_0^{0.3} (1 + 3x^2)^{-2}\, dx$.

(i) Use the trapezium rule with 3 intervals to find an approximation to I, giving the answer correct to 3 decimal places. **[3]**

(ii) For small values of x, $(1 + 3x^2)^{-2} \approx 1 + ax^2 + bx^4$. Find the values of the constants a and b.

Hence, by evaluating $\int_0^{0.3} (1 + ax^2 + bx^4)\, dx$, find a second approximation to I, giving the answer correct to 3 decimal places. **[5]**

Examiner comment

> "Amongst those candidates who demonstrated an understanding of the basic structure of the method, common errors involved working with the wrong number of intervals, not starting from x = 0 and using 0, 1, 2 and 3 as values for x rather than 0, 0.1, 0.2 and 0.3."

> "A 1 1 1 1 common error was to state ( ) = =1+ x2 + x4."

Mark scheme

(i) State or imply correct ordinates $1, 0.94259\ldots, 0.79719\ldots, 0.62000\ldots$ **B1**

Use correct formula or equivalent with $h = 0.1$ and four y values **M1**

Obtain 0.255 with no errors seen **A1 [3]**

(ii) Obtain or imply $a = -6$ **B1**

Obtain x^4 term including correct attempt at coefficient **M1**

Obtain or imply $b = 27$ **A1**

Either Integrate to obtain $x - 2x^3 + \frac{27}{5}x^5$, following their values of a and b **B1***

Obtain 0.259 **B1**

Or Use correct trapezium rule with at least 3 ordinates **M1**

Obtain 0.259 (from 4) **A1 [5]**

Q276 Oct/Nov 2014 | Paper 3 (variant 33) | Question 8

Topics: Differential Equations

The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{1}{5}y^{\frac{1}{2}} \sin\left(\frac{1}{3}x\right).$$

(i) Find the general solution, giving y in terms of x. **[6]**

(ii) Given that $y = 100$ when $x = 0$, find the value of y when $x = 25$. **[3]**

Examiner comment

> "Integrating xsin x proved to be more difficult."

Mark scheme

(i) Sensibly separate variables and attempt integration of at least one side **M1**

Obtain $2y^{\frac{1}{2}} = \ldots$ or equivalent **A1**

Correct integration by parts of $x \sin \frac{1}{3}x$ as far as $ax \cos \frac{1}{3}x \pm \left|b \cos \frac{1}{3}x dx\right|$ **M1**

Obtain $-3x \cos \frac{1}{3}x + \int 3 \cos \frac{1}{3}x dx$ or equivalent **A1**

Obtain $-3x \cos \frac{1}{3}x + 9 \sin \frac{1}{3}x$ or equivalent **A1**

Obtain $y = \left(-\frac{3}{10}x \cos \frac{1}{3}x + \frac{9}{10}\sin \frac{1}{3}x + c\right)^2$ or equivalent **A1 [6]**

(ii) Use $x = 0$ and $y = 100$ to find constant **M1***

Substitute 25 and calculate value of y **DM1***

Obtain 203 **A1 [3]**

Q277 Oct/Nov 2014 | Paper 3 (variant 33) | Question 9

Topics: Indices, Logarithms & Exponents, Numerical Solutions/Iteration

(i) Sketch the curve $y = \ln(x + 1)$ and hence, by sketching a second curve, show that the equation

$$x^3 + \ln(x + 1) = 40$$

has exactly one real root. State the equation of the second curve. **[3]**

(ii) Verify by calculation that the root lies between 3 and 4. **[2]**

(iii) Use the iterative formula

$$x_{n+1} = \sqrt[3]{40 - \ln(x_n + 1)},$$

with a suitable starting value, to find the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

(iv) Deduce the root of the equation

$$(e^y - 1)^3 + y = 40,$$

giving the answer correct to 2 decimal places. **[2]**

Examiner comment

> "The question specifies accuracy to a number of decimal places, but some candidates appeared to be working to that number of significant figures and lost marks as a result."

> "( ) (iv) A small number of candidates spotted the link between x3 +ln(x+1) =40and ey −13 +y = 40, but many either worked through a second set of iterations or made no progress with this part."

Mark scheme

(i) Sketch increasing curve with correct curvature passing through origin, for $x \geq 0$ **B1**

Recognisable sketch of $y = 40 - x^3$, with equation stated, for $x > 0$ **B1**

Indicate in some way the one intersection, dependent on both curves being roughly correct and both existing for some $x < 0$ **B1 [3]**

(ii) Consider signs of $x^3 + \ln(x + 1) - 40$ at 3 and 4 or equivalent or compare values of relevant expressions for $x = 3$ and $x = 4$

M1

Complete argument correctly with correct calculations (-11.6 and 25.6) **A1 [2]**

(iii) Use the iterative formula correctly at least once **M1**

Obtain final answer 3.377 **A1**

Show sufficient iterations to justify accuracy to 3 d.p. or show sign change in interval $(3.3765, 3.3775)$ **A1**

[3]

(iv) Attempt value of $\ln(x+1)$ **M1**

Obtain **1.48 A1 [2]**

Q278 May/June 2013 | Paper 3 (variant 31) | Question 1

Topics: Algebra

Find the quotient and remainder when $2x^2$ is divided by $x+2$. [3]

Examiner comment

> "Another common error was to express the remainder as ."

Mark scheme

Carry out division or equivalent at least as far as two terms of quotient **M1**

Obtain quotient $2x-4$ **A1**

Obtain remainder 8 **A1**

Total: 3

Q279 May/June 2013 | Paper 3 (variant 31) | Question 4

Topics: Algebra, Indices, Logarithms & Exponents

(i) Solve the equation $|4x-1| = |x-3|$. [3]

(ii) Hence solve the equation $|4^{y+1}-1| = |4^y-3|$ correct to 3 significant figures. [3]

Examiner comment

> "This is something that most candidates fail to utilise, namely that where questions are displayed in sections (i), (ii), etc., then there is often a direct link between the individual sections, for example Question 4 (i) and (ii), Question 6 (i) and (ii), Question 9 (i), (b) (ii) and (iii)."

Mark scheme

(i)

EITHER: State or imply non-modular equation $(4x-1)^2 = (x-3)^2$ or pair of linear equations $4x-1 = \pm(x-3)$ **B1**

Solve a three-term quadratic equation or two linear equations **M1**

Obtain $-\frac{2}{3}$ and $\frac{4}{5}$ **A1**

OR: Obtain value $-\frac{2}{3}$ from inspection or solving linear equation **B1**

Obtain value $\frac{4}{5}$ similarly **B2**

Total: 3

(ii)

State or imply at least $4^y = \frac{4}{5}$, following a positive answer from part (i) **B1 FT**

Apply logarithms and use $\log_a b = \frac{\log b}{\log a}$ property **M1**

Obtain -0.161 and no other answer **A1**

Total: 3

Q280 May/June 2013 | Paper 3 (variant 31) | Question 6

Topics: none of the listed topics

The points P and Q have position vectors, relative to the origin O, given by

$$\overrightarrow{OP} = 7\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$$

and

$$\overrightarrow{OQ} = -5\mathbf{i} + \mathbf{j} + \mathbf{k}.$$

The mid-point of PQ is the point A. The plane Π is perpendicular to the line PQ and passes through A.

(i) Find the equation of Π, giving your answer in the form $ax + by + cz = d$. [4]

(ii) The straight line through P parallel to the x-axis meets Π at the point B. Find the distance AB, correct to 3 significant figures. [5]

Examiner comment

> "Question 6 (i) This question proved very difficult for nearly all the candidates, with OA believed usually to be 1 PQ, and the normal to the plane being either the actual line vector or the cross product of OP 2 and OQ."

Mark scheme

(i)

State or imply A is $(1, 4, -2)$ **B1**

State or imply $\overrightarrow{QP} = 12\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$ or equivalent **B1**

Use QP as normal and A as mid-point to find equation of plane **M1**

Obtain $12x + 6y - 6z = 48$ or equivalent **A1**

Total: 4

(ii)

State equation of PB is $\mathbf{r} = 7\mathbf{i} + 7\mathbf{j} - 5\mathbf{k} + \lambda\mathbf{i}$ **B1**

Set up and solve a relevant equation for λ **M1**

Obtain $\lambda = -9$ and hence B is $(-2, 7, -5)$ **A1**

Use correct method to find distance between A and B **M1**

Obtain 5.20 **A1**

Total: 5

Q281 May/June 2013 | Paper 3 (variant 31) | Question 7

Topics: Complex Numbers

(a) Without using a calculator, solve the equation

$$3w + 2iw^* = 17 + 8i,$$

where w^* denotes the complex conjugate of w. Give your answer in the form $a + bi$. [4]

(b) In an Argand diagram, the loci

$$\arg(z - 2i) = \frac{1}{6}\pi \quad \text{and} \quad |z - 3| = |z - 3i|$$

intersect at the point P. Express the complex number represented by P in the form $re^{i\theta}$, giving the exact value of θ and the value of r correct to 3 significant figures. [5]

Examiner comment

> "Question 7 (a) This question proved difficult as few realised that they needed to expressw =u+v i, together with its conjugate w*, and take real and imaginary parts."

Mark scheme

(a)

State or imply $3(a + bi) + 2i(a - bi) = 17 + 8i$ **B1**

Consider real and imaginary parts to obtain two linear equations in a and b **M1***

Solve two simultaneous linear equations for a or b **M1 dep***

Obtain $7 - 2i$ **A1**

Total: 4

(b)

EITHER: Show or imply a triangle with side 2 **B1**

State at least two of the angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{\pi}{12}$ **B1**

State or imply argument is $\frac{\pi}{4}$ **B1**

Use sine rule or equivalent to find r **M1**

Obtain $6.69e^{\frac{1}{4}\pi i}$ **A1**

OR: State $y = x$ **B1**

State $x^2 + (y-2)^2 = \frac{1}{3}y^2$ or equivalent **B1**

State or imply argument is $\frac{\pi}{4}$ **B1**

Solve for x or y **M1**

Obtain $6.69e^{\frac{1}{4}\pi i}$ **A1**

Total: 5

Q282 May/June 2013 | Paper 3 (variant 31) | Question 8

Topics: Integration

(a) Show that

$$\int_2^4 4x \ln x \, dx = 56 \ln 2 - 12.$$

[5]

(b) Use the substitution $u = \sin 4x$ to find the exact value of

$$\int_0^{\frac{1}{24}\pi} \cos^3 4x \, dx.$$

[5]

Examiner comment

> "(ii) Many candidates either failed to substitute correctly for cos2(4x) in terms of u or muddled their substitution of dx, the latter usually resulting in another cos2(4x) appearing in the numerator instead of a cos(4x) cancelling from the numerator and the denominator."

Mark scheme

(a)

Carry out integration by parts and reach $ax^2 \ln x + b\int x \, dx$ **M1***

Obtain $2x^2 \ln x - \int 2x \, dx$ **A1**

Obtain $2x^2 \ln x - x^2$ **A1**

Use limits, having integrated twice **M1 dep***

Confirm given result $56 \ln 2 - 12$ **A1**

Total: 5

(b)

State or imply $\frac{du}{dx} = 4\cos 4x$ **B1**

Carry out complete substitution except limits **M1**

Obtain $\frac{1}{4}\int(1-u^2)\,du$ or equivalent **A1**

Integrate to obtain form $k_1u+k_2u^3$ with non-zero constants k_1, k_2 **M1**

Use appropriate limits to obtain $\frac{11}{96}$ **A1**

Total: 5

Q283 May/June 2013 | Paper 3 (variant 31) | Question 9

Topics: Trigonometry, Integration

(i) Express $4\cos\theta+3\sin\theta$ in the form $R\cos(\theta-\alpha)$, where $R>0$ and $0<\alpha<\frac{1}{2}\pi$. Give the value of α correct to 4 decimal places. [3]

(ii) Hence

(a) solve the equation $4\cos\theta+3\sin\theta=2$ for $0<\theta<2\pi$, [4]

(b) find

$$\int\frac{50}{(4\cos\theta+3\sin\theta)^2}\,d\theta.$$

[3]

Examiner comment

> "Answer: 0.6435 (ii) Again most candidates obtained the answer 1.80, however the other answer proved more difficult, since instead of 2π – cos-1(2/5) + α candidates assumed the answer was 2π– (cos-1(2/5) + α)."

Mark scheme

(i)

State or imply $R=5$ **B1**

Use relevant trigonometry to find α **M1**

Obtain $\alpha=0.6435$ **A1**

Total: 3

(ii)(a)

Carry out appropriate method to find one value in given range **M1**

Obtain **1.80** **A1**

Carry out appropriate method to find second value in given range **M1**

Obtain **5.77** and no other value **A1**

Total: 4

(ii)(b)

Express integrand as $k\sec^2(\theta - \text{their } \alpha)$ for any constant k **M1**

Integrate to obtain result $k\tan(\theta - \text{their } \alpha)$ **A1**

Obtain correct answer $2\tan(\theta - 0.6435)$ **A1**

Total: 3

Q284 May/June 2013 | Paper 3 (variant 31) | Question 10

Topics: Differential Equations, Numerical Solutions/Iteration

Liquid is flowing into a small tank which has a leak. Initially the tank is empty and, t minutes later, the volume of liquid in the tank is V cm^3. The liquid is flowing into the tank at a constant rate of 80 cm^3 per minute. Because of the leak, liquid is being lost from the tank at a rate which, at any instant, is equal to kV cm^3 per minute where k is a positive constant.

(i) Write down a differential equation describing this situation and solve it to show that

$$V = \frac{1}{k}(80 - 80e^{-kt}).$$

[7]

(ii) It is observed that $V = 500$ when $t = 15$, so that k satisfies the equation

$$k = \frac{4 - 4e^{-15k}}{25}.$$

Use an iterative formula, based on this equation, to find the value of k correct to 2 significant figures. Use an initial value of $k = 0.1$ and show the result of each iteration to 4 significant figures. [3]

(iii) Determine how much liquid there is in the tank 20 minutes after the liquid started flowing, and state what happens to the volume of liquid in the tank after a long time. [2]

Examiner comment

> "Those that they found difficult were Question 4 (ii) (a moduli equation requiring substitution), Question 6 (vector equation of a plane), Question 9 (ii) (b) (integration using substitution of an earlier derived relation) and Question 10 (i) (establishing a differential equation and solving it)."

Mark scheme

(i)

State $\frac{dV}{dt} = 80 - kV$ **B1**

Correctly separate variables and attempt integration of one side **M1**

Obtain $a\ln(80 - kV) = t$ or equivalent **M1***

Obtain $-\frac{1}{k}\ln(80 - kV) = t$ or equivalent **A1**

Use $t = 0$ and $V = 0$ to find constant of integration or as limits **M1 dep***

Obtain $-\frac{1}{k}\ln(80-kV) = t - \frac{1}{k}\ln 80$ or equivalent **A1**

Obtain given answer $V = \frac{1}{k}(80 - 80e^{-kt})$ correctly **A1**

Total: 7

(ii)

Use iterative formula correctly at least once **M1**

Obtain final answer **0.14 A1**

Show sufficient iterations to 4 s.f. to justify answer to 2 s.f. or show a sign change in the interval $(0.135, 0.145)$ **A1**

Total: 3

(iii)

State a value between **530** and **540 cm^3** inclusive **B1**

State or imply that volume approaches **569 cm^3**, allowing any value between 567 and 571 inclusive **B1**

Total: 2

Q285 Oct/Nov 2013 | Paper 3 (variant 32) | Question 6

Topics: Differential Equations

A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant.

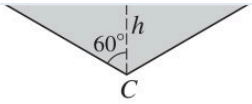
Total: 4

(ii) Solve the differential equation given in part **(i)** and obtain an expression for t in terms of h and H.

Total: 6

(iii) Find the time at which the depth reaches $\frac{1}{2}H$.

Total: 1



A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{\mathrm{d}h}{\mathrm{d}t} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant. [4]

(ii) Solve the differential equation given in part **(i)** and obtain an expression for t in terms of h and H. [6]

(iii) Find the time at which the depth reaches $\frac{1}{2}H$. [1]

Examiner comment

> "Question 6 Part (i) was poorly answered and very few correctly argued proofs were seen."

Mark scheme

(i) B1 State or imply $AB = 2r\cos\theta$ or $AB^2 = 2r^2 - 2r^2\cos(\pi - 2\theta)$

M1 Use correct formula to express the area of sector ABC in terms of r and θ

M1 Use correct area formulae to express the area of a segment in terms of r and θ

A1 State a correct equation in r and θ in any form

A1 Obtain the given answer

[SR: If the complete equation is approached by adding two sectors to the shaded area above BO and OC give the first **M1** as on the scheme, and the second **M1** for using correct area formulae for a triangle AOB or AOC, and a sector AOB or AOC.]

(ii) M1 Use the iterative formula correctly at least once

A1 Obtain final answer 0.95

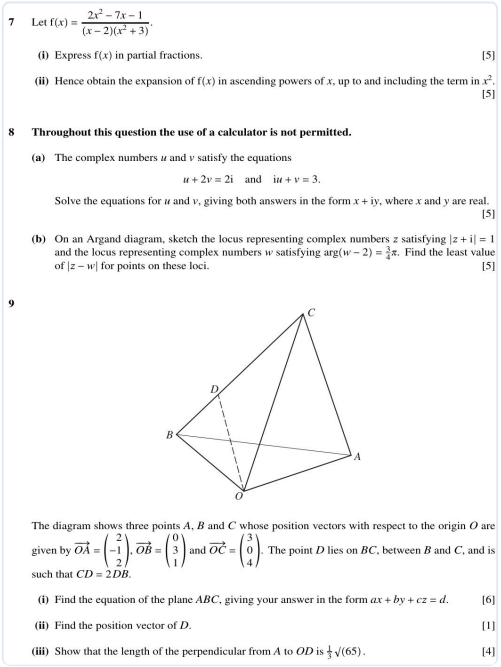
A1 Show sufficient iterations to 4 d.p. to justify 0.95 to 2 d.p., or show there is a sign change in the interval $(0.945, 0.955)$

Total: 3

Q286 Oct/Nov 2013 | Paper 3 (variant 32) | Question 10

Topics: Differential Equations

10



7 Let $f(x) = \frac{2x^2 - 7x - 1}{(x-2)(x^2+3)}$.

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

8 **Throughout this question the use of a calculator is not permitted.**

(a) The complex numbers u and v satisfy the equations

$$u + 2v = 2i \quad \text{and} \quad iu + v = 3.$$

Solve the equations for u and v, giving both answers in the form $x + iy$, where x and y are real. [5]

(b) On an Argand diagram, sketch the locus representing complex numbers z satisfying $|z + i| = 1$ and the locus representing complex numbers w satisfying $\arg(w - 2) = \frac{3}{4}\pi$. Find the least value of $|z - w|$ for points on these loci. [5]

9

The diagram shows three points A, B and C whose position vectors with respect to the origin O are given by $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$. The point D lies on BC, between B and C, and is such that $CD = 2DB$.

(i) Find the equation of the plane ABC, giving your answer in the form $ax + by + cz = d$. [6]

(ii) Find the position vector of D. [1]

(iii) Show that the length of the perpendicular from A to OD is $\frac{1}{3}\sqrt{(65)}$. [4]

A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant. **[4]**

(ii) Solve the differential equation given in part (i) and obtain an expression for t in terms of h and H. **[6]**

(iii) Find the time at which the depth reaches $\frac{1}{4}H$. **[1]**

Examiner comment

> "Question 10 (i) Most candidates found this to be a difficult problem."

> "(iii) There were very few correct solutions."

Mark scheme

10 (i) State or imply $V = \pi h^3$ **B1**

State or imply $\frac{dV}{dt} = -k\sqrt{h}$ **B1**

Use $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$, or equivalent **M1**

Obtain the given equation **A1 [4]**

[The **M1** is only available if $\frac{dV}{dh}$ is in terms of h and has been obtained by a correct method.]

[Allow **B1** for $\frac{dV}{dt} = k\sqrt{h}$ but withhold the final **A1** until the polarity of the constant $\frac{k}{3\pi}$ has been justified.]

(ii) Separate variables and integrate at least one side **M1**

Obtain terms $\frac{2}{5}h^2$ and $-At$, or equivalent **A1**

Use $t = 0, h = H$ in a solution containing terms of the form ah^2 and $bt + c$ **M1**

Use $t = 60, h = 0$ in a solution containing terms of the form ah^2 and $bt + c$ **M1**

Obtain a correct solution in any form, e.g. $\frac{2}{5}h^2 = \frac{1}{150}H^2t + \frac{2}{5}H^2$ **A1**

(iii) Obtain final answer $t = 60\left[1 - \left(\frac{h}{H}\right)^2\right]$, or equivalent **A1 [6]**

(iii) Substitute $h = \frac{1}{2}H$ and obtain answer $t = 49.4$ **B1 [1]**

Q287 May/June 2012 | Paper 3 (variant 31) | Question 1

Topics: Algebra

Solve the equation $|4 - 2x| = 10$, giving your answer correct to 3 significant figures. [3]

Examiner comment

> "Question 1 This question proved difficult for many since, instead of removing the modulus sign and obtaining the equations 2x = 14 and 2x =−6, they tried to expand (4 - 2x)2."

Mark scheme

State or imply $4 - 2x = -10$ and $4 - 2x = 10$ **B1**

Use correct method for solving equation of form $|2x + a| = b$ **M1**

Obtain 3.81 **A1**

Total: 3

Q288 May/June 2012 | Paper 3 (variant 31) | Question 3

Topics: Algebra

The polynomial $p(x)$ is defined by

$$p(x) = x^3 - 3ax + 4a,$$

where a is a constant.

(i) Given that $(x - 2)$ is a factor of $p(x)$, find the value of a. [2]

(ii) When a has this value,

(a) factorise $p(x)$ completely, [3]

(b) find all the roots of the equation $p(x^2) = 0$. [2]

Examiner comment

> "(ii) (a) Again most candidates were successful, although some did not progress beyond (x – 2)(x2 + 2x – 8)."

Mark scheme

(i)

Substitute $x = 2$ and equate to zero, or divide by $x - 2$ and equate constant remainder to zero, or equivalent **M1**

Obtain $a = 4$ **A1**

Total: 2

(ii)(a)

Find further quadratic or linear factor by division, inspection or factor theorem or equivalent **M1**

Obtain $x^2 + 2x - 8$ or $x + 4$ **A1**

State $(x - 2)^2(x + 4)$ or equivalent **A1**

Total: 3

(ii)(b)

State any two of the four or six roots **B1**

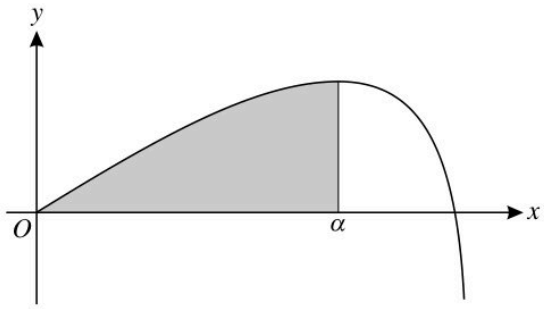
State all roots $\pm\sqrt{2}$, $\pm 2i$, provided two are purely imaginary **B1**

Total: 2

Q289 May/June 2012 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Differentiation, Integration

5



The diagram shows the curve

$$y = 8\sin\tfrac{1}{2}x - \tan\tfrac{1}{2}x$$

for $0 \leqslant x < \pi$. The x-coordinate of the maximum point is α and the shaded region is enclosed by the curve and the lines $x = \alpha$ and $y = 0$.

(i) Show that $\alpha = \frac{2}{3}\pi$. [3]

(ii) Find the exact value of the area of the shaded region. [4]

The diagram shows the curve

$$y = 8\sin\frac{1}{2}x - \tan\frac{1}{2}x$$

for $0 \leq x < \pi$. The x-coordinate of the maximum point is α and the shaded region is enclosed by the curve and the lines $x = \alpha$ and $y = 0$.

(i) Show that $\alpha = \frac{2}{3}\pi$. [3]

(ii) Find the exact value of the area of the shaded region. [4]

Examiner comment

> "Question 5 1 1 (i) Differentiating tan x proved difficult, although those that managed to reach the form bsec2 x 2 2 1 were able to continue to solve for cos x."

> "2 1 (ii) Whilst a few candidates managed to obtain −16cos xusually the sign or the coefficient was 2 1 incorrect, it was rare to see 2ln(cos x)."

Mark scheme

(i)

Differentiate to obtain $4\cos\frac{1}{2}x - \frac{1}{2}\sec^2\frac{1}{2}x$ **B1**

Equate to zero and find value of $\cos\frac{1}{2}x$ **M1**

Obtain $\cos\frac{1}{2}x = \frac{1}{2}$ and confirm $\alpha = \frac{2\pi}{3}$ **A1**

Total: 3

(ii)

Integrate to obtain $-16\cos\frac{1}{2}x$ **B1**

Obtain $+2\ln\left(\cos\frac{1}{2}x\right)$ or equivalent **B1**

Use limits 0 and $\frac{2\pi}{3}$ in $a\cos\frac{1}{2}x + b\ln\left(\cos\frac{1}{2}x\right)$ **M1**

Obtain $8 + 2\ln\frac{1}{2}$ or exact equivalent **A1**

Total: 4

Q290 May/June 2012 | Paper 3 (variant 31) | Question 6

Topics: Differentiation

The equation of a curve is $3x^2 - 4xy + y^2 = 45$.

(i) Find the gradient of the curve at the point $(2, -3)$. [4]

(ii) Show that there are no points on the curve at which the gradient is 1. [3]

Examiner comment

> "Another common error was to still see the constant term of 45 present following differentiation."

Mark scheme

(i)

Obtain $2y\frac{dy}{dx}$ as derivative of y^2 **B1**

Obtain $-4y - 4x\frac{dy}{dx}$ as derivative of $-4xy$ **B1**

Substitute $x = 2$ and $y = -3$ and find value of $\frac{dy}{dx}$, dependent on at least one B1 being earned and $\frac{d}{dx}(45) = 0$ **M1**

Obtain $\frac{12}{7}$ or equivalent **A1**

Total: 4

(ii)

Substitute $\frac{dy}{dx} = 1$ in an expression involving $\frac{dy}{dx}$, x and y and obtain $ay = bx$ **M1**

Obtain $y = x$ or equivalent **A1**

Use $y = x$ in original equation and demonstrate contradiction **A1**

Total: 3

Q291 May/June 2012 | Paper 3 (variant 31) | Question 8

Topics: none of the listed topics

The point P has coordinates $(-1, 4, 11)$ and the line l has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

(i) Find the perpendicular distance from P to l. [4]

(ii) Find the equation of the plane which contains P and l, giving your answer in the form $ax + by + cz = d$, where a, b, c and d are integers. [5]

Examiner comment

> "Question 8 The attempts at this question, apart from the occasional fully correct solution for both parts, were frequently incorrect."

Mark scheme

(i)

Obtain $\pm \begin{pmatrix} 2 \\ -1 \\ -15 \end{pmatrix}$ for vector PA, where A is point on line, or equivalent **B1**

Use scalar product to find cosine of angle between PA and line **M1**

Obtain $\frac{42}{\sqrt{230}\sqrt{14}}$ or equivalent **A1**

Use trigonometry to obtain $\sqrt{104}$ or 10.2 or equivalent **A1**

Alternative methods using foot of perpendicular, vector product, Pythagoras, or triangle area are accepted with equivalent marks.

Total: 4

(ii)

Use scalar product to obtain a relevant equation in a, b, c, e.g. $2a + b + 3c = 0$ or $2a - b - 15c = 0$ **M1**

State two correct equations in a, b and c **A1**

Solve simultaneous equations to obtain one ratio **M1**

Obtain $a : b : c = -3 : 9 : -1$ or equivalent **A1**

Obtain equation $-3x + 9y - z = 28$ or equivalent **A1**

Alternative methods using vector product or two-parameter plane equation are accepted.

Total: 5

Q292 May/June 2012 | Paper 3 (variant 31) | Question 9

Topics: Algebra, Integration

By first expressing $\frac{4x^2+5x+3}{2x^2+5x+2}$ in partial fractions, show that

$$\int_0^4 \frac{4x^2+5x+3}{2x^2+5x+2}\,dx = 8 - \ln 9.$$

[10]

Examiner comment

> "Question 9 B C Many candidates failed to state a correct general form such asA+ + ."

Mark scheme

State or imply form $A + \frac{B}{2x+1} + \frac{C}{x+2}$ **B1**

State or obtain $A = 2$ **B1**

Use correct method for finding B or C **M1**

Obtain $B = 1$ **A1**

Obtain $C = -3$ **A1**

Obtain $2x + \frac{1}{2}\ln(2x+1) - 3\ln(x+2)$ **B3** Deduct B1 for each error or omission.

Substitute limits in expression containing $a\ln(2x+1) + b\ln(x+2)$ **M1**

Show full and exact working to confirm that $8 + \frac{1}{2}\ln 9 - 3\ln 6 + 3\ln 2$, or an equivalent expression, simplifies to given result $8 - \ln 9$ **A1**

Total: 10

Q293 Oct/Nov 2012 | Paper 3 (variant 31) | Question 2

Topics: Indices, Logarithms & Exponents

Solve the equation

$$5^{x-1} = 5^x - 5,$$

giving your answer correct to 3 significant figures. [4]

Examiner comment

> "Common errors included taking ln(5x – 5) to be x In 5 – In 5 and, after letting y = 5X, taking 5x–1 to be y–1."

Mark scheme

EITHER Use laws of indices correctly and solve for 5^x or for 5^{-x} or for 5^{x-1} **M1**

$$\frac{5}{1 - \frac{1}{\sqrt{5}}}$$

Obtain 5^x or for 5^{-x} or for 5^{x-1} in any correct form, e.g. $5^x =$ **A1**

Use correct method for solving $5^y = a$, or $5^{-x} = a$, or $5^{x-1} = a$, where $a \geq 0$ **M1**

Obtain answer $x = 1.14$ **A1**

OR Use an appropriate iterative formula, e.g. $x_{n+1} =$, correctly, at least once **M1**

Obtain answer **1.14 A1**

Show sufficient iterations to at least 3 d.p. to justify **1.14** to 2 d.p., or show there is a sign change in the interval $(1.135, 1.145)$ **A1**

Show there is no other root **A1** [4]

[For the solution $x = 1.14$ with no relevant working give **B1**, and a further **B1** if **1.14** is shown to be the only solution.]

Q294 Oct/Nov 2012 | Paper 3 (variant 31) | Question 5

Topics: Trigonometry, Differentiation, Integration

(i) By differentiating $\frac{1}{\cos x}$, show that if $y = \sec x$ then $\frac{dy}{dx} = \sec x \tan x$. [2]

(ii) Show that

$$\frac{1}{\sec x - \tan x} \equiv \sec x + \tan x.$$

[1]

(iii) Deduce that

$$\frac{1}{(\sec x - \tan x)^2} = 2\sec^2 x - 1 + 2\sec x \tan x.$$

[2]

(iv) Hence show that

$$\int_0^{\frac{\pi}{4}} \frac{1}{(\sec x - \tan x)^2}\, dx = \frac{1}{4}(8\sqrt{2} - \pi).$$

[3]

Examiner comment

> "However, the majority of candidates were challenged by this question."

> "Lack of an overall strategy was a 1–sinx problem in part (iii) and in part (iv) most candidates did not use the result of part (i) to write down the integral of sec X tan x."

Mark scheme

(i) Use correct quotient or chain rule **A1**

Obtain the given answer correctly having shown sufficient working **A1**

(ii) Use a valid method, e.g. multiply numerator and denominator by $\sec x + \tan x$, and a version of Pythagoras to justify the given identity **B1** [1]

(iii) Substitute, expand $(\sec x + \tan x)^2$ and use Pythagoras once **M1**

Obtain given identity **A1** [2]

(iv) Obtain integral $2\tan x - x + 2\sec x$ **B1**

Use correct limits correctly. In an expression of the form $a\tan x + bx + c\sec x$, or equivalent, where $abc \neq 0$ **M1**

Obtain the given answer correctly **A1** [3]

Q295 Oct/Nov 2012 | Paper 3 (variant 31) | Question 6

Topics: Differential Equations

The variables x and y are related by the differential equation

$$x\frac{dy}{dx} = 1 - y^2.$$

When $x = 2$, $y = 0$. Solve the differential equation, obtaining an expression for y in terms of x. [8]

Examiner comment

> "Question 6 This question was poorly answered by most candidates."

> "1–y2 These candidates usually obtained a correct solution in some form, but the removal of logarithms involved in making y the subject proved difficult for some."

Mark scheme

Separate variables correctly and attempt integration of one side **B1**

Obtain term $\ln x$ **B1**

State or imply and use a relevant method to find A or B **M1**

Obtain $A = \frac{1}{2}$, $B = \frac{1}{2}$ **A1**

Integrate and obtain $-\frac{1}{2}\ln(1-y) + \frac{1}{2}\ln(1+y)$, or equivalent **A1**

[If the integral is directly stated as $k_1 \ln$ or $k_2 \ln$ give **M1**, and then **A2** for $k_1 = \frac{1}{2}$ or $k_2 = -\frac{1}{2}$]

Evaluate a constant, or use limits $x = 2, y = 0$ in a solution containing terms $a\ln x$, $b\ln(1-y)$ and $c\ln(1+y)$, where $abc \neq 0$ **M1**

[This M mark is not available if the integral of $1/(1-y^2)$ is initially taken to be of the form $k\ln(1-y^2)$]

Obtain solution in any correct form, e.g. $\frac{1}{2}\ln\frac{x}{2} = \ln x - \ln 2$ **A1**

Rearrange and obtain $y =$, or equivalent, free of logarithms **A1** [8]

Q296 Oct/Nov 2012 | Paper 3 (variant 31) | Question 7

Topics: Indices, Logarithms & Exponents, Differentiation

The equation of a curve is $\ln(xy) - y^3 = 1$.

(i) Show that

$$\frac{dy}{dx} = \frac{y}{x(3y^3 - 1)}.$$

[4]

(ii) Find the coordinates of the point where the tangent to the curve is parallel to the y-axis, giving each coordinate correct to 3 significant figures. [4]

Examiner comment

> "Part (ii) on the other hand was found to be difficult."

Mark scheme

(i) EITHER: State or imply $\frac{x}{xy}\frac{dy}{dx}$ as derivative of $\ln xy$, or equivalent **B1**

State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3, or equivalent **B1**

Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

OR Obtain $xy = \exp(1 + y^3)$ and state or imply $y' + x\frac{dy}{dx}$ as derivative of xy **B1**

State or imply $3y^2\frac{dy}{dx}\exp(1 + y^3)$ as derivative of $(1 + y^3)$ **B1**

Equate derivatives and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1** [4]

(ii) Equate denominator to zero and solve for y **M1***

Obtain $y = 0.693$ only **A1**

Substitute found value in the equation and solve for x **M1**(dep*)

Obtain $x = 5.47$ only **A1** [4]

Q297 Oct/Nov 2012 | Paper 3 (variant 31) | Question 10

Topics: none of the listed topics

10 With respect to the origin O, the points A, B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 2 \\ -1 \\ 7 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix}$$

The plane m is parallel to $\overrightarrow{OC}$ and contains A and B.

(i) Find the equation of m, giving your answer in the form $ax + by + cz = d$. [6]

(ii) Find the length of the perpendicular from C to the line through A and B. [5]

Examiner comment

> "Question 10 In general this question was poorly answered."

> "Yet many candidates did not grasp the geometry of the situation and worked with irrelevant pairs of direction vectors such as AB and AC, or OA and OB."

Mark scheme

10 (i) EITHER Use scalar product of relevant vectors, or subtract point equations to form two equations in a, b, c, e.g. $a - 5 - 4 = 0$ and $a = 3c - 0$ **M1**

State two correct equations in a, b, c **A1**

OR 1: Obtain $b \cdot (-3) : 0, 1$, or equivalent **A1**

OR 1: Attempt to calculate vector product of relevant vectors, e.g. $(1 - 5 - 4) \times (1 - 1 - 3k)$ **M2***

OR 1: Attempt to form 2-parameter equation of the plane with relevant vectors, e.g. $r = 3i - 2j + 4k + s(1 - 5 - 4) + t(1 - 1 - 3k)$ **M2***

OR 2: Attempt to calculate vector product of relevant vectors, e.g. $r = 3i - 2j + 4k$ State 3 equations in x, y, z, and i **A1**

OR 2: Eliminate 2 and u **A1**

OR 2: Obtain equation $3x + z = 13$, or equivalent **M1 (dep*) A1**

Obtain answer $3x + z = 13$, or equivalent **A1 [6]**

(ii) EITHER Find $\overrightarrow{CP}$ for a point P on AB with a parameter t, e.g. $2i + 3j + 7k + t(4i + j + 3k)$ **B1** *

Either: Equate scalar product $\overrightarrow{CP} \cdot \overrightarrow{AB}$ to zero and form an equation in t **M1**

Or 1: Use Pythagoras in triangle $CPA (or CPB)$ and form an equation in t **M1**

Or 2: Use Pythagoras in triangle ABC to find the length of CP **M1**

Solve and obtain correct value of t, e.g. $t = -2$ **A1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 1: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Using a relevant scalar product find the cosine of $\angle CAB$ (or $\angle CBA$) **M1**

Obtain correct value of $\cos A$ or $\cos B$ **A1 [6]**

OR 2: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Using a relevant scalar product find the length of the projection $\overrightarrow{AC}$ (or BC) on $\overrightarrow{AB}$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 3: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Calculate their vector product, e.g. $(-21-3j-7k)\cdot(-4i-j-3k)$ **M1**

Obtain correct product, e.g. $-2i+13j-5k$ **A1**

Divide modulus of the product by the modulus of $\overrightarrow{AB}$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 4: State two of $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$ in component form **B1** *

Use cosine formula in triangle ABC to find $\cos A$ or $\cos B$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1 [5]**

Q298 May/June 2011 | Paper 3 (variant 31) | Question 3

Topics: none of the listed topics

Points A and B have coordinates $(-1, 2, 5)$ and $(2, -2, 11)$ respectively. The plane p passes through B and is perpendicular to AB.

(i) Find an equation of p, giving your answer in the form $ax+by+cz=d$. [3]

(ii) Find the acute angle between p and the y-axis. [4]

Examiner comment

> "Question 3 (i) This proved a difficult question as many candidates did not realise that the vector AB was the normal to the plane and point B was in the plane."

> "(ii) Few candidates could give a correct vector representing the y-axis and many made errors in the evaluation of the scalar product of the vectors OA and OB."

Mark scheme

(i)

Obtain $\begin{pmatrix} 3 \\ -4 \\ 6 \end{pmatrix}$ or its negative as normal to plane **B1**

Form equation of p as $3x - 4y + 6z = k$ or $-3x + 4y - 6z = k$ and use relevant point to find k **M1**

Obtain $3x - 4y + 6z = 80$ or $-3x + 4y - 6z = -80$ **A1**

Total: 3

(ii)

State the direction vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ or equivalent **B1**

Carry out correct process for finding scalar product of two relevant vectors **M1**

Use correct complete process with moduli and scalar product and evaluate $\sin^{-1}$ or $\cos^{-1}$ of result **M1**

Obtain $30.8°$ or 0.538 radians **A1**

Total: 4

Q299 May/June 2011 | Paper 3 (variant 31) | Question 4

Topics: Algebra, Indices, Logarithms & Exponents

The polynomial $f(x)$ is defined by

$$f(x) = 12x^3 + 25x^2 - 4x - 12.$$

(i) Show that $f(-2) = 0$ and factorise $f(x)$ completely. [4]

(ii) Given that

$$12 \times 27^y + 25 \times 9^y - 4 \times 3^y - 12 = 0,$$

state the value of 3^y and hence find y correct to 3 significant figures. [3]

Examiner comment

> "(ii) Few candidates were able to spot the relationship between the roots of f(x) = 0 and the equation in 3 this section."

Mark scheme

(i)

Verify that $-96 + 100 + 8 - 12 = 0$ **B1**

Attempt to find quadratic factor by division by $(x + 2)$, reaching a partial quotient $12x^2 + kx$, inspection or use of an identity **M1**

Obtain $12x^2 + x - 6$ **A1**

State $(x + 2)(4x + 3)(3x - 2)$ **A1**

Total: 4

(ii)

State $3^y = \frac{2}{3}$ and no other value **B1**

Use correct method for finding y from equation of form $3^y = k$, where $k > 0$ **M1**

Obtain -0.369 and no other value **A1**

Total: 3

Q300 May/June 2011 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry, Differential Equations, Numerical Solutions/Iteration

10 The number of birds of a certain species in a forested region is recorded over several ye... t years, the number of birds is N, where N is treated as a continuous variable. The varia... number of birds is modelled by

$$\frac{dN}{dt} = \frac{N(1800 - N)}{3600}.$$

It is given that $N = 300$ when $t = 0$.

(i) Find an expression for N in terms of t. [9]

(ii) According to the model, how many birds will there be after a long time? [1]

The diagram shows a circle with centre O and radius 10 cm. The chord AB divides the circle into two regions whose areas are in the ratio $1 : 4$ and it is required to find the length of AB. The angle AOB is θ radians.

(i) Show that $\theta = \frac{2}{5}\pi + \sin\theta$. [3]

(ii) Showing all your working, use an iterative formula, based on the equation in part (i), with an initial value of 2.1, to find θ correct to 2 decimal places. Hence find the length of AB in centimetres correct to 1 decimal place. [5]

Examiner comment

> "(ii) The iteration work was usually sound, but many candidates failed to give the final answer to the requested number of decimal places once convergence had been established."

Mark scheme

(i)

State or imply area of segment is $\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$ or $50\theta - 50\sin\theta$ **B1**

Attempt to form equation from area of segment $= \frac{1}{5}$ of area of circle, or equivalent **M1**

Confirm given result $\theta = \frac{2\pi}{5} + \sin\theta$ **A1**

Total: 3

(ii)

Use iterative formula correctly at least once **M1**

Obtain value for θ of **2.11 A1**

Show sufficient iterations to justify value of θ or show sign change in interval $(2.105, 2.115)$ **A1**

Use correct trigonometry to find an expression for the length of AB **M1** e.g. $20 \sin 1.055$ or $\sqrt{200 - 200\cos 2.11}$

Hence **17.4 A1**

Iterations: $2.1 \to 2.1198 \to 2.1097 \to 2.1149 \to 2.1122$

Total: 5

Q301 May/June 2011 | Paper 3 (variant 31) | Question 9

Topics: Trigonometry, Integration

(i) Prove the identity

$$\cos 4\theta + 4\cos 2\theta \equiv 8\cos^4\theta - 3.$$

[4]

(ii) Hence

(a) solve the equation $\cos 4\theta + 4\cos 2\theta = 1$ for $-\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi$, [3]

(b) find the exact value of $\int_0^{\frac{1}{4}\pi} \cos^4\theta \, d\theta$. [3]

Examiner comment

> "(ii) Whilst many candidates knew to use their result from part (i) in part (ii) they were often unable to 1 successfully convert 8cos4 θ – 3 = 1 to cos4θ= ."

Mark scheme

(i)

Express $\cos 4\theta$ as $2\cos^2 2\theta - 1$ or $\cos^2 2\theta - \sin^2 2\theta$ or $1 - 2\sin^2 2\theta$ **B1**

Express $\cos 4\theta$ in terms of $\cos\theta$ **M1**

Obtain $8\cos^4\theta - 8\cos^2\theta + 1$ **A1**

Use $\cos 2\theta = 2\cos^2\theta - 1$ to obtain given answer $8\cos^4\theta - 3$ **A1** AG

Total: 4

(ii)(a)

State or imply $\cos^4\theta = \frac{1}{2}$ **B1**

Obtain 0.572 **B1**

Obtain -0.572 **B1**

Total: 3

(ii)(b)

Integrate and obtain form $k_1\theta + k_2 \sin 4\theta + k_3 \sin 2\theta$ **M1**

Obtain $\frac{3}{8}\theta + \frac{1}{32}\sin 4\theta + \frac{1}{4}\sin 2\theta$ **A1**

Obtain $\frac{3\pi}{32} + \frac{1}{4}$ following completely correct work **A1**

Total: 3

Q302 May/June 2010 | Paper 3 (variant 31) | Question 1

Topics: Algebra

Solve the inequality $|x + 3a| > 2|x - 2a|$, where a is a positive constant. [4]

Examiner comment

> "Only a a few candidates who reached x = and x = 7a were able to combine them into the correct final solution."

Mark scheme

EITHER: State or imply non-modular inequality $(x + 3a)^2 > 4(x - 2a)^2$, or corresponding quadratic equation, or pair of linear equations $x + 3a = \pm 2(x - 2a)$ **B1**

Make reasonable solution attempt at a 3-term quadratic, or solve two linear equations **M1**

Obtain critical values $x = \frac{1}{3}a$ and $x = 7a$ **A1**

State answer $\frac{1}{3}a < x < 7a$ **A1**

OR: Obtain the critical value $x = 7a$ from a graphical method, or by inspection, or by solving a linear equation or inequality **B1**

Obtain the critical value $x = \frac{1}{3}a$ similarly **B2**

State answer $\frac{1}{3}a < x < 7a$ **B1**

Do not condone $\leq$ for $<$. Accept 0.33 for $\frac{1}{3}$.

Total: 4

Q303 May/June 2010 | Paper 3 (variant 31) | Question 4

Topics: Trigonometry, Integration

(i) Using the expansions of $\cos(3x - x)$ and $\cos(3x + x)$, prove that

$$\frac{1}{2}(\cos 2x - \cos 4x) \equiv \sin 3x \sin x.$$

[3]

(ii) Hence show that

$$\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \sin 3x \sin x \, dx = \frac{1}{8}\sqrt{3}.$$

[3]

Examiner comment

> "Question 4 Most candidates could state the required expansions, but too many then did not realise that this meant that they had the cos 2x and cos 4x required for the next part."

Mark scheme

(i)

State correct expansion of $\cos(3x - x)$ or $\cos(3x + x)$ **B1**

Substitute expansions in $\frac{1}{2}(\cos 2x - \cos 4x)$, or equivalent **M1**

Simplify and obtain the given identity correctly **A1**

Total: 3

(ii)

Obtain integral $\frac{1}{4}\sin 2x - \frac{1}{8}\sin 4x$ **B1**

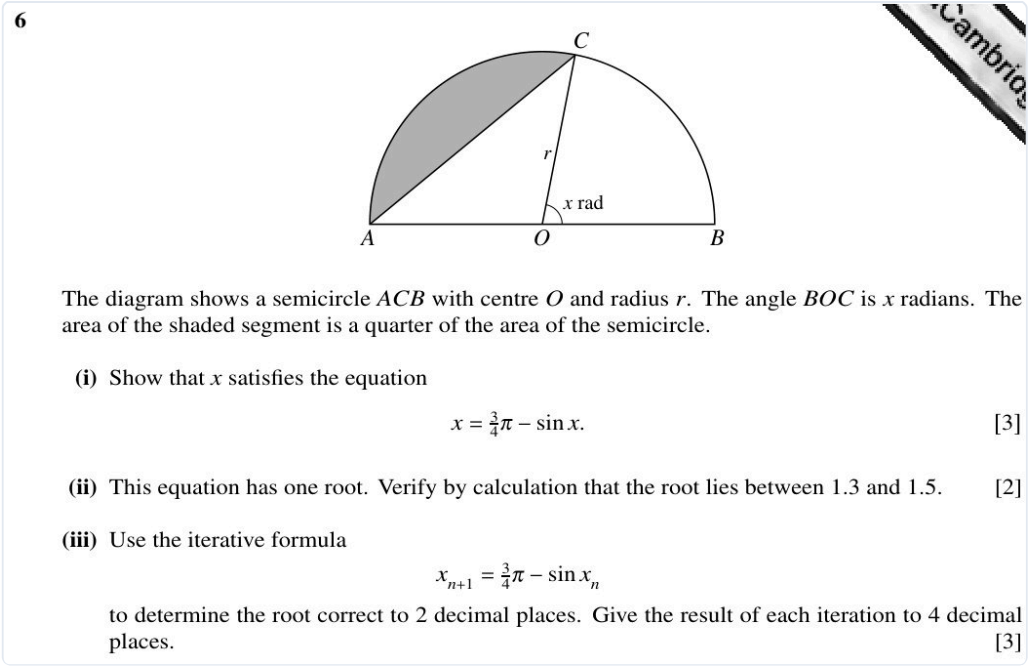
Substitute limits correctly in an integral of the form $a \sin 2x + b \sin 4x$ **M1**

Obtain given answer following full, correct and exact working **A1**

Total: 3

Q304 May/June 2010 | Paper 3 (variant 31) | Question 6

Topics: Trigonometry, Numerical Solutions/Iteration



The diagram shows a semicircle ACB with centre O and radius r. The angle BOC is x radians. The area of the shaded segment is a quarter of the area of the semicircle.

(i) Show that x satisfies the equation

$$x = \frac{3}{4}\pi - \sin x.$$

[3]

(ii) This equation has one root. Verify by calculation that the root lies between 1.3 and 1.5. [2]

(iii) Use the iterative formula

$$x_{n+1} = \frac{3}{4}\pi - \sin x_n$$

to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "Furthermore, relating a θ in their sector formula to something usually involving (x – π), which was the approach that was usually adopted, proved difficult."

Mark scheme

(i)

Using the formulae $\frac{1}{2}r^2\theta$ and $\frac{1}{2}r^2\sin\theta$, or equivalent, form an equation **M1**

Obtain a correct equation in r and x and/or $x/2$ in any form **A1**

Obtain the given equation correctly **A1**

Total: 3

(ii)

Consider the sign of $x - \frac{3}{4}\pi + \sin x$ at $x = 1.3$ and $x = 1.5$, or equivalent **M1**

Complete the argument with correct calculations **A1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer 1.38 **A1**

Show sufficient iterations to at least 4 d.p. to justify its accuracy to 2 d.p., or show there is a sign change in the interval $(1.375, 1.385)$ **A1**

Total: 3

Q305 May/June 2010 | Paper 3 (variant 31) | Question 7

Topics: Complex Numbers

The complex number $2 + 2i$ is denoted by u.

(i) Find the modulus and argument of u. [2]

(ii) Sketch an Argand diagram showing the points representing the complex numbers 1, i and u. Shade the region whose points represent the complex numbers z which satisfy both the inequalities $|z - 1| \leq |z - i|$ and $|z - u| \leq 1$. [4]

(iii) Using your diagram, calculate the value of $|z|$ for the point in this region for which $\arg z$ is least. [3]

Examiner comment

> "Part (iii) of this question was something that candidates found difficult."

Mark scheme

(i)

Obtain modulus $\sqrt{8}$ **B1**

Obtain argument $\frac{1}{4}\pi$ or $45°$ **B1**

Total: 2

(ii)

Show 1, i and u in relatively correct positions on an Argand diagram **B1**

Show the perpendicular bisector of the line joining 1 and i **B1**

Show a circle with centre u and radius 1 **B1**

Shade the correct region **B1**

Total: 4

(iii)

State or imply relevance of the appropriate tangent from O to the circle **B1 FT**

Carry out complete strategy for finding $|z|$ for the critical point **M1**

Obtain answer $\sqrt{7}$ **A1**

Total: 3

Q306 May/June 2010 | Paper 3 (variant 31) | Question 10

Topics: none of the listed topics

The lines l and m have vector equations

$$\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + s(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

and

$$\mathbf{r} = 4\mathbf{i} + 6\mathbf{j} + \mathbf{k} + t(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})$$

respectively.

(i) Show that l and m intersect. [4]

(ii) Calculate the acute angle between the lines. [3]

(iii) Find the equation of the plane containing l and m, giving your answer in the form $ax + by + cz = d$. [5]

Examiner comment

> "Part (iii) again saw many candidates fail to choose the correct vectors from which to find relevant simultaneous equations or the vector product."

Mark scheme

(i)

Express general point of l or m in component form, e.g. $(1 + s, 1 - s, 1 + 2s)$ or $(4 + 2t, 6 + 2t, 1 + t)$ **B1**

Equate at least two corresponding pairs of components and solve for s or t **M1**

Obtain $s = -1$ or $t = -2$ **A1**

Verify that all three component equations are satisfied **A1**

Total: 4

(ii)

Carry out correct process for evaluating the scalar product of the direction vectors of l and m **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result **M1**

Obtain answer $74.2°$ or 1.30 radians **A1**

Total: 3

(iii)

EITHER: Use scalar product to obtain $a - b + 2c = 0$ and $2a + 2b + c = 0$ **B1**

Solve and obtain one ratio, e.g. $a : b$ **M1**

Obtain $a : b : c = 5 : -3 : -4$, or equivalent **A1**

Substitute coordinates of a relevant point and values for a, b and c in general equation of plane and evaluate d **M1**

Obtain answer $5x - 3y - 4z = -2$, or equivalent **A1**

OR: Attempt to calculate vector product of direction vectors of l and m **M1**

Obtain two correct components of the product **A1**

Obtain correct product, e.g. $-5\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ **A1**

Form a plane equation and use coordinates of a relevant point to calculate d **M1**

Obtain answer $-5x + 3y + 4z = 2$, or equivalent **A1**

Total: 5

Q307 Oct/Nov 2010 | Paper 3 (variant 31) | Question 2

Topics: Indices, Logarithms & Exponents

Solve the equation

$$\ln(1 + x^2) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. **[4]**

Examiner comment

> "A common error was to start by replacing ln(1 + x2) with ln 1 + ln x2, and sometimes a further error led to a quadratic in In x."

Mark scheme

M1 Use law for the logarithm of a power, a quotient, or a product correctly at least once

M1 Use $\ln e = 1$ or $e = \exp(1)$

A1 Obtain a correct equation free of logarithms, e.g. $1 + x^2 = ex^2$

A1 Solve and obtain answer $x = 0.763$ only

Total: 4

[For the solution $x = 0.763$ with no relevant working give B1, and a further B1 if 0.763 is shown to be the only root.]

[Treat the use of logarithms to base 10 with answer 0.333 only, as a misread.]

[SR: Allow iteration, giving B1 for an appropriate formula, e.g. $x_{n+1} = \exp((\ln(1 + x_n^2) - 1)/2)$, **M1** for using it correctly once, **A1** for 0.763, and **A1** for showing the equation has no other root but 0.763.]

Q308 Oct/Nov 2010 | Paper 3 (variant 31) | Question 3

Topics: Trigonometry

Solve the equation

$$\cos(\theta + 60^\circ) = 2\sin\theta,$$

giving all solutions in the interval $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Examiner comment

> "The most common errors arose in the algebraic work leading to the value of tan θ."

Mark scheme

M1 Attempt use of $\cos(A + B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$

M1 Use trig formula to obtain an equation in $\tan\theta$ (or $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain $\tan\theta = 1/(4 + \sqrt{3})$ or equivalent (or find $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain answer $\theta = 9.9^\circ$

A1 Obtain $\theta = 189.9^\circ$, and no others in the given interval

Total: 5

[Ignore answers outside the given interval. Treat answers in radians as a misread (0.173, 3.31).]

[The other solution methods are *via* $\cos\theta = \pm(4 + \sqrt{3})/\sqrt{1 + (4 + \sqrt{3})^2}$ or $\sin\theta = \pm 1/\sqrt{1 + (4 + \sqrt{3})^2}$.]

Q309 Oct/Nov 2010 | Paper 3 (variant 31) | Question 5

Topics: Integration

Let $I = \displaystyle\int_0^{\pi/6} \frac{x^2}{\sqrt{4 - x^2}}\, dx$.

(i) Using the substitution $x = 2\sin\theta$, show that

$$I = \int_0^{\pi/6} 4\sin^2\theta\, d\theta. \quad **[3]**$$

(ii) Hence find the exact value of I. **[4]**

Examiner comment

> "The most common errors in faulty attempts were (a) replacing (4−4sin2θ) with 2 – 2 sinθ, (b) replacing dx by dθ, and (c) omitting a justification of the change in limits."

Mark scheme

(i)

B1 State or imply $dx = 2\cos\theta d\theta$, or $\frac{dx}{d\theta} = 2\cos\theta$, or equivalent

M1 Substitute for x and dx throughout the integral

A1 Obtain the given answer correctly, having changed limits and shown sufficient working

Total: 3

(ii)

B1 Replace integrand by $2 - 2\cos 2\theta$, or equivalent

B1√ Obtain integral $2\theta - \sin 2\theta$, or equivalent

M1 Substitute limits correctly in an integral of the form $a\theta + b\sin 2\theta$, where ab\ⓑ0

A1 Obtain answer $\frac{1}{3}\pi - \frac{\sqrt{3}}{2}$ or exact equivalent

Total: 4

[The f.t. is on integrands of the form $a + c\cos 2\theta$, where acⓑ0.]

Q310 Oct/Nov 2010 | Paper 3 (variant 31) | Question 7

Topics: none of the listed topics

With respect to the origin O, the points A and B have position vectors given by $\vec{OA} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 4\mathbf{j}$. The point P lies on the line AB and OP is perpendicular to AB.

(i) Find a vector equation for the line AB. **[1]**

(ii) Find the position vector of P. **[4]**

(iii) Find the equation of the plane which contains AB and which is perpendicular to the plane OAB, giving your answer in the form $ax + by + cz = d$. **[4]**

Examiner comment

> "In this passage of work common errors included ' −2(2 − 2λ) = −4 − 4λ ' and 1 '12λ + 2 = 0, ∴λ= '."

Mark scheme

(i) State correct equation in any form, e.g. $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \lambda(2\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})$

B1 [1]

(ii) EITHER: Equate a relevant scalar product to zero and form an equation in λ

M1 OR 1: Equate derivative of OP^2 (or OP) to zero and form an equation in λ

M1 OR 2: Use Pythagoras in OAP or OBP and form an equation in λ

M1 State a correct equation in any form

A1 Solve and obtain $\lambda = -\frac{1}{6}$ or equivalent

A1 Obtain final answer $\vec{OP} = \frac{2}{3}\mathbf{i} + \frac{5}{3}\mathbf{j} + \frac{7}{3}\mathbf{k}$, or equivalent

A1 [4]

(iii) EITHER: State or imply $\vec{OP}$ is a normal to the required plane

M1 State normal vector $2\mathbf{i} + 5\mathbf{i} + 7\mathbf{k}$, or equivalent

A1√ Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d

M1 Obtain answer $2x + 5y + 7z = 26$, or equivalent

A1 OR 1: Find a vector normal to plane AOB and calculate its vector product with a direction vector for the line AB

M1* **A1** Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d

M1(dep*) Obtain answer $2x + 5y + 7z = 26$, or equivalent

A1 OR 2: Set up and solve simultaneous equations in a, b, c derived from zero scalar products of $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ with (i) a direction vector for line AB, (ii) a normal to plane OAB

M1* **A1** Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d

M1(dep*) Obtain answer $2x + 5y + 7z = 26$, or equivalent

A1 OR 3: With Q on plane, use Pythagoras in OPQ to form an equation in x, y and z

M1* **A1√** Form a correct equation

M1(dep*) Reduce to linear form

A1 Obtain answer $2x + 5y + 7z = 26$, or equivalent

A1 OR 4: Find a vector normal to plane AOB and form a 2-parameter equation with relevant vectors, e.g., $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \lambda(2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}) + \mu(8\mathbf{i} - 6\mathbf{j} + 2\mathbf{k})

∗∗ M1 ∗∗∗∗∗ A1 ∗∗ State three correct equations in x,y,z,\lambda **and** \mu

∗∗ M1(dep∗) ∗∗ Eliminate \lambda **and** \mu **∗∗ A1 ∗∗ Obtain answer** 2x + 5y + 7z = 26$, or equivalent

A1 [4]

Q311 Oct/Nov 2010 | Paper 3 (variant 32) | Question 2

Topics: Indices, Logarithms & Exponents

Solve the equation

$$\ln(1 + x^2) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. **[4]**

Examiner comment

> "A common error was to start by replacing ln(1 + x2) with ln 1 + ln x2, and sometimes a further error led to a quadratic in 1 In x."

Mark scheme

M1 Use law for the logarithm of a power, a quotient, or a product correctly at least once

M1 Use $\ln e = 1$ or $e = \exp(1)$

A1 Obtain a correct equation free of logarithms, e.g. $1 + x^2 = ex^2$

A1 Solve and obtain answer $x = 0.763$ only

Total: 4

[For the solution $x = 0.763$ with no relevant working give B1, and a further B1 if 0.763 is shown to be the only root.]

[Treat the use of logarithms to base 10 with answer 0.333 only, as a misread.]

[SR: Allow iteration, giving B1 for an appropriate formula, e.g. $x_{n+1} = \exp((\ln(1+x_n^2)-1)/2)$, **M1** for using it correctly once, **A1** for 0.763, and **A1** for showing the equation has no other root but 0.763.]

Q312 Oct/Nov 2010 | Paper 3 (variant 32) | Question 3

Topics: Trigonometry

Solve the equation

$$\cos(\theta + 60^\circ) = 2\sin\theta,$$

giving all solutions in the interval $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Examiner comment

> "The most common errors arose in the algebraic work leading to the value of tan θ."

Mark scheme

M1 Attempt use of $\cos(A+B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$

M1 Use trig formula to obtain an equation in $\tan\theta$ (or $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain $\tan\theta = 1/(4+\sqrt{3})$ or equivalent (or find $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain answer $\theta = 9.9^\circ$

A1 Obtain $\theta = 189.9^\circ$, and no others in the given interval

Total: 5

[Ignore answers outside the given interval. Treat answers in radians as a misread (0.173, 3.31).]

[The other solution methods are *via* $\cos\theta = \pm(4+\sqrt{3})/\sqrt{1+(4+\sqrt{3})^2}$ or $\sin\theta = \pm 1/\sqrt{1+(4+\sqrt{3})^2}$.]

Q313 Oct/Nov 2010 | Paper 3 (variant 32) | Question 5

Topics: Integration

Let $I = \displaystyle\int_0^{\pi/6} \frac{x^2}{\sqrt{4-x^2}}\,dx$.

(i) Using the substitution $x = 2\sin\theta$, show that

$$I = \int_0^{\pi/6} 4\sin^2\theta\,d\theta. \quad **[3]**$$

(ii) Hence find the exact value of I. **[4]**

Examiner comment

> "The most common errors in faulty attempts were (a) replacing (4–4sin2θ) with 2 – 2 sinθ, (b) replacing dx by dθ, and (c) omitting a justification of the change in limits."

Mark scheme

(i)

B1 State or imply $dx = 2\cos\theta d\theta$, or $\frac{dx}{d\theta} = 2\cos\theta$, or equivalent

M1 Substitute for x and dx throughout the integral

A1 Obtain the given answer correctly, having changed limits and shown sufficient working

Total: 3

(ii)

B1 Replace integrand by $2 - 2\cos 2\theta$, or equivalent

B1√ Obtain integral $2\theta - \sin 2\theta$, or equivalent

M1 Substitute limits correctly in an integral of the form $a\theta + b\sin 2\theta$, where $ab \neq 0$

A1 Obtain answer $\frac{1}{3}\pi - \frac{\sqrt{3}}{2}$ or exact equivalent

Total: 4

[The f.t. is on integrands of the form $a + c\cos 2\theta$, where $ac \neq 0$.]

Q314 Oct/Nov 2010 | Paper 3 (variant 32) | Question 10

Topics: Differential Equations

10 A certain substance is formed in a chemical reaction. The mass of substance formed t seconds after the start of the reaction is x grams. At any time the rate of formation of the substance is proportional to $(20 - x)$. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = 1$.

(i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.05(20 - x).$$

[2]

(ii) Find, in any form, the solution of this differential equation. [5]

(iii) Find x when $t = 10$, giving your answer correct to 1 decimal place. [2]

(iv) State what happens to the value of x as t becomes very large. [1]

Examiner comment

> "Only a few candidates answered part (iv) correctly."

Mark scheme

(i) State or imply $\frac{dx}{dt} = k(20 - x)$ **B1**

Show that $k = 0.05$ **B1** [2]

(ii) Separate variables correctly and integrate both sides **B1**

Obtain term $-\ln(20 - x)$, or equivalent **B1**

Obtain term $\frac{1}{20}t$, or equivalent **B1**

Evaluate a constant or use limits $t = 0, x = 0$ in a solution containing terms $a\ln(20 - x)$ and bt **M1***

Obtain correct answer in any form, e.g. $\ln 20 - \ln(20 - x) = \frac{1}{20}t$ **A1** [5]

(iii) Substitute $t = 10$ and calculate x **M1(dep*)$**

Obtain answer $x = 7.9$ **A1** [2]

(iv) State that x approaches 20 **B1** [1]

Q315 Oct/Nov 2010 | Paper 3 (variant 33) | Question 1

Topics: Algebra

1 Expand $(1 + 2x)^{-3}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. [3]

Examiner comment

> "A few candidates were able to write down an expression involving $\binom{-3}{1}\binom{-3}{2}$ unsimplified binomial coefficients and but they did not gain any credit until they expanded these coefficients."

Mark scheme

Obtain $1 - 6x$ **B1**

State correct unsimplified x^2 term. Binomial coefficients must be expanded. **M1**

Obtain $\ldots + 24x^2$ **A1** [3]

Q316 Oct/Nov 2010 | Paper 3 (variant 33) | Question 2

Topics: Differentiation

2 The parametric equations of a curve are

$$x = \frac{t}{2t + 3}, \quad y = e^{-2t}.$$

Find the gradient of the curve at the point for which $t = 0$. [5]

Examiner comment

> "The use of product or quotient rule to dt dx dt obtain or proved to be more difficult."

> "Although many candidates did not state the rule that they were dt dx using, it was often clear that the rule was being misused, with confusion between terms and sign errors."

> "Many candidates did not recognise the expression as a quotient at dx −2t all – they treated the numerator as a constant and obtained = for their derivative."

> "This was a particular problem dt for candidates who had used the product rule, for whom the incorrect statement ( ) 1 2t +3 2t +3 2 = – was a common error."

Mark scheme

Use of correct quotient or product rule to differentiate x or t **M1**

Obtain correct $\frac{3}{(2t+3)^2}$ or unsimplified equivalent **A1**

Obtain $-2e^{-2t}$ for derivative of y **B1**

Use $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ or equivalent **M1**

Obtain -6 **cwo A1** [5]

Alternative:

Eliminate parameter and attempt differentiation $\left(y = e^{\frac{-6x}{1-2x}}\right)$ **B1**

Use correct quotient or product rule **M1**

Use chain rule **M1**

Obtain $\frac{dy}{dx} = \frac{-6}{(1-2x)^2}e^{\frac{-6x}{1-2x}}$ **A1**

Obtain -6 **cwo A1** [5]

Q317 Oct/Nov 2010 | Paper 3 (variant 33) | Question 4

Topics: Differentiation, Integration

4 It is given that $f(x) = 4\cos^2 3x$.

(i) Find the exact value of $f'\left(\frac{1}{5}\pi\right)$. [3]

(ii) Find $\int f(x)\,dx$. [3]

Examiner comment

> "The most common errors by this method were to omit a factor of 3 in the differentiation or to lose it from the trigonometric term resulting in an expression of the form kcos3xsinx."

> "Several candidates struggled to make any progress with the integral, including some who had started part (i) by using the correct double angle form."

Mark scheme

(i) Obtain derivative of form $k\cos 3x \sin 3x$, any constant k **M1**

Obtain $-24\cos 3x \sin 3x$ or unsimplified equivalent **A1**

Obtain $-6\sqrt{3}$ or exact equivalent **A1** [3]

(ii) Express integrand in the form $a + b\cos 6x$, where $ab \neq 0$ **M1**

Obtain $2 + 2\cos 6x$ **o.e. A1**

Obtain $2x + \frac{1}{3}\sin 6x$ or equivalent, condoning absence of $+c$, ft on a, b **A1√** [3]

Q318 Oct/Nov 2010 | Paper 3 (variant 33) | Question 7

Topics: Integration, Numerical Solutions/Iteration

7 (i) Given that $\int_1^a \frac{\ln x}{x^2}\,dx = \frac{2}{3}$, show that $a = \frac{3}{2}(1 + \ln a)$. [5]

(ii) Use an iteration formula based on the equation $a = \frac{3}{2}(1 + \ln a)$ to find the value of a correct to 2 decimal places. Use an initial value of 4 and give the result of each iteration to 4 decimal places. [3]

Examiner comment

> "Where the process was not properly 5( ) understood, the most common error was to see calculation of values for a = 1+lnn for integer n 3 values of n."

Mark scheme

(i) Attempt integration by parts **M1**

Obtain $-x^{-1}\ln x + \int \frac{1}{x^2}\,dx$, $\frac{x\ln x - x}{x^2} + 2\int \frac{\ln x}{x^2}\,dx - 2\int \frac{1}{x^2}\,dx$ or equivalent **A1**

Obtain $-x^{-1}\ln x - x^{-1}$ or equivalent **A1**

Use limits correctly, equate to $\frac{2}{3}$ and attempt rearrangement to obtain a in terms of $\ln a$ **M1**

Obtain given answer $a = \frac{3}{2}(1 + \ln a)$ correctly **A1** [5]

(ii) Use valid iterative formula correctly at least once **M1**

Obtain final answer 3.96 **A1**

Show sufficient iterations to > 4 dp to justify accuracy to 2 dp or show sign change in interval $(3.955, 3.965)$ **A1** [3]

$[4 \to 3.9772 \to 3.9676 \to 3.9636 \to 3.9619]$

SR: Use of $a_{n+1} = e^{(3a_n - 1)}$ to obtain 0.50 also earns $3/3$.

Q319 Oct/Nov 2010 | Paper 3 (variant 33) | Question 8

Topics: Trigonometry

8 (i) Express $(\sqrt{6})\cos\theta + (\sqrt{10})\sin\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give the value of α correct to 2 decimal places. [3]

(ii) Hence, in each of the following cases, find the smallest positive angle θ which satisfies the equation

(a) $(\sqrt{6})\cos\theta + (\sqrt{10})\sin\theta = -4$, [2]

(b) $(\sqrt{6})\cos\frac{\theta}{2} + (\sqrt{10})\sin\frac{\theta}{2} = 3$. [4]

Examiner comment

> "The most common error was to see α=52.2° rather than the 2 decimal places asked for in the question."

Mark scheme

(i) Obtain or imply $R = 4$ **B1**

Use appropriate trigonometry to find α **M1**

Obtain $\alpha = 52.24$ or better from correct work **A1** [3]

(ii) (a) State or imply $\theta - \alpha = \cos^{-1}(-4 \div R)$ **M1**

Obtain 232.2 or better **A1** [2]

(b) Attempt at least one value using $\cos^{-1}(3 \div R)$ **M1**

Obtain one correct value e.g. $\pm 41.41^\circ$ **A1**

Use $\frac{1}{2}\theta - \alpha = \cos^{-1}\left(\frac{3}{R}\right)$ to find θ **M1**

Obtain 21.7 **A1** [4]

Q320 Oct/Nov 2010 | Paper 3 (variant 33) | Question 9

Topics: Differential Equations

9 A biologist is investigating the spread of a weed in a particular region. At time t weeks after the start of the investigation, the area covered by the weed is A m^2. The biologist claims that the rate of increase of A is proportional to $\sqrt{2A - 5}$.

(i) Write down a differential equation representing the biologist's claim. [1]

(ii) At the start of the investigation, the area covered by the weed was 7 m^2 and, 10 weeks later, the area covered was 27 m^2. Assuming that the biologist's claim is correct, find the area covered 20 weeks after the start of the investigation. [9]

Examiner comment

> "The most common error in the integration was the mistaken use of ( ) logarithms to obtain a term in ln 2A−5 ."

Mark scheme

(i) State $\frac{dA}{dt} = k\sqrt{2A - 5}$ **B1** [1]

(ii) Separate variables correctly and attempt integration of each side **M1**

Obtain $(2A - 5)^{\frac{1}{2}} = \ldots$ or equivalent **A1**

Obtain $= kt$ or equivalent **A1**

Use $t = 0$ and $A = 7$ to find value of arbitrary constant **M1**

Obtain $C = 3$ or equivalent **A1**

Use $t = 10$ and $A = 27$ to find k **M1**

Obtain $k = 0.4$ or equivalent **A1**

Substitute $t = 20$ and values for C and k to find value of A **M1**

Obtain **63 cwo A1** [9]

Q321 Oct/Nov 2009 | Paper 3 (variant 32) | Question 3

Topics: Differentiation

The equation of a curve is $x^3 - x^2y - y^3 = 3$.

(i) Find $\frac{dy}{dx}$ in terms of x and y. [4]

(ii) Find the equation of the tangent to the curve at the point $(2, 1)$, giving your answer in the form $ax + by + c = 0$. [2]

Examiner comment

> "Question 3 (i) The implicit differentiation of y3 was often correct, but applying the product rule to x2y caused more dy problems, and many candidates made a sign error at this stage, with −2xy +x2 being a very dx dy common error."

Mark scheme

(i) State $2xy + x^2\frac{dy}{dx}$ as derivative of x^2y **B1**

State $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain answer $\frac{3x^2-2xy}{x^2+3y^2}$, or equivalent **A1** [4]

(ii) Find gradient of tangent at $(2, 1)$ and form equation of tangent **M1**

Obtain answer $8x - 7y - 9 = 0$, or equivalent **A1** [2]

Total: 6

Q322 Oct/Nov 2009 | Paper 3 (variant 32) | Question 5

Topics: Algebra

The polynomial $2x^3 + ax^2 + bx - 4$, where a and b are constants, is denoted by $p(x)$. The result of differentiating $p(x)$ with respect to x is denoted by $p'(x)$. It is given that $(x + 2)$ is a factor of $p(x)$ and of $p'(x)$.

(i) Find the values of a and b. [5]

(ii) When a and b have these values, factorise $p(x)$ completely. [3]

Examiner comment

> "The standard of presentation in solutions to this problem was disappointing, with many candidates forming correct expressions by substituting x = −2 but not immediately equating their expressions to zero."

Mark scheme

(i) Substitute $x = -2$, equate to zero and state a correct equation, e.g. $-16 + 4a - 2b - 4 = 0$ **B1**

Differentiate $p(x)$, substitute $x = -2$ and equate to zero **M1**

Obtain a correct equation, e.g. $24 - 4a + b = 0$ **A1**

Solve for a or for b **M1**

Obtain $a = 7$ and $b = 4$ **A1** [5]

(ii) EITHER: State or imply $(x + 2)^2$ is a factor **B1**

Attempt division by $(x + 2)^2$ reaching a quotient $2x + k$ or use inspection with unknown factor $cx + d$ reaching $c = 2$ or $d = -1$ **M1**

Obtain factorisation $(x + 2)^2(2x - 1)$ **A1**

OR: Attempt division by $(x + 2)$ **M1**

Obtain quadratic factor $2x^2 + 3x - 2$ **A1**

Obtain factorisation $(x + 2)(x + 2)(2x - 1)$ **A1** [3]

[The M1 is earned if division reaches a partial quotient of $2x^2 + kx$, or if inspection has an unknown factor of $2x^2 + ex + f$ and an equation in e and/or f, or if two coefficients with the correct moduli are stated without working.]

Total: 8

Q323 Oct/Nov 2009 | Paper 3 (variant 32) | Question 6

Topics: Integration

(i) Use the substitution $x = 2\tan\theta$ to show that

$$\int_0^2 \frac{8}{(4 + x^2)^2}\, dx = \int_0^{\pi/4} \cos^2\theta\, d\theta.$$

[4]

(ii) Hence find the exact value of

$$\int_0^2 \frac{8}{(4 + x^2)^2}\, dx.$$

[4]

Examiner comment

> "Question 6 (i) This part of the question proved to be particularly difficult for many candidates, either through algebraic errors or through errors in completing the substitution."

> “Common errors included using x2 = 2 tan2 θ, and simply replacing dx by dθ.”

> “(ii) There were many correct answers to this part, but many candidates did not appreciate that the first part of the question assisted them with the second and produced false answers, often involving a ∫ logarithm.”

Mark scheme

(i) State or imply $\frac{dx}{d\theta} = 2\sec^2\theta$ or $dx = 2\sec^2\theta\, d\theta$ **B1**

Substitute for x and dx throughout **M1**

Obtain any correct form in terms of θ **A1**

Obtain the given form correctly (including the limits) **A1** [4]

(ii) Use $\cos 2\theta$ formula, replacing integrand by $a + b\cos 2\theta$, where $ab \neq 0$ **M1***

Integrate and obtain $\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta$ **A1**

Use limits $\theta = 0$ and $\theta = \frac{1}{4}\pi$ **M1(dep*)**

Obtain answer $\frac{1}{8}(\pi + 2)$, or exact equivalent **A1** [4]

Total: 8

Q324 Oct/Nov 2009 | Paper 3 (variant 32) | Question 8

Topics: Algebra

(i) Express $\frac{1+x}{(1-x)(2+x^2)}$ in partial fractions. [5]

(ii) Hence obtain the expansion of $\frac{1+x}{(1-x)(2+x^2)}$ in ascending powers of x, up to and including the term in x^2. [5]

Examiner comment

> “Common errors included 3 correct deduction that 3A = 2, but then concluding that A = or starting with an initial form 2 omitting one of the terms in the linear numerator.”

> “The term 2x–1 ( ) caused many more difficulties.”

> “Common errors included confusion over how to deal with 32+x2 2x–1 the constants in the numerator resulting in a starting point of ( ), use of the incorrect form 92+x2 –1 □ x2 □ (2 + x2)–1 = 2□14+ □ , and errors in multiplying their expansion by (2x – 1).”

Mark scheme

(i) State or imply partial fractions are of the form $\frac{A}{1-x} + \frac{Bx+C}{2+x^2}$ **B1**

Use a relevant method to determine a constant **M1**

Obtain $A = \frac{1}{3}$, $B = \frac{1}{3}$ and $C = \frac{1}{3}$ **A1 + A1 + A1** [5]

Total: 5

Q325 Oct/Nov 2009 | Paper 3 (variant 32) | Question 9

Topics: Differential Equations

The temperature of a quantity of liquid at time t is θ. The liquid is cooling in an atmosphere whose temperature is constant and equal to A. The rate of decrease of θ is proportional to the temperature difference $(\theta - A)$. Thus θ and t satisfy the differential equation

$$\frac{d\theta}{dt} = -k(\theta - A),$$

where k is a positive constant.

(i) Find, in any form, the solution of this differential equation, given that $\theta = 4A$ when $t = 0$. [5]

(ii) Given also that $\theta = 3A$ when $t = 1$, show that $k = \ln \frac{3}{2}$. [1]

(iii) Find θ in terms of A when $t = 2$, expressing your answer in its simplest form. [3]

Examiner comment

> "Common errors included incorrect or non-existent attempts to separate the variables in the given equation, or omission of a constant of ☐ 3A ☐ integration."

Mark scheme

(i) Separate variables correctly **B1**

Integrate and obtain term $\ln(\theta - A)$, or equivalent **B1**

Integrate and obtain term $-kt$, or equivalent **B1**

Use $\theta = 4A$, $t = 0$ to determine a constant, or as limits **M1**

Obtain correct answer in any form, e.g. $\ln(\theta - A) = -kt + \ln 3A$, with no errors seen **A1** [5]

(ii) Substitute $\theta = 3A$, $t = 1$ and justify the given statement **B1** [1]

(iii) Substitute $t = 2$ and solve for θ in terms of A **M1**

Remove logarithms **M1**

Obtain answer $\theta = \frac{9}{4}A$, or equivalent, with no errors seen **A1** [3]

[The M marks are only available if the solution to part (i) contains terms $a\ln(\theta - A)$ and $br.t.]

Total: 9

Q326 Oct/Nov 2017 | Paper 3 (variant 33) | Question 2

Topics: Indices, Logarithms & Exponents

2 Two variable quantities x and y are believed to satisfy an equation of the form $y = C(a^x)$, where C and a are constants. An experiment produced four pairs of values of x and y. The table below gives the corresponding values of x and $\ln y$.

x	0.9	1.6	2.4	3.2
$\ln y$	1.7	1.9	2.3	2.6

By plotting $\ln y$ against x for these four pairs of values and drawing a suitable straight line, estimate the values of C and a. Give your answers correct to 2 significant figures. **[5]**

Examiner comment

Mark scheme

\section*{Question 2}

\begin{tabular}{|l|l|l|}

Answer & Marks & Guidance \

Plot the four points and draw straight line & **B1** & \

State or imply that $\ln y = \ln C + x \ln a$ & **B1** & \

Carry out a completely correct method for finding $\ln C$ or $\ln a$ & **M1** & \

Obtain answer $C = 3.7$ & **A1** & \

Obtain answer $a = 1.5$ & **A1** & \

& 5 & \

\end{tabular}

Total: 5